HAP-Z1ES

SERVICE MANUAL

US Model
Canadian Model
AEP Model
UK Model

Ver. 1.0 2013.10

Note:
Be sure to keep your PC used for service and checking of this unit always updated with the latest version of your anti-virus software.
In case a virus affected unit was found during service, contact your Service Headquarters.

SPECIFICATIONS

Playback Specifications

Frequency response: 2 Hz - 80 kHz (–3 dB)
Dynamic range: 105 dB or higher
THD: 0.0015% or less

Network section

Wired LAN

1000BASE-T/100BASE-TX/10BASE-T

Wireless LAN

Compatible standards: IEEE 802.11 b/g/n
Frequency band/channel: 2.4 GHz band, channels 1-11 (models for the USA and Canada), channels 1-13 (models for Europe)

HDD section

Capacity

1 TB (*)

* Some portions of the capacity are used for data management. Therefore, the capacity a user can use is less than 1 TB.

Supported playback format

DSD (DSF, DSDIFF), LPCM (WAV, AIFF), FLAC, ALAC, ATRAC Advanced Lossless, ATRAC, MP3, AAC, WMA (2 channels)

Jack section

Output jacks

LINE OUT UNBALANCED
Output level: 2.0 Vrms (50 kilohms)
Impedance: 10 kilohms or higher

LINE OUT BALANCED
Output level: 2.0 Vrms (50 kilohms)
Impedance: 600 ohms or higher

EXT port

Type A USB, Hi-Speed USB, for connecting an external hard disk drive

IR REMOTE OUT jack

for connecting the monaural mini-plug cable (supplied) or IR blaster (supplied)

General and others

Power requirements

Models for the USA and Canada: AC 120 V 60 Hz
Models for Europe: AC 230 V 50/60 Hz

Power consumption

On: 35 W
During standby mode (when [Network Standby] is set to [Off]): 0.3 W
During standby mode (when [Network Standby] is set to [On] and a wired LAN is used): 2.6 W
During standby mode (when [Network Standby] is set to [On] and a wireless LAN is used): 2.8 W

Dimensions (approx.) (w/h/d)

430 mm × 130 mm × 390 mm (17 in. × 5 1/8 in. × 15 3/8 in.) including projecting parts and controls

Mass (approx.)

14.5 kg (32 lbs 0 oz)

Supplied accessories

Remote control (1)
R03 (size-AAA) batteries (2)
AC power cord (mains lead) (1)
LAN cable (1)
Audio cord (1)
IR blaster (1)
Monaural mini-plug cable (1)

Design and specifications are subject to change without notice.

HDD AUDIO PLAYER

9-893-887-01
2013J33-1

Sony Corporation
Published by Sony Techno Create Corporation

SONY®

Trademarks

- ATRAC, ATRAC3, ATRAC3plus, ATRAC Advanced Lossless and their logos are trademarks of Sony Corporation.
- 12 TONE ANALYSIS and its logo are trademarks of Sony Corporation.
- "DSEE" is a registered trademark of Sony Corporation.
- SensMe™ and the SensMe™ logo are trademarks or registered trademarks of Sony Mobile Communications AB.
- Microsoft, Windows, Windows Vista and Windows Media are either trademarks or registered trademarks of Microsoft Corporation in the United States and/or other countries.
  This product is protected by certain intellectual property rights of Microsoft Corporation. Use or distribution of such technology outside of this product is prohibited without a license from Microsoft or an authorized Microsoft subsidiary.
- MPEG Layer-3 audio coding technology and patents licensed from Fraunhofer IIS and Thomson.
- Mac, Mac OS and iTunes are trademarks of Apple Inc., registered in the U.S. and other countries.
- iPhone, iPod touch and iPad are trademarks of Apple Inc., registered in the U.S. and other countries.
- IOS is a trademark or registered trademark of Cisco in the U.S. and other countries and is used under license.
- Google Play, Android, YouTube and their respective logos are trademarks of Google Inc.
- Wi-Fi®, Wi-Fi Protected Access® and Wi-Fi Alliance® are registered marks of the Wi-Fi Alliance.
- Wi-Fi CERTIFIED™, WPA™, WPA2™ and Wi-Fi Protected Setup™ are marks of the Wi-Fi Alliance.

gracenote.

- Gracenote®, Gracenote logo and logotype, and the "Powered by Gracenote" logo are either a registered trademark or a trademark of Gracenote, Inc. in the United States and/or other countries.

JASRAC®

T-1380009

All other trademarks and registered trademarks are trademarks or registered trademarks of their respective holders. In this Help, TM and ® marks are not specified.

NOTES ON CHIP COMPONENT REPLACEMENT

- Never reuse a disconnected chip component.
- Notice that the minus side of a tantalum capacitor may be damaged by heat.

FLEXIBLE CIRCUIT BOARD REPAIRING

- Keep the temperature of soldering iron around 270 °C during repairing.
- Do not touch the soldering iron on the same conductor of the circuit board (within 3 times).
- Be careful not to apply force on the conductor when soldering or unsoldering.

SAFETY CHECK-OUT

After correcting the original service problem, perform the following safety check before releasing the set to the customer:
Check the antenna terminals, metal trim, "metallized" knobs, screws, and all other exposed metal parts for AC leakage.
Check leakage as described below.

LEAKAGE TEST

The AC leakage from any exposed metal part to earth ground and from all exposed metal parts to any exposed metal part having a return to chassis, must not exceed 0.5 mA (500 microamperes.).
Leakage current can be measured by any one of three methods.

1. A commercial leakage tester, such as the Simpson 229 or RCA WT-540A. Follow the manufacturers' instructions to use these instruments.
2. A battery-operated AC milliammeter. The Data Precision 245 digital multimeter is suitable for this job.
3. Measuring the voltage drop across a resistor by means of a VOM or battery-operated AC voltmeter. The "limit" indication is 0.75 V, so analog meters must have an accurate low-voltage scale. The Simpson 250 and Sanwa SH-63Trd are examples of a passive VOM that is suitable. Nearly all battery operated digital multimeters that have a 2 V AC range are suitable. (See Fig. A)

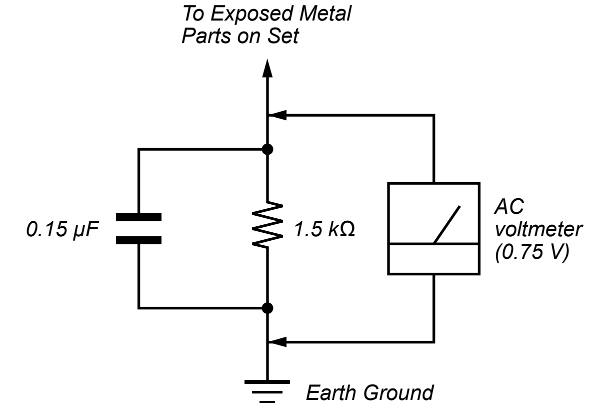


Fig. A. Using an AC voltmeter to check AC leakage.

SAFETY-RELATED COMPONENT WARNING!

COMPONENTS IDENTIFIED BY MARK ⚠ OR DOTTED LINE WITH MARK ⚠ ON THE SCHEMATIC DIAGRAMS AND IN THE PARTS LIST ARE CRITICAL TO SAFE OPERATION. REPLACE THESE COMPONENTS WITH SONY PARTS WHOSE PART NUMBERS APPEAR AS SHOWN IN THIS MANUAL OR IN SUPPLEMENTS PUBLISHED BY SONY.

ATTENTION AU COMPOSANT AYANT RAPPORT À LA SÉCURITÉ!

LES COMPOSANTS IDENTIFIÉS PAR UNE MARQUE ⚠ SUR LES DIAGRAMMES SCHÉMATIQUES ET LA LISTE DES PIÈCES SONT CRITIQUES POUR LA SÉCURITÉ DE FONCTIONNEMENT. NE REMPLACER CES COMPOSANTS QUE PAR DES PIÈCES SONY DONT LES NUMÉROS SONT DONNÉS DANS CE MANUEL OU DANS LES SUPPLÉMENTS PUBLIÉS PAR SONY.

TABLE OF CONTENTS

1. SERVICING NOTES 4

2. DISASSEMBLY
2-1. Disassembly Flow 6
2-2. Case Top, Fuse (F901, F902, F1401, F1402) 7
2-3. Case L Block, Case R Block 7
2-4. Plate Bottom Outer 8
2-5. Plate Bottom, Cable Tie 9
2-6. Front Panel Block 10
2-7. WLAN/BT COMBO Card (WBC1) 11
2-8. U-COM Board 11
2-9. KEY JOG Board 12
2-10. SPIRITOSO-ANT-R Board 12
2-11. SPIRITOSO-ANT-L Board 13
2-12. SPIRITOSO-LCD Board 13
2-13. LED Board 14
2-14. LCD Panel (LCD1) 14
2-15. HDD Block 15
2-16. HDD (HDD1) 15
2-17. Power Transformer (T1, T2)-1 16
2-18. Power Transformer (T1, T2)-2 17
2-19. FPGA DSP Board 18
2-20. MAIN Board Block 19
2-21. MAIN Board 19
2-22. APOWER Board 20
2-23. DCDC Board 21
2-24. IO Board Block 22
2-25. IO Board 22
2-26. STANBY Board 23
2-27. AC Inlet (2P) (AC1) 23
2-28. AUDIO Board 24

3. TEST MODE 25

4. ELECTRICAL CHECK 33

5. DIAGRAMS
5-1. Block Diagram
- HDD/USB/LAN/FPGA/DSP Section - 35
5-2. Block Diagram - AUDIO Section - 36
5-3. Block Diagram - MEMORY/PANEL Section - 37
5-4. Block Diagram - POWER SUPPLY Section - 38
5-5. Printed Wiring Boards - MAIN Section - 40
5-6. Schematic Diagram - MAIN Section (1/9) - 41
5-7. Schematic Diagram - MAIN Section (2/9) - 42
5-8. Schematic Diagram - MAIN Section (3/9) - 43
5-9. Schematic Diagram - MAIN Section (4/9) - 44
5-10. Schematic Diagram - MAIN Section (5/9) - 45
5-11. Schematic Diagram - MAIN Section (6/9) - 46
5-12. Schematic Diagram - MAIN Section (7/9) - 47
5-13. Schematic Diagram - MAIN Section (8/9) - 48
5-14. Schematic Diagram - MAIN Section (9/9) - 49
5-15. Printed Wiring Board - IO Board - 50
5-16. Schematic Diagram - IO Board - 51
5-17. Printed Wiring Board - FPGA DSP Board - 52
5-18. Schematic Diagram - FPGA DSP Board (1/3) - 53
5-19. Schematic Diagram - FPGA DSP Board (2/3) - 54
5-20. Schematic Diagram - FPGA DSP Board (3/3) - 55
5-21. Printed Wiring Board - AUDIO Board - 56
5-22. Schematic Diagram - AUDIO Board - 57
5-23. Printed Wiring Board - U-COM Board - 58
5-24. Schematic Diagram - U-COM Board - 59
5-25. Printed Wiring Boards
- AUUBOUT/SPIRITOSO-LCD Boards - 60
5-26. Schematic Diagram
- AUUBOUT/SPIRITOSO-LCD Boards - 61
5-27. Printed Wiring Boards - KEY/LED Section - 62
5-28. Schematic Diagram - KEY/LED Section - 63
5-29. Printed Wiring Board - APOWER Board - 64
5-30. Schematic Diagram - APOWER Board - 65
5-31. Printed Wiring Boards
- DCDC/FAN-CONNECT Boards - 66
5-32. Schematic Diagram
- DCDC/FAN-CONNECT Boards - 67
5-33. Printed Wiring Board - STANBY Board - 68
5-34. Schematic Diagram - STANBY Board - 69

6. EXPLODED VIEWS
6-1. Overall Section 95
6-2. Plate Bottom Outer Section 96
6-3. LCD Section 97
6-4. Front Panel Section 98
6-5. Power Transformer Section 99
6-6. HDD Section 100
6-7. MAIN Board Section 101
6-8. AUDIO Board Section 102
6-9. Chassis Section 103

7. ELECTRICAL PARTS LIST 104

Accessories are given in the last of the electrical parts list.

SECTION 1
SERVICING NOTES

UNLEADED SOLDER

Boards requiring use of unleaded solder are printed with the lead-free mark (LF) indicating the solder contains no lead.

(**Caution:** Some printed circuit boards may not come printed with the lead free mark due to their particular size)

: LEAD FREE MARK

Unleaded solder has the following characteristics.

- Unleaded solder melts at a temperature about 40 °C higher than ordinary solder.
  Ordinary soldering irons can be used but the iron tip has to be applied to the solder joint for a slightly longer time.
  Soldering irons using a temperature regulator should be set to about 350 °C.
  Caution: The printed pattern (copper foil) may peel away if the heated tip is applied for too long, so be careful!
- Strong viscosity
  Unleaded solder is more viscous (sticky, less prone to flow) than ordinary solder so use caution not to let solder bridges occur such as on IC pins, etc.
- Usable with ordinary solder
  It is best to use only unleaded solder but unleaded solder may also be added to ordinary solder.

ABOUT THE SPIRITOSO-ANT-L BOARD AND SPIRITOSO-ANT-R BOARD

SPIRITOSO-ANT-L and SPIRITOSO-ANT-R boards are not covered in the servicing.

NOTE OF REPLACING THE IC101, IC301, IC302, IC501 AND IC602 ON THE MAIN BOARD

IC101, IC301, IC302, IC501 and IC602 on the MAIN board cannot exchange with single. When these parts are damaged, exchange the complete mounted board.

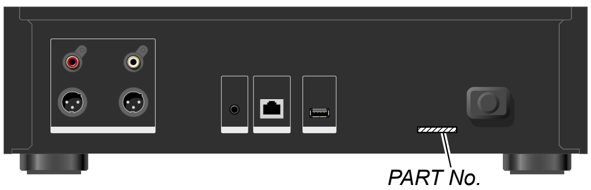
NOTE OF REPLACING THE IC001 AND IC601 ON THE FPGA DSP BOARD

IC001 and IC601 on the FPGA DSP board cannot exchange with single. When these parts are damaged, exchange the complete mounted board.

DESTINATION IDENTIFICATION

Distinguish by Part No. on the rear side of a main unit.

– Rear Panel –



Destination	Part No.
US and Canadian models	4-466-800-1□
AEP and UK models	4-466-800-2□

NOTE OF REPLACING THE HARD DISK DRIVE (Ref. No. HDD1)

When the hard disk drive (Ref. No. HDD1) is replaced, be sure to carry out the "Factory Reset" according to the following procedure.

Resetting to the factory default (Factory Reset)

You can reset the HDD AUDIO PLAYER to the factory default settings. All settings you have configured and all of the content stored on the internal hard disk drive will be deleted. (Sample tracks saved on the internal hard disk drive by default remain.)

1. Select [Settings] - [System Settings] in the Home screen.
2. Select [Factory Reset].
3. Select [Yes].
   The Factory Reset waiting screen appears.
4. When the Factory Reset complete screen appears, select [OK].
   When the Factory Reset is complete, the HDD AUDIO PLAYER restarts automatically.

Note

- If you perform the Factory Reset function while playing tracks, playback of the music files stops.
- If you perform the Factory Reset function while transferring music files, transfer of the music files stops.
- If you want to re-transfer music files that were transferred to the HDD AUDIO PLAYER before you performed the Factory Reset function, select the backup folder and check "Watch" in the Contents Settings window of HAP Music Transfer. Go to the Transfer Settings window and select [Clear] for [Transferred Files List] to delete the transfer history, and then automatically or manually transfer the files.

NOTE OF REPLACING THE COMPLETE MAIN BOARD

When the complete MAIN board is replaced, the following settings that is recorded on the MAIN board are cleared, and returned to their initial values.

- Network Settings
- Language, Brightness, Auto Standby, Sleep and Software Update Notification of System Settings
- Amp Control Settings
- DSEE and Volume Normalization of Audio Settings

Take the following step.

Procedure:

1. At the time of operation check after replacing the complete MAIN board, set the language same as the language settings that a customer was set, and returns the this unit to the customer.
2. Display the "1.Version Info" of the service DIAG mode on the liquid crystal display, and check that "3. Region" and "4. Model" accord with the customer's unit.
   (Refer to "4-1. Version Info" on page 25)
3. Check the version of the "1. uCON ver." and "2. MPU ver." of "1.Version Info" is displayed on the liquid crystal display, and when it is not the latest version, carry out the "Network Update" of the "Settings", and update to the latest version.

Note: After replacing the complete MAIN board, it is not necessary to carry out the "Factory Reset".
Be careful to carry out the "Factory Reset", and not to delete the contents in the hard disk drive (Ref. No. HDD1).

4. After replacing the complete MAIN board, an initial setting screen is displayed on the liquid crystal display at the time of the first start up, but is not displayed on the next time when it has been started up once.
   When returning the unit that repair was completed to the customer print and cut out the following explanation, and hand it over with the unit.

-CUT-

The MAC address for the wired LAN has been changed by this repair.
Customers who use the MAC address filtering function of the connected access point equipment should set it again.

Also, the following settings have been cleared and returned to their initial values.
Please set the following settings again.

- Network Settings
- Language, Brightness, Auto Standby, Sleep and Software Update Notification of System Settings
- Amp Control Settings
- DSEE and Volume Normalization of Audio Settings

Note: There are two kinds of MAC addresses, wired and wireless.
Only wired MAC addresses are changed by this repair

-CUT-

CHECKING METHOD OF NETWORK CONNECTION

It is necessary to check the network connection, when replacing the complete MAIN board, WLAN/BT COMBO card (Ref. No. WBC1) or coaxial (U.FL connector) cable (Ref. No. WR1, WR2). Check the connection of wireless and wired LAN, according to the following method.

If the WLAN/BT COMBO card has been changed, when returning the unit to the customer, print and cut out the following explanation, and hand it over with the unit.

-CUT-

The MAC address for the wireless LAN has been changed by this repair.
Customers who use the MAC address filtering function of the connected access point equipment should set it again.

Note: There are two kinds of MAC addresses, wired and wireless.
Only wireless MAC addresses are changed by this repair.

-CUT-

1. Checking Method of Wireless LAN Connection

Necessary Equipment:

- Access point supporting WPS

Procedure:

1. Press the [I/⏻] key to turn the power on.
2. Press the [HOME] key to the display the "Home" menu on the liquid crystal display.
3. Rotate the selector knob to select "Settings" and press the [ENTER] key.
4. Rotate the selector knob to select "Network Settings" and press the [ENTER] key.
5. Rotate the selector knob to select "Internet Settings" and press the [ENTER] key.
6. Rotate the selector knob to select "Wireless Setup" and press the [ENTER] key.
7. Rotate the selector knob to select "Wi-Fi Protected Setup (WPS)" and press the [ENTER] key.
8. After "Start" is displayed on the liquid crystal display, and press the [ENTER] key.
9. Press the [WPS] key on the access point.
10. After "Next" is displayed on the liquid crystal display, press the [ENTER] key.
11. When check of wireless LAN connection is completed, "Wireless Connection: OK" and "Internet Access: OK"and is displayed on the liquid crystal display.
12. Press the [ENTER] key, and press the [I/⏻] key to turn the power off.

Note: Refer to the help guide about details of the checking method.

2. Checking method of wired LAN connection

Necessary Equipment:

- Router
- LAN cable

Procedure:

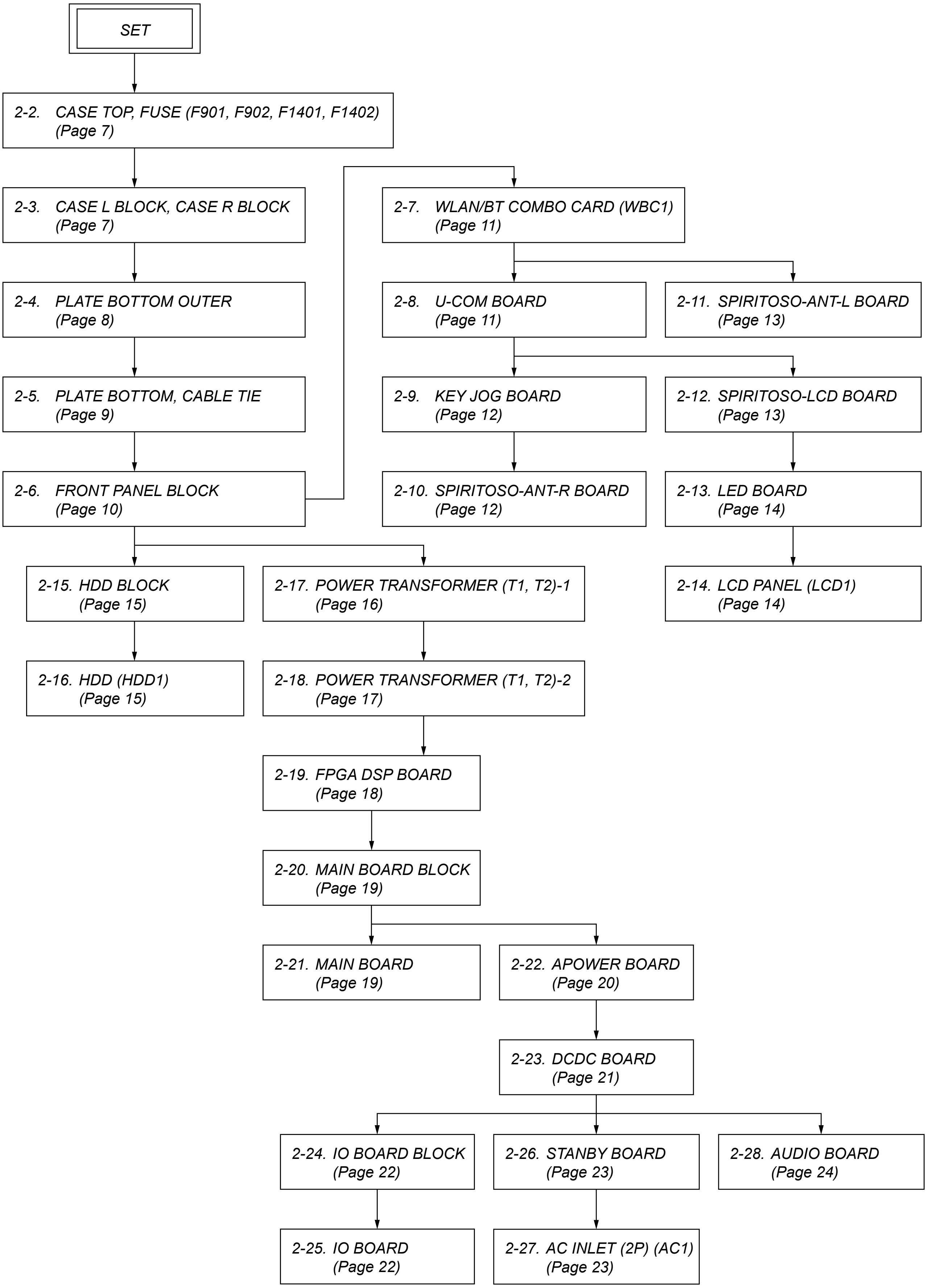
1. Connect the main unit to the router with the network LAN cable.
2. Press the [HOME] key to the display the "Home" menu on the liquid crystal display.
3. Rotate the selector knob to select "Settings" and press the [ENTER] key.
4. Rotate the selector knob to select "Network Settings" and press the [ENTER] key.
5. Rotate the selector knob to select "Internet Settings" and press the [ENTER] key.
6. Rotate the selector knob to select "Wired Setup" and press the [ENTER] key.
7. Rotate the selector knob to select "Auto" and press the [ENTER] key.
8. After "Next" is displayed on the liquid crystal display, press the [ENTER] key.
9. Rotate the selector knob to select "Save & Connect" and press the [ENTER] key.
10. When wired LAN connection is completed, "Physical Connection: OK" and "Internet Access: OK"and is displayed on the liquid crystal display.
11. Press the [ENTER] key, and press the [I/⏻] key to turn the power off.

Note: Refer to the help guide about details of the checking method.

SECTION 2
DISASSEMBLY

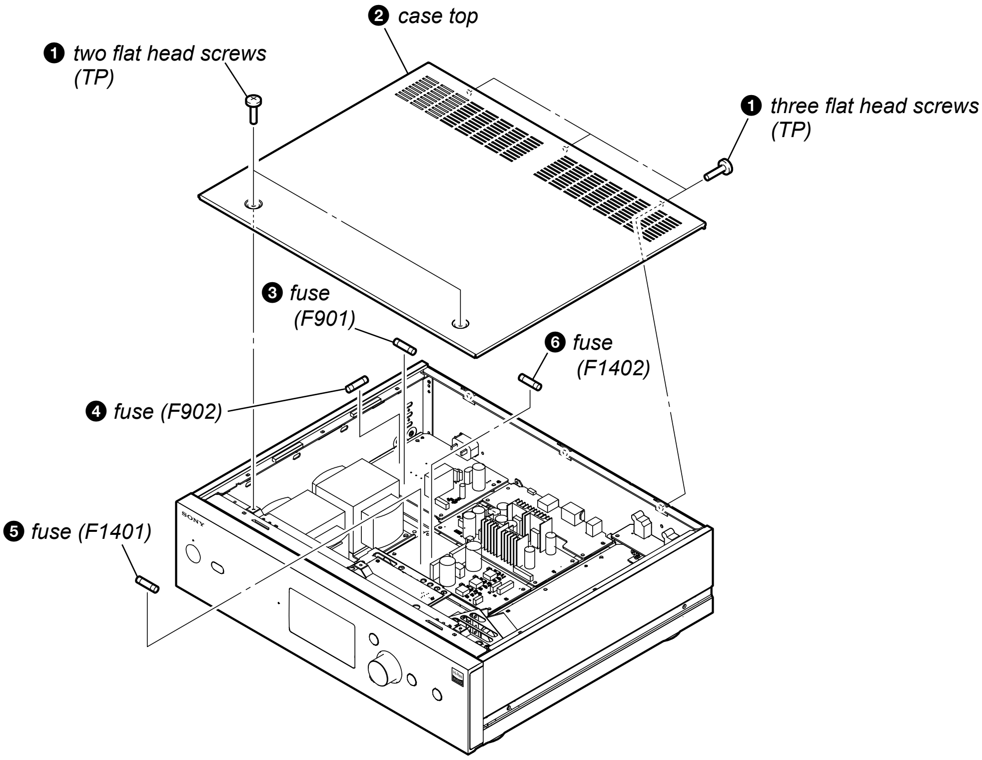
- This set can be disassembled in the order shown below.

2-1. DISASSEMBLY FLOW

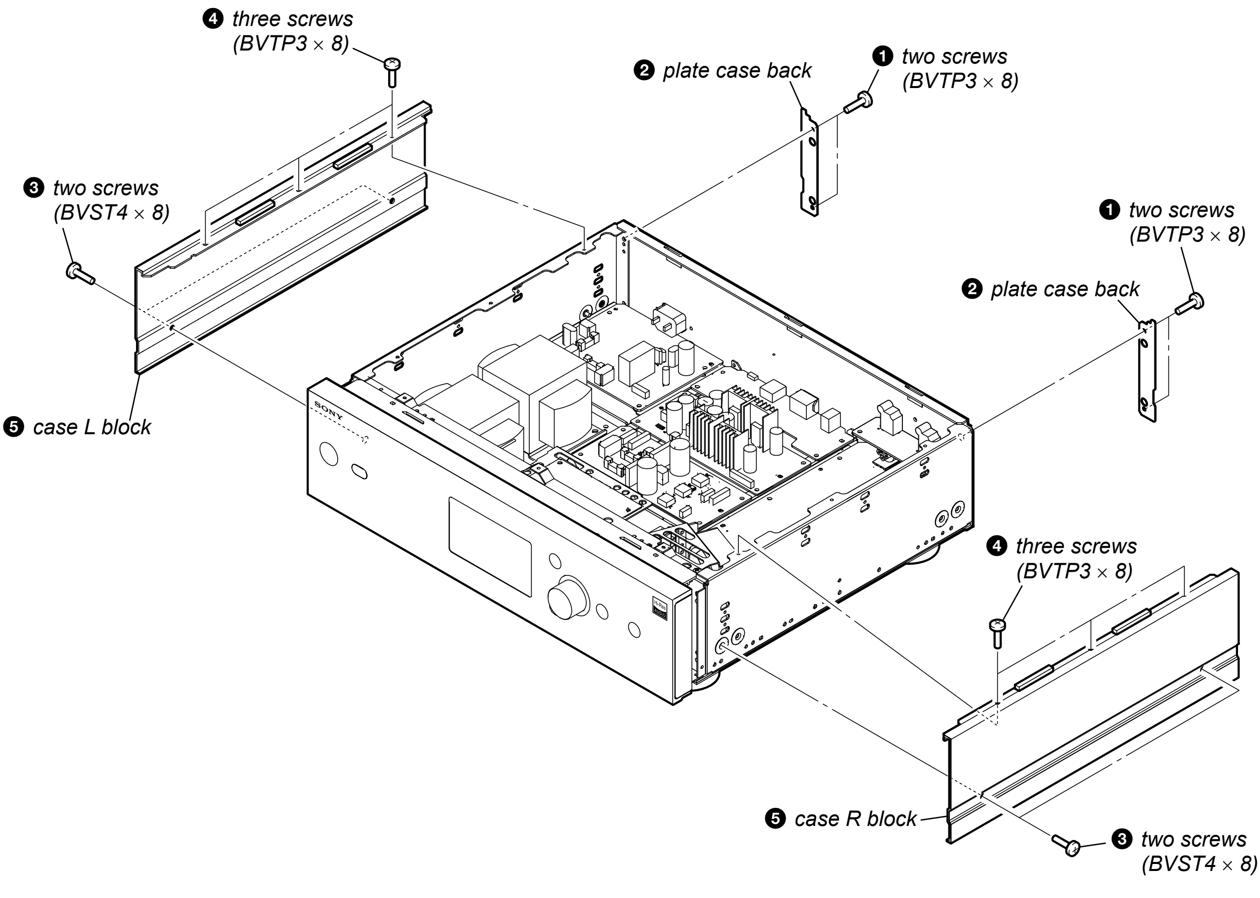


Note: Follow the disassembly procedure in the numerical order given.

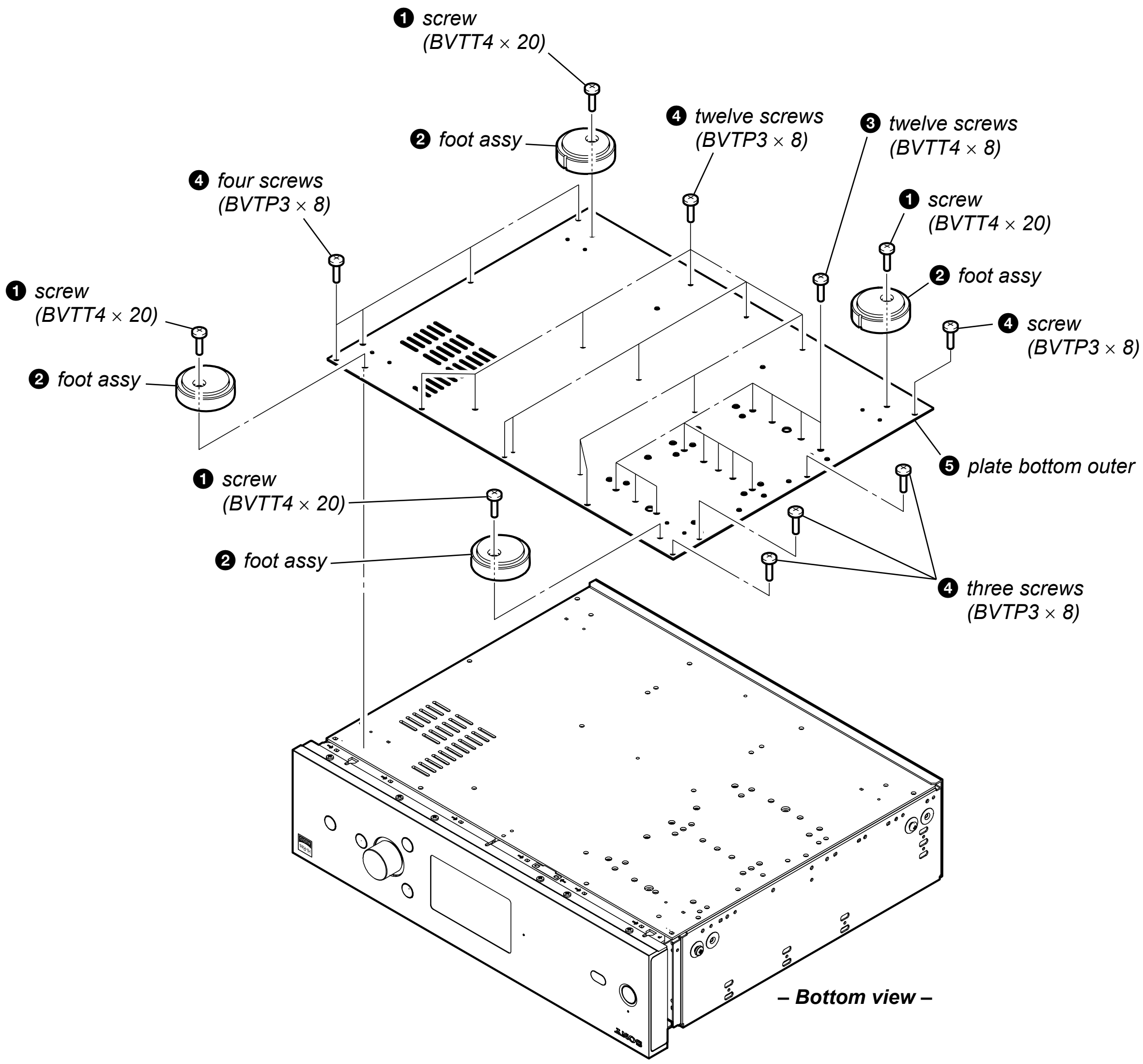
2-2. CASE TOP, FUSE (F901, F902, F1401, F1402)



2-3. CASE L BLOCK, CASE R BLOCK

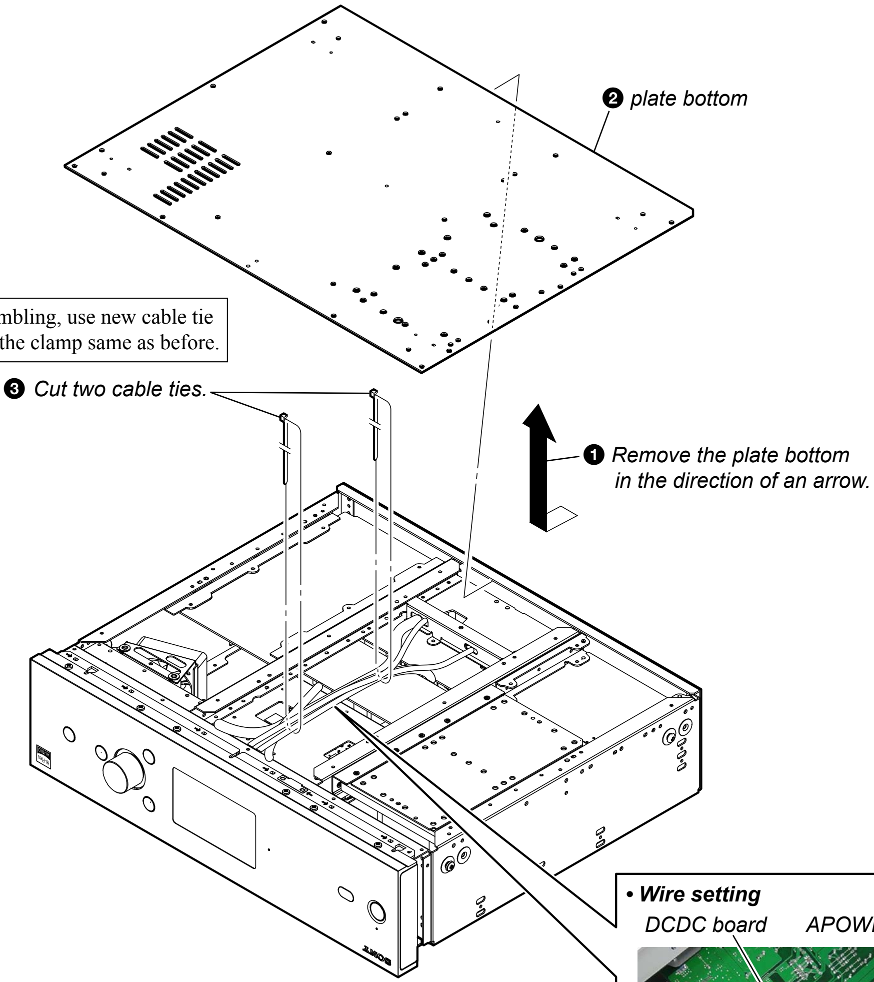


2-4. PLATE BOTTOM OUTER



2-5. PLATE BOTTOM, CABLE TIE

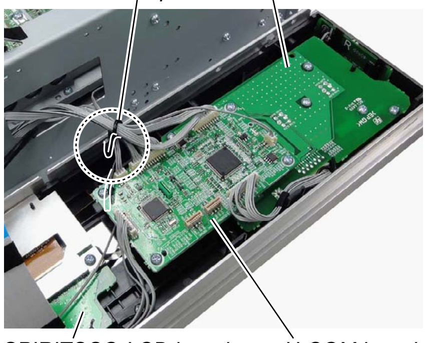
Note: In reassembling, use new cable tie to fasten the clamp same as before.



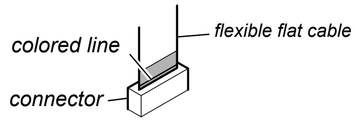
– Bottom view –

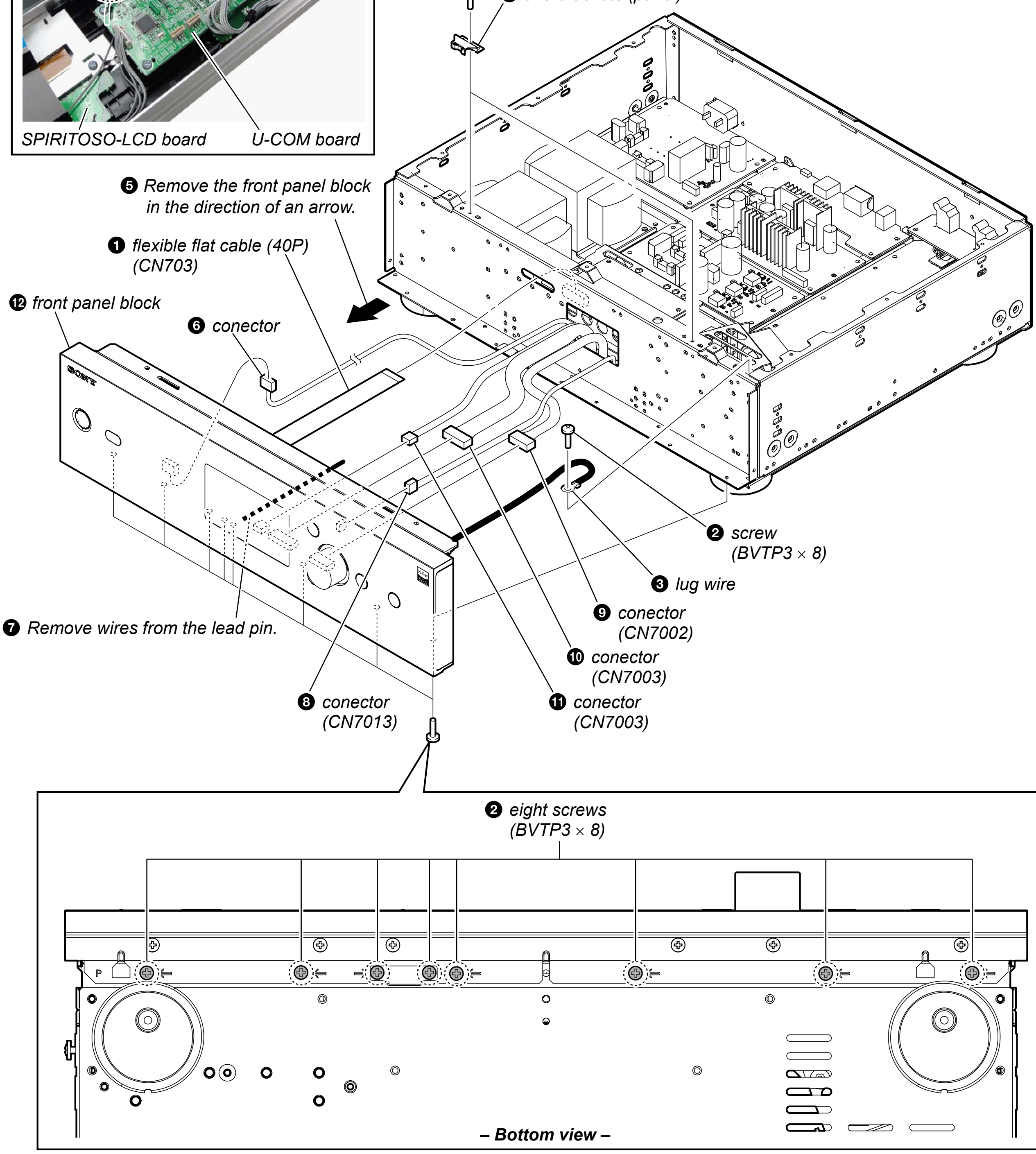
2-6. FRONT PANEL BLOCK

• Wire setting



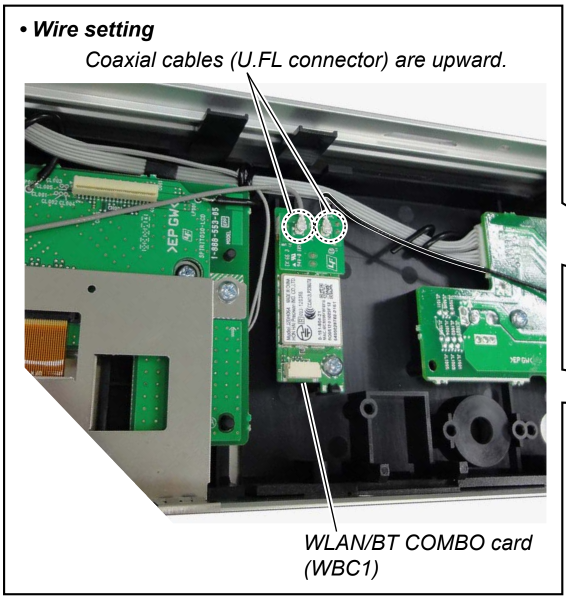
Note: When installing the flexible flat cable, ensure the colored line.
No slanting after insertion.

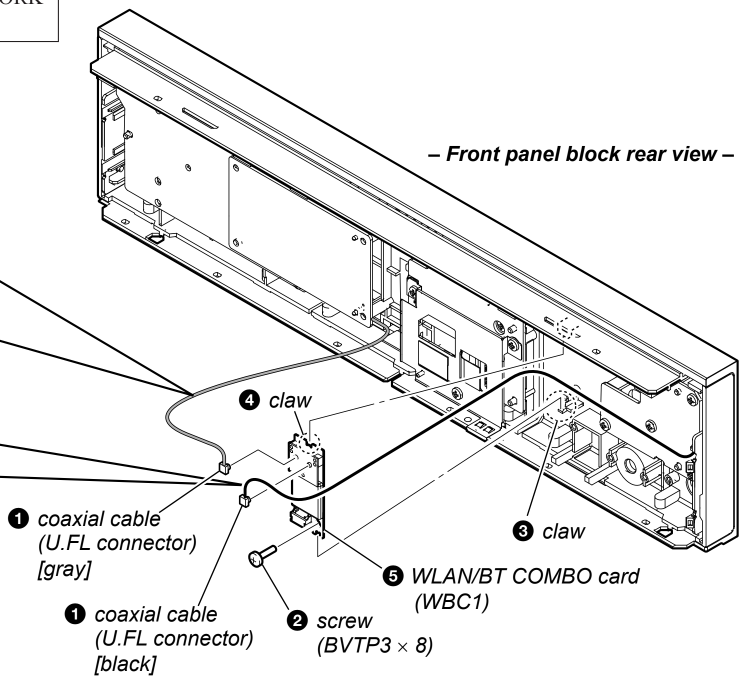




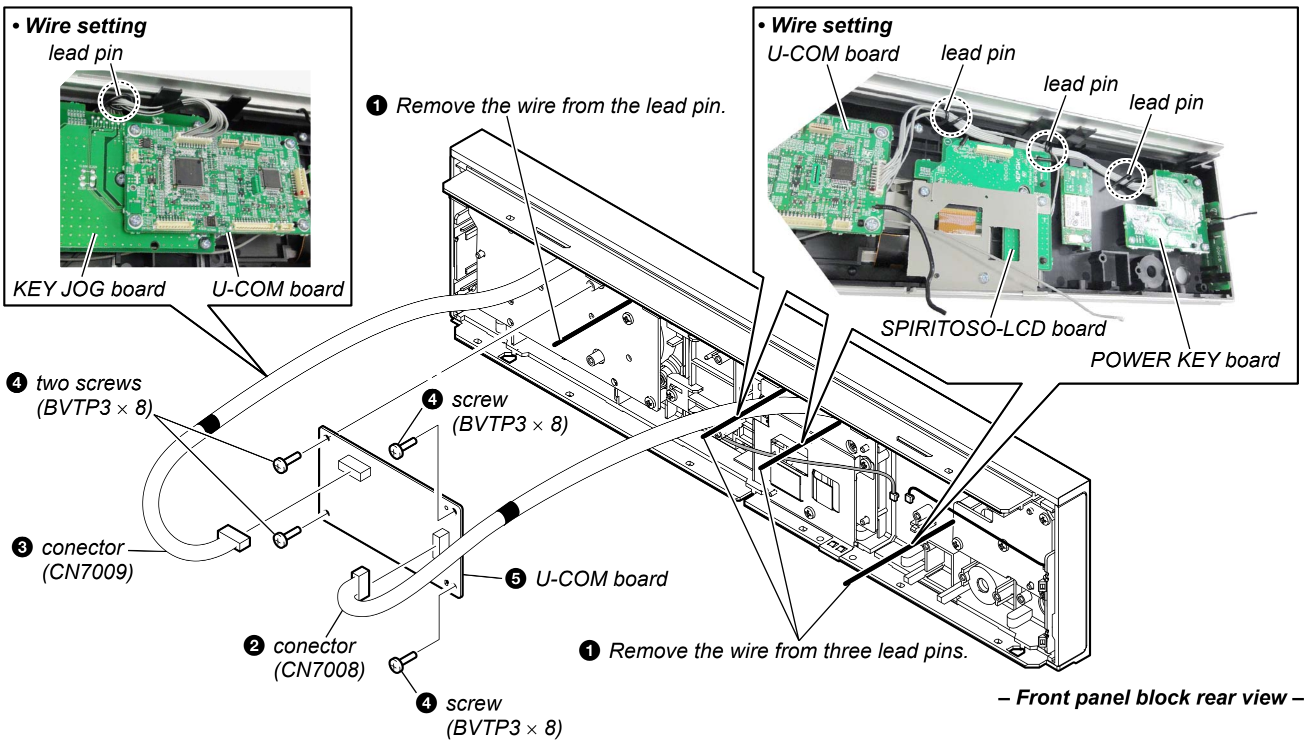
2-7. WLAN/BT COMBO CARD (WBC1)

Note: When the WLAN/BT COMBO card (Ref. No. WBC1) and coaxial (U.FL connector) cables (Ref. No. WR1, WR2) are replaced, refer to "CHECKING METHOD OF NETWORK CONNECTION" on page 5.

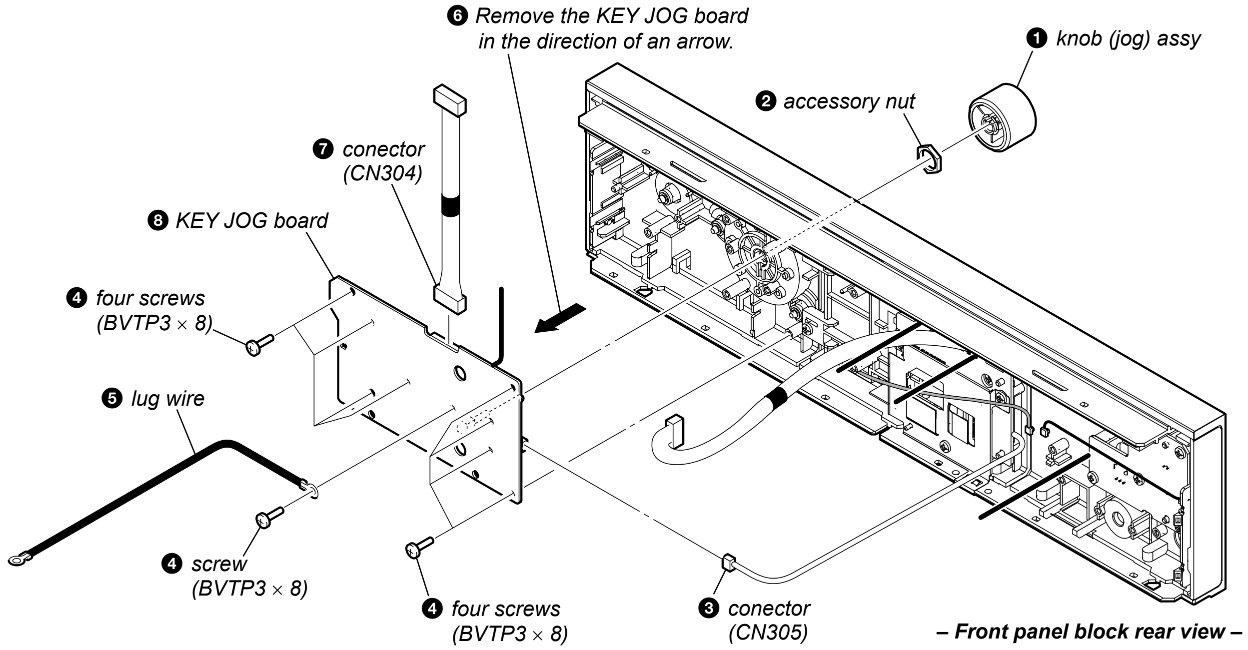




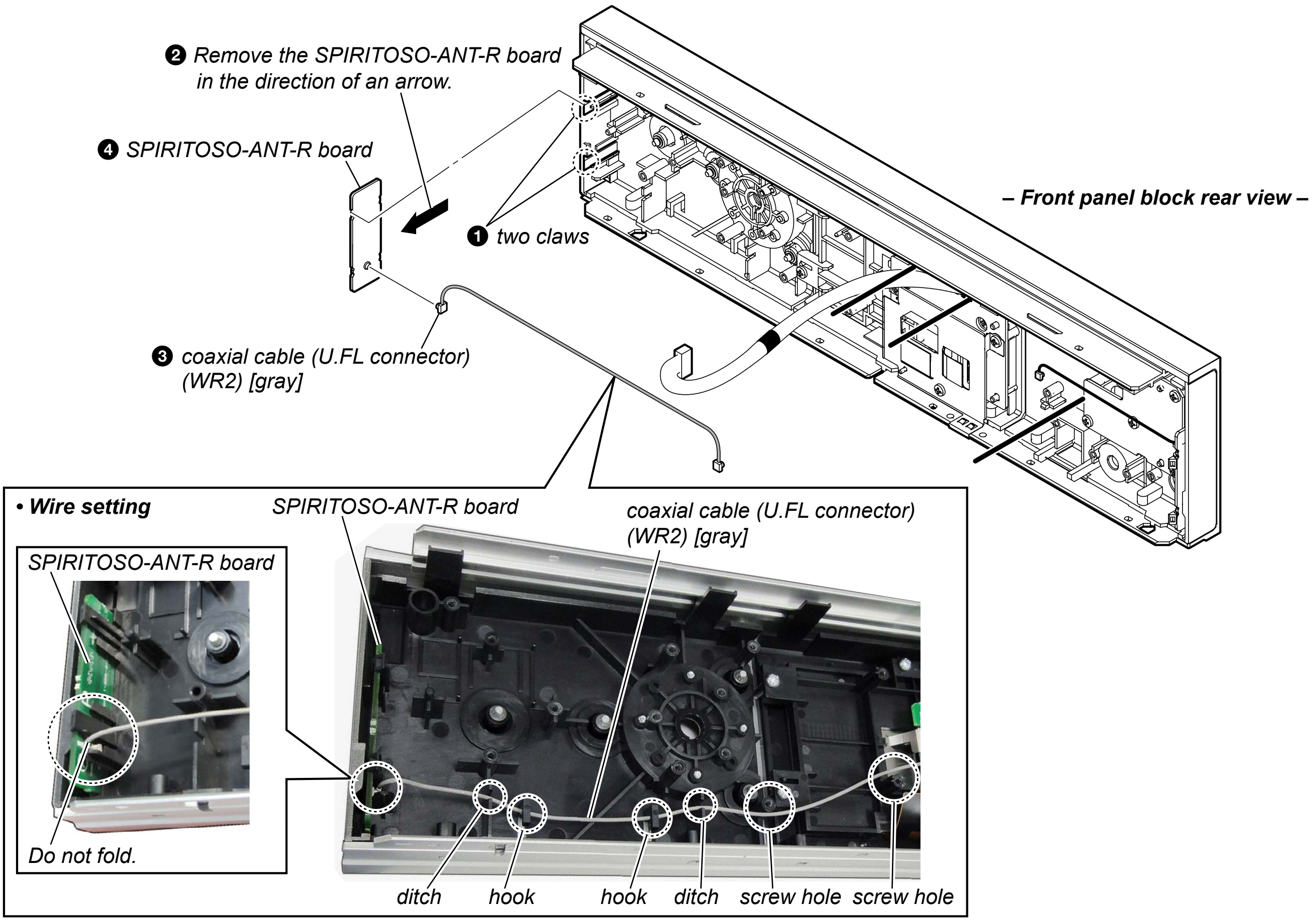
2-8. U-COM BOARD



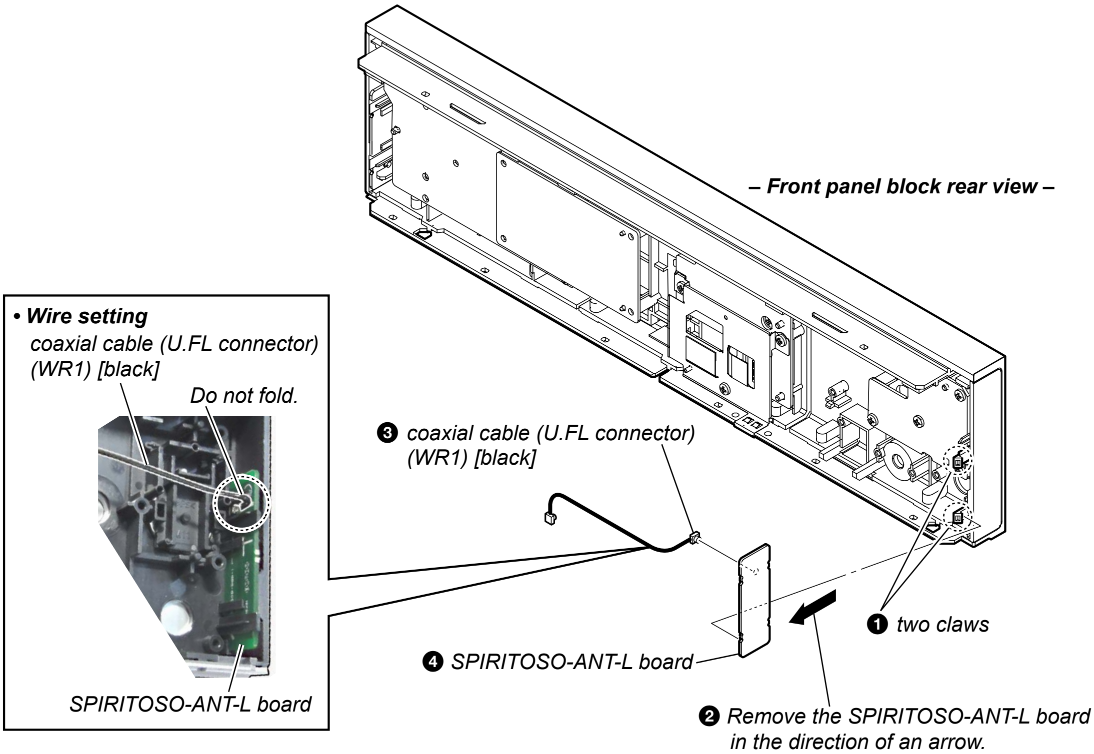
2-9. KEY JOG BOARD



2-10. SPIRITOSO-ANT-R BOARD

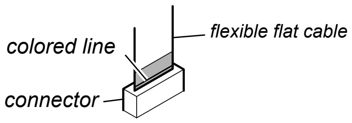


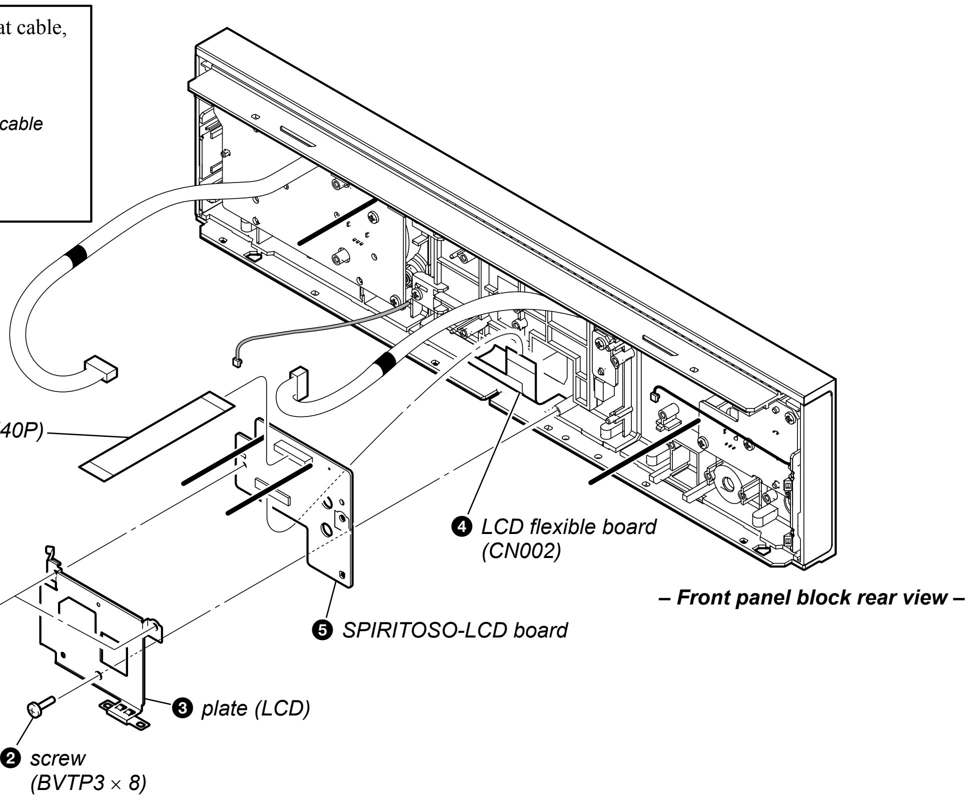
2-11. SPIRITOSO-ANT-L BOARD



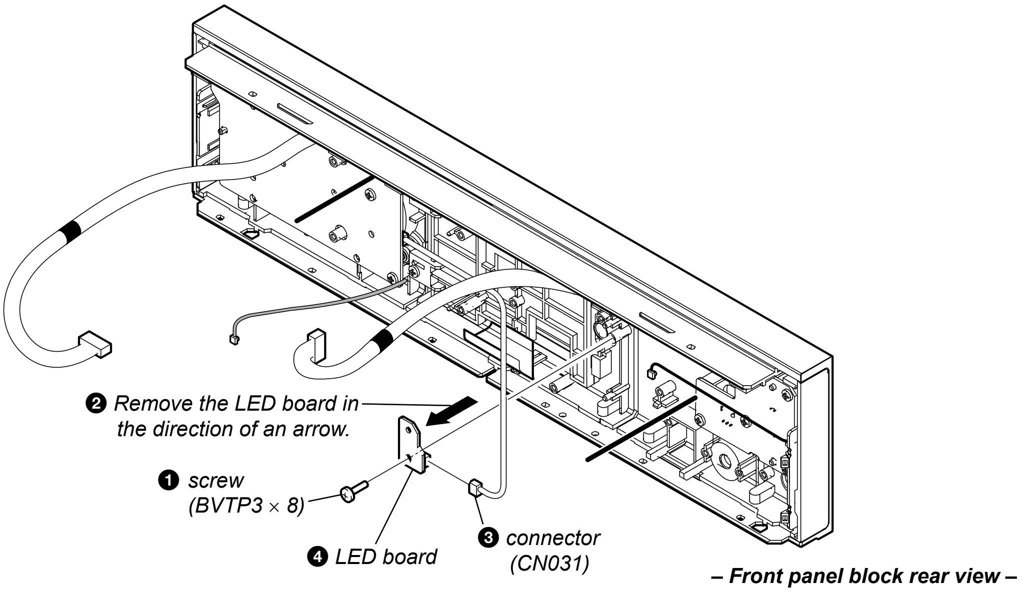
2-12. SPIRITOSO-LCD BOARD

Note: When installing the flexible flat cable,
ensure the colored line.
No slanting after insertion.

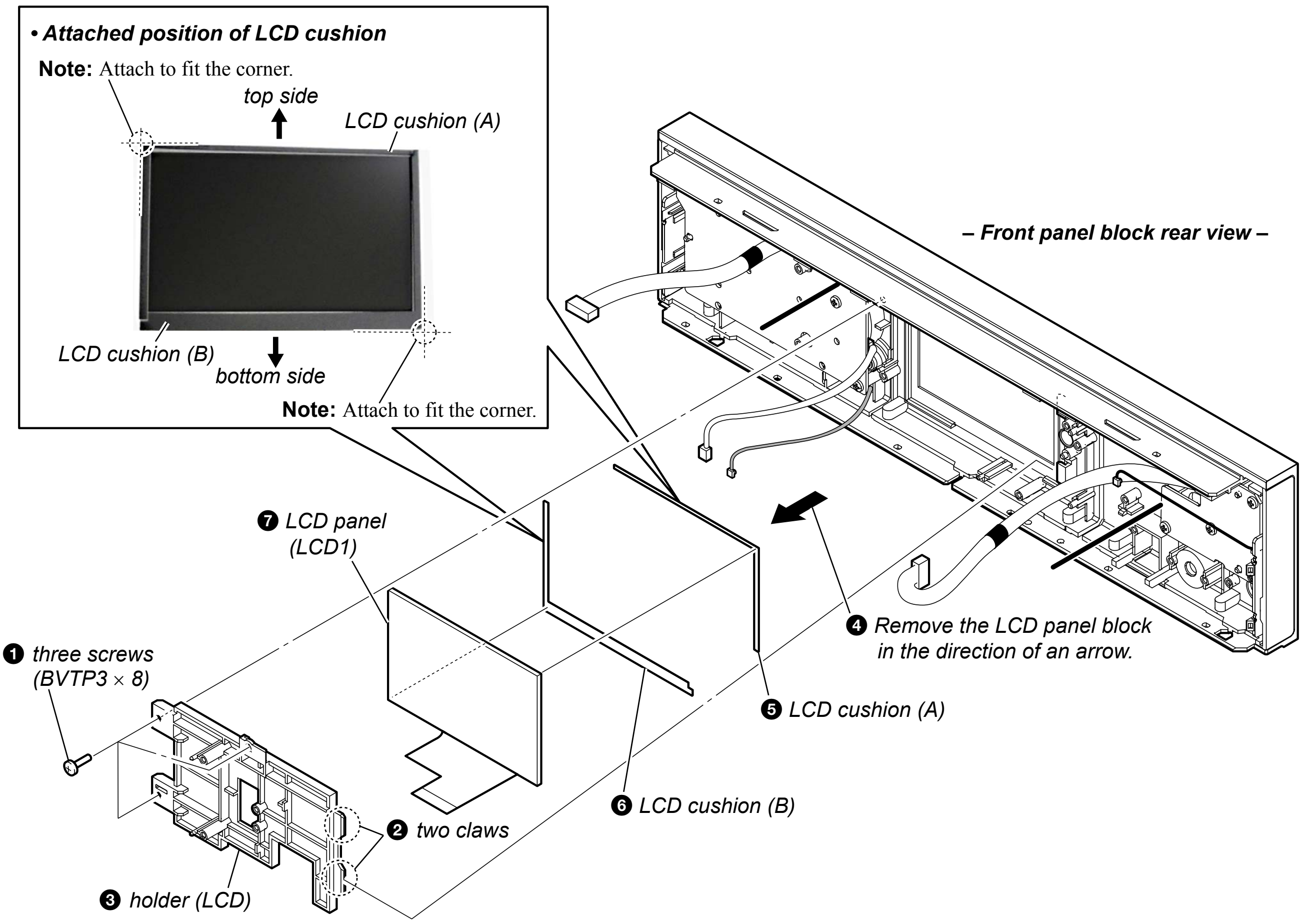




2-13. LED BOARD



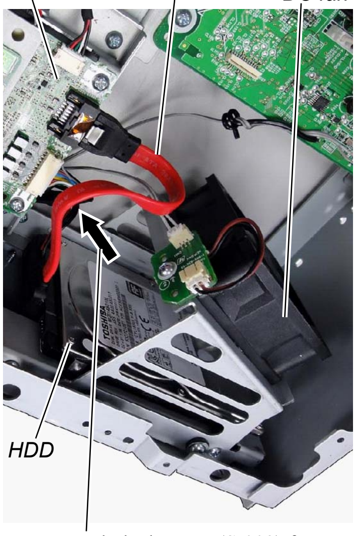
2-14. LCD PANEL (LCD1)



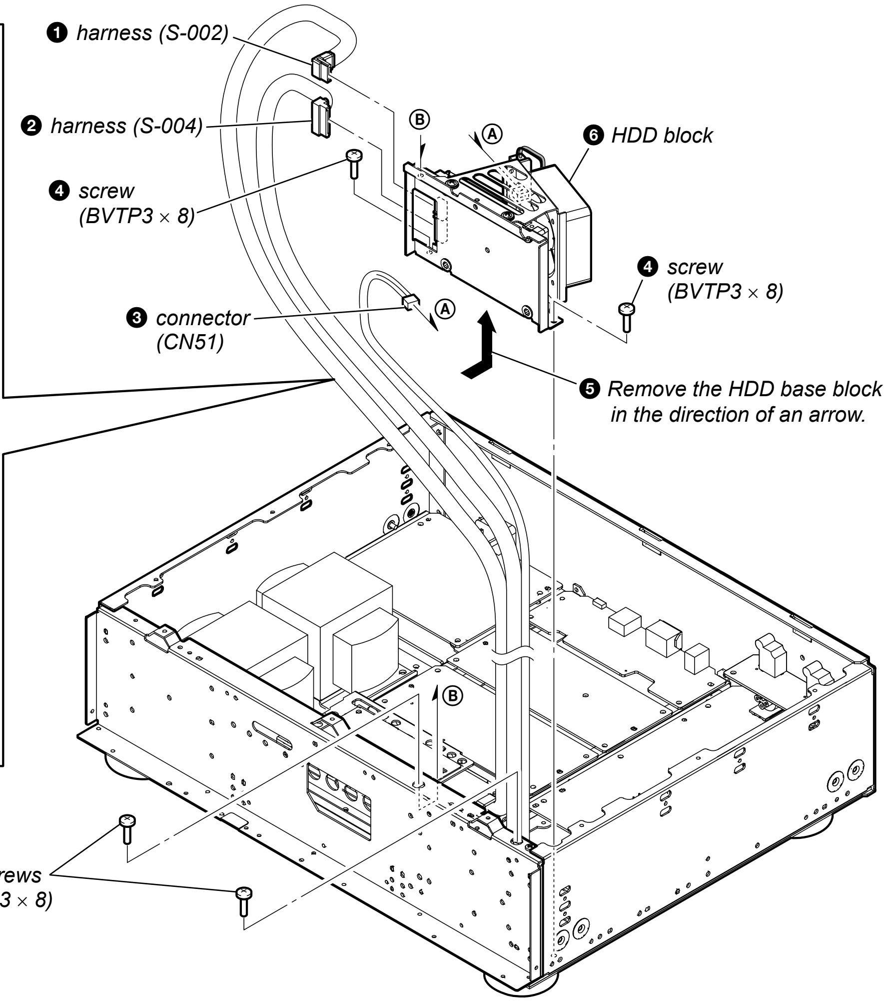
2-15. HDD BLOCK

• Wire setting

Note 1: Check that the harness (S-002) is not twisted.



Note 2: Push the harness (S-002) from HDD in the direction of an arrow as not to touch the DC fan.

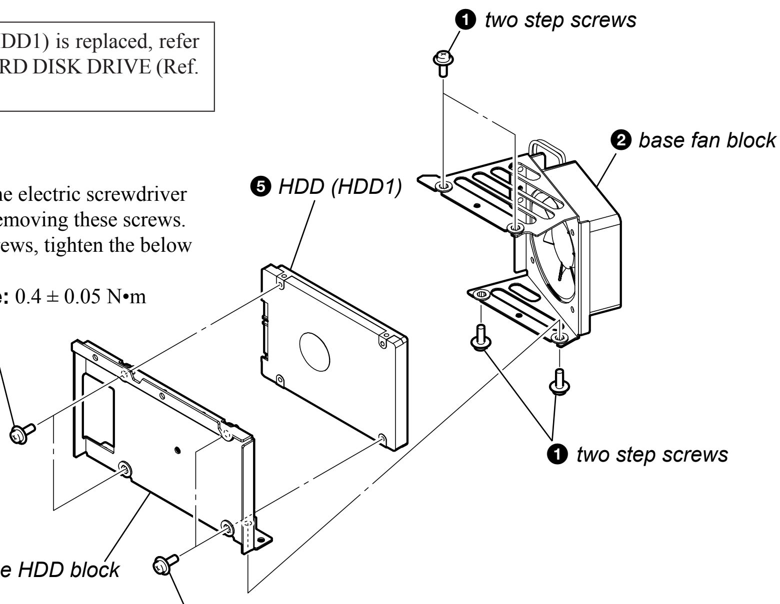


2-16. HDD (HDD1)

Note 1: When the hard disk drive (Ref. No. HDD1) is replaced, refer to "NOTE OF REPLACING THE HARD DISK DRIVE (Ref. No. HDD1)" on page 4.

❸ *two step screws*

Note 2: No problem using the electric screwdriver when installing or removing these screws. When tightening screws, tighten the below torque value.
tightening torque: 0.4 ± 0.05 N•m

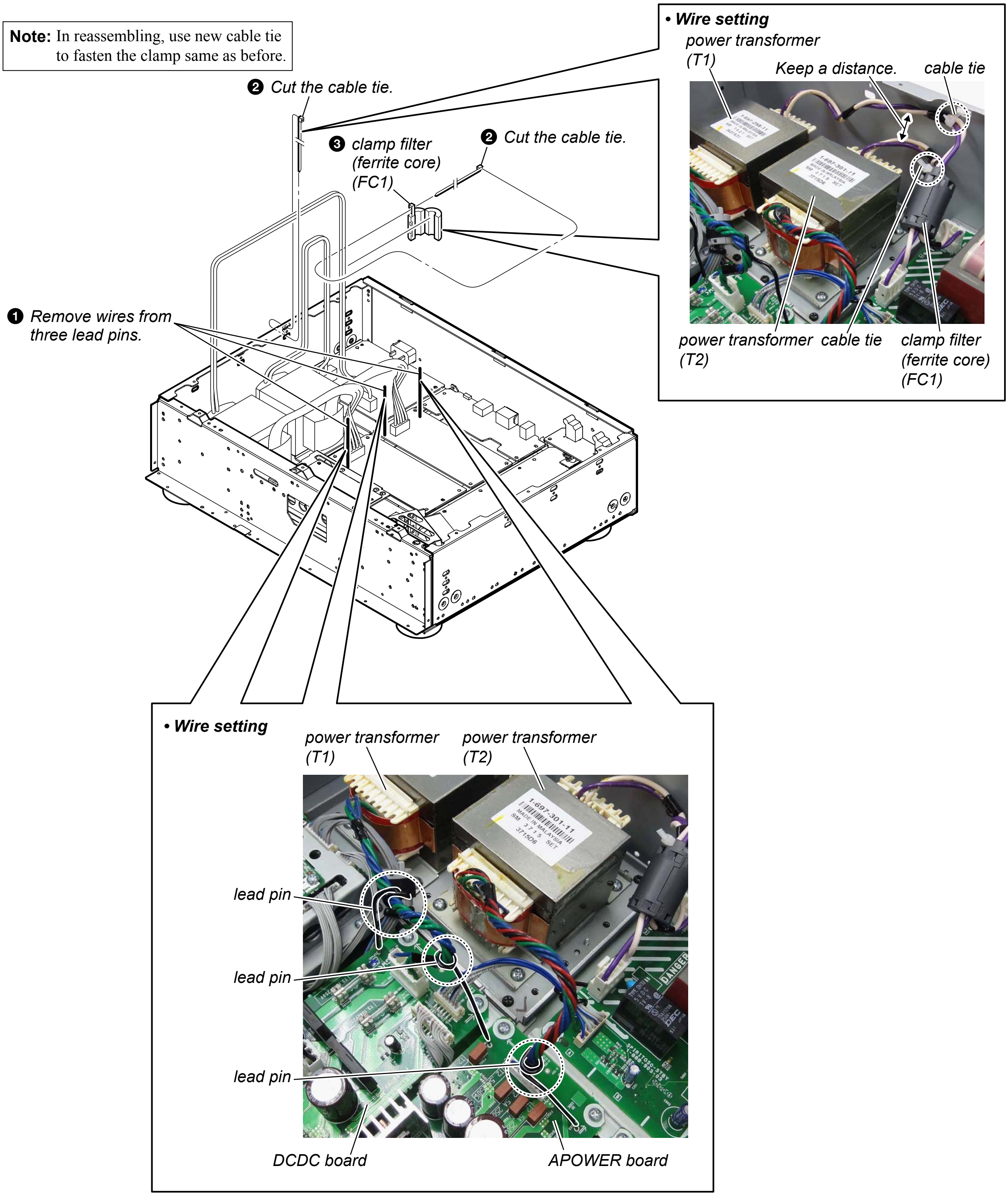


❸ *two step screws*

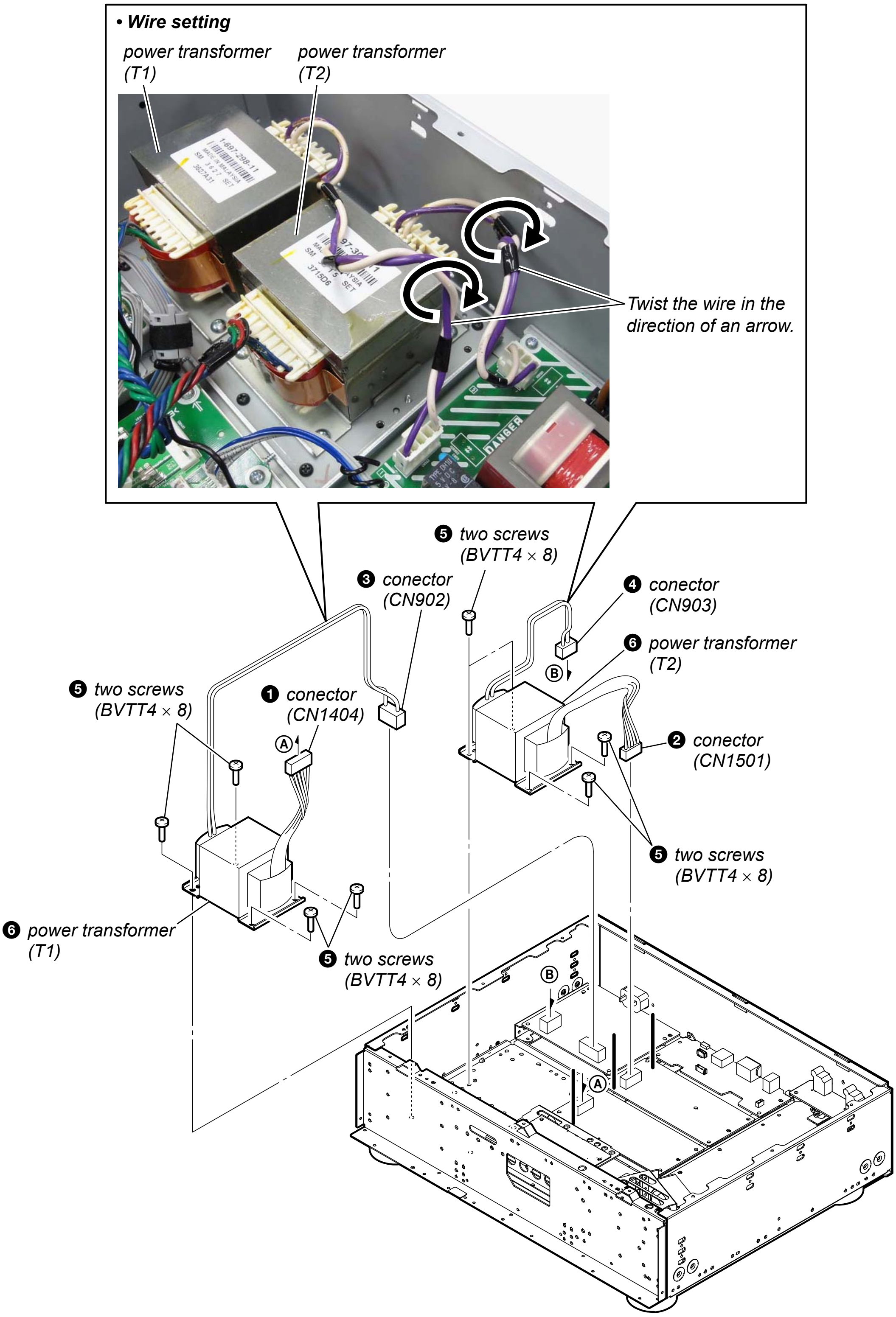
Note 2: No problem using the electric screwdriver when installing or removing these screws. When tightening screws, tighten the below torque value.
tightening torque: 0.4 ± 0.05 N•m

2-17. POWER TRANSFORMER (T1, T2)-1

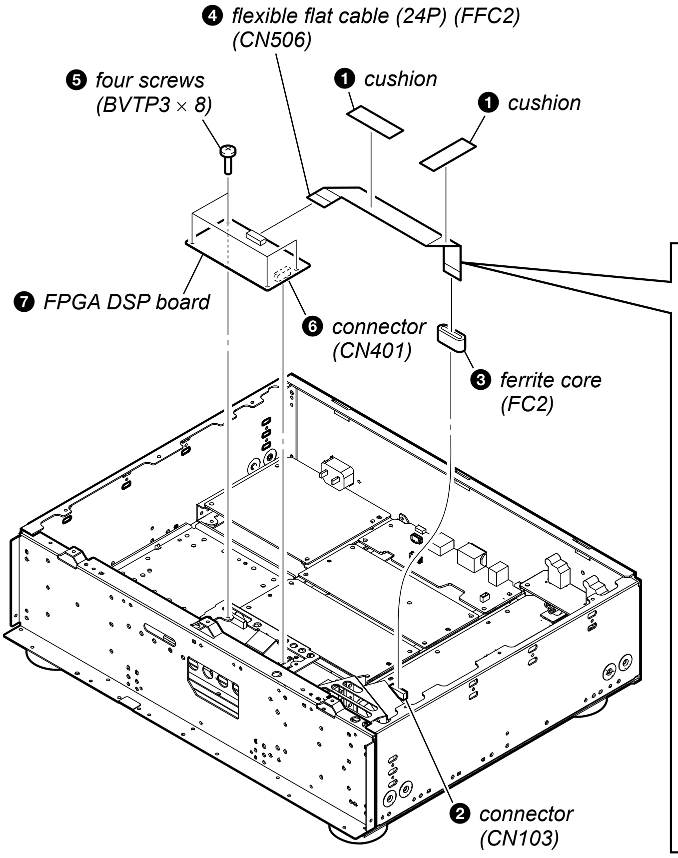
Note: In reassembling, use new cable tie to fasten the clamp same as before.



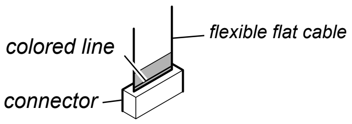
2-18. POWER TRANSFORMER (T1, T2)-2



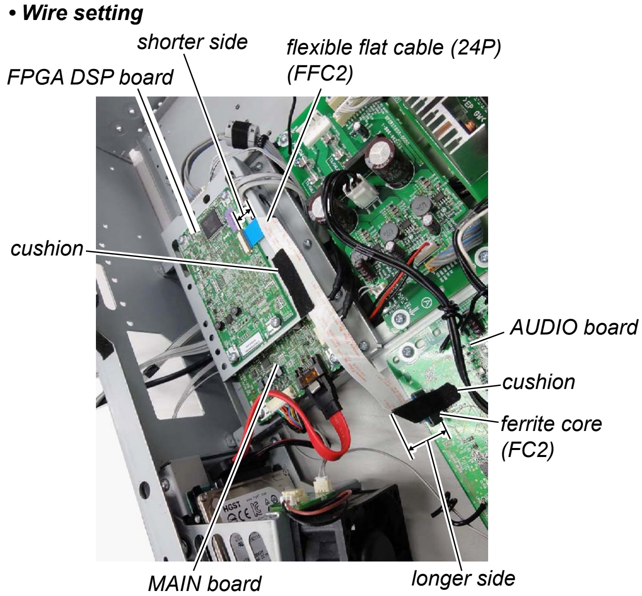
2-19. FPGA DSP BOARD



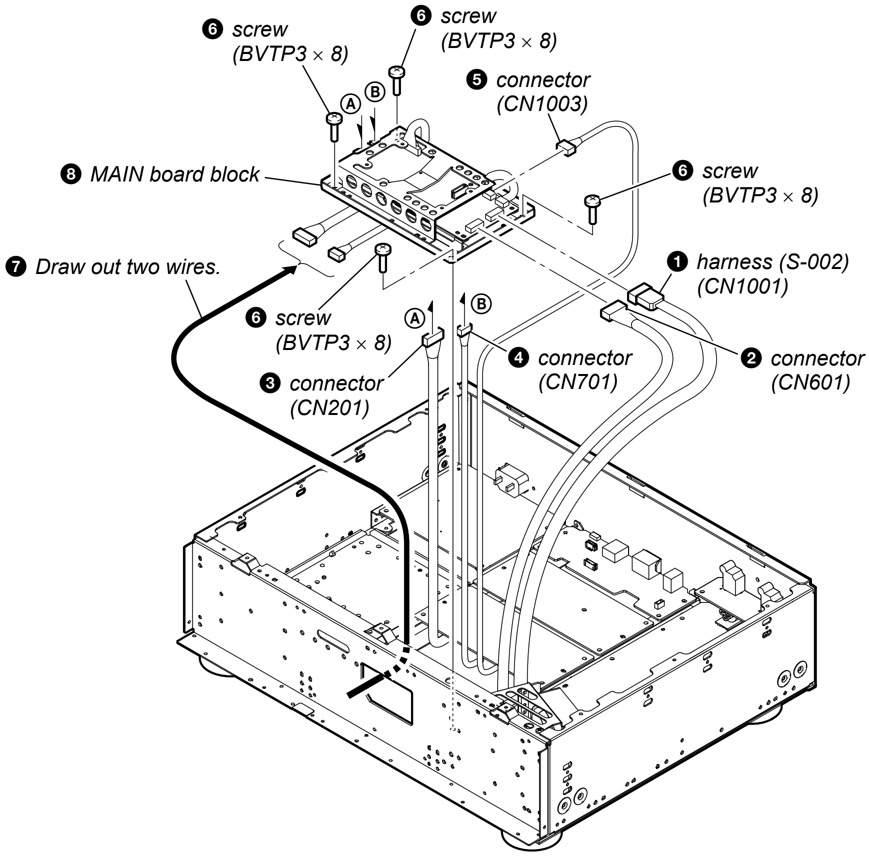
Note: When installing the flexible flat cable, ensure the colored line.
No slanting after insertion.



• Wire setting

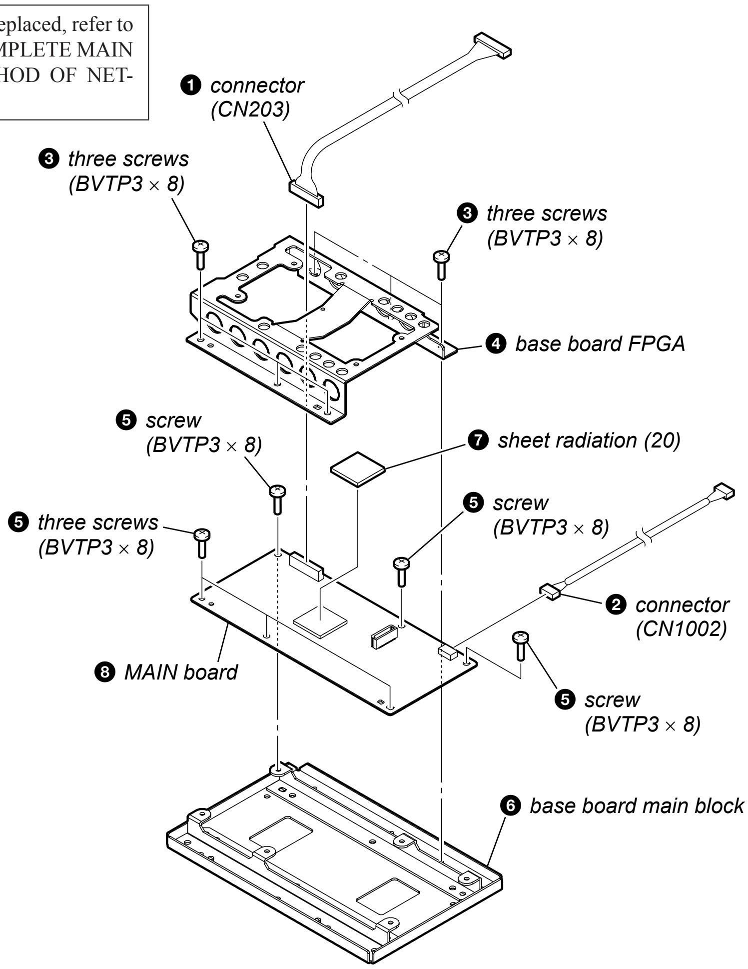


2-20. MAIN BOARD BLOCK



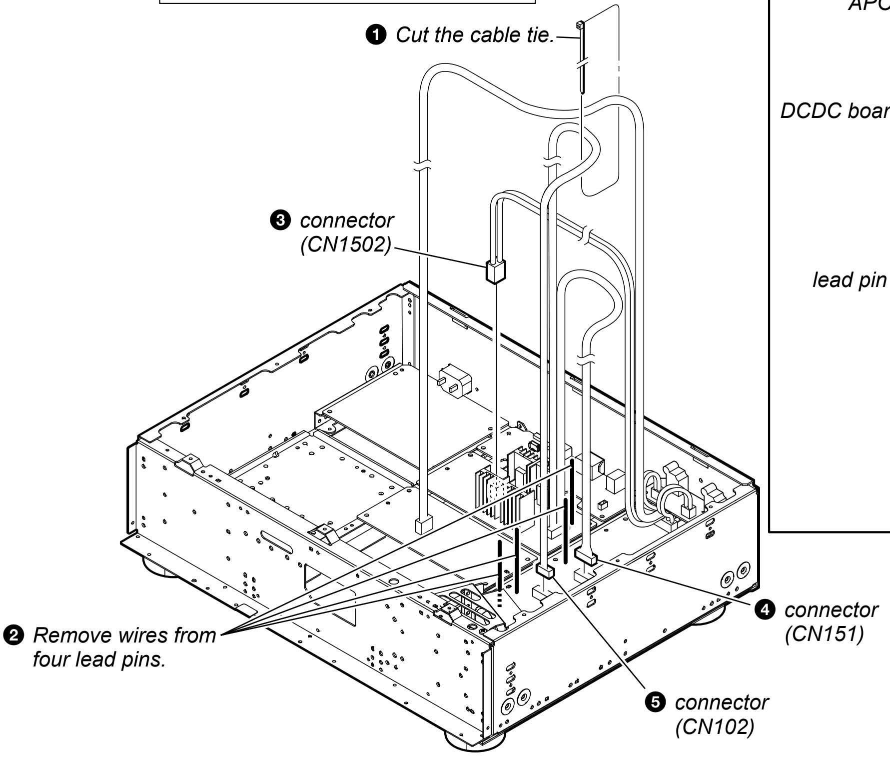
2-21. MAIN BOARD

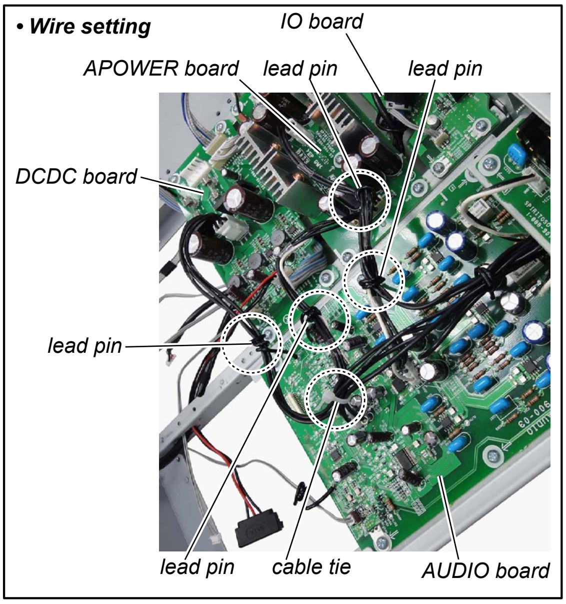
Note: When the complete MAIN board is replaced, refer to "NOTE OF REPLACING THE COMPLETE MAIN BOARD" and "CHECKING METHOD OF NETWORK CONNECTION" on page 5.

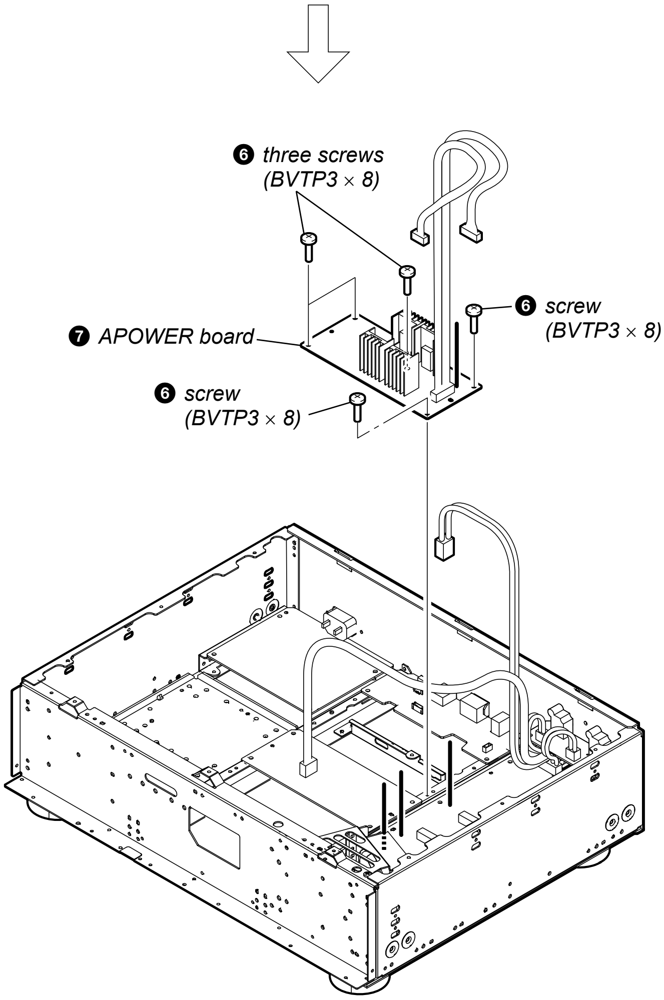


2-22. APOWER BOARD

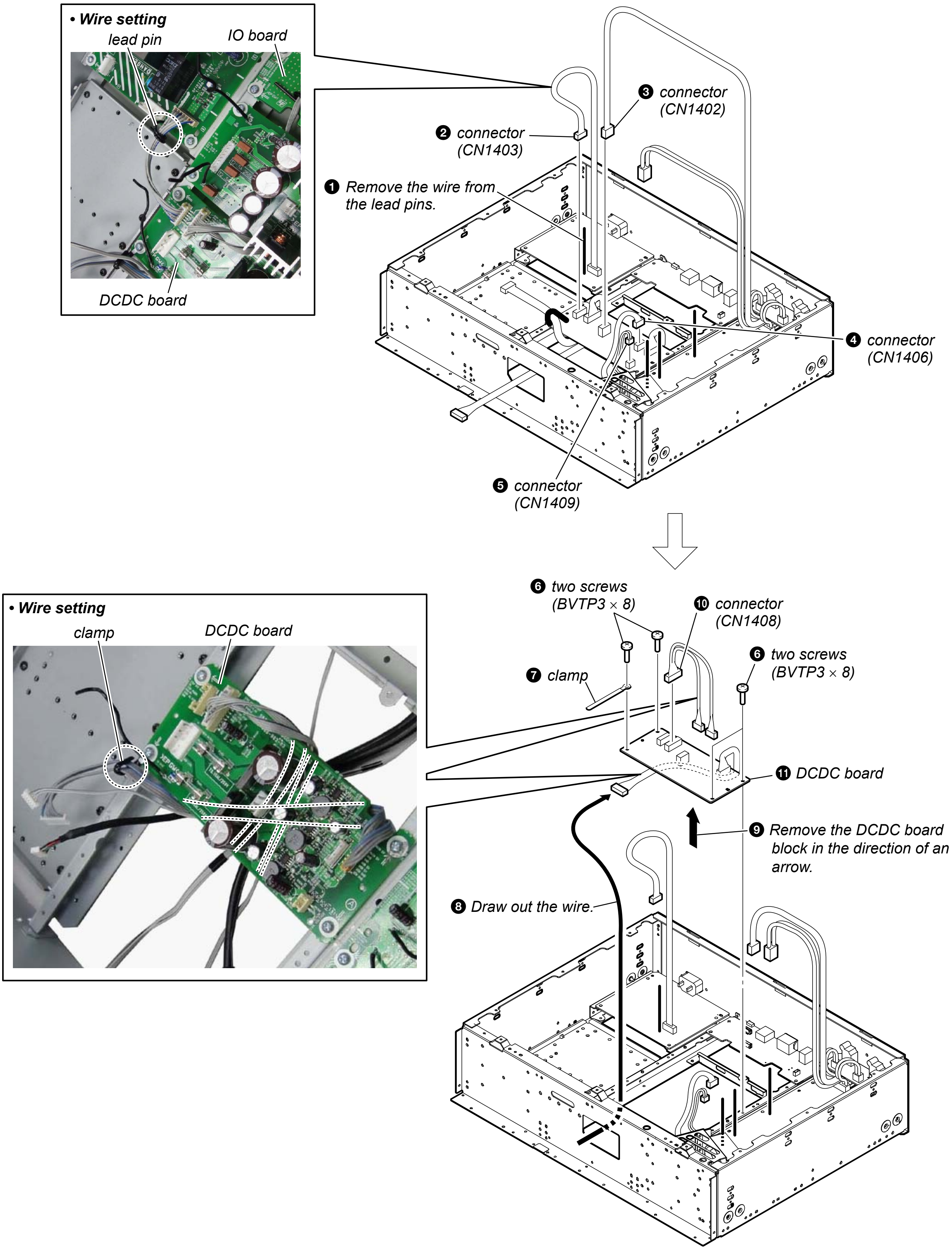
Note: In reassembling, use new cable tie to fasten the clamp same as before.



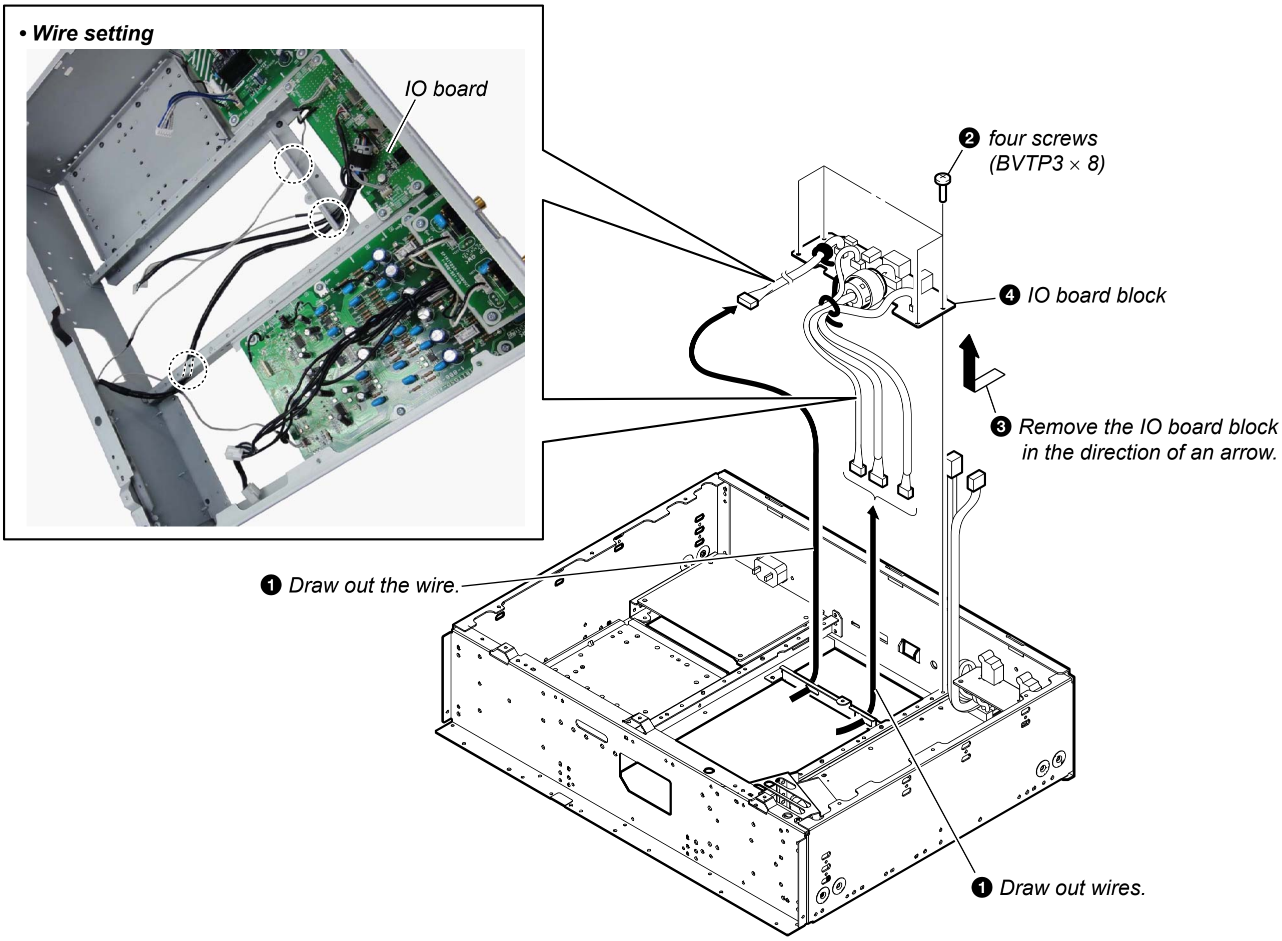




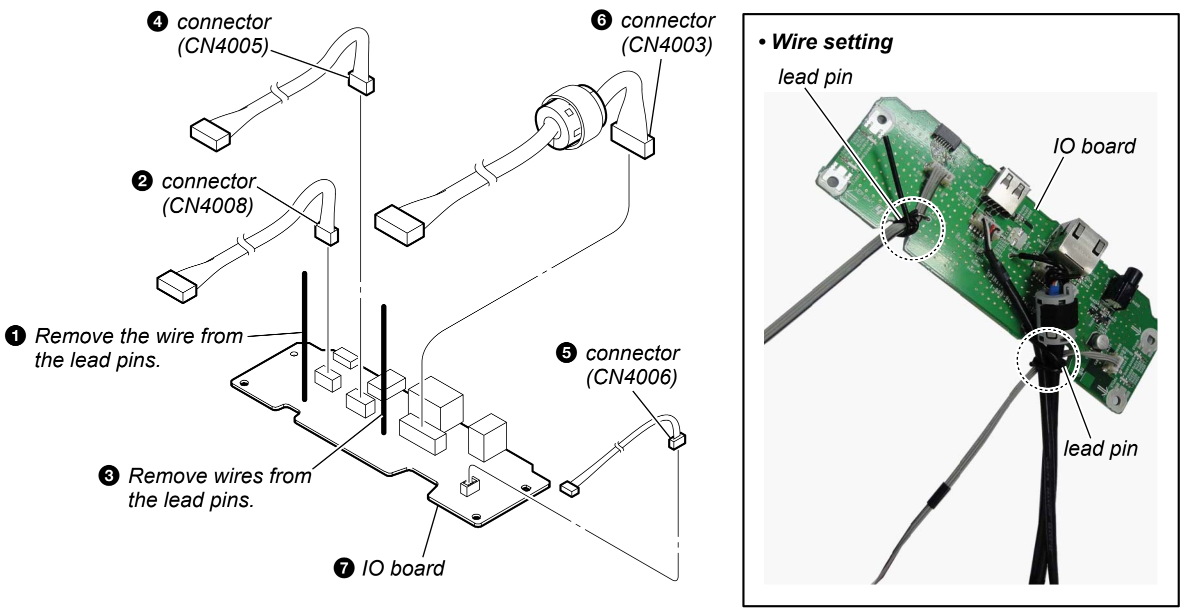
2-23. DCDC BOARD



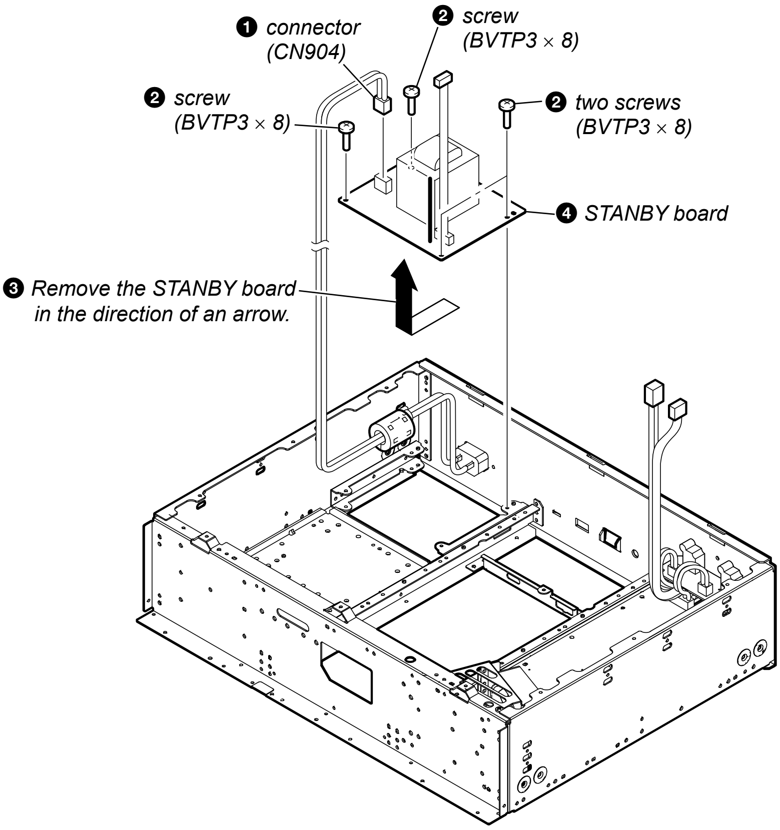
2-24. IO BOARD BLOCK



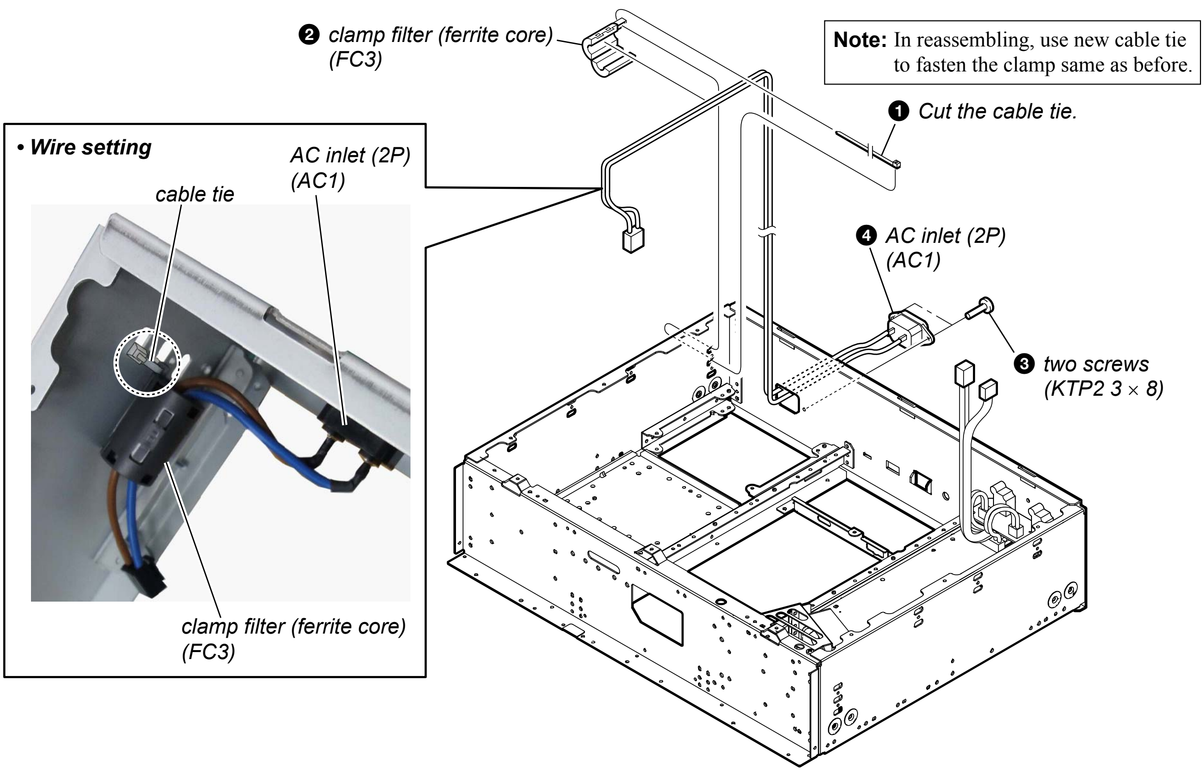
2-25. IO BOARD



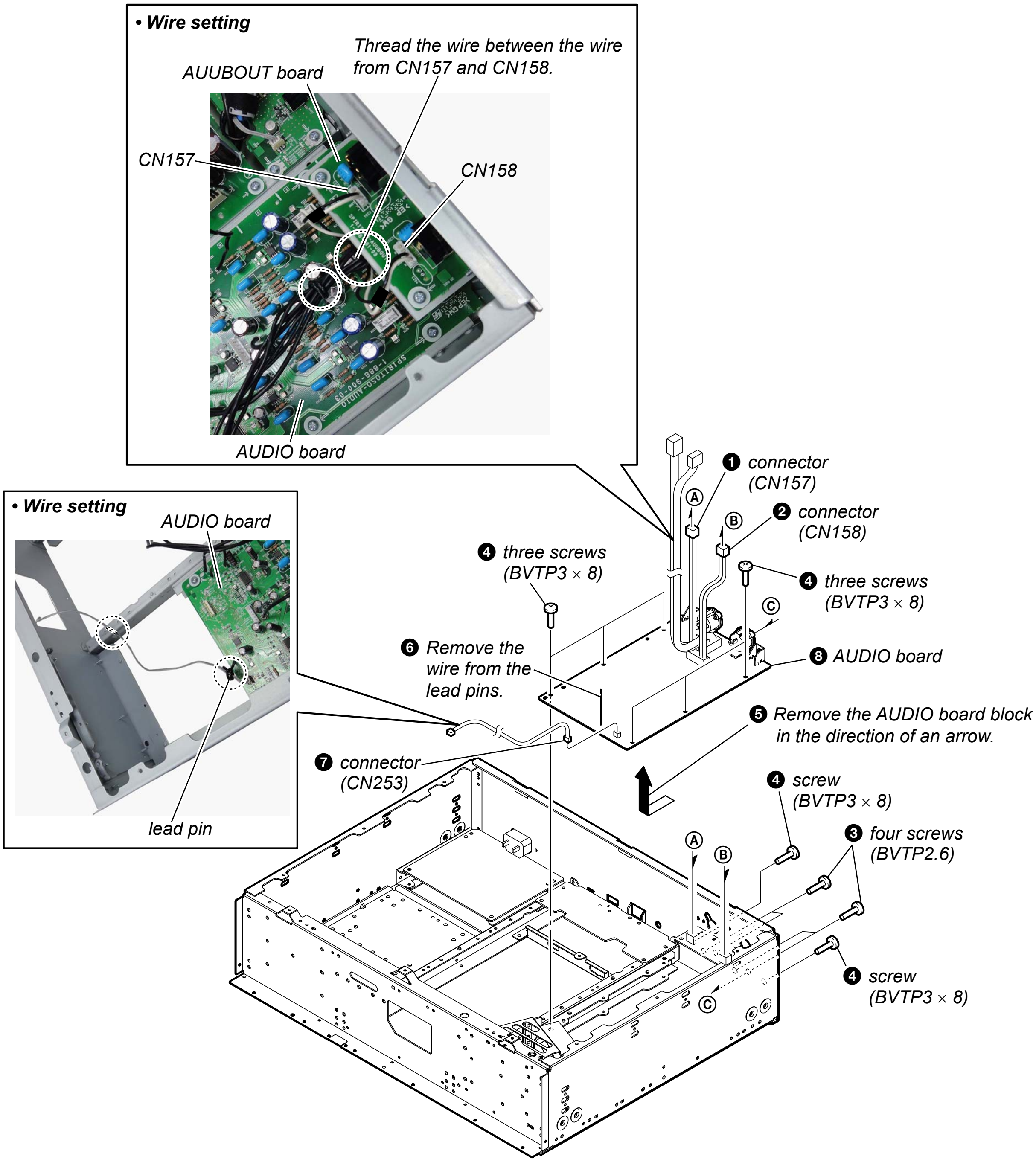
2-26. STANBY BOARD



2-27. AC INLET (2P) (AC1)



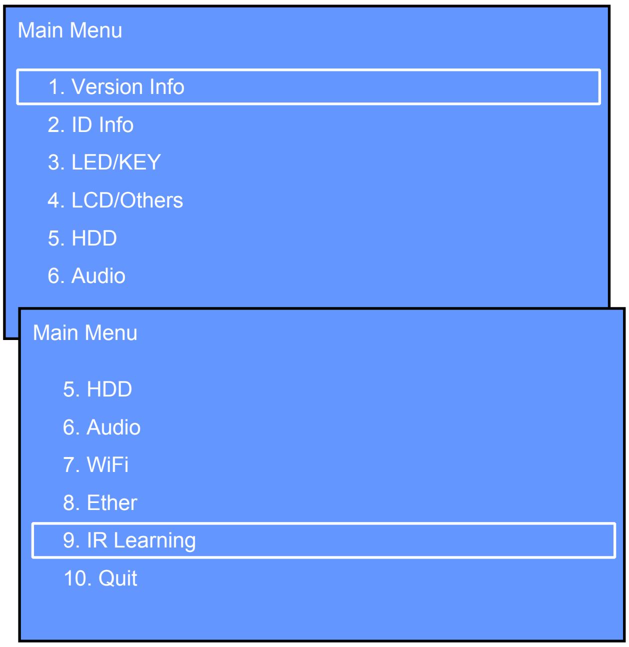
2-28. AUDIO BOARD



SECTION 3
TEST MODE

1. Entering Method of The Service Diag Mode

Procedure:
1. Connect the power cord to an outlet.
2. After becoming the standby state, while holding down two keys of the [HOME] and [BACK], press two keys in order of the [▶II] → [I/⏻].
3. Enter the service DIAG mode, “Main Menu” is displayed on the liquid crystal display.



2. Operation Method of The Service DIAG Mode

It can operate the service DIAG mode with the key of the main unit or remote commander (RM-ANU183). A key having no particular description in the text, indicates the main unit key

Key Assign

Operation	Main unit key	Remote commander (RM-ANU183) key
Moves to the cursor and selects the item	Selector	⏮/⏭
Enters the item	ENTER	▶II
Returns to the main menu	HOME	–
Returns to the previous menu	BACK	–

3. Releasing Method of The Service DIAG Mode

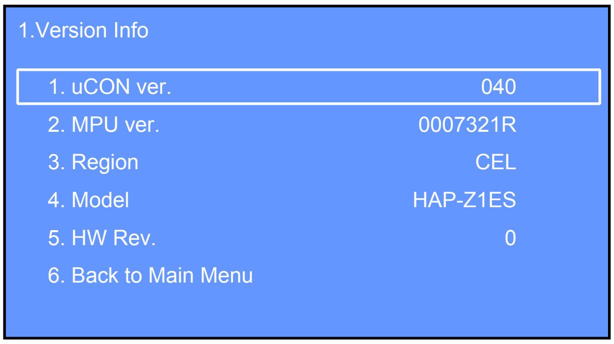
- Press the [I/⏻] button (Except the state of “3-3.KEY Check”, “5-5.SMART Short Test” and “5-6.SMART Long Test”).
- Select the “10. QUIT” on the main menu.

4. Operation of The Service DIAG Mode

4-1. Version Info

It can display the version information.

Procedure:
1. Enter the service DIAG mode.
   (Refer to “1. Entering Method of The Service DIAG Mode”)
2. Rotate the selector knob to select the “1. Version Info”.
3. Press the [ENTER] key to enter the “1. Version Info” mode.

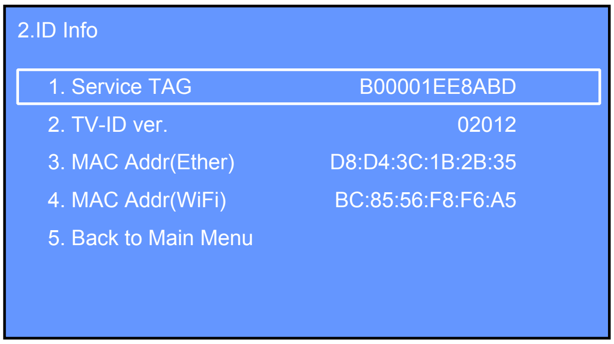


(Displayed characters/values in the above figure are example)

4-2. ID Info

It can display the ID information.

Procedure:
1. Enter the service DIAG mode.
   (Refer to “1. Entering Method of The Service DIAG Mode”)
2. Rotate the selector knob to select the “2. ID Info”.
3. Press the [ENTER] key to enter the “2. ID Info” mode.

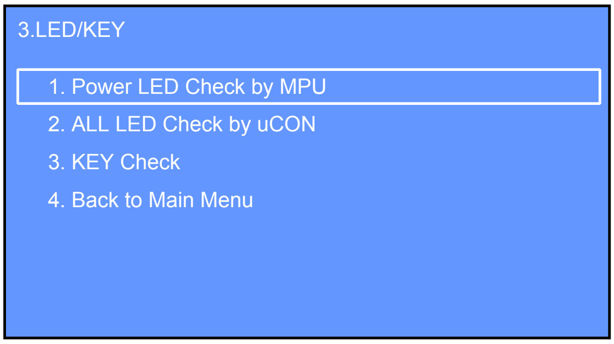


(Displayed characters/values in the above figure are example)

4-3. LED/KEY

Procedure:

1. Enter the service DIAG mode.
   (Refer to “1. Entering Method of The Service DIAG Mode” on page 25)
2. Rotate the selector knob to select the “3. LED/KEY”.
3. Press the [ENTER] key to enter the “3. LED/KEY” mode.



4-3-1. Power LED Check by MPU

It can check the [I/⏻] LED.

Procedure:

1. Enter the “3. LED/KEY” mode.
   (Refer to “4-3. LED/KEY”)
2. Rotate the selector knob to select the “1. Power LED Check by MPU”.
3. Press the [ENTER] key to enter the “1. Power LED Check by MPU” mode.
4. The [I/⏻] LED continues blinking.

4-3-2. ALL LED Check by uCON

It can check two LEDs of the [I/⏻] and [DSEE].

Procedure:

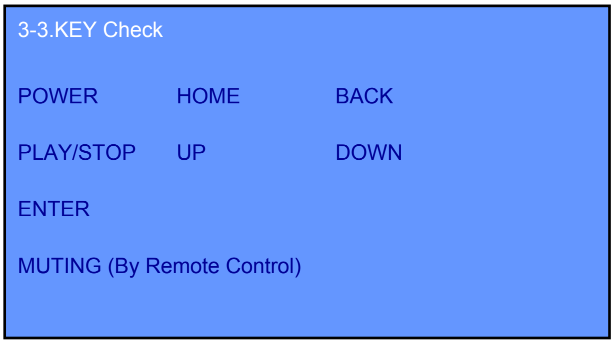
1. Enter the “3.LED/KEY” mode.
   (Refer to “4-3. LED/KEY”)
2. Rotate the selector knob to select the “2. ALL LED Check by uCON”.
3. Press the [ENTER] key to enter the “2. ALL LED Check by uCON” mode.
4. Two LEDs of the [I/⏻] and [DSEE] continues blinking.

4-3-3. KEY Check

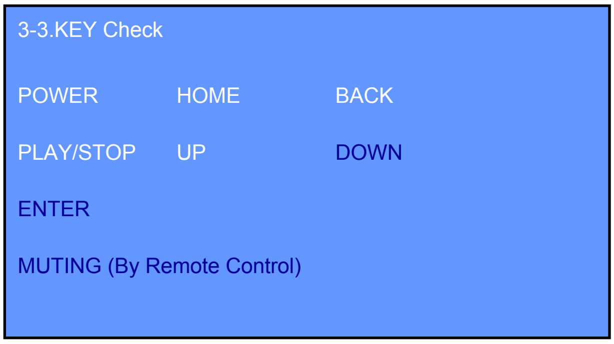
It can check the keys and knobs.

Procedure:

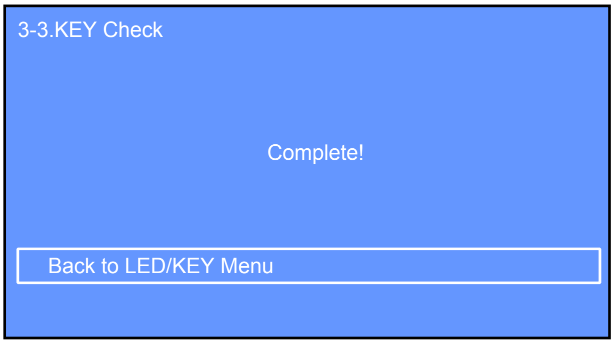
1. Enter the “3. LED/KEY” mode.
   (Refer to “4-3. LED/KEY”)
2. Rotate the selector knob to select the “3. KEY Check”.
3. Press the [ENTER] key to enter the “3. KEY Check” mode.



4. Each time a key on the main unit or remote commander is pressed and each time a knob on the main unit is rotated, character on the liquid crystal display corresponding to the keys and knobs turns the white color.



5. When all character on the liquid crystal display corresponding to the keys and knobs turns the white color, following screen is displayed on the liquid crystal display.



4-4. LCD/Others

Procedure:

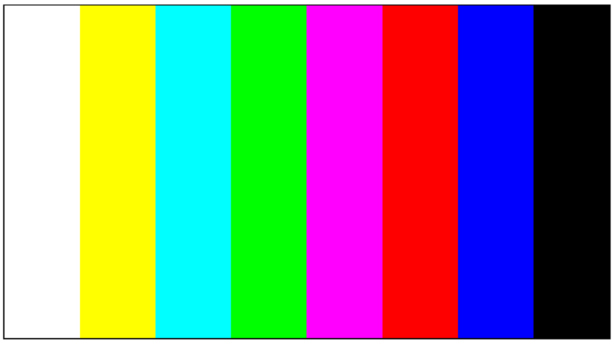
1. Enter the service DIAG mode.
   (Refer to "1. Entering Method of The Service DIAG Mode" on page 25)
2. Rotate the selector knob to select the "4. LCD/Others".
3. Press the [ENTER] key to enter the "4. LCD/Others" mode.

4-4-1. LCD Color Bar

It can check the liquid crystal display.

Procedure:

1. Enter the "4. LCD/Others" mode.
   (Refer to "4-4. LCD/Others")
2. Rotate the selector knob to select the "1. LCD Color Bar".
3. Press the [ENTER] key to enter the "1. LCD Color Bar" mode.
4. The color bar is displayed on the liquid crystal display.

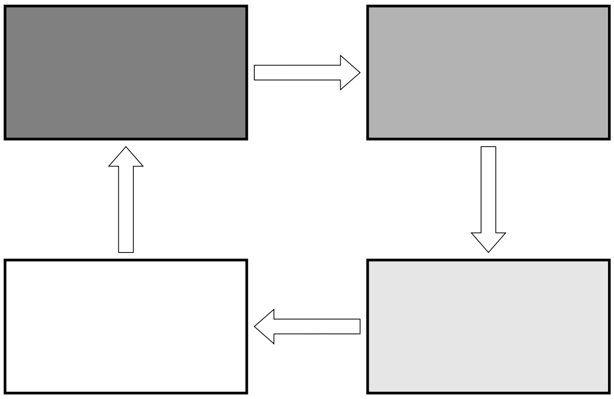


4-4-2. LCD Back light Check

It can check the liquid crystal display back light.

Procedure:

1. Enter the "4. LCD/Others" mode.
   (Refer to "4-4. LCD/Others")
2. Rotate the selector knob to select the "2. LCD Back light Check".
3. Press the [ENTER] key to enter the "2. LCD Back light Check.
4. The back light continues changing on the white screen as follows.

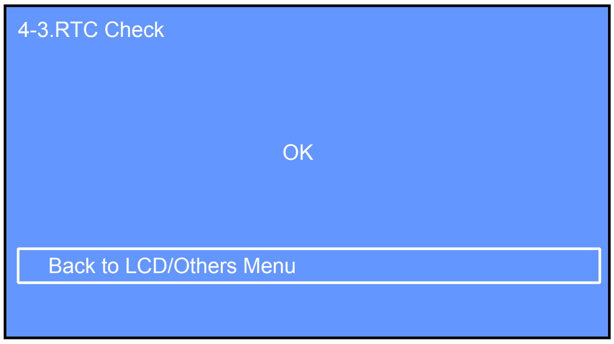


4-4-3. RTC Check

It can check the real time clock.

Procedure:

1. Enter the "4. LCD/Others" mode.
   (Refer to "4-4. LCD/Others")
2. Rotate the selector knob to select the "3. RTC Check".
3. Press the [ENTER] key to enter the "3. RTC Check" mode.
4. The result of real time clock check "OK" or "NG" is displayed on the liquid crystal display.



4-4-4. FAN Control Check

It can change the fan motor speed.

Procedure:

1. Enter the "4. LCD/Others" mode.
   (Refer to "4-4. LCD/Others" on page 27)
2. Rotate the selector knob to select the "4. FAN Control Check".
3. Press the [ENTER] key to enter the "4. FAN Control Check" mode.

4-4.FAN Control Check

1. fan off (HAP-Z1ES only)
2. fan slow
3. fan mid (HAP-S1 only)
4. fan high
5. Back to LCD/Others Menu

Note: "3. fan mid (HAP-S1 only)" cannot use in this unit.

4. Rotate the selector knob to select the "1. fan off (HAP-Z1ES only)" and press the [ENTER] key.
5. The fan motor is stopped.
6. Rotate the selector knob to select the "2. fan slow" and press the [ENTER] key.
7. The fan motor is changed to slow speed.
8. Rotate the selector knob to select the "4. fan high" and press the [ENTER] key.
9. The fan motor is changed to high speed.

4-5. HDD

It can display the hard disk drive information.

Procedure:

1. Enter the service DIAG mode.
   (Refer to "1. Entering Method of The Service DIAG Mode" on page 25)
2. Rotate the selector knob to select the "5. HDD".
3. Press the [ENTER] key to enter the "5. HDD" mode.

Note 1: When the hard disk drive is damaged, the information (model name, serial number and firmware revision) of hard disk drive is not displayed on the liquid crystal display.
In addition, when the cable connected to the hard disk drive is disconnected or broken, the information of hard disk drive is not displayed on the liquid crystal display.
When the information of hard disk drive is not displayed, check the cable connected to the hard disk drive. When the cable is broken, replace it and enter the "5. HDD" mode again.

5.HDD

1. Model TOSHIBA MQ01ABD100
2. Serial 5307C10GT
3. Firmware Rev. AM001Y
4. SMART Data Read(Need USB Memory)
5. SMART Short Test (2min)
6. SMART Long Test (2-4 hours)

5.HDD

2. Serial 5307C10GT
3. Firmware Rev. AM001Y
4. SMART Data Read(Need USB Memory)
5. SMART Short Test (2min)
6. SMART Long Test (2-4 hours)
7. Back to Main Menu

(Displayed characters/values in the above figure are example)

Note 2: Refer to "HDD CHECK" for the "5. SMART Short Test (2min)" mode and "6. SMART Long Test (2-4 hours)" mode on page 33.

4. Connect a USB memory (FAT32 formatted) to the USB connector on the main unit.
5. Rotate the selector knob to select the "4. SMART Data Read(Need USB Memory)" and press the [ENTER] key.
   (When a USB memory is not connected to the USB connector on the main unit, "Failed to save USB memory" is displayed on the liquid crystal display. But the SMART data is displayed on the liquid crystal display)

5-4.SMART Data Read

XXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXX
XXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXX
XXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXX
XXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXX
XXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXX
XXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXX
XXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXX
XXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXX
XXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXX
XXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXX

The data was saved to USB memory

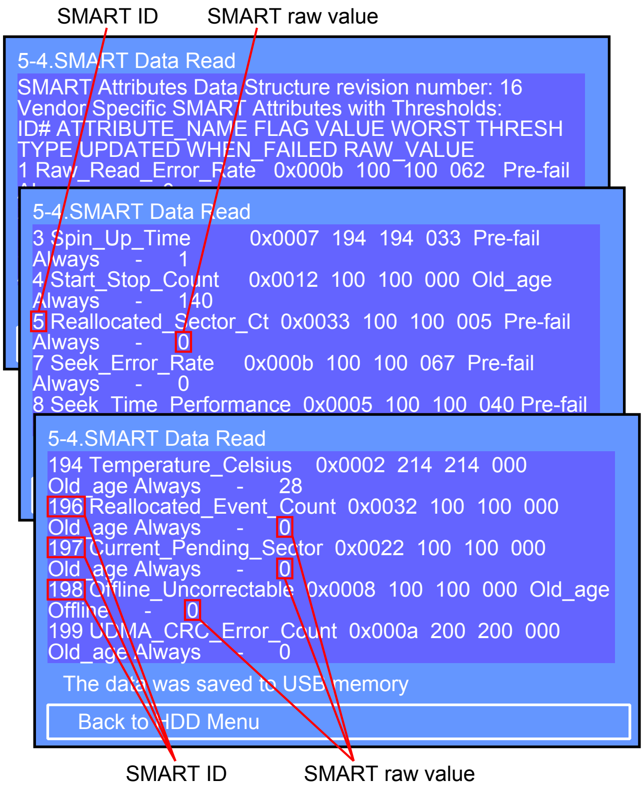
Back to HDD Menu

XXX...: SMART data

6. The SMART data is saved as a text file in the USB memory. (The file name is “hap_smart_data_xxxxx_yyyyy.log (xxxxx: hard disk drive model name, yyyyy: hard disk drive serial number))

Note 3: This file uses Linux newline code “LF”. You should use “Internet Explorer” or “text editor” that supports “LF” to open this file.

7. Rotate the selector knob to scroll the SMART data on the liquid crystal display, and SMART ID is displayed on the liquid crystal display.



(Displayed characters/values in the above figure are example)

8. Check the SMART raw value of following SMART IDs is displayed on the liquid crystal display.

ID	Name
5	Reallocated sector count
196	Reallocated event count
197	Current pending sector count
198	Offline scan uncorrectable sector count

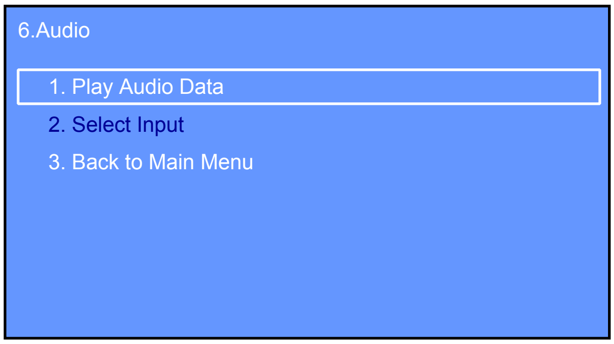
9. If at least one of these ID’s SMART raw value is not “0”, SMART data check is NG.

Note 4: When the SMART data check is NG, replace the hard disk drive.

4-6. Audio

Procedure:

1. Enter the service DIAG mode.
   (Refer to “1. Entering Method of The Service DIAG Mode” on page 25)
2. Rotate the selector knob to select the “6. Audio”.
3. Press the [ENTER] key to enter the “6. Audio” mode.



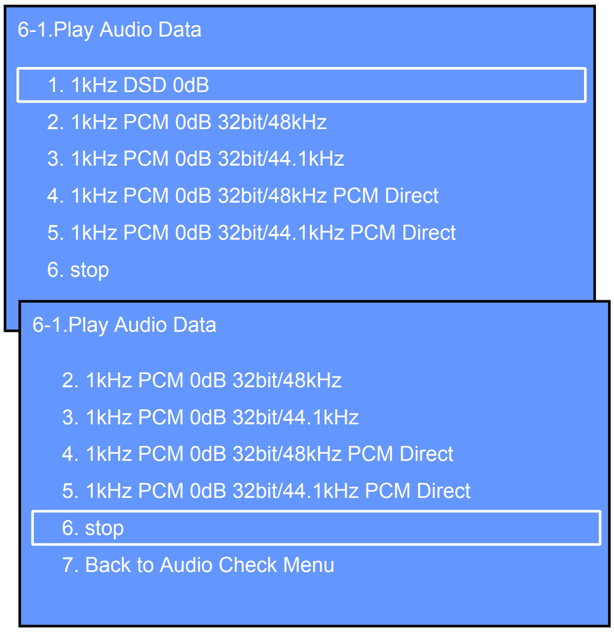
Note: “2. Select Input” cannot use in this unit.

4-6-1. Play Audio Data

It can play back various audio data.

Procedure:

1. Enter the “6. Audio” mode.
   (Refer to “4-6. Audio”)
2. Rotate the selector knob to select the “1. Play Audio Data”.
3. Press the [ENTER] key to enter the “1. Play Audio Data” mode.



4. Select the audio data by rotating the selector knob and press the [ENTER] key.

Note: “1. kHz DSD 0dB” does not play back continuously, and stops once after a few seconds.

5. The selected audio data is played back.
6. To stop the audio data, rotate the selector knob to select the “6. stop” and press the [ENTER] key.

4-7. WiFi

Procedure:

1. Set the SSID and KEY of the connecting access point with "SONY_HAP_TEST".
2. Enter the service DIAG mode.
   (Refer to "1. Entering Method of The Service DIAG Mode" on page 25)
3. Rotate the selector knob to select the "7. WiFi".
4. Press the [ENTER] key to enter the "7. WiFi" mode.

7.WiFi
1. Network/Module Info
2. Ping/RSSI
3. Back to Main Menu

4-7-1. Network/Module Info

It can display the wireless network and module information.

Procedure:

1. Enter the "7. WiFi" mode.
   (Refer to "4-7. WiFi")
2. Rotate the selector knob to select the "1. Network/module Info".
3. Press the [ENTER] key to enter the "1. Network/module Info" mode.

Note: "1. IP Address", "2. Subnet Mask" and "3. Default Gateway" supports only DHCP.

7-1.Network/Module Info	
1. IP Address	172.17.100.27
2. Subnet Mask	255.255.0.0
3. Default Gateway	192.168.10.254
4. SSID	SONY_HAP_TEST
5. KEY	SONY_HAP_TEST
6. Module S/N	201305310100116

7-1.Network/Module Info	
3. Default Gateway	172.17.0.200
4. SSID	SONY_HAP_TEST
5. KEY	SONY_HAP_TEST
6. Module S/N	201305310100116
7. HW ver.	19
8. Back to WiFi Menu	

(Displayed characters/values in the above figure are example)

4-7-2. Ping/RSSI

It can display the results indicating whether or not the gateway responded to the PING and RSSI value.

Note: This mode supports only DHCP.

Procedure:

1. Enter the "7. WiFi" mode.
   (Refer to "4-7. WiFi")
2. Rotate the selector knob to select the "1. Network/module Info".
3. Press the [ENTER] key to enter the "1. Network/module Info" mode.
4. The results indicating whether or not the gateway responded to the PING and RSSI value is displayed on the liquid crystal display.
   (When the gateway does not respond to the PING, "Error!! Can't get gateway-ip" is displayed on the liquid crystal display)

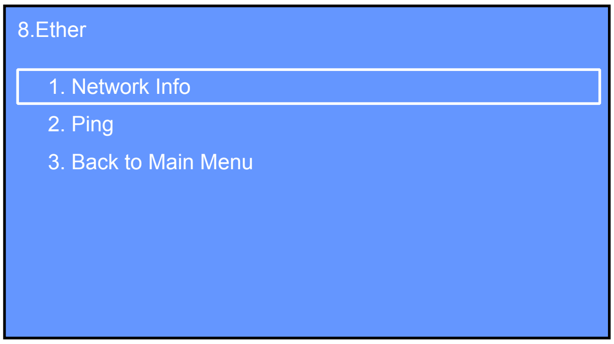
7-2.Ping/RSSI
Antenna 1
Ping to Gateway (192.168.10.254) OK
RSSI : -70
Antenna 2
Ping to Gateway (192.168.10.254) OK
RSSI : -59
Back to WiFi Menu

(Displayed characters/values in the above figure are example)

4-8. Ether

Procedure:

1. Connect the main unit to the router with the network LAN cable.
2. Enter the service DIAG mode.
   (Refer to "1. Entering Method of The Service DIAG Mode" on page 25)
3. Rotate the selector knob to select the "8. Ether".
4. Press the [ENTER] key to enter the "8. Ether" mode.



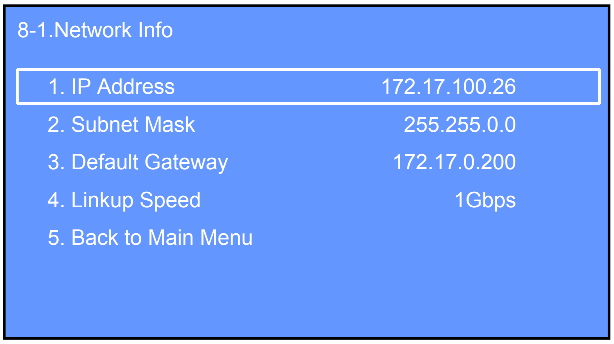
4-8-1. Network Info

It can display the wired network information.

Procedure:

1. Enter the "8. Ether" mode.
   (Refer to "4-8. Ether")
2. Rotate the selector knob to select the "1. Network Info".
3. Press the [ENTER] key to enter the "1. Network Info" mode.

Note: "1. IP Address", "2. Subnet Mask" and "3. Default Gateway" supports only DHCP.



(Displayed characters/values in the above figure are example)

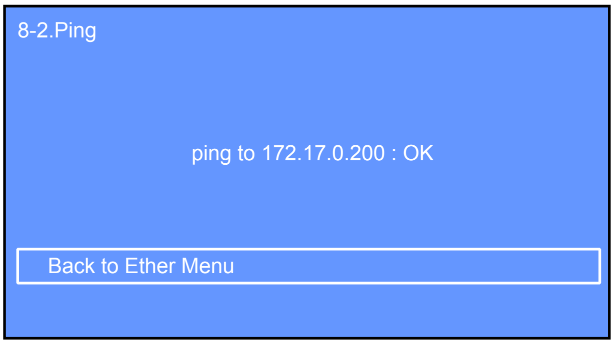
4-8-2. Ping

It can display the results indicating whether or not the gateway responded to the PING.

Note: This mode supports only DHCP.

Procedure:

1. Enter the "8. Ether" mode.
   (Refer to "4-8. Ether")
2. Rotate the selector knob to select the "2. Ping".
3. Press the [ENTER] key to enter the "2. Ping" mode.
4. The results indicating whether or not the gateway responded to the PING is displayed on the liquid crystal display.
   (When the gateway does not respond to the PING, "Error!! Can't get gateway-ip" is displayed on the liquid crystal display)

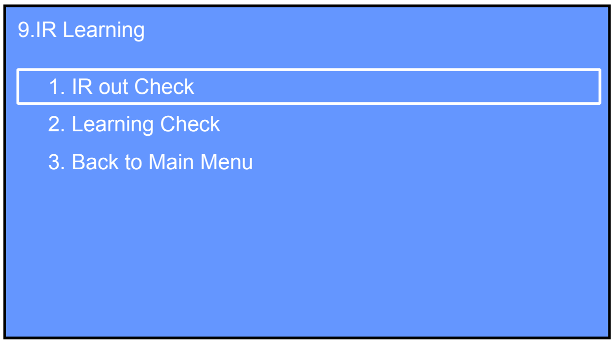


(Displayed characters/values in the above figure are example)

4-9. IR Learning

Procedure:

1. Enter the service DIAG mode.
   (Refer to “1. Entering Method of The Service DIAG Mode” on page 25)
2. Rotate the selector knob to select the “9. IR Learning”.
3. Press the [ENTER] key to enter the “9. IR Learning” mode.

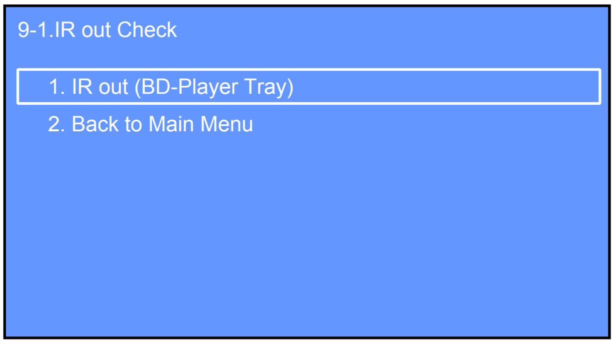


4-9-1. IR out Check

It can check the IR code output.

Procedure:

1. Prepare the BD player (model name example: BDP-S350), and install it so that the remote control receiver faces the front side of the main unit.
2. Enter the “9. IR Learning” mode.
   (Refer to “4-9. IR Learning”)
3. Rotate the selector knob to select the “1. IR out Check”.
4. Press the [ENTER] key to enter the “1. IR out Check” mode.



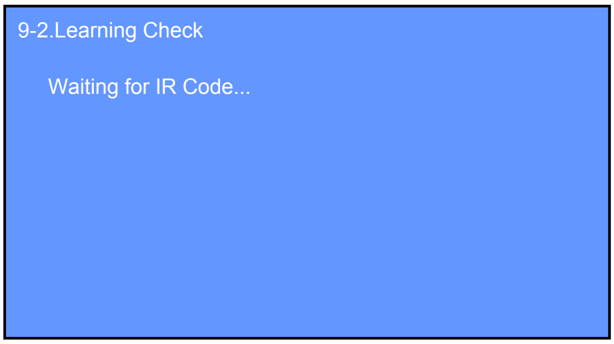
5. Rotate the selector knob to select the “1. IR out (BD-Player Tray)” and press the [ENTER] key.
6. BD-player tray open/close code is outputted once, tray of BD-player is opened.

4-9-2. Learning Check

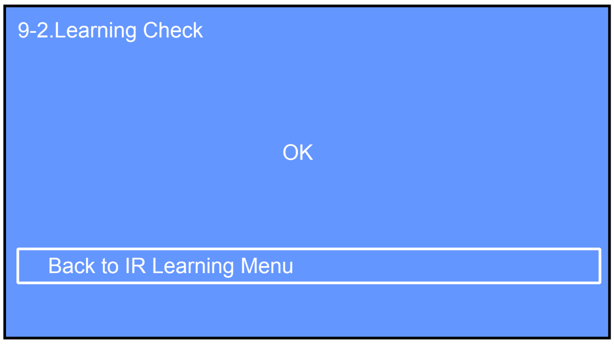
It can check the learning function.

Procedure:

1. Enter the “9. IR Learning” mode.
   (Refer to “4-9. IR Learning”)
2. Rotate the selector knob to select the “2. Learning Check”.
3. Press the [ENTER] key to enter the “2. Learning Check” mode.



4. Send the remote control code that you want memorized to the remote control receiver of the main unit.
   (Hold the remote commander 2 – 3 cm away from the remote control receiver, point the remote commander toward the remote control receiver, and then press and hold the remote commander key)
5. The result of remote control code receive “OK” or “NG” is displayed on the liquid crystal display.



4-10. Quit

Rotate the selector knob to select the “10. Quit” and press the [ENTER] key, service DIAG mode is released.

SECTION 4
ELECTRICAL CHECK

HDD CHECK

It can check the result of hard disk drive test.

1. Operation of The SMART Long Test

The SMART long test is mode to perform the overall test of hard disk drive.
When checking the hard disk drive, perform the SMART long test.

Procedure:

1. Enter the service DIAG mode.
   (Refer to "1. Entering method of the service DIAG mode" on page 25)
2. Rotate the selector knob to select the "5. HDD".
3. Press the [ENTER] key to enter the "5. HDD" mode.

```
5.HDD
 1. Model                TOSHIBA MQ01ABD100
 2. Serial                       5307C10GT
 3. Firmware Rev.                   AM001Y
 4. SMART Data Read(Need USB Memory)
 5. SMART Short Test (2min)
 6. SMART Long Test (2-4 hours)
```

```
5.HDD
 2. Serial                       5307C10GT
 3. Firmware Rev.                   AM001Y
 4. SMART Data Read(Need USB Memory)
 5. SMART Short Test (2min)
 6. SMART Long Test (2-4 hours)
 7. Back to Main Menu
```

(Displayed characters/values in the above figure are example)

4. Rotate the selector knob to select the "6. SMART Long Test (2-4 hours)".
5. Press the [ENTER] key to enter the "6. SMART Long Test (2-4 hours)" mode.

Note 1: Two keys of the [HOME] and [BACK] is invalid during SMART long test.

```
5-6.SMART Long Test
    SMART Long Test is running...
    This test needs about 2-4 hours.
    50% of test remaing.

    Abort
```

(Displayed characters/values in the above figure are example)

6. Press the [ENTER] key to abort the SMART long test.
7. When the result of test is OK, following screen is displayed on the liquid crystal display.

```
5-6.SMART Long Test
    SMART Test is Succeded.

    Back to HDD Menu
```

Note 2: When "SMART Test is Failed." is displayed on the liquid crystal display, refer to "3. Checking Method of The SMART Test Result" on page 34.

2. Operation of The SMART Short Test

The SMART short test is mode to perform the simple test of hard disk drive in a short time.

Procedure:

1. Enter the "5. HDD" mode.
   (Refer to "1. Operation of The SMART Long Test")
2. Rotate the selector knob to select the "5. SMART Short Test (2min)".
3. Press the [ENTER] key to enter the "5. SMART Short Test (2min)" mode.

Note 1: Two keys of the [HOME] and [BACK] is invalid during SMART short test.

```
5-5.SMART Short Test
    SMART Short Test is running...
    This test needs about 2 minutes.
    50% of test remaing.

    Abort
```

(Displayed characters/values in the above figure are example)

4. Press the [ENTER] key to abort the SMART short test.
5. When the result of test is OK, following screen is displayed on the liquid crystal display.

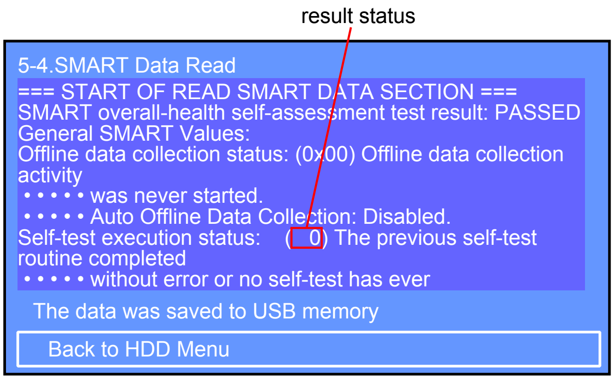
```
5-5.SMART Short Test
    SMART Test is Succeded.

    Back to HDD Menu
```

Note 2: When "SMART Test is Failed." is displayed on the liquid crystal display, refer to "3. Checking Method of The SMART Test Result" on page 34.

3. Checking Method of The SMART Test Result

Procedure:

1. After the SMART test is completed, press the [ENTER] key or [BACK] key to return to the “5. HDD” menu.
2. Rotate the selector knob to select the “4. SMART Data Read(Need USB Memory)” and press the [ENTER] key.
3. Rotate the selector knob to scroll the SMART data on the liquid crystal display, and “START OF READ SMART DATA SECTION” is displayed on the liquid crystal display.



(Displayed characters/values in the above figure are example)

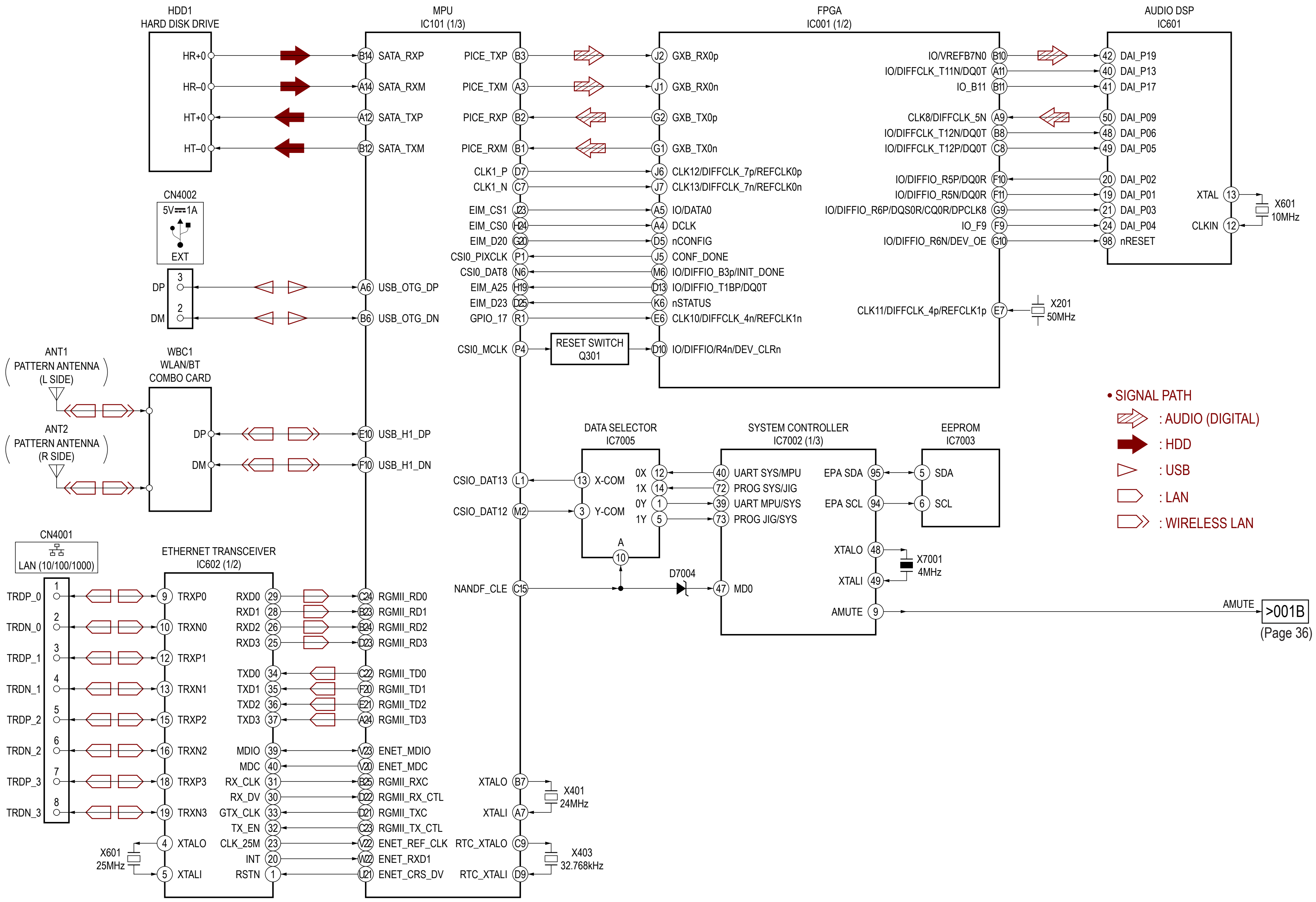
4. Read the result status value is displayed on the liquid crystal display, and check the error contents at the following table.

- Result status = XXh

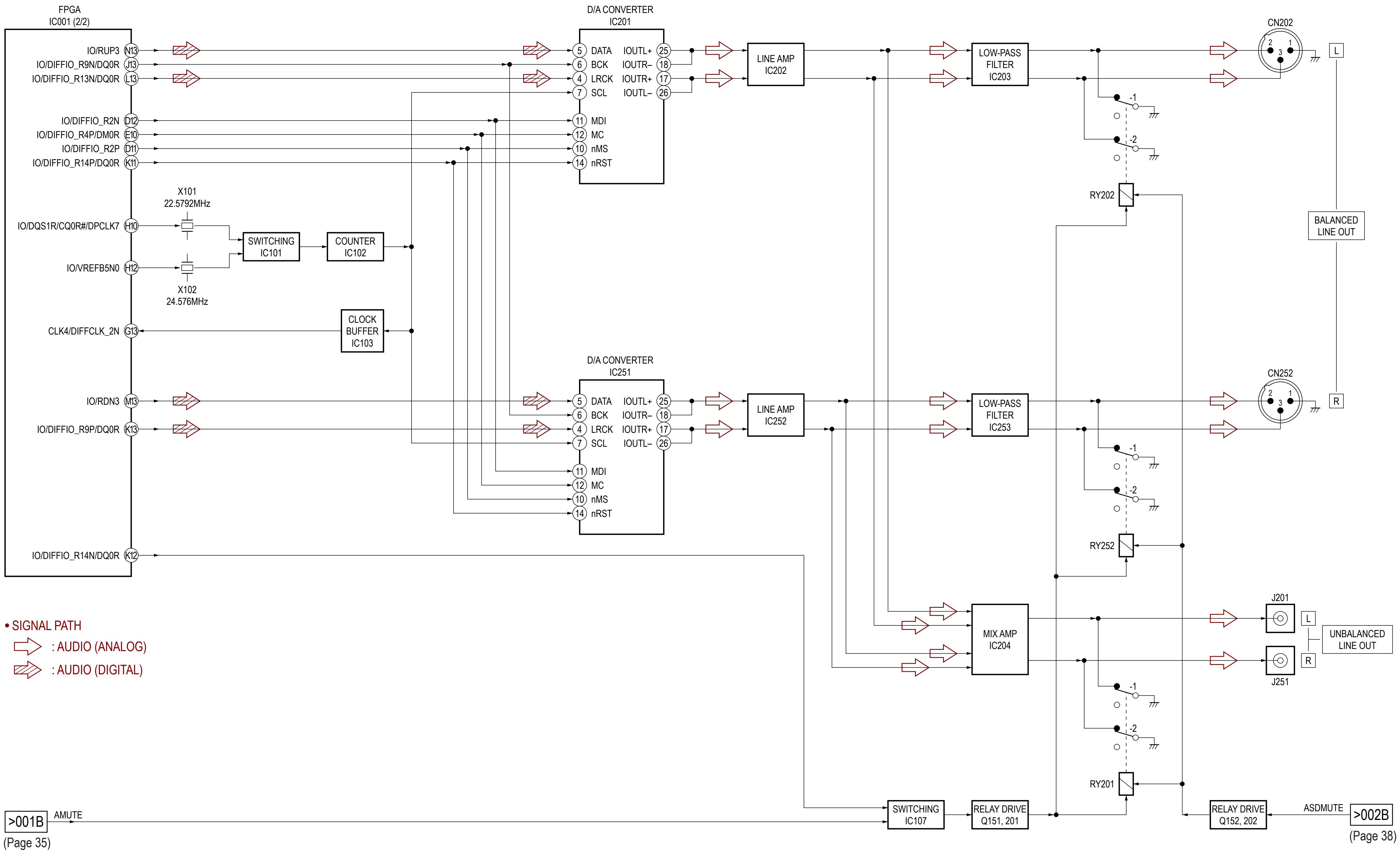
Result status	Description
0Xh	The previous self-test routine completed without error or no self-test has ever been run
1Xh	The self-test routine was aborted by the host
2Xh	The self-test routine was interrupted by the host with a hardware or software reset
3Xh	A fatal error or unknown test error occurred while the device was executing its self-test routine and the device was unable to complete the self-test routine
4Xh	The previous self-test completed having a test element that failed and the test element that failed is not known.
5Xh	The previous self-test completed having the electrical element of the test failed
6Xh	The previous self-test completed having the servo and/or seek test element of the test failed
7Xh	The previous self-test completed having the read element of the test failed
8Xh	The previous self-test completed having a test element that failed and the device is suspected of having handling damage
9Xh – EXh	Reserved
FXh	Self-test routine in progress

SECTION 5
DIAGRAMS

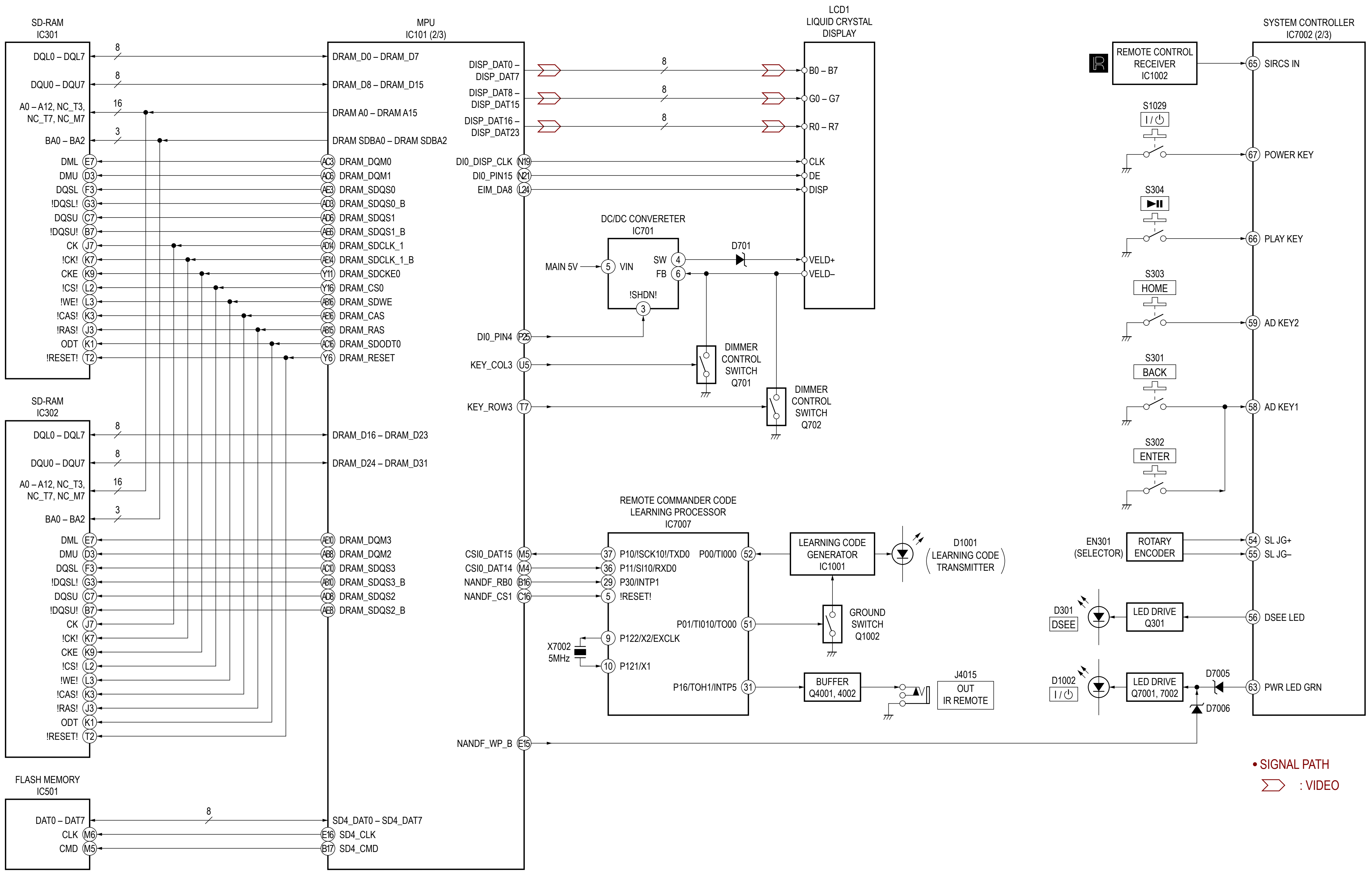
5-1. BLOCK DIAGRAM - HDD/USB/LAN/FPGA/DSP Section -



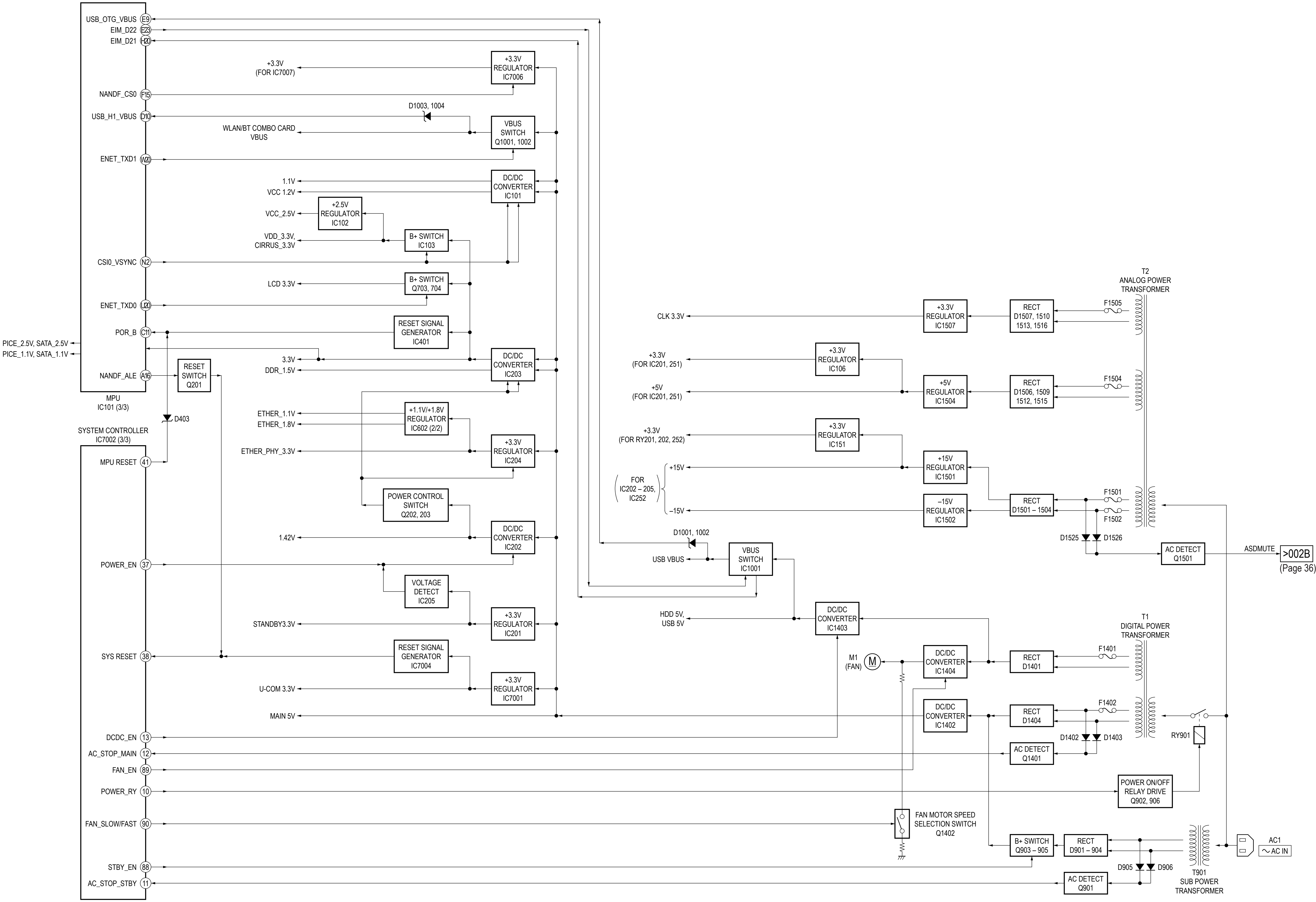
5-2. BLOCK DIAGRAM - AUDIO Section -



5-3. BLOCK DIAGRAM - MEMORY/PANEL Section -



5-4. BLOCK DIAGRAM - POWER SUPPLY Section -



THIS NOTE IS COMMON FOR PRINTED WIRING BOARDS AND SCHEMATIC DIAGRAMS.
(In addition to this, the necessary note is printed in each block.)

For Printed Wiring Boards.

Note:

- ○——— : Parts extracted from the component side.
- ——— : Parts extracted from the conductor side.
- △ : Internal component.
- ▬ : Pattern from the side which enables seeing.
  (The other layers' patterns are not indicated.)

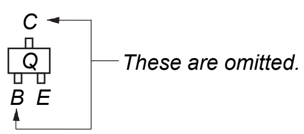
Caution:

Pattern face side: (Conductor Side)	Parts on the pattern face side seen from the pattern face are indicated.
Parts face side: (Component Side)	Parts on the parts face side seen from the parts face are indicated.

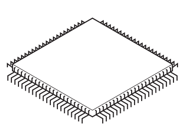
Caution:

Pattern face side: (SIDE B)	Parts on the pattern face side seen from the pattern face are indicated.
Parts face side: (SIDE A)	Parts on the parts face side seen from the parts face are indicated.

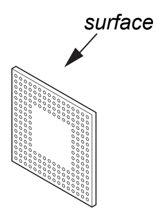
- FPGA DSP and MAIN boards are multi-layer printed board. However, the patterns of intermediate layers have not been included in diagrams.
- Indication of transistor.



- Lead layouts



Lead layout of conventional IC



CSP (Chip Size Package)

- Abbreviation
  CND : Canadian model

Note: When the complete MAIN board is replaced, refer to "NOTE OF REPLACING THE COMPLETE MAIN BOARD" and "CHECKING METHOD OF NETWORK CONNECTION" on page 5.

For Schematic Diagrams.

Note:

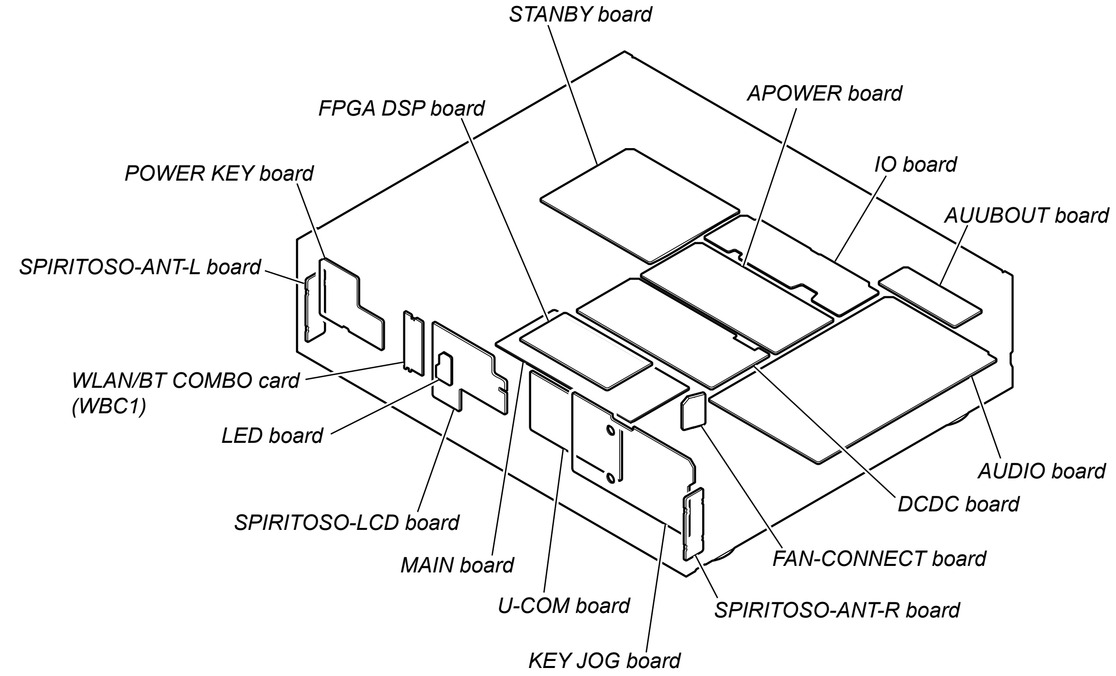
- All capacitors are in μF unless otherwise noted. (p: pF) 50 WV or less are not indicated except for electrolytics and tantalums.
- All resistors are in Ω and 1/4 W or less unless otherwise specified.
- △ : Internal component.
- : Nonflammable resistor.
- ▭ : Panel designation.

Note:	**Note:**
The components identified by mark ⚠ or dotted line with mark ⚠ are critical for safety. Replace only with part number specified.	Les composants identifiés par une marque ⚠ sont critiques pour la sécurité. Ne les remplacer que par une pièce portant le numéro spécifié.

- ▬▬▬ : B+ Line.
- ▬ ▬ ▬ : B– Line.
- Voltages and waveforms are dc with respect to ground under no-signal conditions.
  no mark: POWER ON
- Voltages are taken with VOM (Input impedance 10 MΩ).
  Voltage variations may be noted due to normal production tolerances.
- Waveforms are taken with a oscilloscope.
  Voltage variations may be noted due to normal production tolerances.
- Circled numbers refer to waveforms.
- Signal path.
  ⇨ : AUDIO (ANALOG)
  ⇨ : AUDIO (DIGITAL)
  ➡ : HDD
  ▷ : USB
  ▭ : LAN
  ▭» : WIRELESS LAN
  ⪢ : VIDEO
- The voltage and waveform of CSP (chip size package) cannot be measured, because its lead layout is different from that of conventional IC.

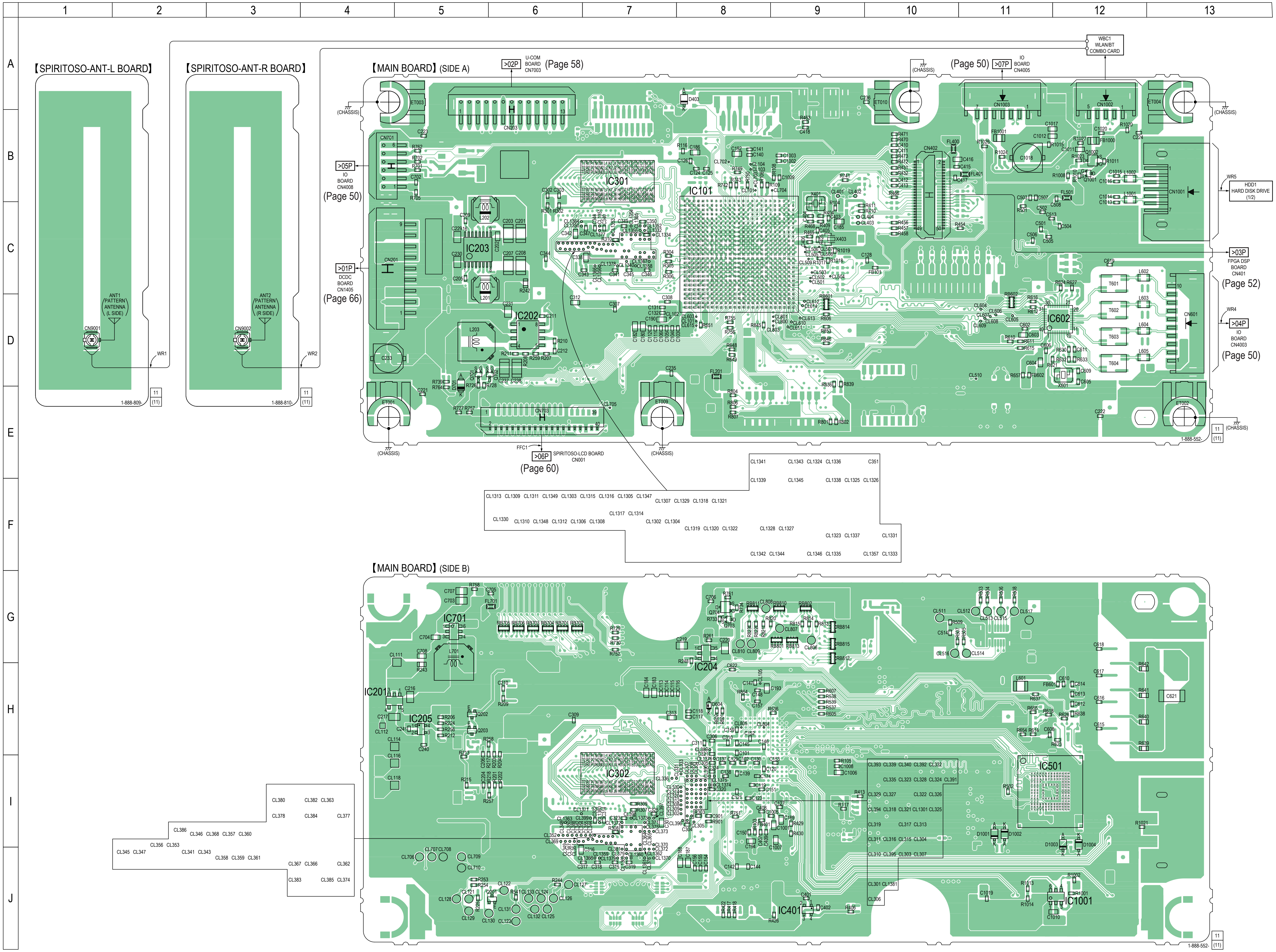
Note: When the complete MAIN board is replaced, refer to "NOTE OF REPLACING THE COMPLETE MAIN BOARD" and "CHECKING METHOD OF NETWORK CONNECTION" on page 5.

- **Circuit Boards Location**



5-5. PRINTED WIRING BOARDS - MAIN Section -

• See page 39 for Circuit Boards Location. • : Uses unleaded solder.



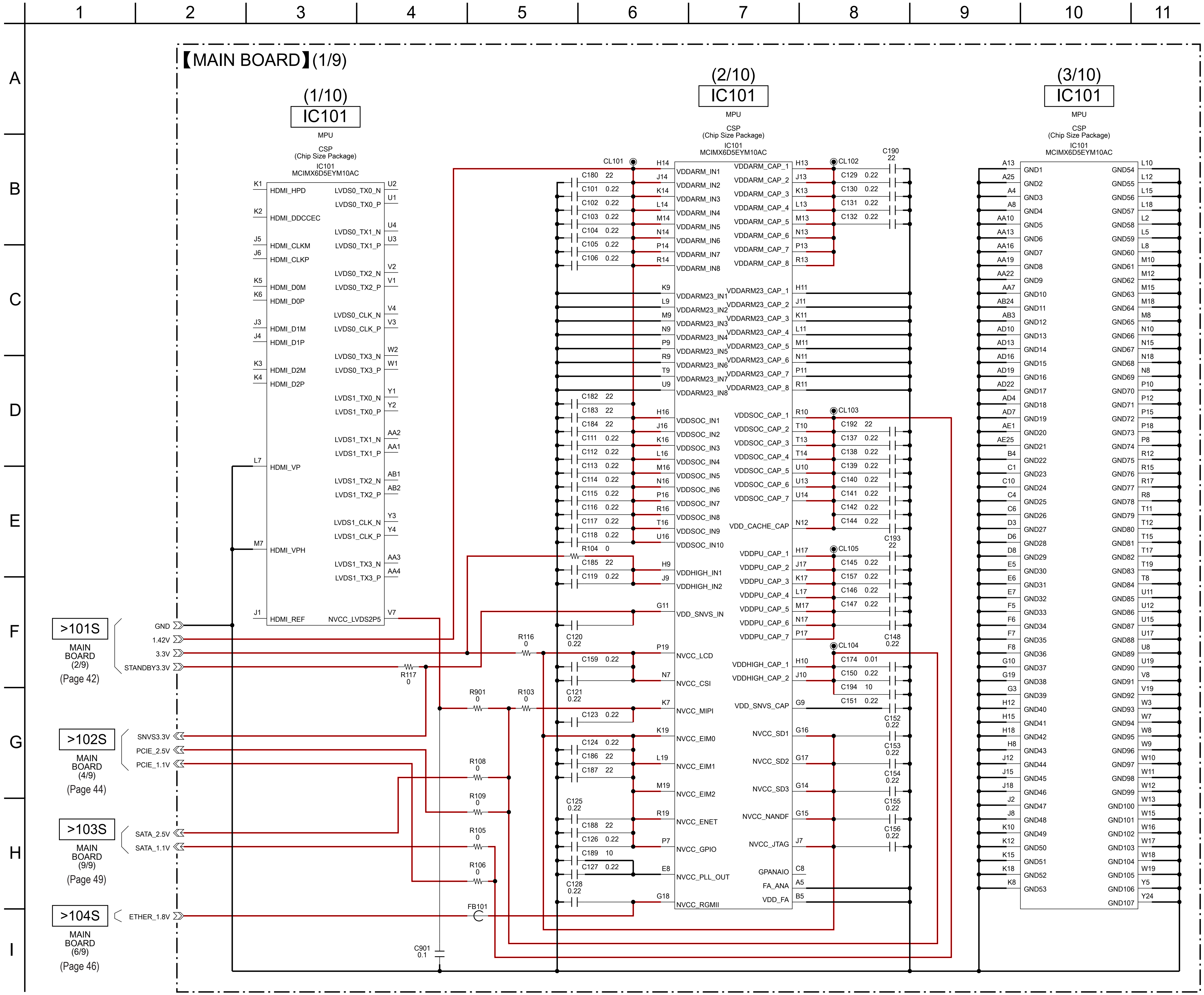
Note 1: The SPIRITOSO-ANT-L and SPIRITOSO-ANT-R boards are not covered in the servicing.

Note 2: IC101 on the MAIN board cannot exchange with single. When this part is damaged, exchange the complete mounted board.

Note 3: When the WLAN/BT COMBO card (Ref. No. WBC1) and coaxial (U.FL connector) cables (Ref. No. WR1, WR2) are replaced, refer to “CHECKING METHOD OF NETWORK CONNECTION” on page 5.

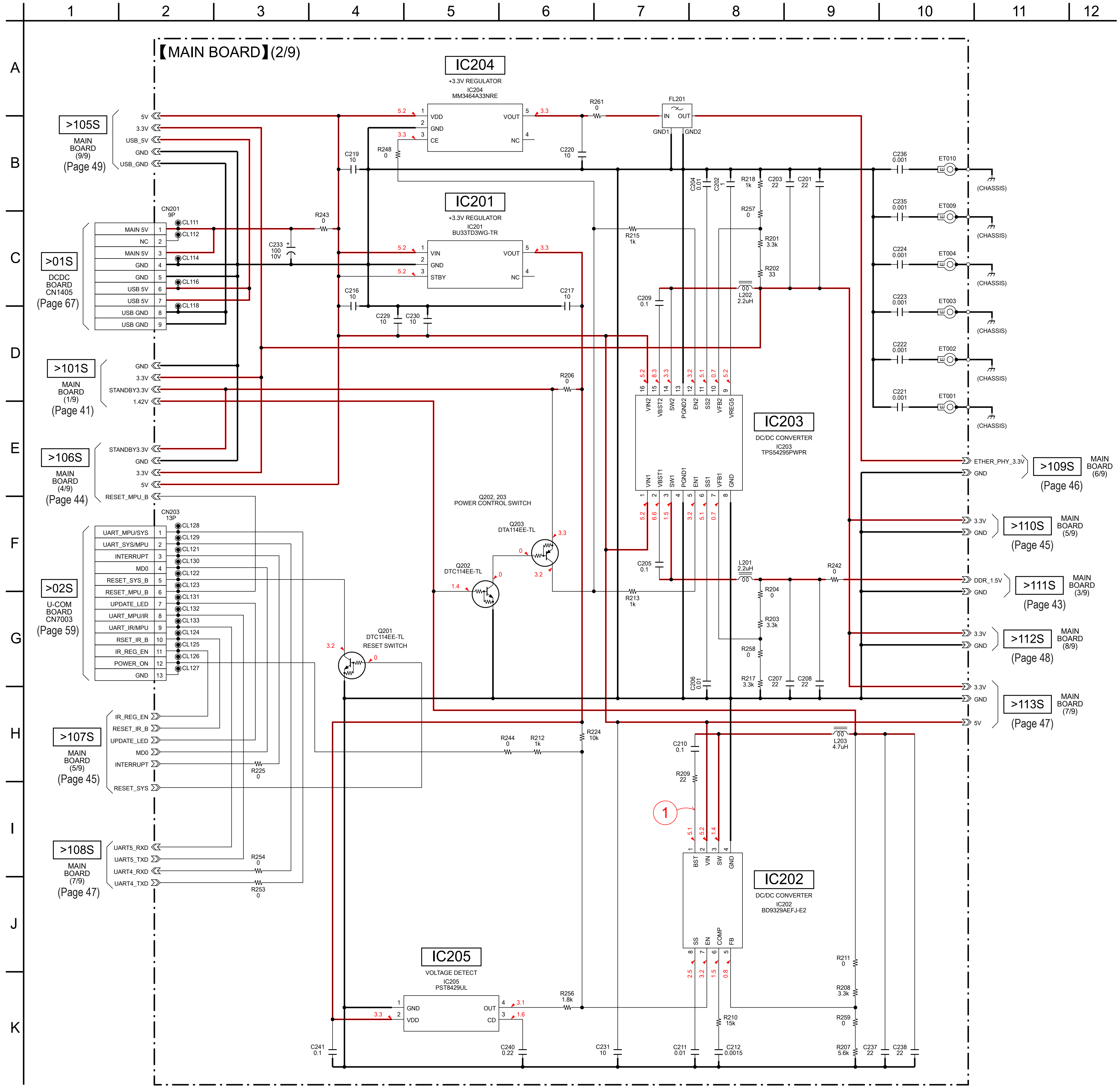
Note 4: When the hard disk drive (Ref. No. HDD1) is replaced, refer to “NOTE OF REPLACING THE HARD DISK DRIVE (Ref. No. HDD1)” on page 4.

5-6. SCHEMATIC DIAGRAM - MAIN Section (1/9) - • See page 75 for IC Pin Function Description.

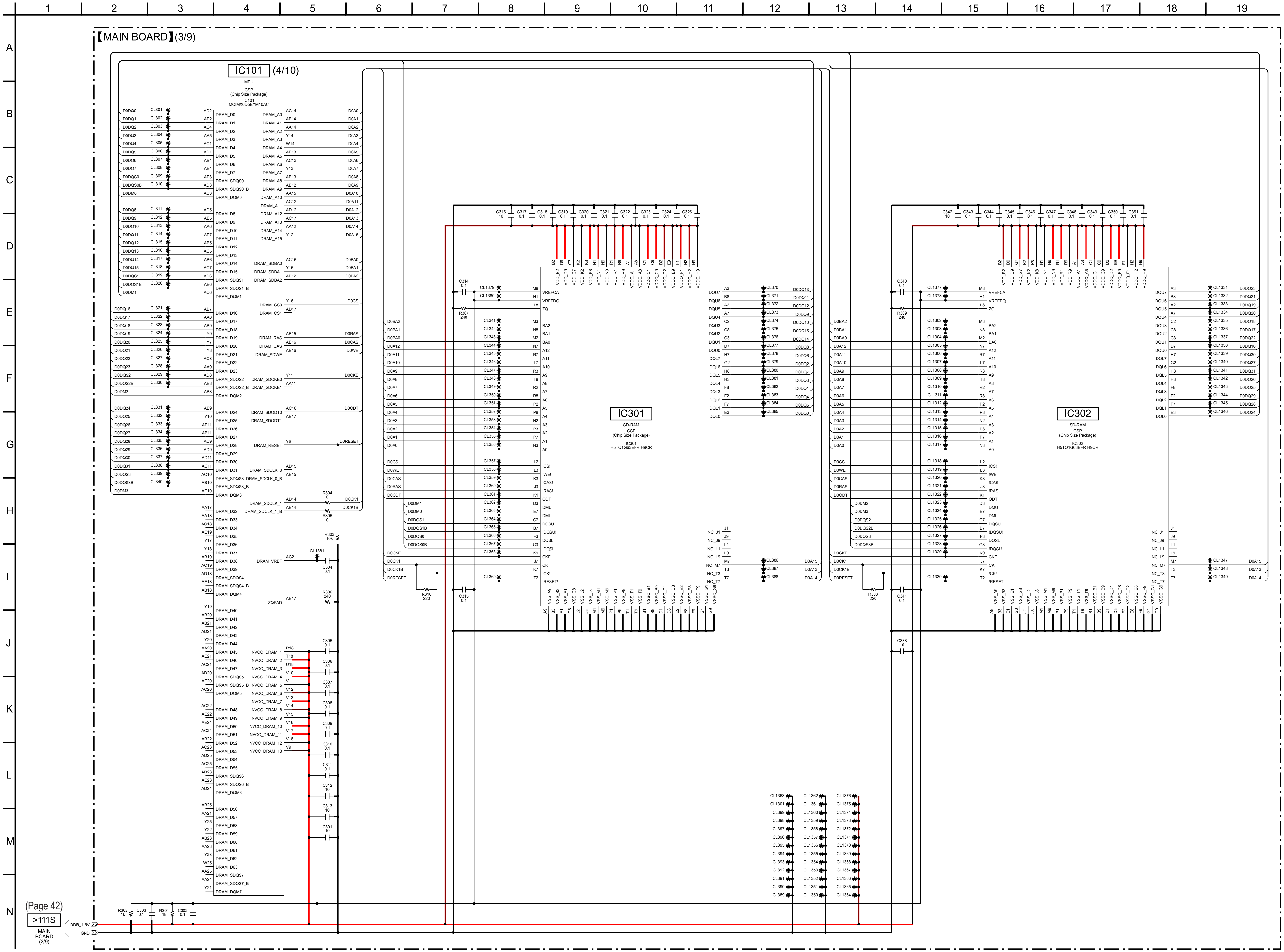


Note: IC101 on the MAIN board cannot exchange with single. When this part is damaged, exchange the complete mounted board.

5-7. SCHEMATIC DIAGRAM - MAIN Section (2/9) - • See page 70 for Waveforms. • See page 70 for IC Block Diagrams.

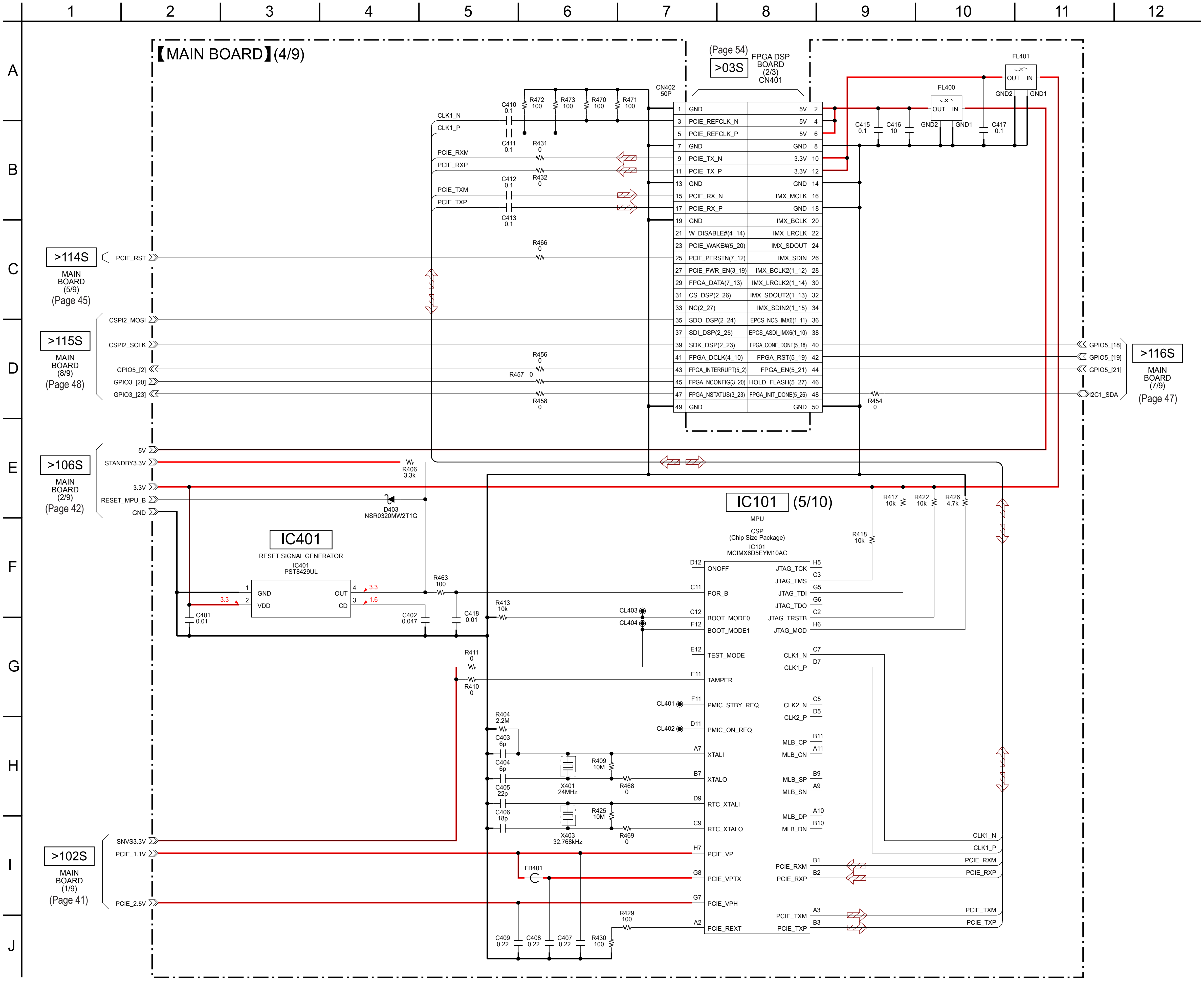


5-8. SCHEMATIC DIAGRAM - MAIN Section (3/9) - • See page 75 for IC Pin Function Description.



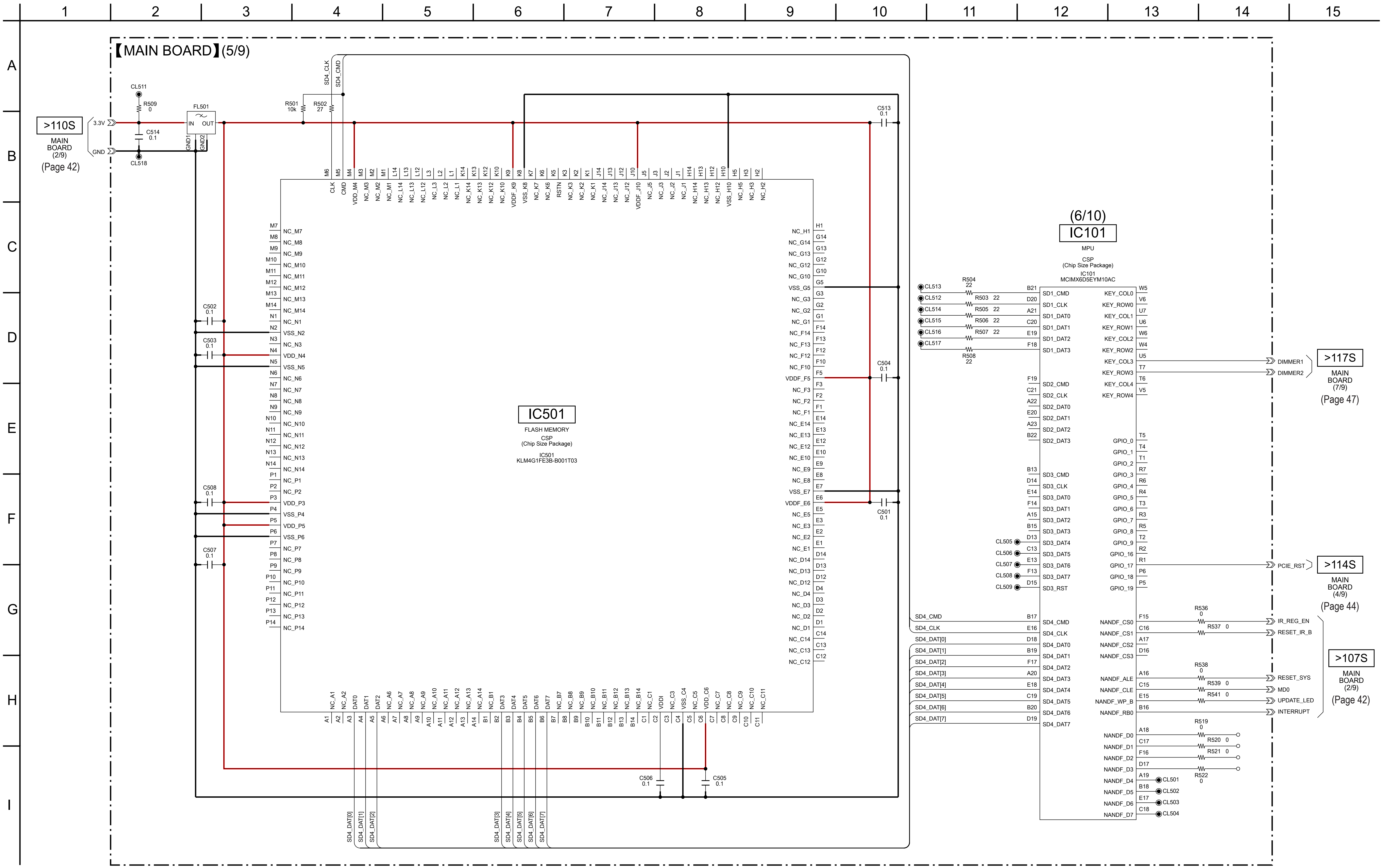
Note: IC101, IC301 and IC302 on the MAIN board cannot exchange with single. When these parts are damaged, exchange the complete mounted board.

5-9. SCHEMATIC DIAGRAM - MAIN Section (4/9) - See page 70 for IC Block Diagrams. • See page 75 for IC Pin Function Description.



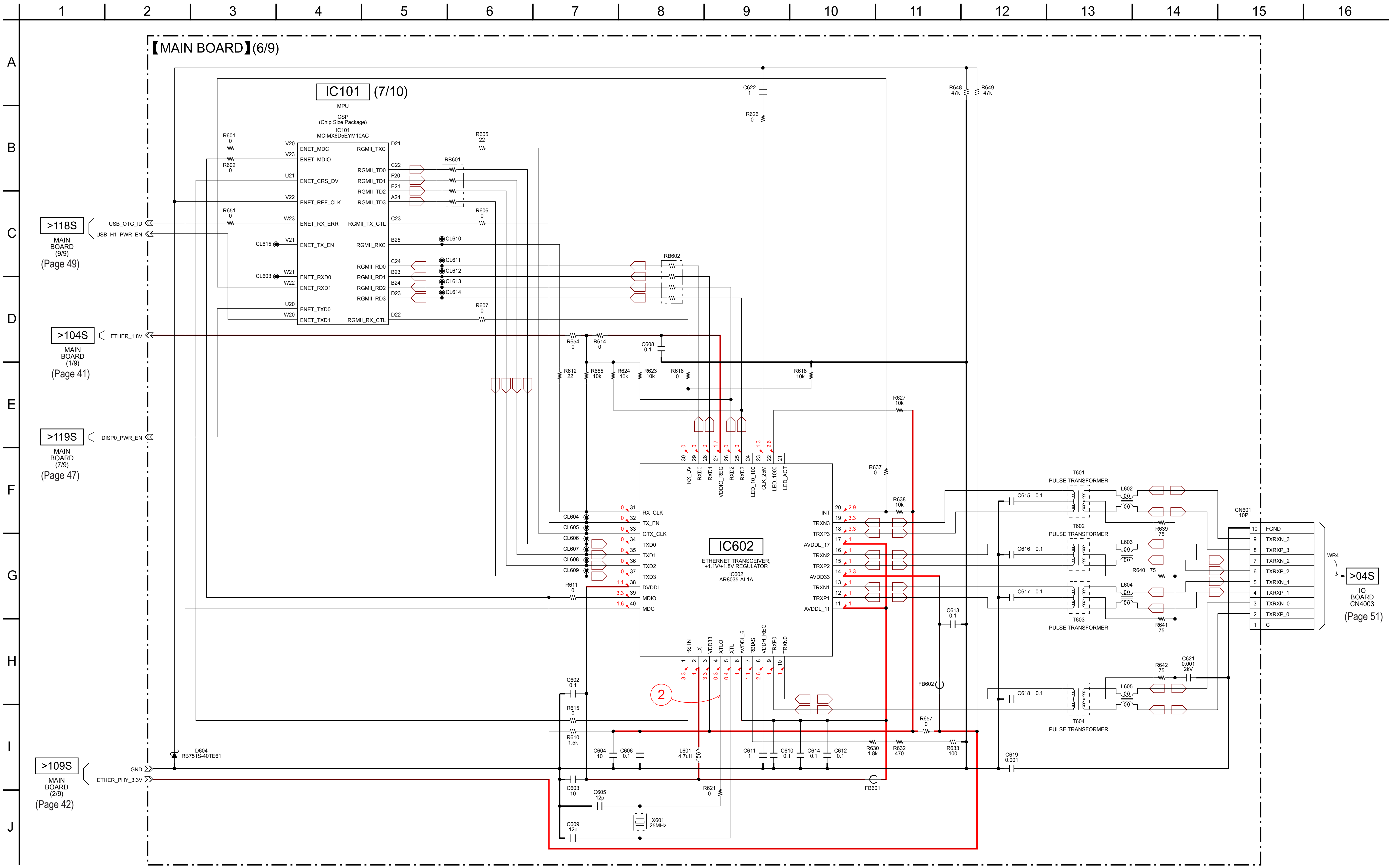
Note: IC101 on the MAIN board cannot exchange with single. When this part is damaged, exchange the complete mounted board.

5-10. SCHEMATIC DIAGRAM - MAIN Section (5/9) - • See page 75 for IC Pin Function Description.



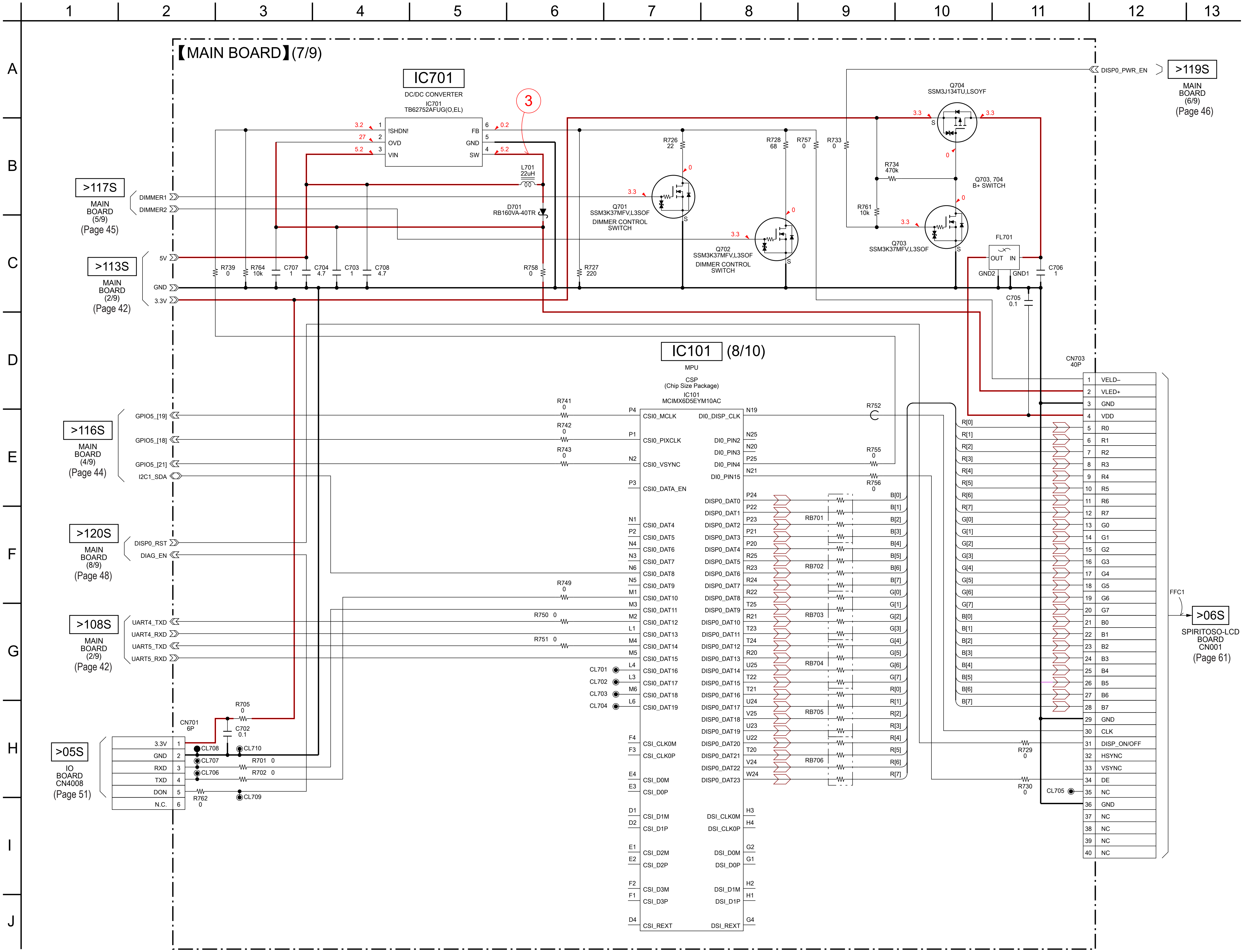
Note: IC101 and IC501 on the MAIN board cannot exchange with single. When these parts are damaged, exchange the complete mounted board.

5-11. SCHEMATIC DIAGRAM - MAIN Section (6/9) - • See page 70 for Waveforms. • See page 75 for IC Pin Function Description.



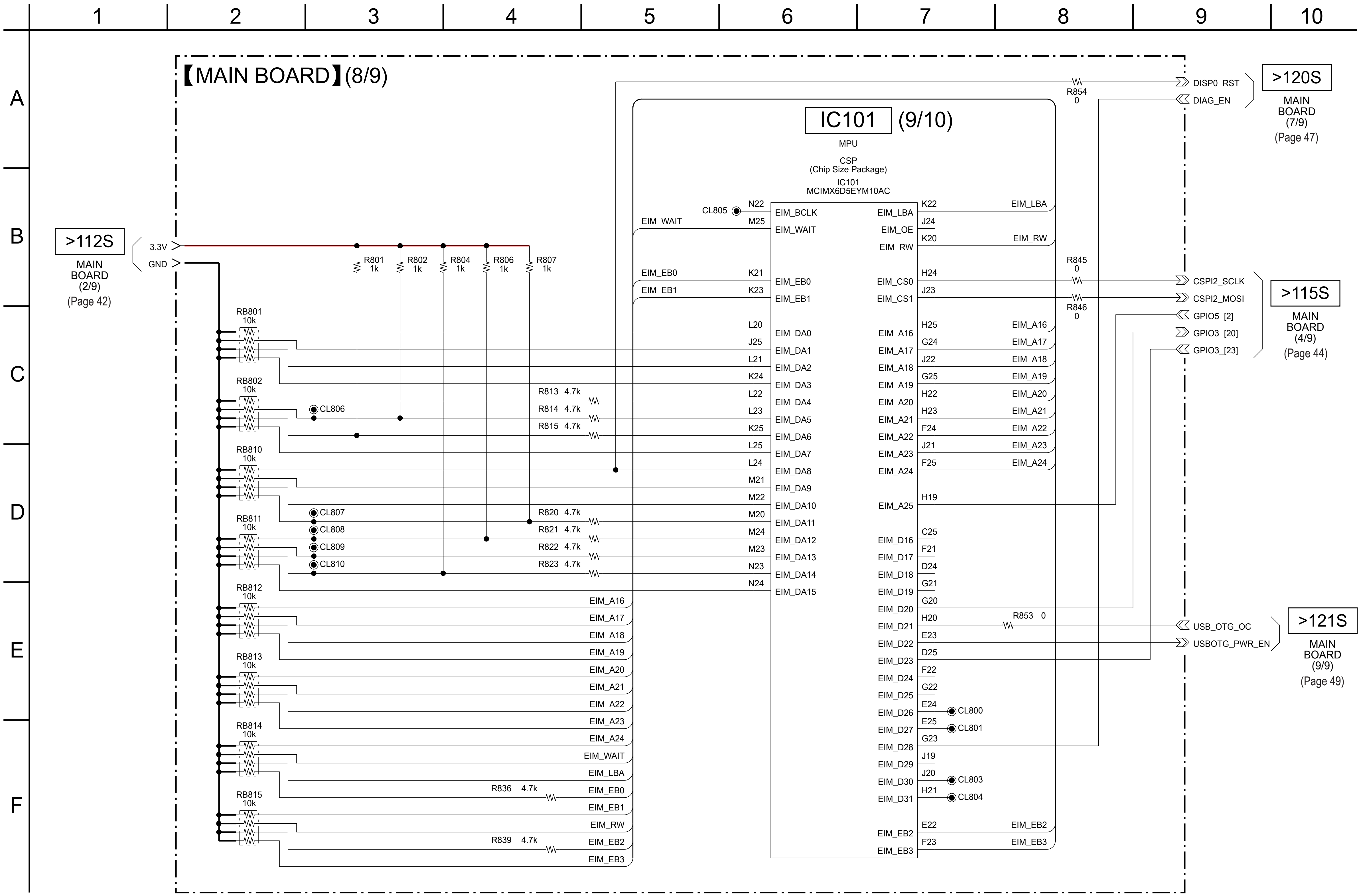
Note: IC101 and IC602 on the MAIN board cannot exchange with single. When these parts are damaged, exchange the complete mounted board.

5-12. SCHEMATIC DIAGRAM - MAIN Section (7/9) - • See page 70 for Waveforms. • See page 75 for IC Pin Function Description.



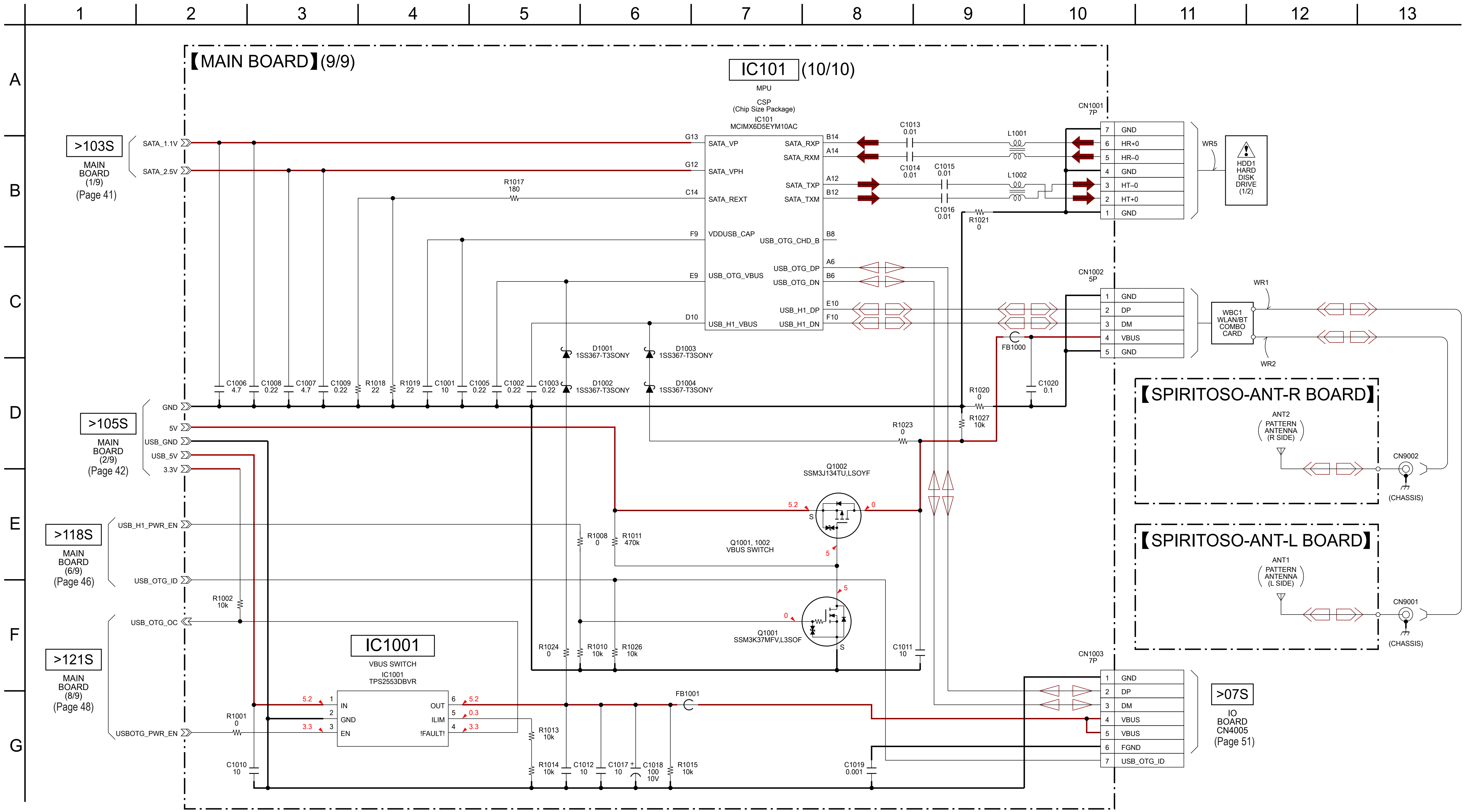
Note: IC101 on the MAIN board cannot exchange with single. When this part is damaged, exchange the complete mounted board.

5-13. SCHEMATIC DIAGRAM - MAIN Section (8/9) - • See page 75 for IC Pin Function Description.



Note: IC101 on the MAIN board cannot exchange with single. When this part is damaged, exchange the complete mounted board.

5-14. SCHEMATIC DIAGRAM - MAIN Section (9/9) - • See page 70 for IC Block Diagrams. • See page 75 for IC Pin Function Description.



Note 1: The SPIRITOSO-ANT-L and SPIRITOSO-ANT-R boards are not covered in the servicing.

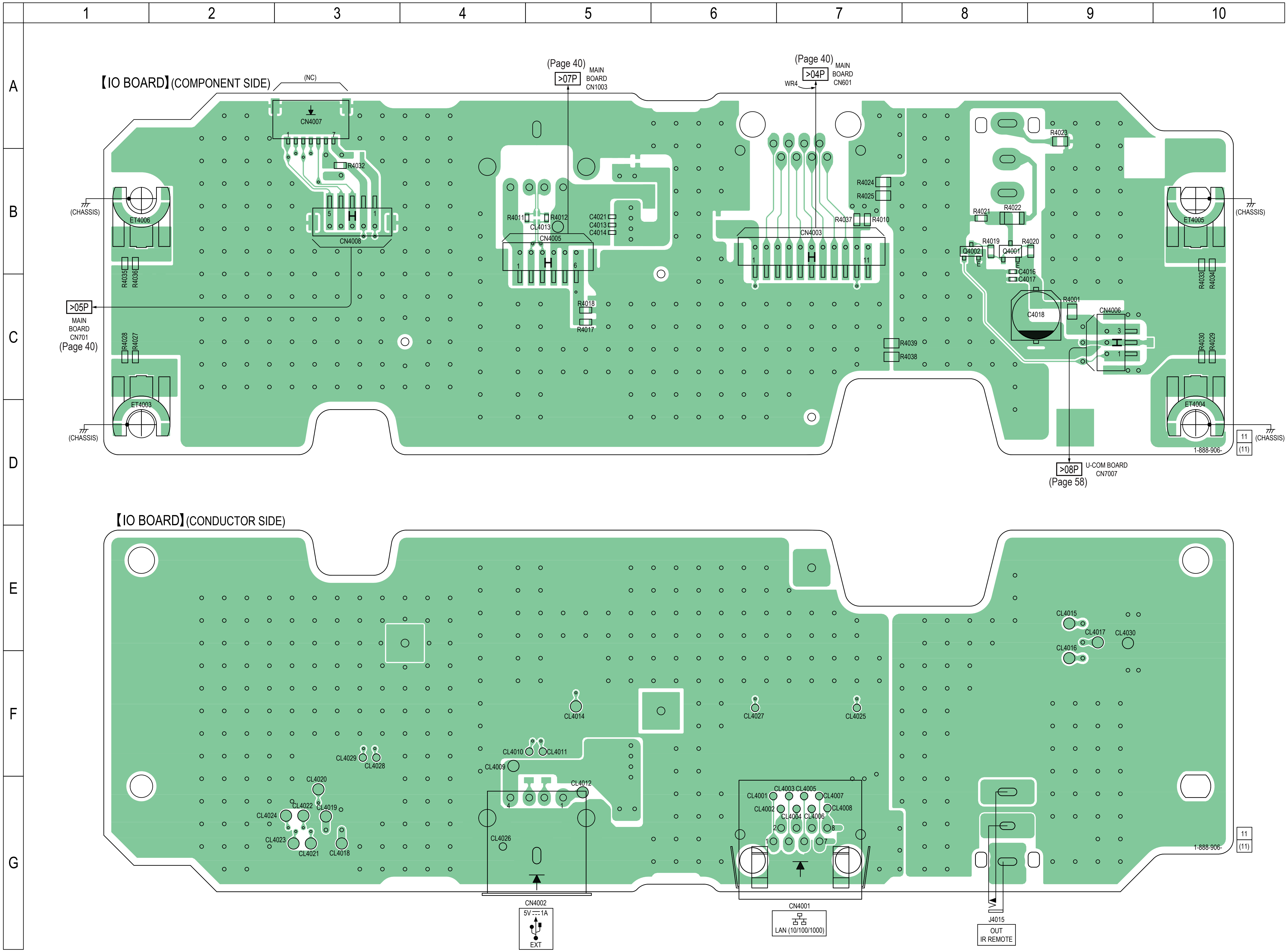
Note 2: IC101 on the MAIN board cannot exchange with single. When this part is damaged, exchange the complete mounted board.

Note 3: When the WLAN/BT COMBO card (Ref. No. WBC1) and coaxial (U.FL connector) cables (Ref. No. WR1, WR2) are replaced, refer to "CHECKING METHOD OF NETWORK CONNECTION" on page 5.

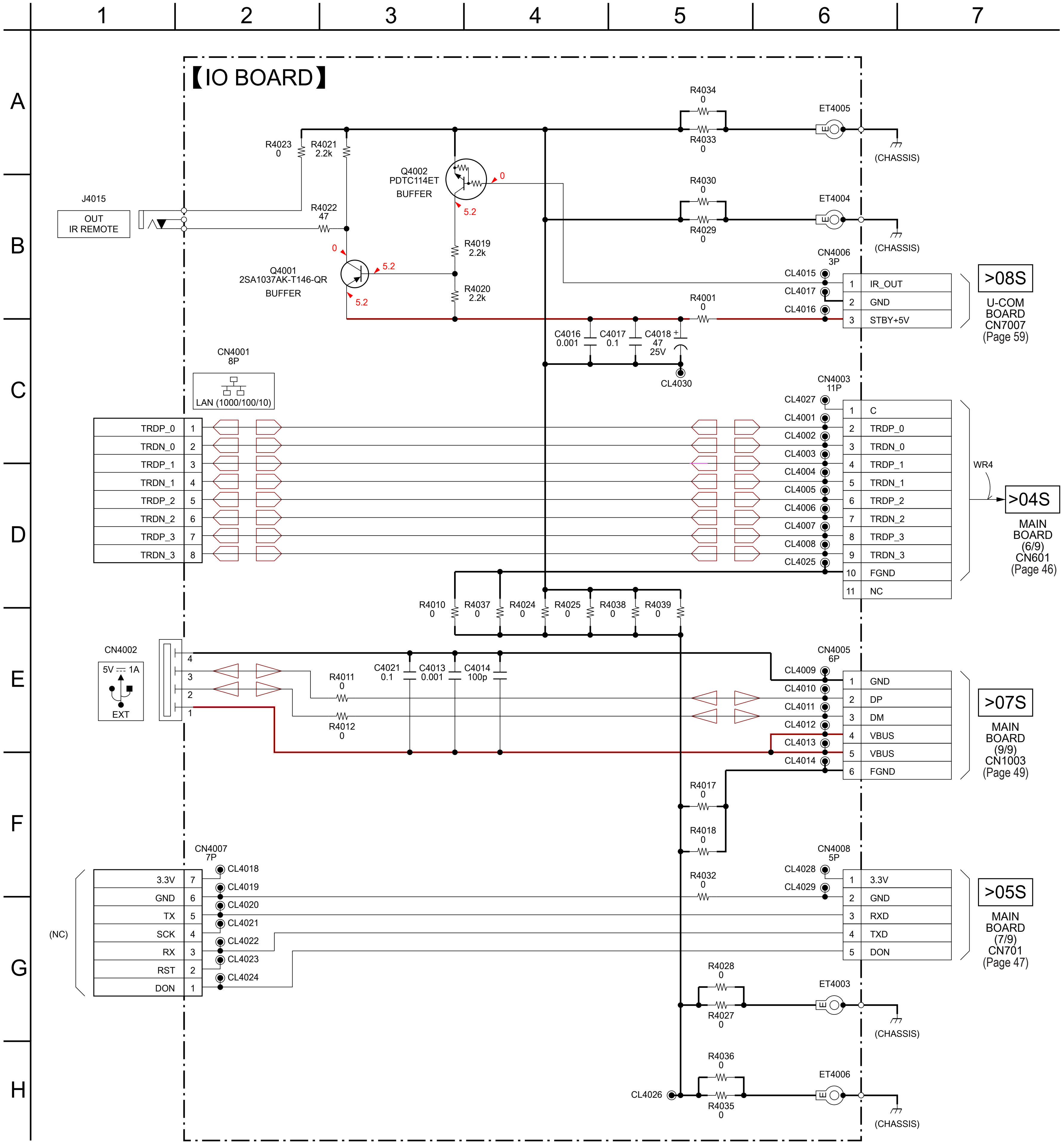
Note 4: When the hard disk drive (Ref. No. HDD1) is replaced, refer to "NOTE OF REPLACING THE HARD DISK DRIVE (Ref. No. HDD1)" on page 4.

5-15. PRINTED WIRING BOARD - IO Board -

• See page 39 for Circuit Boards Location. • : Uses unleaded solder.



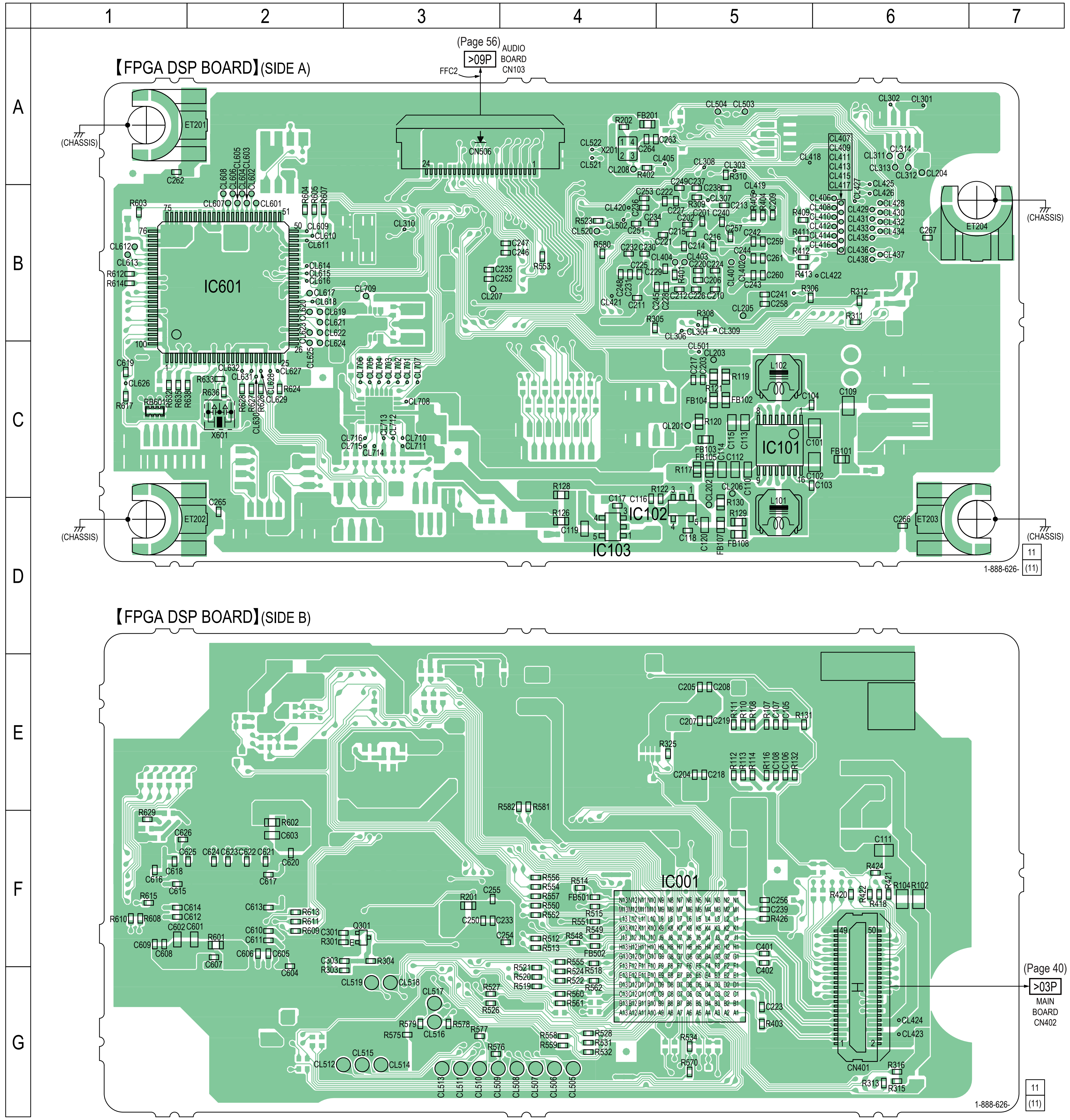
5-16. SCHEMATIC DIAGRAM - IO Board -



5-17. PRINTED WIRING BOARD - FPGA DSP Board -

• See page 39 for Circuit Boards Location.

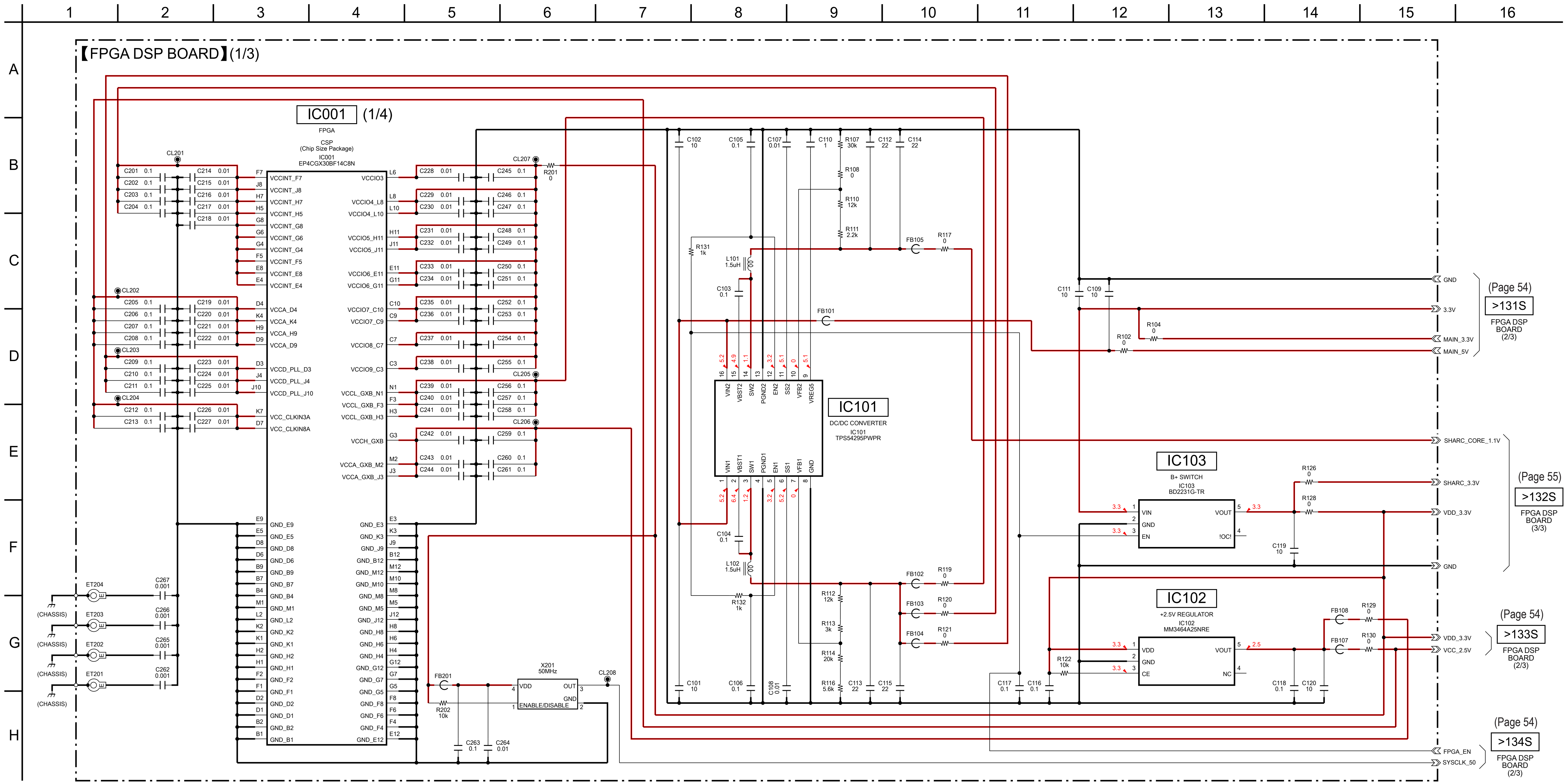
• : Uses unleaded solder.



Note: IC001 and IC601 on the FPGA DSP board cannot exchange with single. When these parts are damaged, exchange the complete mounted board.

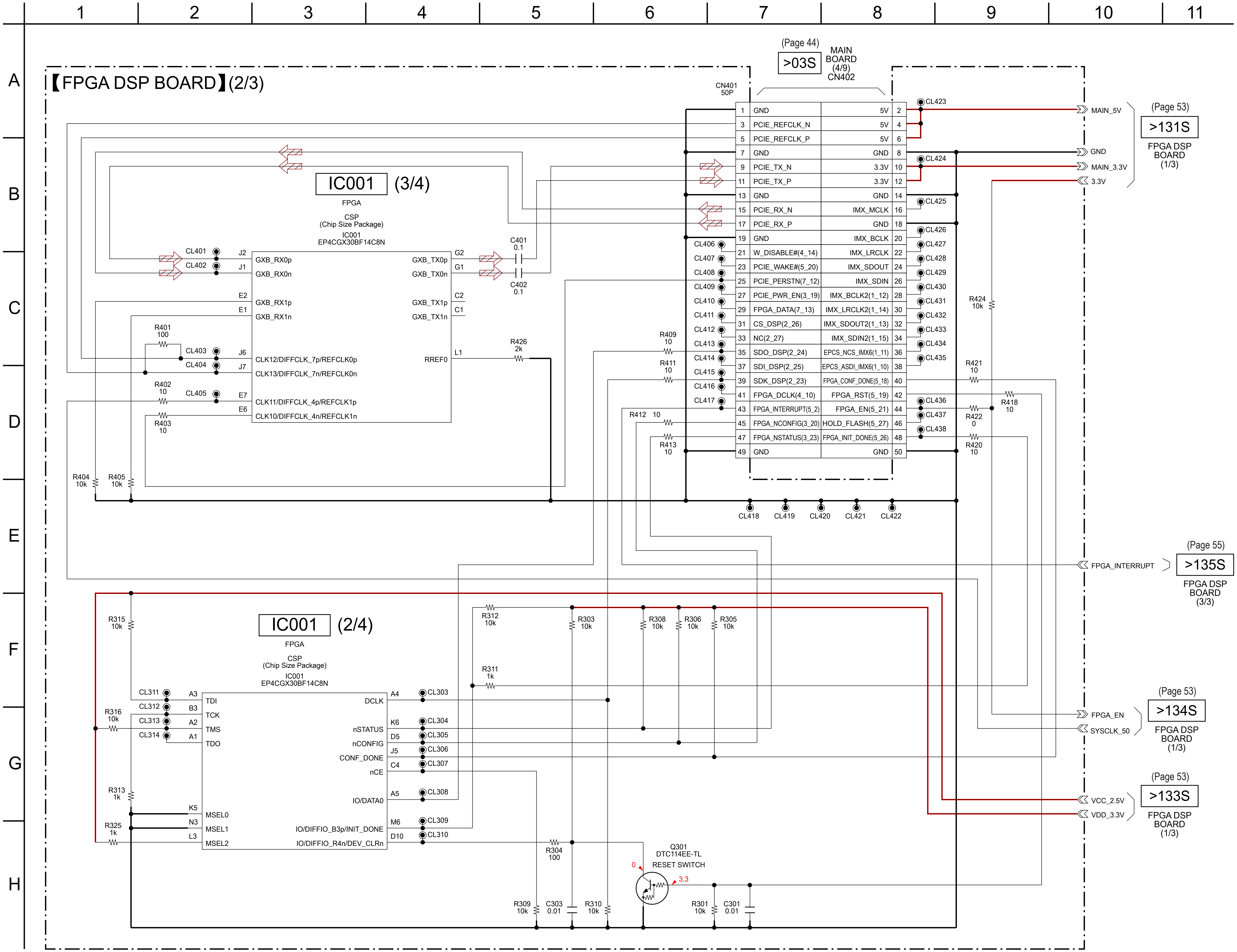
5-18. SCHEMATIC DIAGRAM - FPGA DSP Board (1/3) -

• See page 70 for IC Block Diagrams. • See page 75 for IC Pin Function Description.



Note: IC001 on the FPGA DSP board cannot exchange with single. When this part is damaged, exchange the complete mounted board.

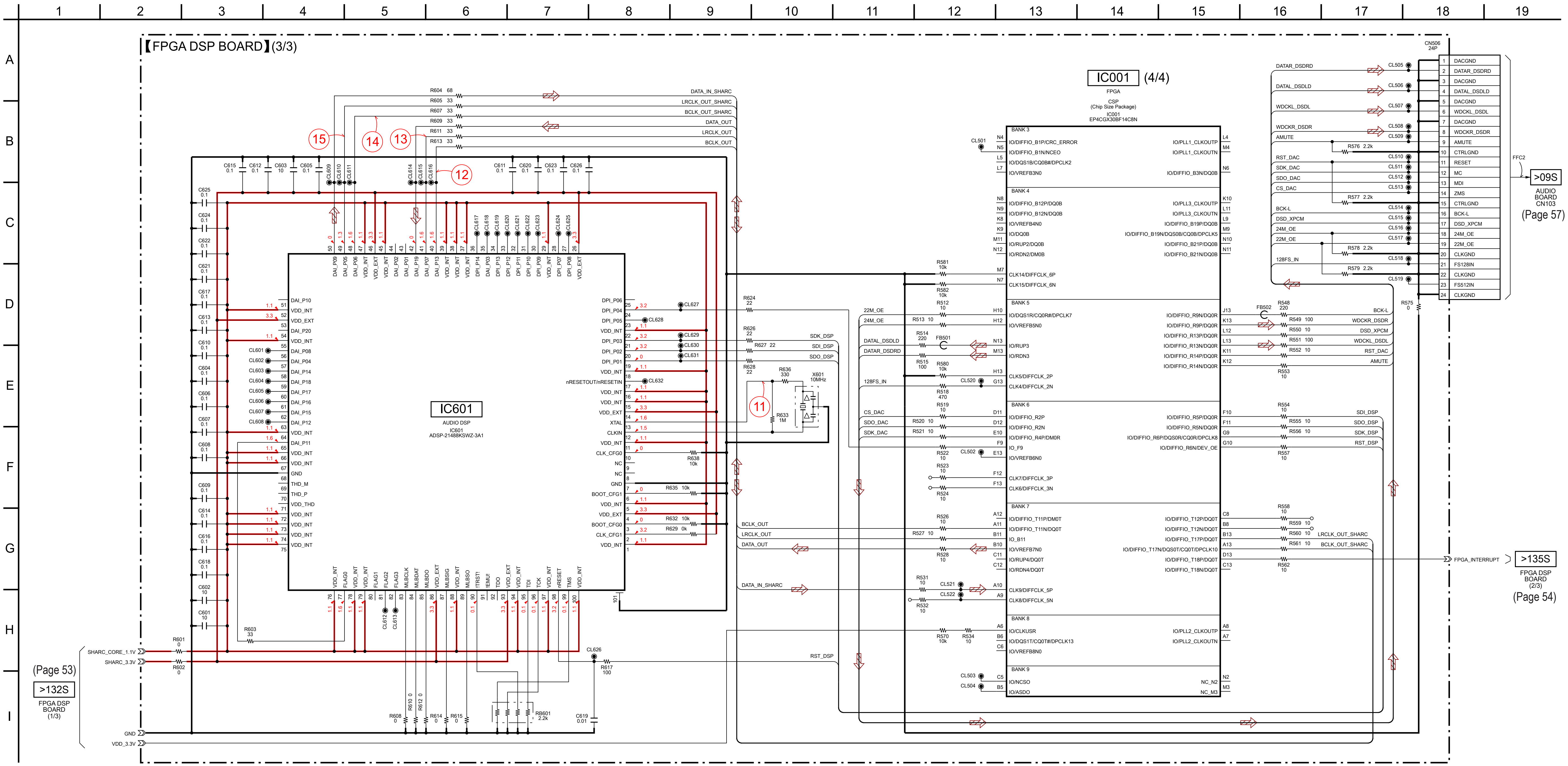
5-19. SCHEMATIC DIAGRAM - FPGA DSP Board (2/3) - • See page 75 for IC Pin Function Description.



Note: IC001 on the FPGA DSP board cannot exchange with single. When this part is damaged, exchange the complete mounted board.

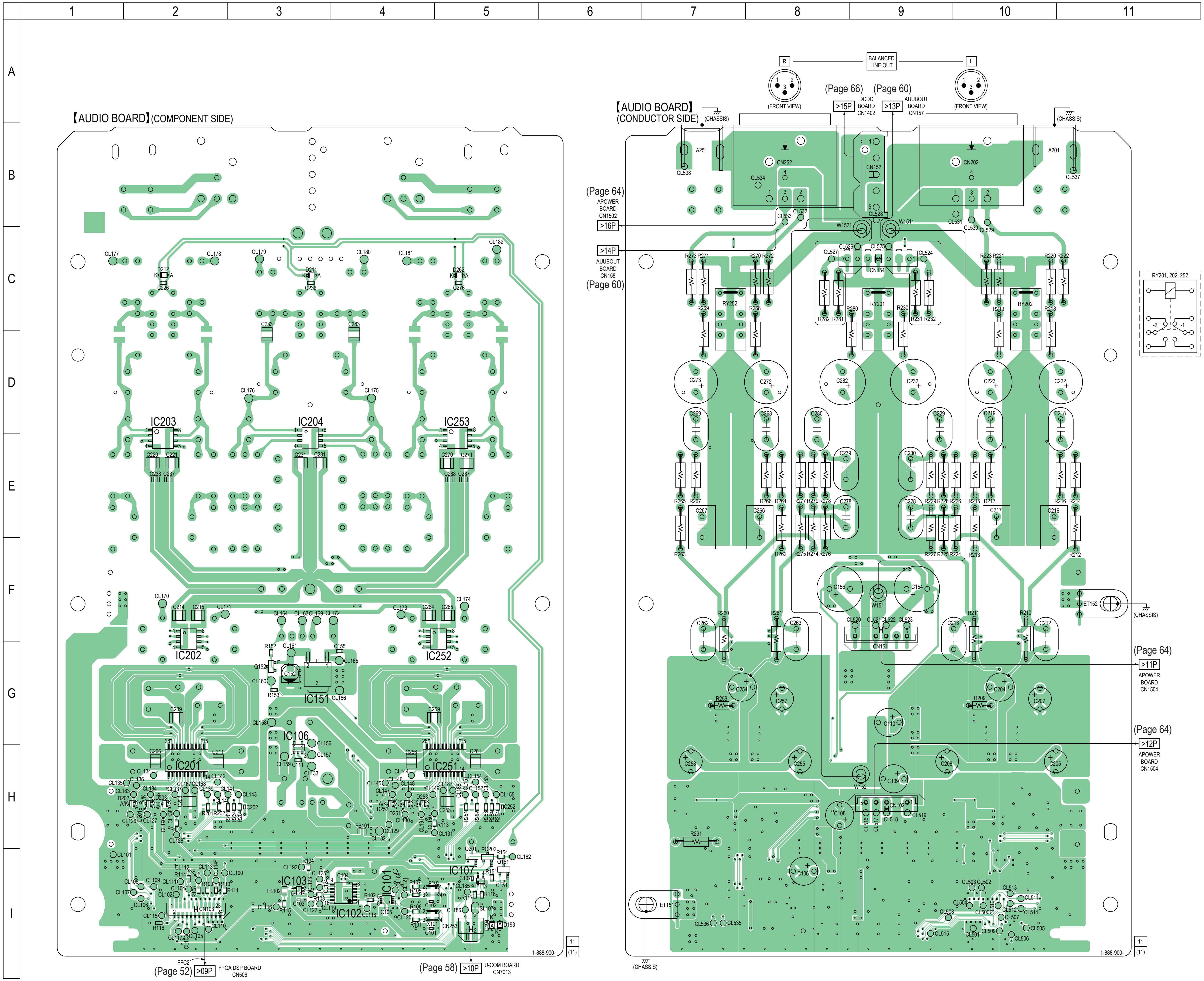
5-20. SCHEMATIC DIAGRAM - FPGA DSP Board (3/3) -

• See page 70 for Waveforms. • See page 75 for IC Pin Function Description.

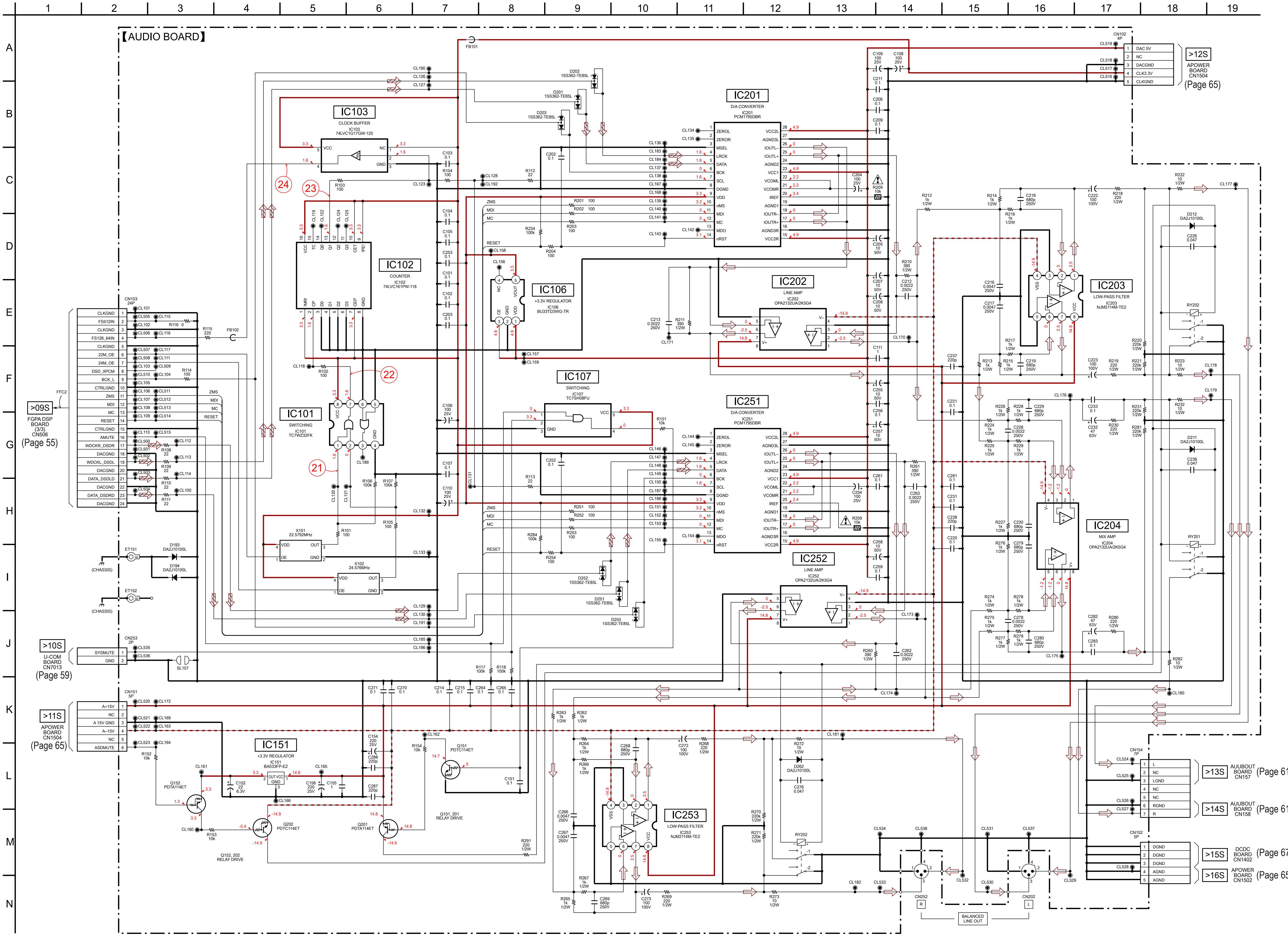


Note: IC001 and IC601 on the FPGA DSP board cannot exchange with single. When these parts are damaged, exchange the complete mounted board.

5-21. PRINTED WIRING BOARD - AUDIO Board - • See page 39 for Circuit Boards Location. • : Uses unleaded solder.

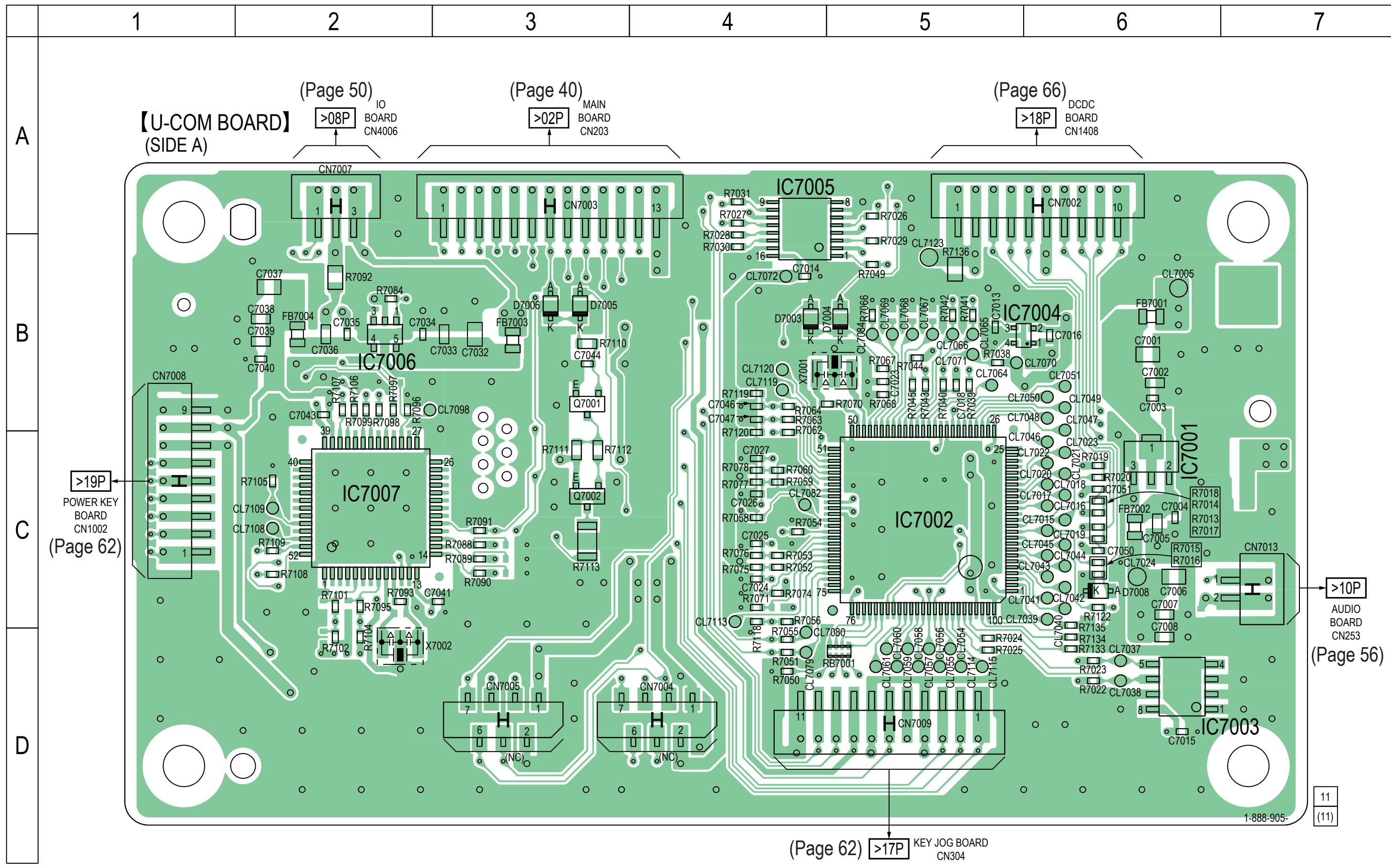


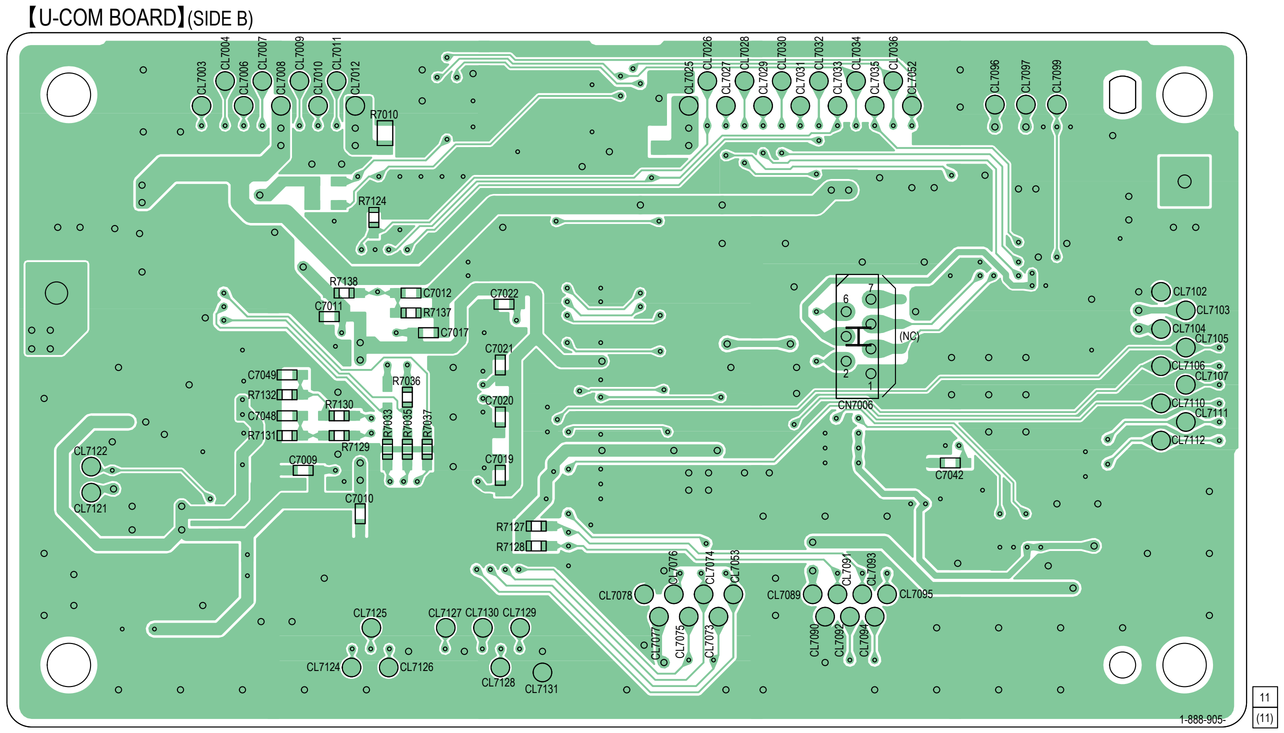
5-22. SCHEMATIC DIAGRAM - AUDIO Board - • See page 70 for Waveforms. • See page 70 for IC Block Diagrams.



5-23. PRINTED WIRING BOARD - U-COM Board -

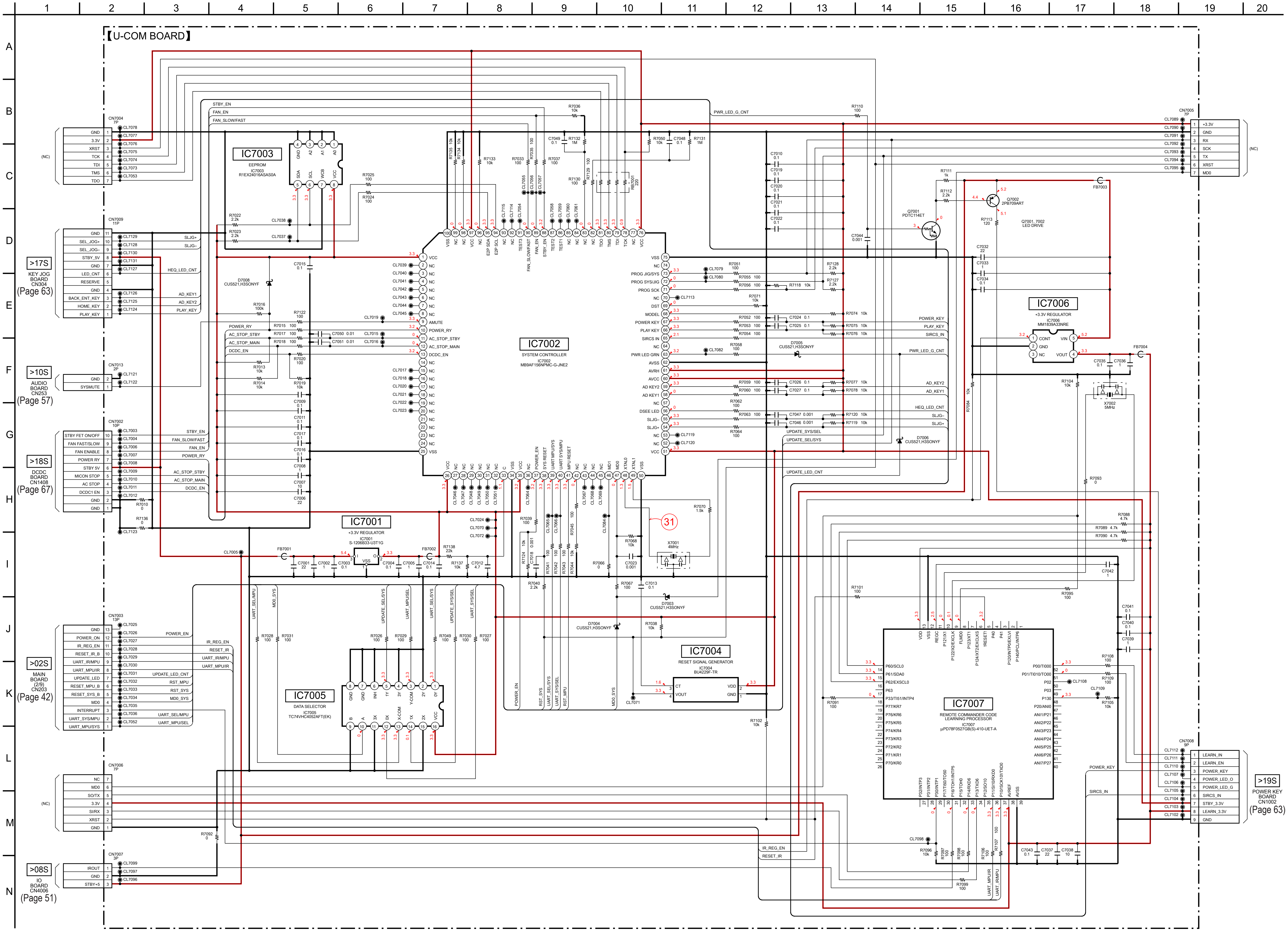
• See page 39 for Circuit Boards Location. • : Uses unleaded solder.



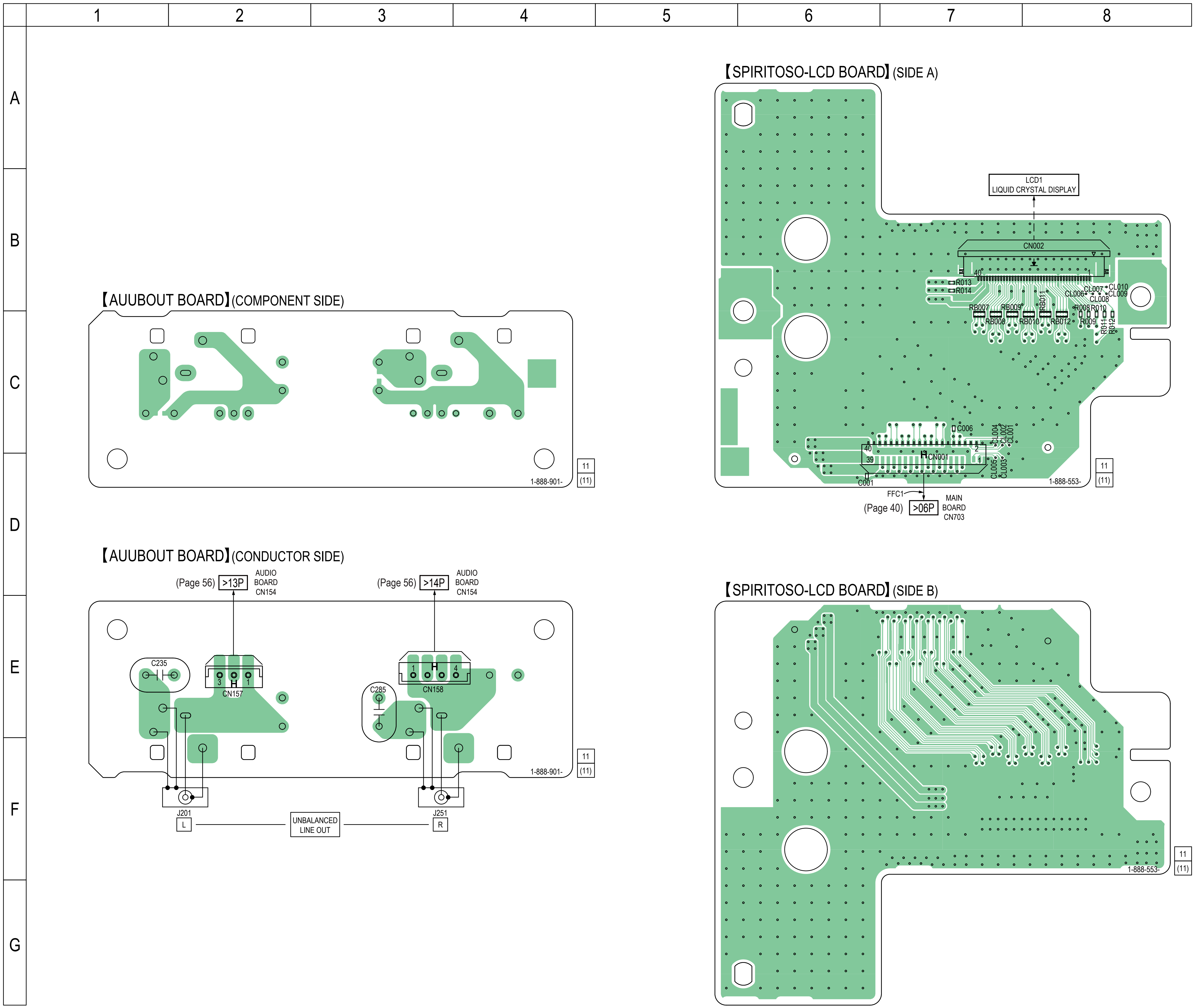


5-24. SCHEMATIC DIAGRAM - U-COM Board -

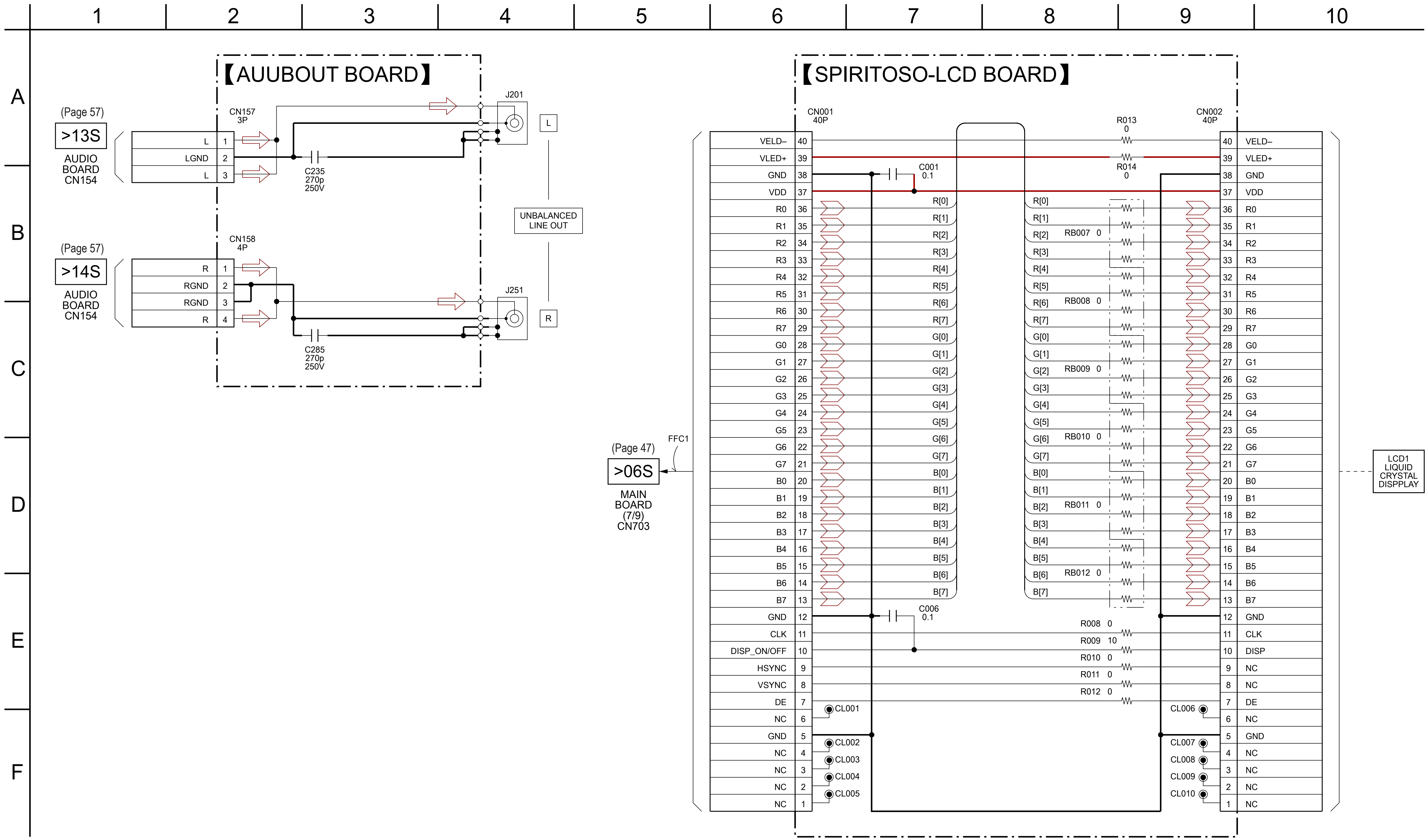
• See page 70 for Waveforms. • See page 70 for IC Block Diagrams. • See page 75 for IC Pin Function Description.



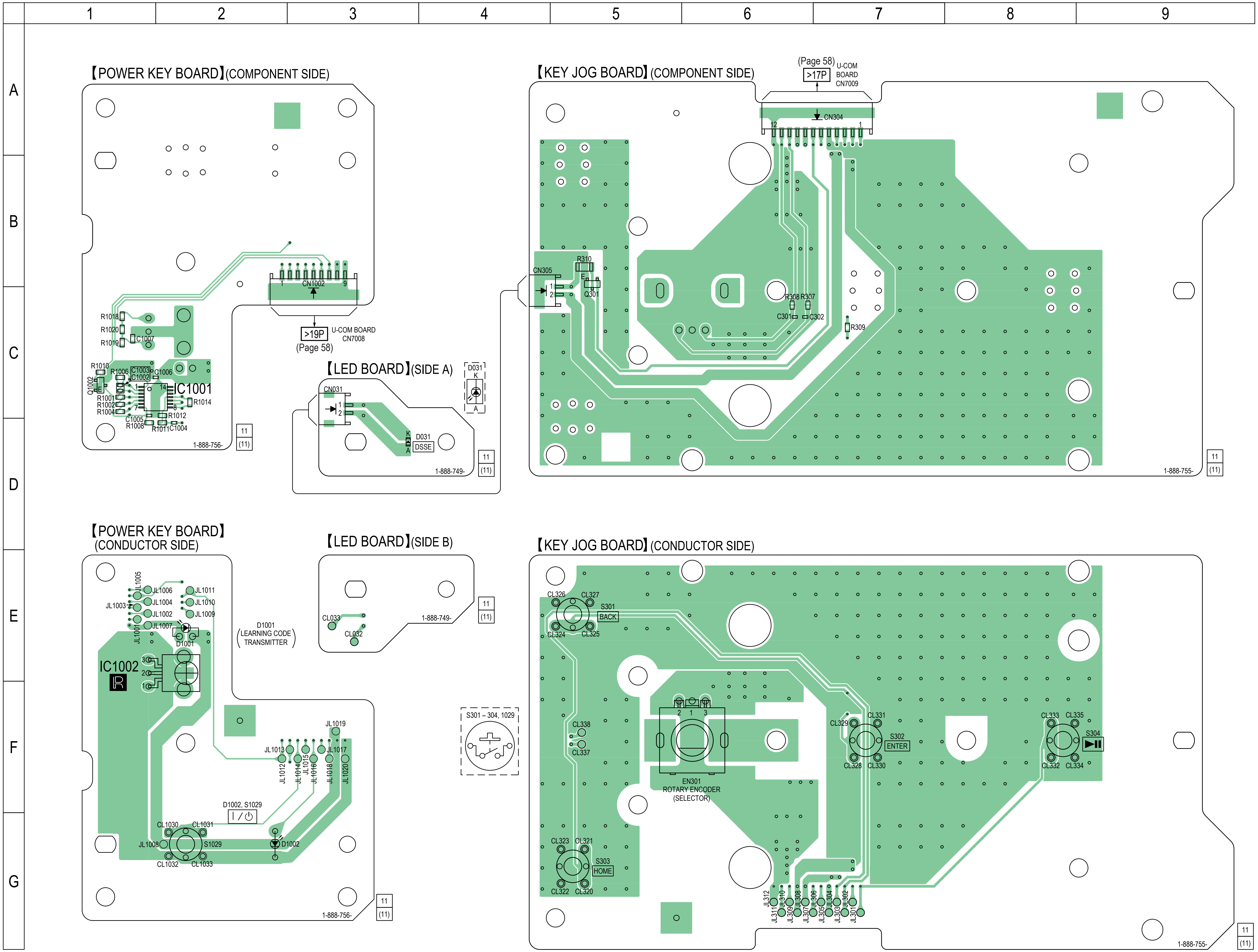
5-25. PRINTED WIRING BOARDS - AUUBOUT/SPIRITOSO-LCD Boards - • See page 39 for Circuit Boards Location. • : Uses unleaded solder.



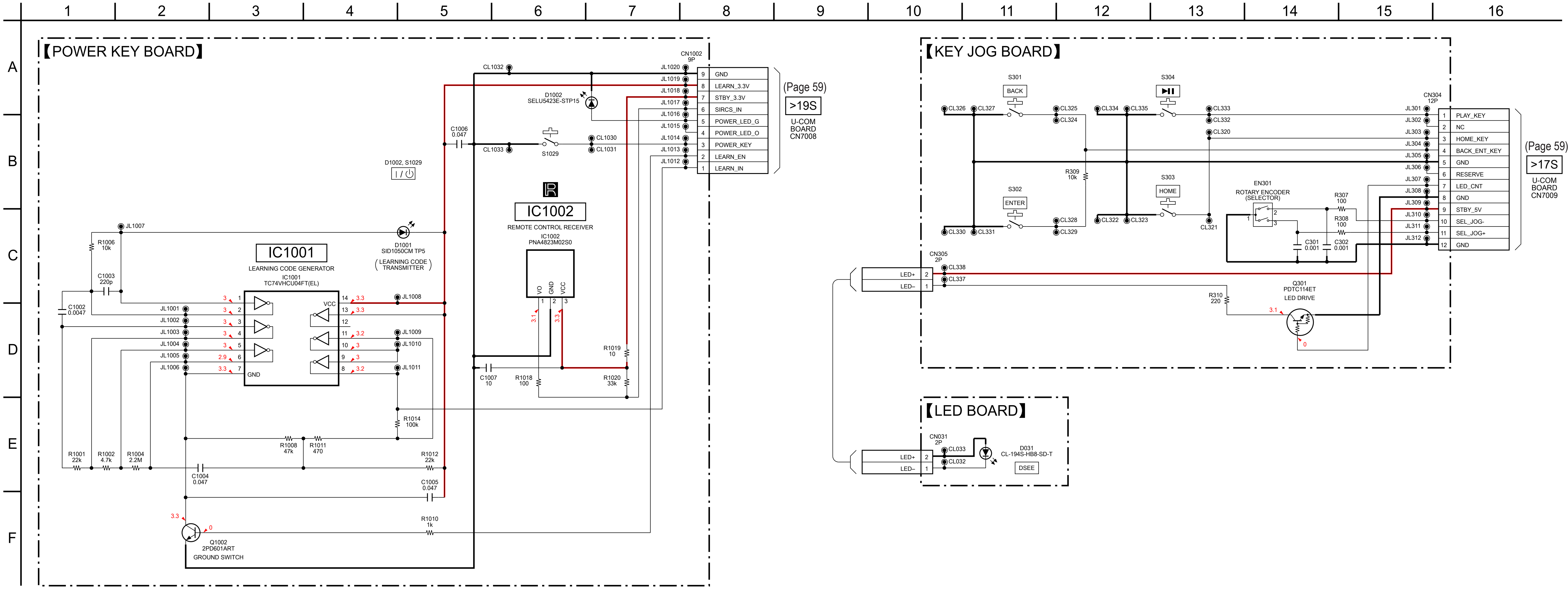
5-26. SCHEMATIC DIAGRAM - AUUBOUT/SPIRITOSO-LCD Boards -



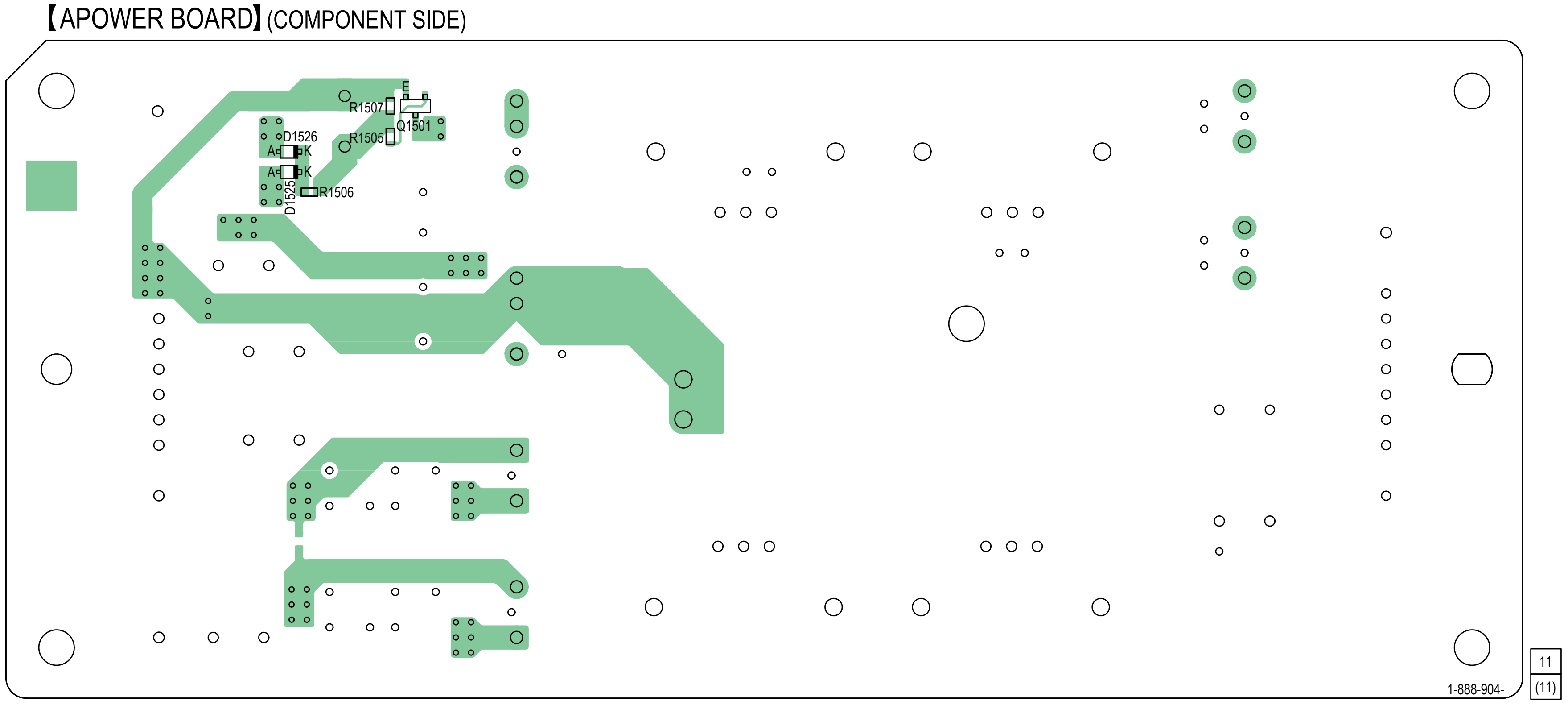
5-27. PRINTED WIRING BOARDS - KEY/LED Section - • See page 39 for Circuit Boards Location. • : Uses unleaded solder.

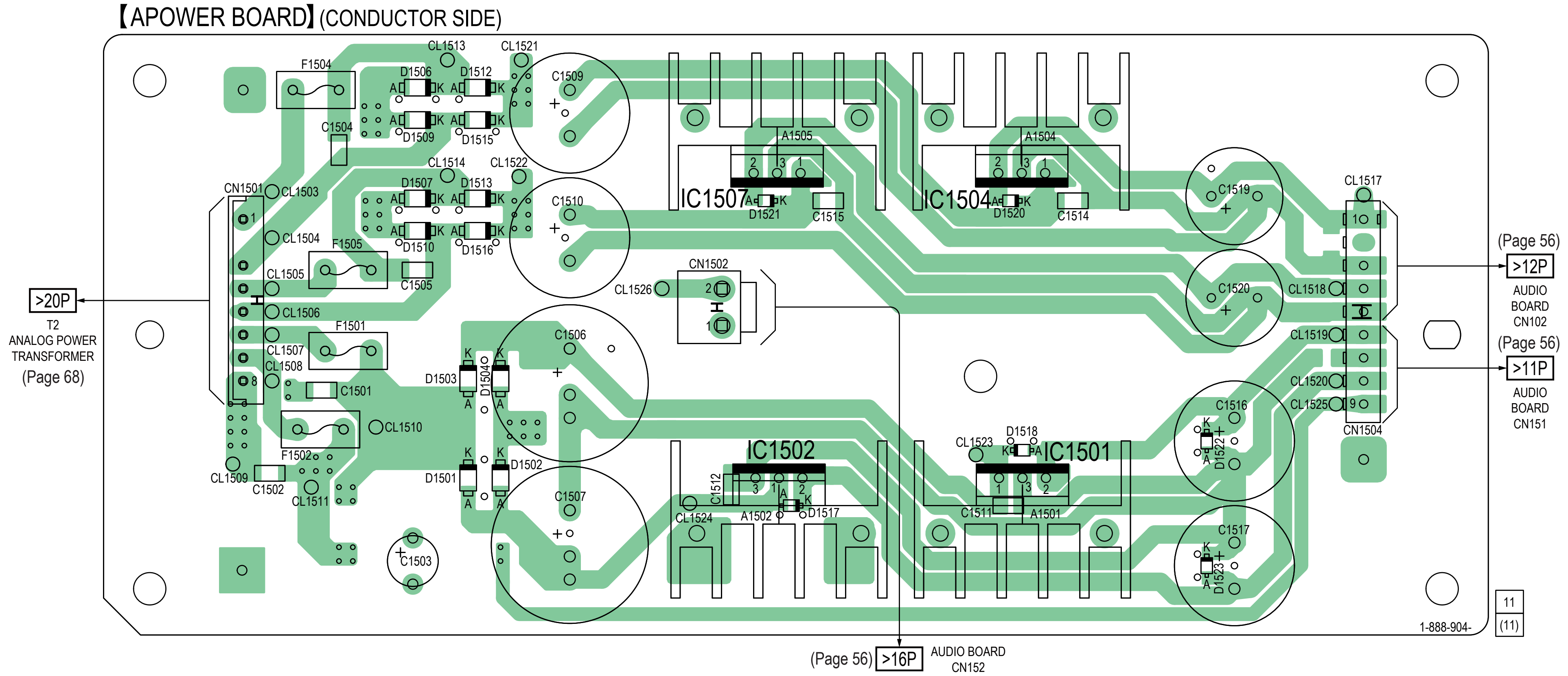


5-28. SCHEMATIC DIAGRAM - KEY/LED Section -

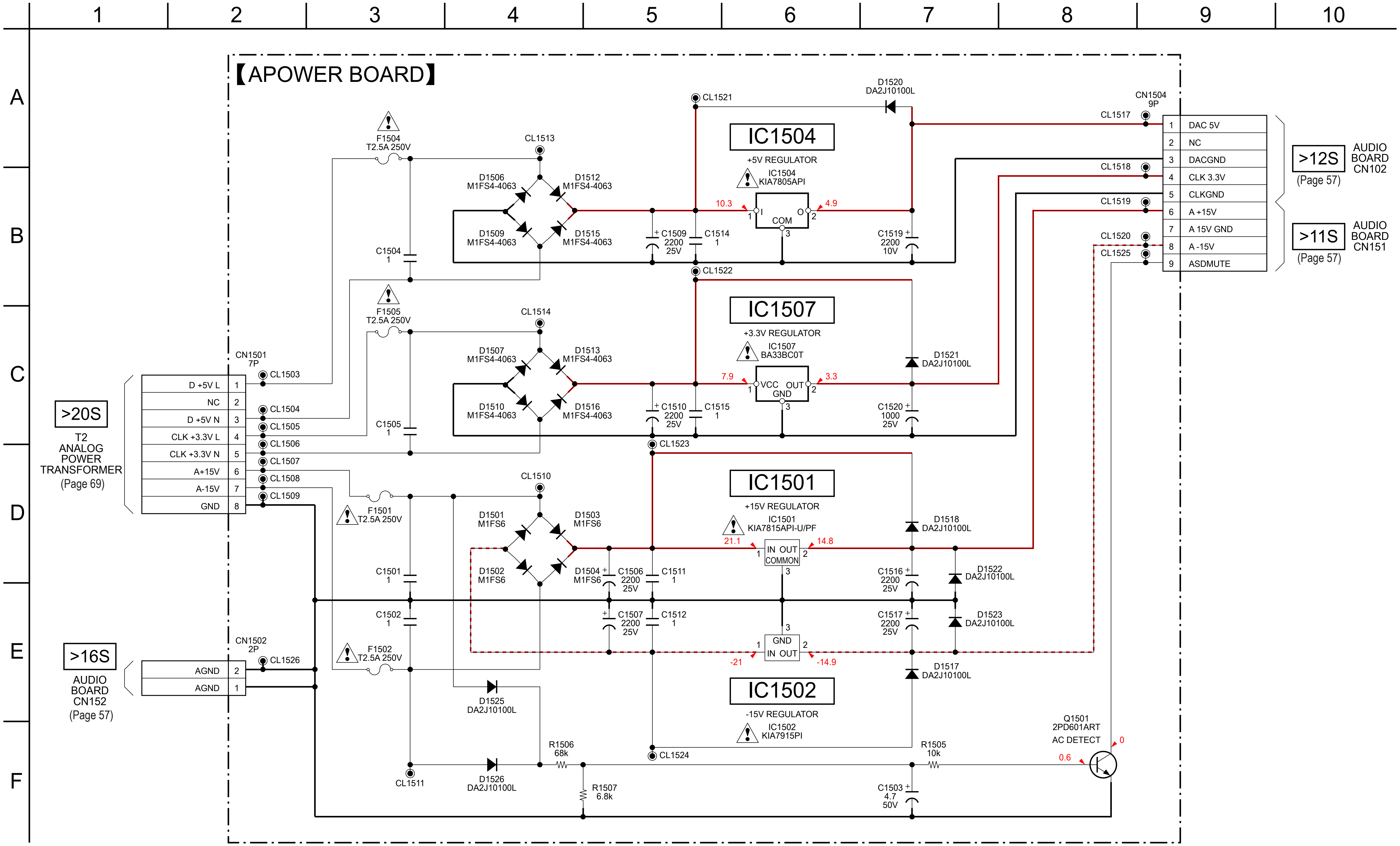


5-29. PRINTED WIRING BOARD - APOWER Board - • See page 39 for Circuit Boards Location. • : Uses unleaded solder.

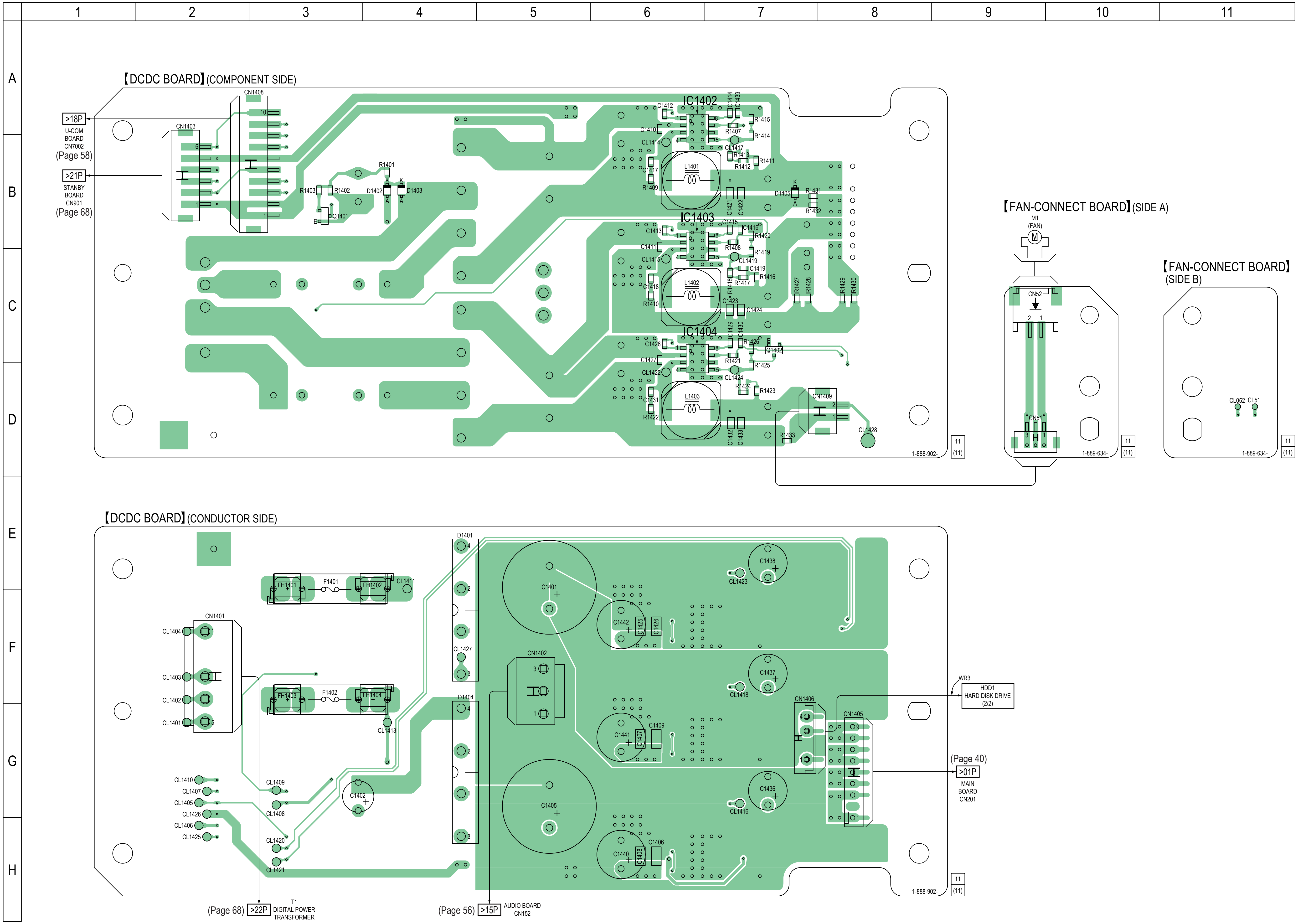




5-30. SCHEMATIC DIAGRAM - APOWER Board -

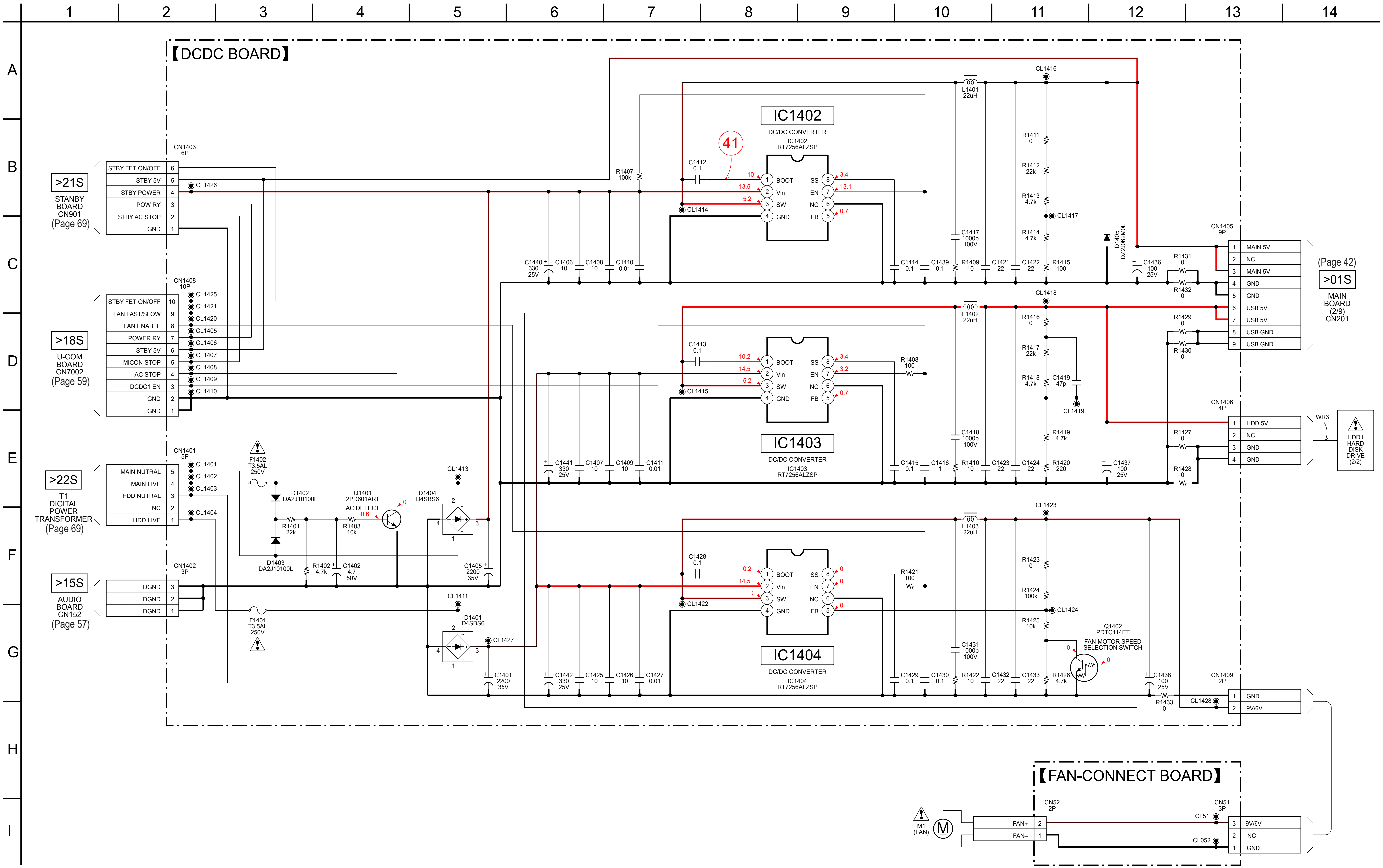


5-31. PRINTED WIRING BOARDS - DCDC/FAN-CONNECT Boards - • See page 39 for Circuit Boards Location. • : Uses unleaded solder.



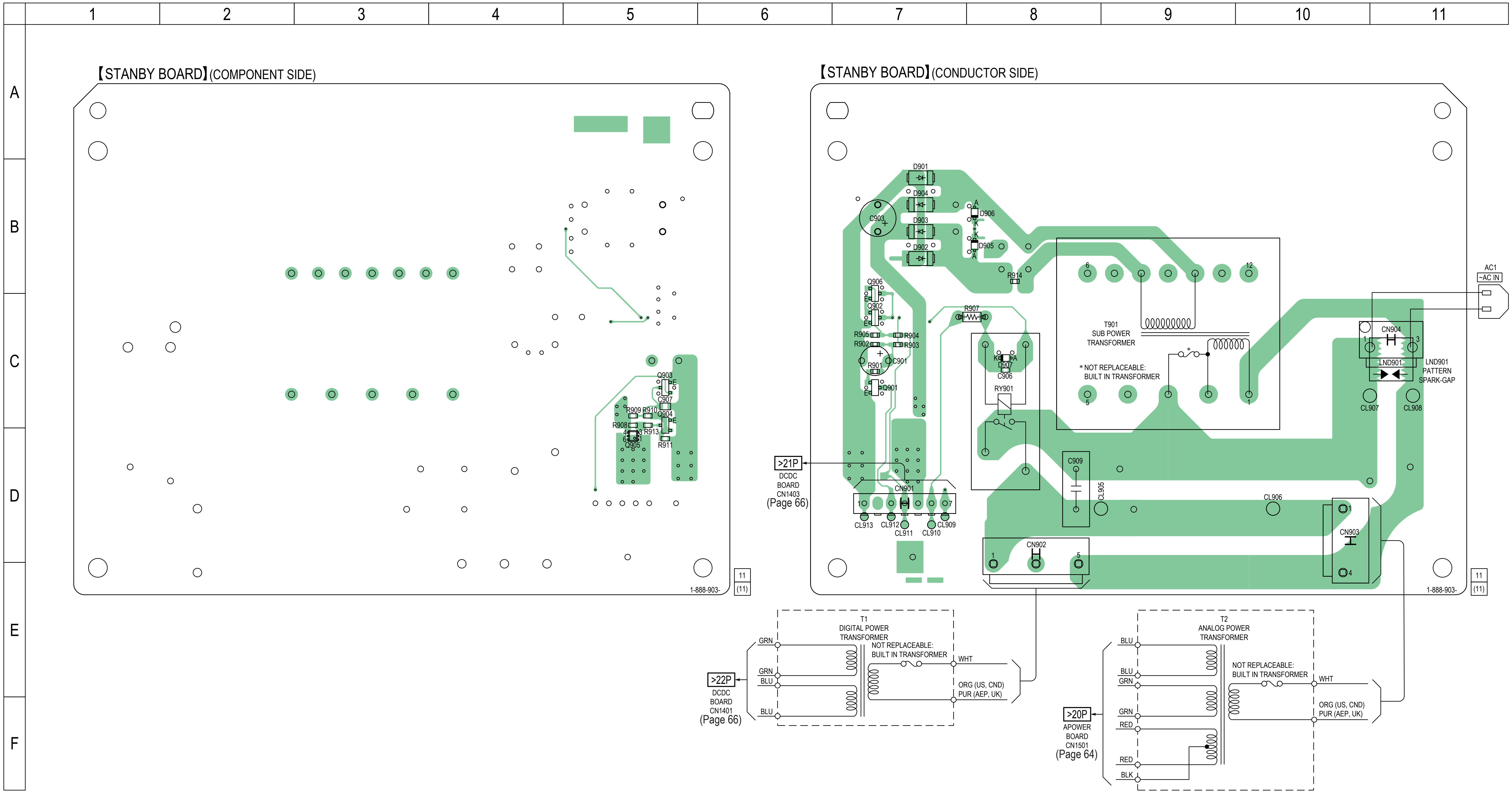
Note: When the hard disk drive (Ref. No. HDD1) is replaced, refer to "NOTE OF REPLACING THE HARD DISK DRIVE (Ref. No. HDD1)" on page 4.

5-32. SCHEMATIC DIAGRAM - DCDC/FAN-CONNECT Boards - • See page 70 for Waveforms. • See page 70 for IC Block Diagrams.

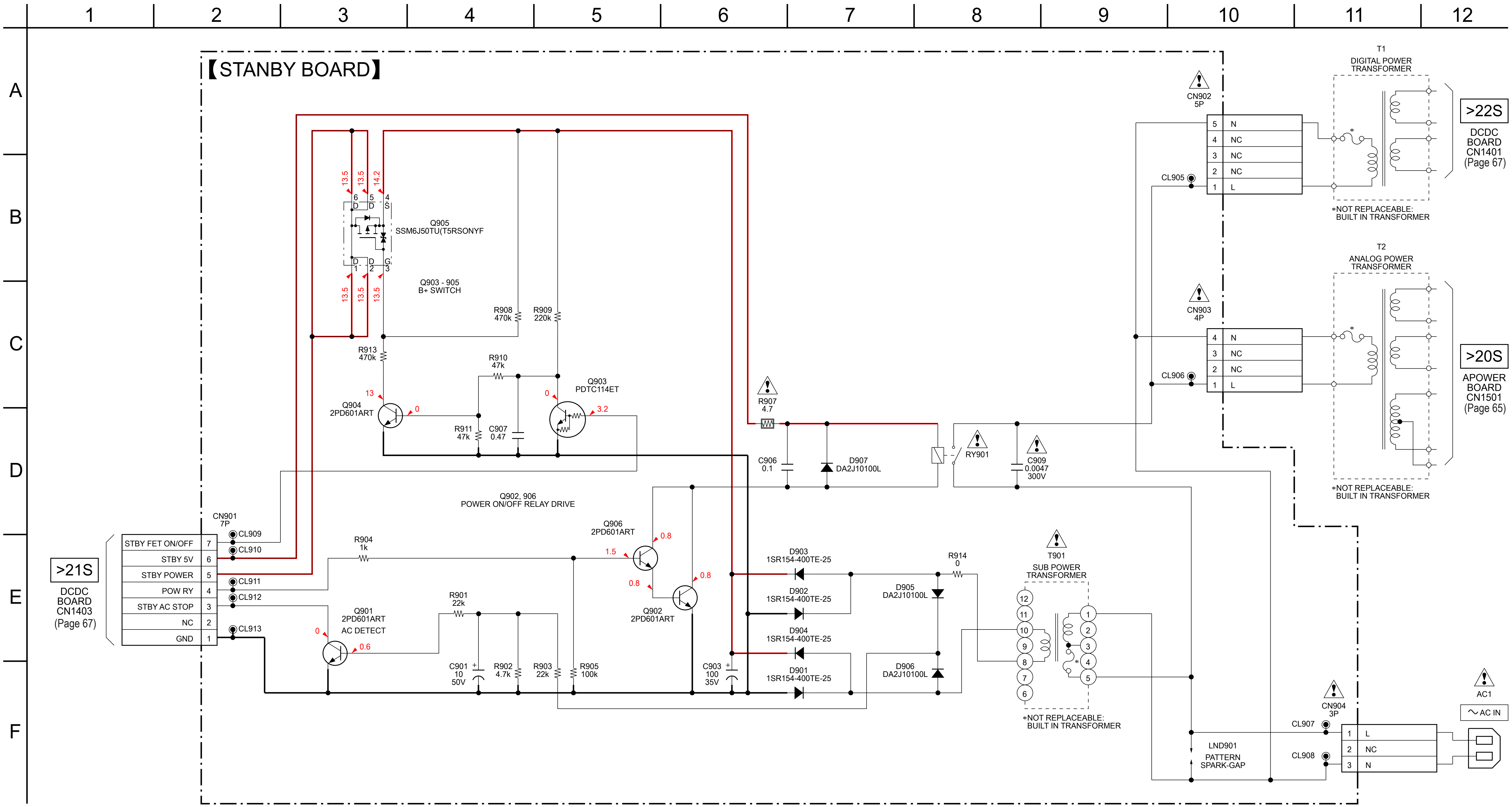


Note: When the hard disk drive (Ref. No. HDD1) is replaced, refer to "NOTE OF REPLACING THE HARD DISK DRIVE (Ref. No. HDD1)" on page 4.

5-33. PRINTED WIRING BOARD - STANBY Board - • See page 39 for Circuit Boards Location. • : Uses unleaded solder.



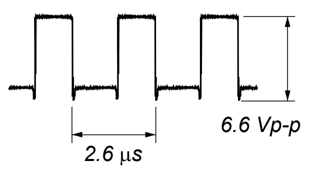
5-34. SCHEMATIC DIAGRAM - STANBY Board -



• Waveforms

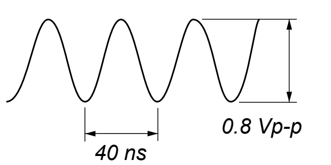
– MAIN Board –

① IC202 ① (BST)



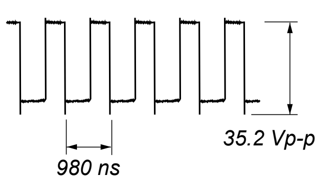
2 V/DIV, 1 µs/DIV

② IC602 ④ (XTLO)



500 mV/DIV, 10 ns/DIV

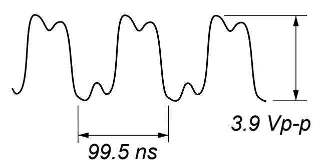
③ IC701 ④ (SW)



10 V/DIV, 500 ns/DIV

– FPGA DSP Board –

⑪ IC601 ⑬ (XTAL)

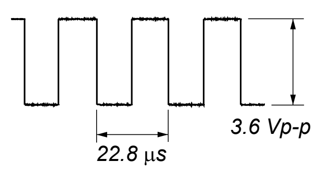


2 V/DIV, 50 ns/DIV

⑫ IC601 ㊵ (DAI-P13)

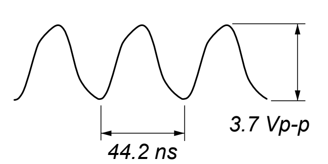
2 V/DIV, 200 ns/DIV

⑬ IC601 ㊶ (DAI-P07)



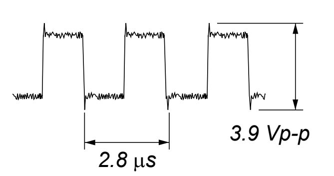
2 V/DIV, 10 µs/DIV

⑭ IC601 ㊽ (DAI-P06)



2 V/DIV, 20 ns/DIV

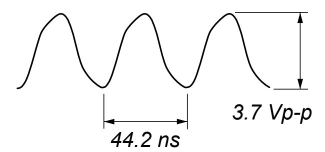
⑮ IC601 ㊾ (DAI-P05)



2 V/DIV, 1 µs/DIV

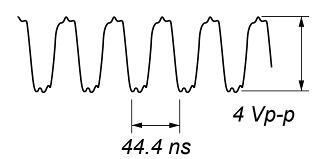
– AUDIO Board –

㉑ IC101 ①



2 V/DIV, 20 ns/DIV

㉒ IC101 ⑦



2 V/DIV, 20 ns/DIV

㉓ IC102 ⑬ (Q1)

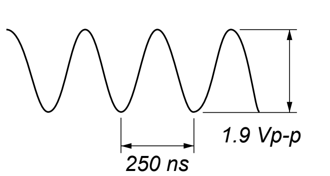
2 V/DIV, 100 ns/DIV

㉔ IC103 ④

2 V/DIV, 100 ns/DIV

– U-COM Board –

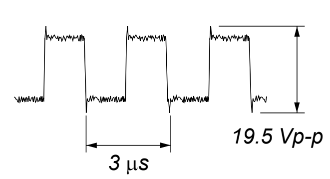
㉛ IC7002 ㊽ (XTALO)



1 V/DIV, 100 ns/DIV

– DCDC Board –

㊶ IC1402 ① (BOOT)

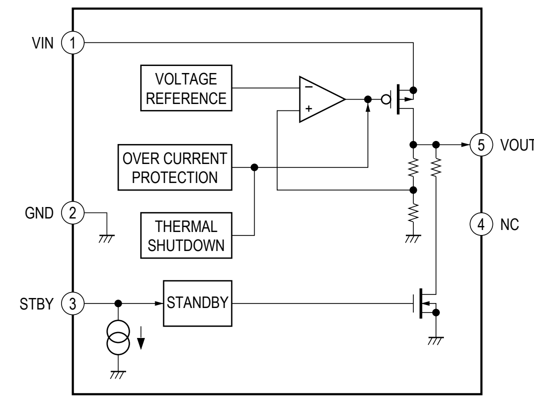


10 V/DIV, 2 µs/DIV

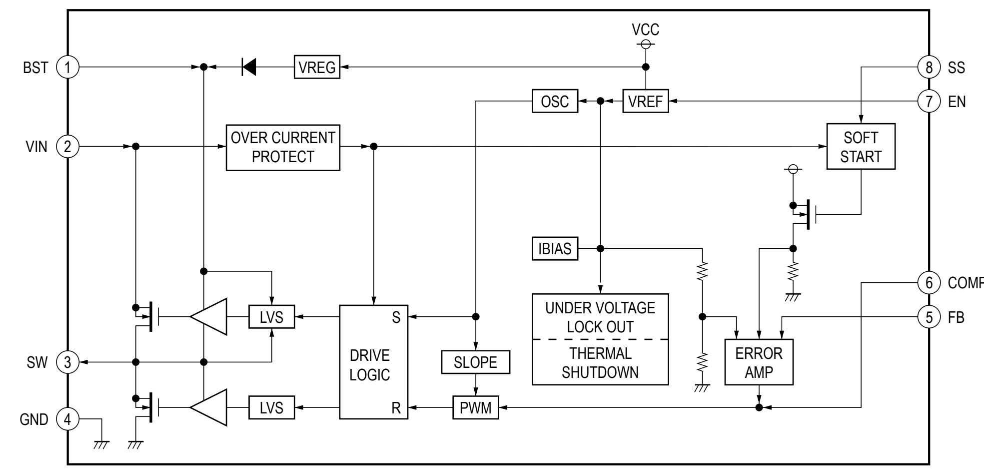
• IC Block Diagrams

– MAIN Board –

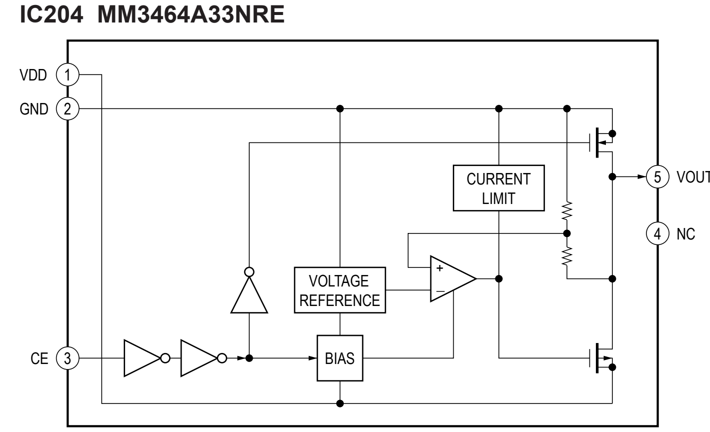
IC201 BU33TD3WG-TR



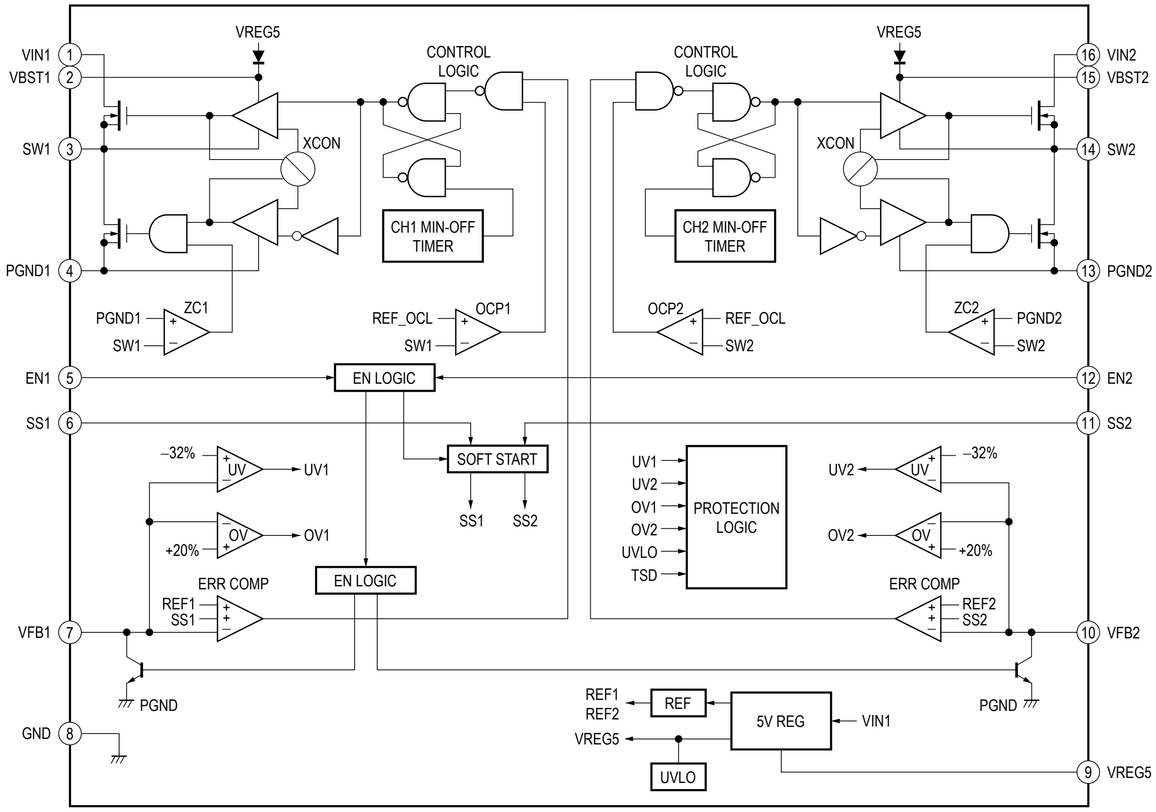
IC202 BD9329AEFJ-E2



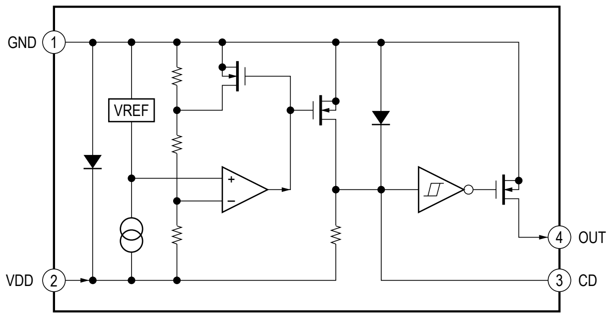
IC204 MM3464A33NRE



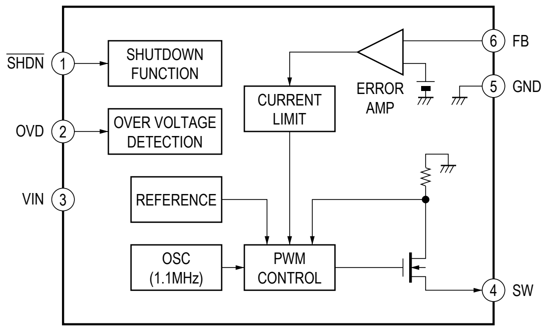
IC203 TPS54295PWPR



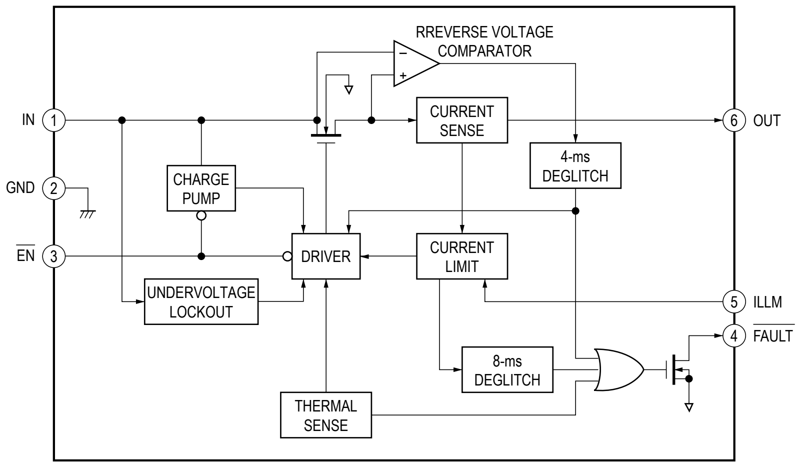
IC205, 401 PST8429UL



IC701 TB62752AFUG (O, EL)

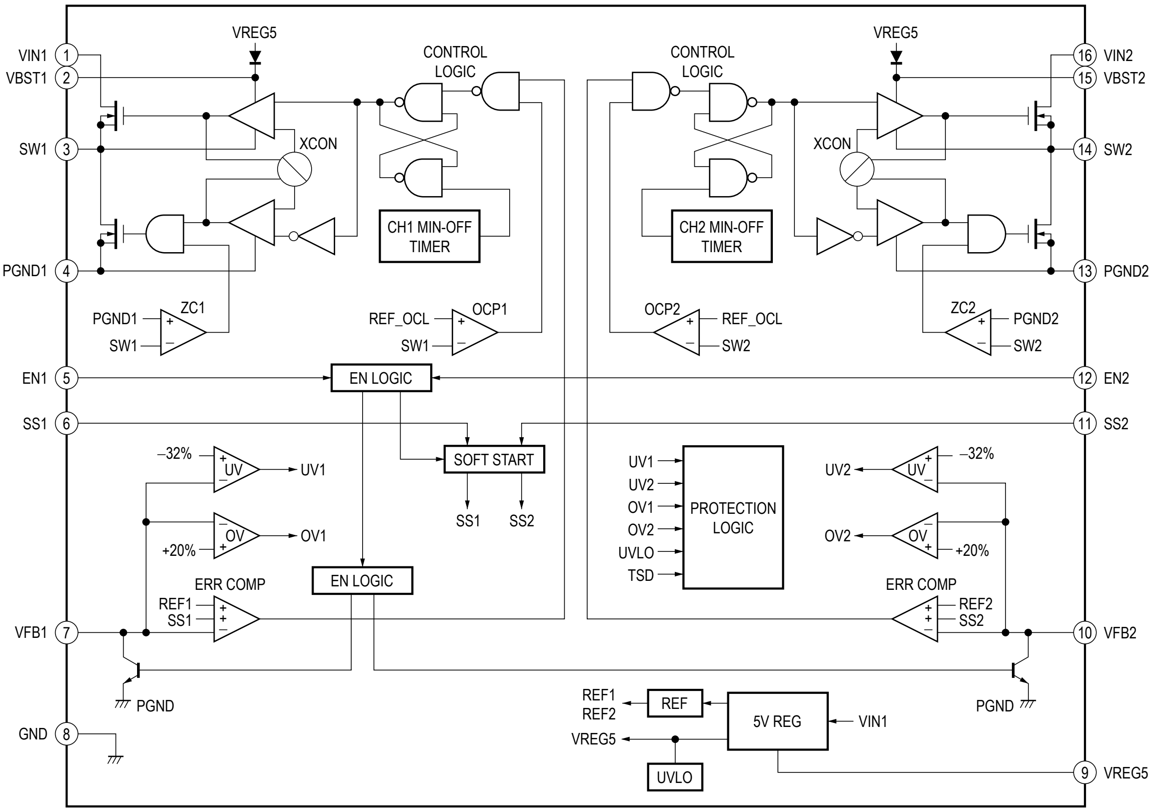


IC1001 TPS2553DBVR

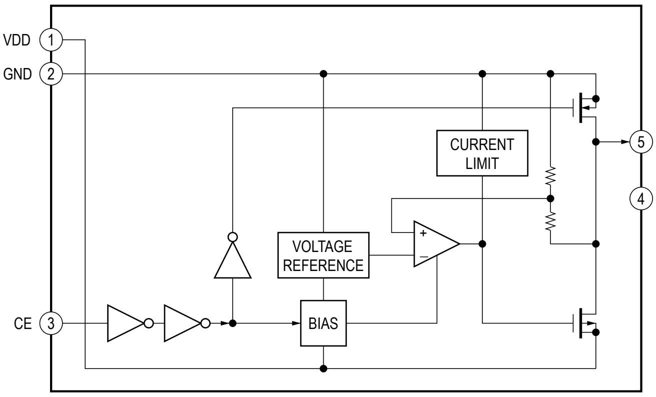


– FPGA DSP Board –

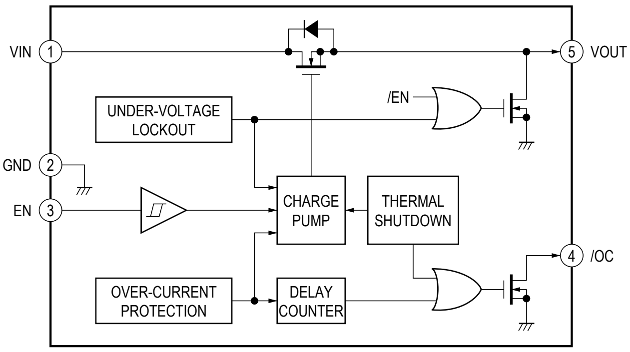
IC101 TPS54295PWPR



IC102 MM3464A25NRE

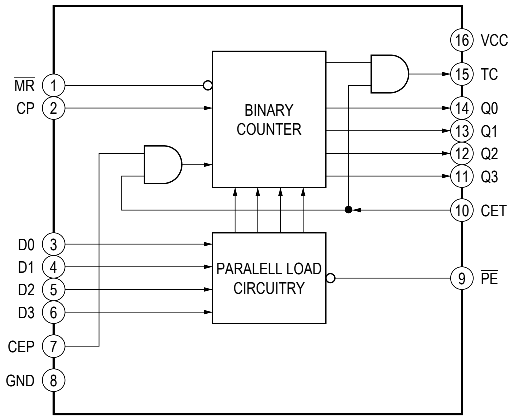


IC103 BD2231G-TR

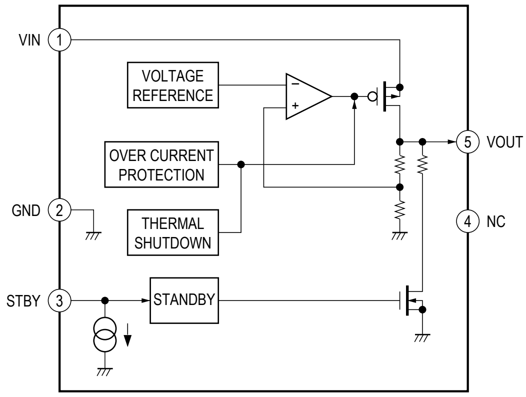


– AUDIO Board –

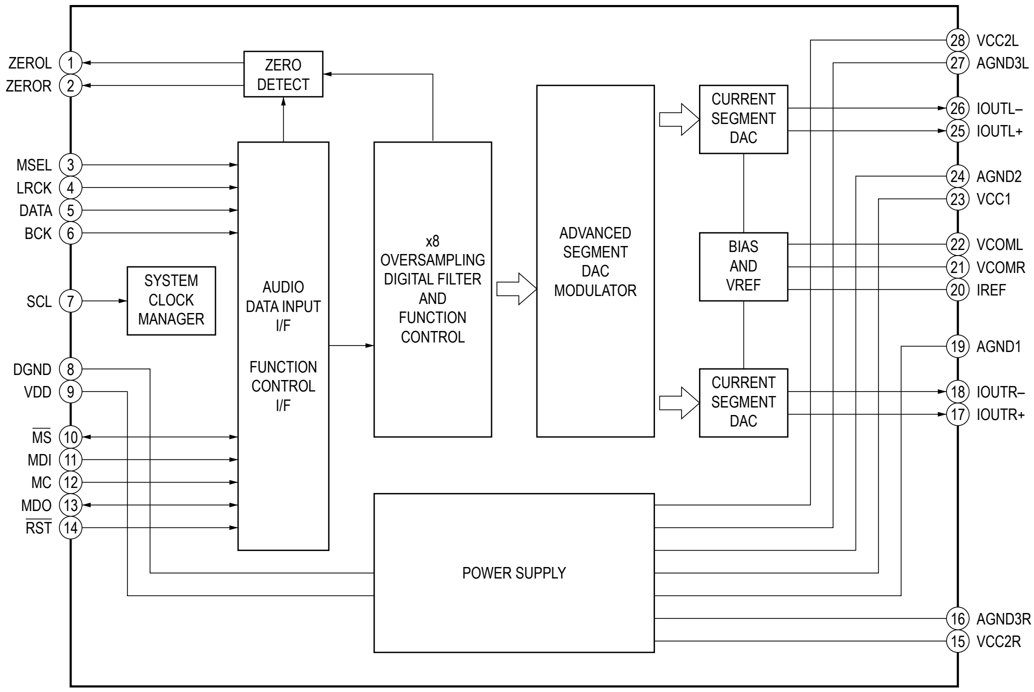
IC102 74LVC161PW-118



IC106 BU33TD3WG-TR

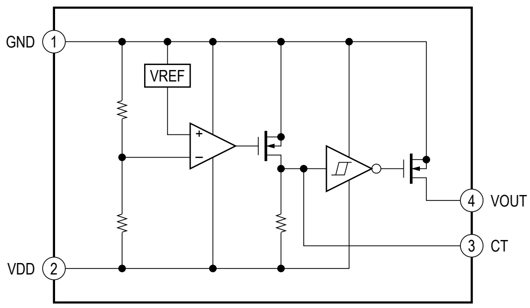


IC201, 251 PCM1795DBR

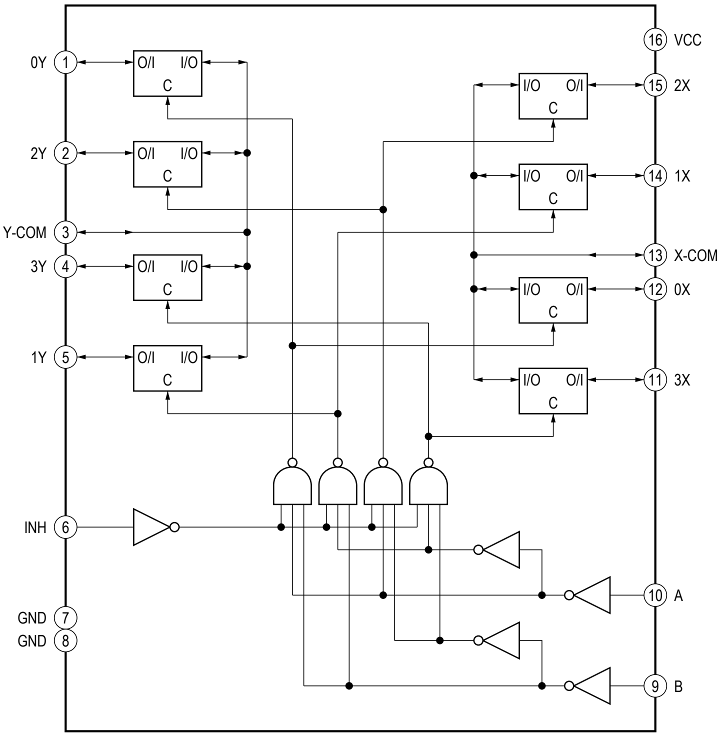


– U-COM Board –

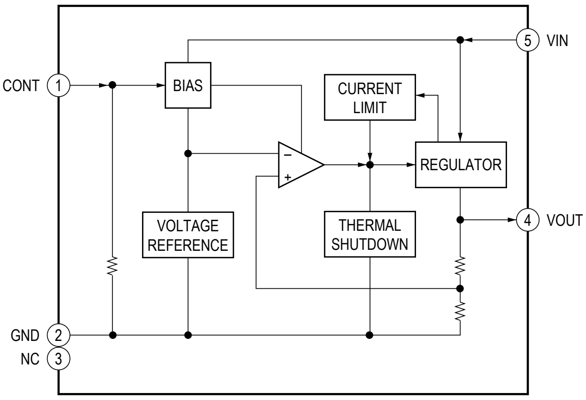
IC7004 BU4229F-TR



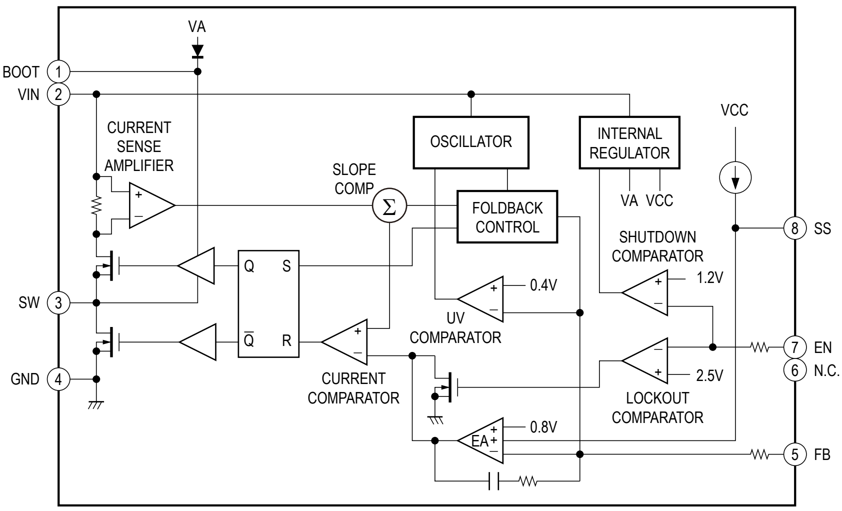
IC7005 TC74VHC4052AFT (EK)



IC7006 MM1839A33NRE



– DCDC Board –
IC1402 – 1404 RT7256ALZSP



- **IC Pin Function Description**

MAIN BOARD IC101 MCIMX6D5EYM10AC (MPU)

Pin No.	Pin Name	I/O	Description
A2	PCIE_REXT	-	Terminal for the Impedance calibration
A3	PCIE_TXM	O	Transmit data (negative) output to the FPGA
A4	GND3	-	Ground terminal
A5	FA_ANA	-	Not used
A6	USB_OTG_DP	I/O	Two-way USB data (positive) with the USB connector
A7	XTALI	I	System clock input terminal (24 MHz)
A8	GND4	-	Ground terminal
A9	MLB_SN	-	Not used
A10	MLB_DP	-	Not used
A11	MLB_CN	-	Not used
A12	SATA_TXP	O	Transmit data (positive) output to the hard disk drive
A13	GND1	-	Ground terminal
A14	SATA_RXM	I	Receive data (negative) input from the hard disk drive
A15	SD3_DAT2	I/O	Two-way data bus terminal Not used
A16	NANDF_ALE	O	Reset signal output to the system controller "H": reset
A17	NANDF_CS2	-	Not used
A18	NANDF_D0	I/O	Two-way data bus terminal Not used
A19	NANDF_D4	-	Not used
A20	SD4_DAT3	I/O	Two-way data bus with the flash memory
A21	SD1_DAT0	I/O	Two-way data bus terminal Not used
A22, A23	SD2_DAT0, SD2_DAT2	-	Not used
A24	RGMII_TD3	O	RGMII transmit data output to the ethernet transceiver
A25	GND2	-	Ground terminal
B1	PCIE_RXM	I	Receive data (negative) input from the FPGA
B2	PCIE_RXP	I	Receive data (positive) input from the FPGA
B3	PCIE_TXP	O	Transmit data (positive) output to the FPGA
B4	GND22	-	Ground terminal
B5	VDD_FA	-	Not used
B6	USB_OTG_DN	I/O	Two-way USB data (negative) with the USB connector
B7	XTALO	O	System clock output terminal (24 MHz)
B8	USB_OTG_CHD_B	-	Not used
B9	MLB_SP	-	Not used
B10	MLB_DN	-	Not used
B11	MLB_CP	-	Not used
B12	SATA_TXM	O	Transmit data (negative) output to the hard disk drive
B13	SD3_CMD	-	Not used
B14	SATA_RXP	I	Receive data (positive) input from the hard disk drive
B15	SD3_DAT3	-	Not used
B16	NANDF_RB0	O	Interrupt signal output to the remote commander code learning processor
B17	SD4_CMD	O	Command signal output to the flash memory
B18	NANDF_D5	-	Not used
B19, B20	SD4_DAT1, SD4_DAT6	I/O	Two-way data bus with the flash memory
B21	SD1_CMD	-	Not used
B22	SD2_DAT3	-	Not used
B23, B24	RGMII_RD1, RGMII_RD2	I	RGMII receive data input from the ethernet transceiver
B25	RGMII_RXC	I	RGMII receive clock signal input from the ethernet transceiver
C1	GND23	-	Ground terminal
C2	JTAG_TRSTB	-	Not used
C3	JTAG_TMS	-	Not used
C4	GND25	-	Ground terminal
C5	CLK2_N	-	Not used
C6	GND26	-	Ground terminal
C7	CLK1_N	O	Clock signal (negative) output to the FPGA
C8	GPANAIO	-	Not used

Pin No.	Pin Name	I/O	Description
C9	RTC_XTALO	O	Real time clock output terminal (32.768 kHz)
C10	GND24	-	Ground terminal
C11	POR_B	I	Reset signal input from reset signal generator and system controller "L": reset
C12	BOOT_MODE0	I	Boot mode setting terminal Fixed at "L"
C13	SD3_DAT5	-	Not used
C14	SATA_REXT	-	Terminal for the impedance calibration
C15	NANDF_CLE	O	Update signal output to the system controller "H": update
C16	NANDF_CS1	O	Reset signal output to the remote commander code learning processor "L": reset
C17	NANDF_D1	I/O	Two-way data bus terminal Not used
C18	NANDF_D7	-	Not used
C19	SD4_DAT5	I/O	Two-way data bus with the flash memory
C20	SD1_DAT1	I/O	Two-way data bus terminal Not used
C21	SD2_CLK	-	Not used
C22	RGMII_TD0	O	RGMII transmit data output to the ethernet transceiver
C23	RGMII_TX_CTL	O	RGMII transmit data enable signal output to the ethernet transceiver
C24	RGMII_RD0	I	RGMII receive data input from the ethernet transceiver
C25	EIM_D16	-	Not used
D1	CSI_D1M	-	Not used
D2	CSI_D1P	-	Not used
D3	GND27	-	Ground terminal
D4	CSI_REXT	-	Not used
D5	CLK2_P	-	Not used
D6	GND28	-	Ground terminal
D7	CLK1_P	O	Clock signal (positive) output to the FPGA
D8	GND29	-	Ground terminal
D9	RTC_XTALI	I	Real time clock input terminal (32.768 kHz)
D10	USB_H1_VBUS	I	Power supply of the USB interface for the WLAN/BT COMBO card
D11	PMIC_ON_REQ	O	Not used
D12	ONOFF	-	Not used
D13	SD3_DAT4	-	Not used
D14	SD3_CLK	-	Not used
D15	SD3_RST	-	Not used
D16	NANDF_CS3	-	Not used
D17	NANDF_D3	-	Not used
D18, D19	SD4_DAT0, SD4_DAT7	I/O	Two-way data bus with the flash memory
D20	SD1_CLK	O	Clock signal output terminal Not used
D21	RGMII_TXC	O	RGMII transmit clock signal output to the ethernet transceiver
D22	RGMII_RX_CTL	I	RGMII receive data valid signal input from the ethernet transceiver
D23	RGMII_RD3	I	RGMII receive data input from the ethernet transceiver
D24	EIM_D18	-	Not used
D25	EIM_D23	I	Status signal input from the FPGA
E1	CSI_D2M	-	Not used
E2	CSI_D2P	-	Not used
E3	CSI_D0P	-	Not used
E4	CSI_D0M	-	Not used
E5 to E7	GND30 to GND32	-	Ground terminal
E8	NVCC_PLL_OUT	-	Power supply terminal for the PLL output Not used
E9	USB_OTG_VBUS	I	Power supply of the USB interface for the USB connector
E10	USB_H1_DP	I/O	Two-way USB data (positive) with the WLAN/BT COMBO card
E11	TAMPER	I	Mode setting terminal Fixed at "H"
E12	TEST_MODE	I	Mode setting terminal Not used
E13	SD3_DAT6	-	Not used
E14	SD3_DAT0	-	Not used
E15	NANDF_WP_B	O	LED drive signal output terminal "H": LED on
E16	SD4_CLK	O	Clock signal output to the flash memory
E17	NANDF_D6	-	Not used
E18	SD4_DAT4	I/O	Two-way data bus with the flash memory
E19	SD1_DAT2	I/O	Two-way data bus terminal Not used
E20	SD2_DAT1	-	Not used

Pin No.	Pin Name	I/O	Description
E21	RGMII_TD2	O	RGMII transmit data output to the ethernet transceiver
E22	EIM_EB2	I	Boot mode setting terminal Fixed at "L"
E23	EIM_D22	O	VBUS power on/off control signal output terminal for the USB connector "H": power on
E24, E25	EIM_D26, EIM_D27	-	Not used
F1	CSI_D3P	-	Not used
F2	CSI_D3M	-	Not used
F3	CSI_CLK0P	-	Not used
F4	CSI_CLK0M	-	Not used
F5 to F8	GND33 to GND36	-	Ground terminal
F9	VDDUSB_CAP1	O	Internal regulator output terminal
F10	USB_H1_DN	I/O	Two-way USB data (negative) with the WLAN/BT COMBO card
F11	PMIC_STBY_REQ	O	Not used
F12	BOOT_MODE1	I	Boot mode setting terminal Fixed at "H"
F13	SD3_DAT7	-	Not used
F14	SD3_DAT1	-	Not used
F15	NANDF_CS0	O	Power on/off control signal output terminal for the learning circuit power "H": power on
F16	NANDF_D2	-	Not used
F17	SD4_DAT2	I/O	Two-way data bus with the flash memory
F18	SD1_DAT3	-	Not used
F19	SD2_CMD	-	Not used
F20	RGMII_TD1	O	RGMII transmit data output to the ethernet transceiver
F21, F22	EIM_D17, EIM_D24	-	Not used
F23	EIM_EB3	I	Boot mode setting terminal Fixed at "L"
F24, F25	EIM_A22, EIM_A24	I	Boot mode setting terminal Fixed at "L"
G1	DSI_D0P	-	Not used
G2	DSI_D0M	-	Not used
G3	GND39	-	Ground terminal
G4	DSI_REXT	-	Not used
G5	JTAG_TDI	-	Not used
G6	JTAG_TDO	-	Not used
G7	PCIE_VPH	-	Power supply terminal for the PCIe interface (+2.5V)
G8	PCIE_VPTX	-	Power supply terminal for the PCIe interface (+1.1V)
G9	VDD_SNVS_CAP	O	Internal regulator output terminal
G10	GND37	-	Ground terminal
G11	VDD_SNVS_IN	-	Power supply terminal for the SNVS Regulator (+3.3V)
G12	SATA_VPH	-	Power supply terminal for the SATA interface (+2.5V)
G13	SATA_VP	-	Power supply terminal for the SATA interface (+1.1V)
G14	NVCC_SD3	O	Internal regulator output terminal
G15	NVCC_NANDF	O	Internal regulator output terminal
G16, G17	NVCC_SD1, NVCC_SD2	O	Internal regulator output terminal
G18	NVCC_RGMII	-	Power supply terminal for the ENET interface (+1.8V)
G19	GND38	-	Ground terminal
G20	EIM_D20	O	Configuration signal output to the FPGA
G21, G22	EIM_D19, EIM_D25	-	Not used
G23	EIM_D28	I	Diag mode enable signal input terminal
G24, G25	EIM_A17, EIM_A19	I	Boot mode setting terminal Fixed at "L"
H1	DSI_D1P	-	Not used
H2	DSI_D1M	-	Not used
H3	DSI_CLK0M	-	Not used
H4	DSI_CLK0P	-	Not used
H5	JTAG_TCK	-	Not used
H6	JTAG_MOD	-	Not used
H7	PCIE_VP	-	Power supply terminal for the PCIe interface (+1.1V)
H8	GND43	-	Ground terminal
H9	VDDHIGH_IN1	-	Power supply terminal for the +2.5V regulator (+3.3V)
H10	VDDHIGH_CAP1	O	Internal regulator output terminal (+2.5V)
H11	VDDARM23_CAP1	O	Internal regulator output terminal Not used

Pin No.	Pin Name	I/O	Description
H12	GND40	-	Ground terminal
H13	VDDARM_CAP1	O	Internal regulator output terminal
H14	VDDARM_IN1	-	Power supply terminal for the cores regulator (+1.42V)
H15	GND41	-	Ground terminal
H16	VDDSOC_IN1	-	Power supply terminal for the SOC and PU regulators (+1.42V)
H17	VDDPU_CAP1	O	Internal regulator output terminal
H18	GND42	-	Ground terminal
H19	EIM_A25	I	Interrupt signal input from the FPGA
H20	EIM_D21	I	VBUS power over current detection signal input terminal for the USB connector
H21	EIM_D31	-	Not used
H22, H23	EIM_A20, EIM_A21	I	Boot mode setting terminal Fixed at “L”
H24	EIM_CS0	O	Serial data transfer clock signal output to the FPGA
H25	EIM_A16	I	Boot mode setting terminal Fixed at “L”
J1	HDMI_REF	-	Not used
J2	GND47	-	Ground terminal
J3	HDMI_D1M	-	Not used
J4	HDMI_D1P	-	Not used
J5	HDMI_CLKM	-	Not used
J6	HDMI_CLKP	-	Not used
J7	NVCC_JTAG	O	Internal regulator output terminal
J8	GND48	-	Ground terminal
J9	VDDHIGH_IN2	-	Power supply terminal for the +2.5V regulator (+3.3V)
J10	VDDHIGH_CAP2	O	Internal regulator output terminal (+2.5V)
J11	VDDARM23_CAP2	O	Internal regulator output terminal Not used
J12	GND44	-	Ground terminal
J13	VDDARM_CAP2	O	Internal regulator output terminal
J14	VDDARM_IN2	-	Power supply terminal for the cores regulator (+1.42V)
J15	GND45	-	Ground terminal
J16	VDDSOC_IN2	-	Power supply terminal for the SOC and PU regulators (+1.42V)
J17	VDDPU_CAP2	O	Internal regulator output terminal
J18	GND46	-	Ground terminal
J19, J20	EIM_D29, EIM_D30	-	Not used
J21, J22	EIM_A23, EIM_A18	I	Boot mode setting terminal Fixed at “L”
J23	EIM_CS1	O	Configuration data output to the FPGA
J24	EIM_OE	-	Not used
J25	EIM_DA1	I	Boot mode setting terminal Fixed at “L”
K1	HDMI_HPD	-	Not used
K2	HDMI_DDCCEC	-	Not used
K3	HDMI_D2M	-	Not used
K4	HDMI_D2P	-	Not used
K5	HDMI_D0M	-	Not used
K6	HDMI_D0P	-	Not used
K7	NVCC_MIPI	-	Power supply terminal for the MIPI interface (+2.5V)
K8	GND53	-	Ground terminal
K9	VDDARM23_IN1	-	Power supply terminal for the cores regulator Not used
K10	GND49	-	Ground terminal
K11	VDDARM23_CAP3	O	Internal regulator output terminal Not used
K12	GND50	-	Ground terminal
K13	VDDARM_CAP3	O	Internal regulator output terminal
K14	VDDARM_IN3	-	Power supply terminal for the cores regulator (+1.42V)
K15	GND51	-	Ground terminal
K16	VDDSOC_IN3	-	Power supply terminal for the SOC and PU regulators (+1.42V)
K17	VDDPU_CAP3	O	Internal regulator output terminal
K18	GND52	-	Ground terminal
K19	NVCC_EIM0	-	Power supply terminal for the EMI interface (+3.3V)
K20	EIM_RW	I	Boot mode setting terminal Fixed at “L”
K21	EIM_EB0	I	Boot mode setting terminal Fixed at “L”
K22	EIM_LBA	I	Boot mode setting terminal Fixed at “L”
K23	EIM_EB1	I	Boot mode setting terminal Fixed at “L”

Pin No.	Pin Name	I/O	Description
K24	EIM_DA3	I	Boot mode setting terminal Fixed at “L”
K25	EIM_DA6	I	Boot mode setting terminal Fixed at “H”
L1	CSI0_DAT13	I	Serial data or program data input from the system controller
L2	GND58	-	Ground terminal
L3, L4	CSI0_DAT17, CSI0_DAT16	-	Not used
L5	GND59	-	Ground terminal
L6	CSI0_DAT19	-	Not used
L7	HDMI_VP	-	Power supply terminal for the HDMI interface Not used
L8	GND60	-	Ground terminal
L9	VDDARM23_IN2	-	Power supply terminal for the cores regulator Not used
L10	GND54	-	Ground terminal
L11	VDDARM23_CAP4	O	Internal regulator output terminal Not used
L12	GND55	-	Ground terminal
L13	VDDARM_CAP4	O	Internal regulator output terminal
L14	VDDARM_IN4	-	Power supply terminal for the cores regulator (+1.42V)
L15	GND56	-	Ground terminal
L16	VDDSOC_IN4	-	Power supply terminal for the SOC and PU regulators (+1.42V)
L17	VDDPU_CAP4	O	Internal regulator output terminal
L18	GND57	-	Ground terminal
L19	NVCC_EIM1	-	Power supply terminal for the EMI interface (+3.3V)
L20 to L22	EIM_DA0, EIM_DA2, EIM_DA4	I	Boot mode setting terminal Fixed at “L”
L23	EIM_DA5	I	Boot mode setting terminal Fixed at “H”
L24	EIM_DA8	O	Reset signal output to the liquid crystal display “L”: reset
L25	EIM_DA7	I	Boot mode setting terminal Fixed at “L”
M1	CSI0_DAT10	O	Transmit data output terminal
M2	CSI0_DAT12	O	Serial data or program data output to the system controller
M3	CSI0_DAT11	I	Receive data input terminal
M4	CSI0_DAT14	O	Serial data output to the remote commander code learning processor
M5	CSI0_DAT15	I	Serial data input from the remote commander code learning processor
M6	CSI0_DAT18	-	Not used
M7	HDMI_VPH	-	Power supply terminal for the HDMI interface Not used
M8	GND65	-	Ground terminal
M9	VDDARM23_IN3	-	Power supply terminal for the cores regulator Not used
M10	GND61	-	Ground terminal
M11	VDDARM23_CAP5	O	Internal regulator output terminal Not used
M12	GND62	-	Ground terminal
M13	VDDARM_CAP5	O	Internal regulator output terminal
M14	VDDARM_IN5	-	Power supply terminal for the cores regulator (+1.42V)
M15	GND63	-	Ground terminal
M16	VDDSOC_IN5	-	Power supply terminal for the SOC and PU regulators (+1.42V)
M17	VDDPU_CAP5	O	Internal regulator output terminal
M18	GND64	-	Ground terminal
M19	NVCC_EIM2	-	Power supply terminal for the EMI interface (+3.3V)
M20	EIM_DA11	I	Boot mode setting terminal Fixed at “H”
M21 to M23	EIM_DA9, EIM_DA10, EIM_DA13	I	Boot mode setting terminal Fixed at “L”
M24	EIM_DA12	I	Boot mode setting terminal Fixed at “H”
M25	EIM_WAIT	I	Boot mode setting terminal Fixed at “L”
N1	CSI0_DAT4	-	Not used
N2	CSI0_VSYNC	O	Power on/off control signal output terminal for the FPGA power “H”: power on
N3 to N5	CSI0_DAT7, CSI0_DAT6, CSI0_DAT9	-	Not used
N6	CSI0_DAT8	I	INIT_DONE signal input from the FPGA
N7	NVCC_CSI	I	Power supply terminal for the camera sensor interface (+3.3V)
N8	GND69	-	Ground terminal
N9	VDDARM23_IN4	-	Power supply terminal for the cores regulator Not used
N10	GND66	-	Ground terminal

Pin No.	Pin Name	I/O	Description
N11	VDDARM23_CAP6	O	Internal regulator output terminal Not used
N12	VDD_CACHE_CAP	O	Internal regulator output terminal (+1.1V)
N13	VDDARM_CAP6	O	Internal regulator output terminal
N14	VDDARM_IN6	-	Power supply terminal for the cores regulator (+1.42V)
N15	GND67	-	Ground terminal
N16	VDDSOC_IN6	-	Power supply terminal for the SOC and PU regulators (+1.42V)
N17	VDDPU_CAP6	O	Internal regulator output terminal
N18	GND68	-	Ground terminal
N19	DI0_DISP_CLK	O	Clock signal output to the liquid crystal display
N20	DI0_PIN3	-	Not used
N21	DI0_PIN15	O	Data enable signal output to the liquid crystal display
N22	EIM_BCLK	-	Not used
N23	EIM_DA14	I	Boot mode setting terminal Fixed at “L”
N24	EIM_DA15	I	Boot mode setting terminal Fixed at “H”
N25	DI0_PIN2	-	Not used
P1	CSI0_PIXCLK	I	CONF_DONE signal input from the FPGA
P2	CSI0_DAT5	-	Not used
P3	CSI0_DATA_EN	-	Not used
P4	CSI0_MCLK	O	Reset signal output to the FPGA “L”: reset
P5, P6	GPIO_19, GPIO_18	-	Not used
P7	NVCC_GPIO	-	Power supply terminal for the GPIO interface (+3.3V)
P8	GND74	-	Ground terminal
P9	VDDARM23_IN5	-	Power supply terminal for the cores regulator Not used
P10	GND70	-	Ground terminal
P11	VDDARM23_CAP7	O	Internal regulator output terminal Not used
P12	GND71	-	Ground terminal
P13	VDDARM_CAP7	O	Internal regulator output terminal
P14	VDDARM_IN7	-	Power supply terminal for the cores regulator (+1.42V)
P15	GND72	-	Ground terminal
P16	VDDSOC_IN7	-	Power supply terminal for the SOC and PU regulators (+1.42V)
P17	VDDPU_CAP7	O	Internal regulator output terminal
P18	GND73	-	Ground terminal
P19	NVCC_LCD	-	Power supply terminal for the LCD interface (+3.3V)
P20 to P24	DISP0_DAT4, DISP0_DAT3, DISP0_DAT1, DISP0_DAT2, DISP0_DAT0	O	RGB signal (blue) output to the liquid crystal display
P25	DI0_PIN4	O	Liquid crystal display backlight on/off control signal output terminal “H”: backlight on
R1	GPIO_17	O	Reset signal output to the PCIe transceiver “L”: reset
R2 to R7	GPIO_16, GPIO_7, GPIO_5, GPIO_8, GPIO_4, GPIO_3	-	Not used
R8	GND78	-	Ground terminal
R9	VDDARM23_IN6	-	Power supply terminal for the cores regulator Not used
R10	VSSSOC_CAP1	O	Internal regulator output terminal (+1.1V)
R11	VDDARM23_CAP8	O	Internal regulator output terminal Not used
R12	GND75	-	Ground terminal
R13	VDDARM_CAP8	O	Internal regulator output terminal
R14	VDDARM_IN8	-	Power supply terminal for the cores regulator (+1.42V)
R15	GND76	-	Ground terminal
R16	VDDSOC_IN8	-	Power supply terminal for the SOC and PU regulators (+1.42V)
R17	GND77	-	Ground terminal
R18	NVCC_DRAM1	-	Power supply terminal for the DDR interface (+1.5V)
R19	NVCC_ENET	-	Power supply terminal for the ENET interface (+3.3V)
R20 to R22	DISP0_DAT13, DISP0_DAT10, DISP0_DAT8	O	RGB signal (green) output to the liquid crystal display
R23 to R25	DISP0_DAT6, DISP0_DAT7, DISP0_DAT5	O	RGB signal (blue) output to the liquid crystal display

Pin No.	Pin Name	I/O	Description
T1 to T5	GPIO_2, GPIO_9, GPIO_6, GPIO_1, GPIO_0	-	Not used
T6	KEY_COL4	-	Not used
T7	KEY_ROW3	O	Liquid crystal display dimmer control signal output terminal
T8	GND84	-	Ground terminal
T9	VDDARM23_IN7	-	Power supply terminal for the cores regulator Not used
T10	VSSSOC_CAP2	O	Internal regulator output terminal (+1.1V)
T11, T12	GND79, GND80	-	Ground terminal
T13, T14	VSSSOC_CAP3, VSSSOC_CAP4	O	Internal regulator output terminal (+1.1V)
T15	GND81	-	Ground terminal
T16	VDDSOC_IN9	-	Power supply terminal for the SOC and PU regulators (+1.42V)
T17	GND82	-	Ground terminal
T18	NVCC_DRAM2	-	Power supply terminal for the DDR interface (+1.5V)
T19	GND83	-	Ground terminal
T20, T21	DISP0_DAT21, DISP0_DAT16	O	RGB signal (red) output to the liquid crystal display
T22 to T25	DISP0_DAT15, DISP0_DAT11, DISP0_DAT12, DISP0_DAT9	O	RGB signal (green) output to the liquid crystal display
U1	LVDS0_TX0_P	-	Not used
U2	LVDS0_TX0_N	-	Not used
U3	LVDS0_TX1_P	-	Not used
U4	LVDS0_TX1_N	-	Not used
U5	KEY_COL3	O	Liquid crystal display dimmer control signal output terminal
U6	KEY_ROW1	-	Not used
U7	KEY_COL1	-	Not used
U8	GND89	-	Ground terminal
U9	VDDARM23_IN8	-	Power supply terminal for the cores regulator Not used
U10	VSSSOC_CAP5	O	Internal regulator output terminal (+1.1V)
U11, U12	GND85, GND86	-	Ground terminal
U13, U14	VSSSOC_CAP6, VSSSOC_CAP7	O	Internal regulator output terminal (+1.1V)
U15	GND87	-	Ground terminal
U16	VDDSOC_IN10	-	Power supply terminal for the SOC and PU regulators (+1.42V)
U17	GND88	-	Ground terminal
U18	NVCC_DRAM3	-	Power supply terminal for the DDR interface (+1.5V)
U19	GND90	-	Ground terminal
U20	ENET_TXD0	O	Power on/off control signal output terminal for the liquid crystal display "H": power on
U21	ENET_CRS_DV	O	Reset signal output to the ethernet transceiver "L": reset
U22 to U24	DISP0_DAT20, DISP0_DAT19, DISP0_DAT17	O	RGB signal (red) output to the liquid crystal display
U25	DISP0_DAT14	O	RGB signal (green) output to the liquid crystal display
V1	LVDS0_TX2_P	-	Not used
V2	LVDS0_TX2_N	-	Not used
V3	LVDS0_CLK_P	-	Not used
V4	LVDS0_CLK_N	-	Not used
V5, V6	KEY_ROW4, KEY_ROW0	-	Not used
V7	NVCC_LVDS2P5	-	Power supply terminal for the LVDS display interface (+2.5V)
V8	GND91	-	Ground terminal
V9 to V18	NVCC_DRAM13, NVCC_DRAM4 to NVCC_DRAM12	-	Power supply terminal for the DDR interface (+1.5V)
V19	GND92	-	Ground terminal
V20	ENET_MDC	O	Management data clock signal output to the ethernet transceiver
V21	ENET_TX_EN	-	Not used
V22	ENET_REF_CLK	I	25 MHz clock signal input from the ethernet transceiver
V23	ENET_MDIO	I/O	Two-way management data bus with the ethernet transceiver

Pin No.	Pin Name	I/O	Description
V24, V25	DISP0_DAT22, DISP0_DAT18	O	RGB signal (red) output to the liquid crystal display
W1	LVDS0_TX3_P	-	Not used
W2	LVDS0_TX3_N	-	Not used
W3	GND93	-	Ground terminal
W4	KEY_ROW2	-	Not used
W5, W6	KEY_COL0, KEY_COL2	-	Not used
W7 to W13	GND94 to GND100	-	Ground terminal
W14	DRAM_A4	O	Address signal output to the SD-RAM
W15 to W19	GND101 to GND105	-	Ground terminal
W20	ENET_TXD1	O	VBUS power on/off control signal output terminal for the WLAN/BT COMBO card “H”: power on
W21	ENET_RXD0	-	Not used
W22	ENET_RXD1	I	Interrupt signal input from the ethernet transceiver
W23	ENET_RX_ER	O	ID signal output terminal for the USB OTG Not used
W24	DISP0_DAT23	O	RGB signal (red) output to the liquid crystal display
W25	DRAM_D63	-	Not used
Y1	LVDS1_TX0_N	-	Not used
Y2	LVDS1_TX0_P	-	Not used
Y3	LVDS1_CLK_N	-	Not used
Y4	LVDS1_CLK_P	-	Not used
Y5	GND106	-	Ground terminal
Y6	DRAM_RESET	O	Reset signal output to the SD-RAM “L”: reset
Y7 to Y10	DRAM_D20, DRAM_D21, DRAM_D19, DRAM_D25	I/O	Two-way data bus with the SD-RAM
Y11	DRAM_SDCKE0	O	Clock enable signal output to the SD-RAM
Y12 to Y14	DRAM_A15, DRAM_A7, DRAM_A3	O	Address signal output to the SD-RAM
Y15	DRAM_SDBA1	O	Bank address signal output to the SD-RAM
Y16	DRAM_CS0	O	Chip select signal output to the SD-RAM
Y17 to Y20	DRAM_D36, DRAM_D37, DRAM_D40, DRAM_D44	-	Not used
Y21	DRAM_DQM7	-	Not used
Y22, Y23	DRAM_D59, DRAM_D62	-	Not used
Y24	GND107	-	Ground terminal
Y25	DRAM_D58	-	Not used
AA1	LVDS1_TX1_P	-	Not used
AA2	LVDS1_TX1_N	-	Not used
AA3	LVDS1_TX3_N	-	Not used
AA4	LVDS1_TX3_P	-	Not used
AA5, AA6	DRAM_D3, DRAM_D10	I/O	Two-way data bus with the SD-RAM
AA7	GND10	-	Ground terminal
AA8, AA9	DRAM_D17, DRAM_D23	I/O	Two-way data bus with the SD-RAM
AA10	GND5	-	Ground terminal
AA11	DRAM_SDCKE1	-	Not used
AA12	DRAM_A14	O	Address signal output to the SD-RAM
AA13	GND6	-	Ground terminal
AA14, AA15	DRAM_A2, DRAM_A10	O	Address signal output to the SD-RAM
AA16	GND7	-	Ground terminal
AA17, AA18	DRAM_D32, DRAM_D33	-	Not used

Pin No.	Pin Name	I/O	Description
AA19	GND8	-	Ground terminal
AA20, AA21	DRAM_D45, DRAM_D57	-	Not used
AA22	GND9	-	Ground terminal
AA23	DRAM_D61	-	Not used
AA24	DRAM_SDQS7_B	-	Not used
AA25	DRAM_SDQS7	-	Not used
AB1	LVDS1_TX2_N	-	Not used
AB2	LVDS1_TX2_P	-	Not used
AB3	GND12	-	Ground terminal
AB4 to AB7	DRAM_D6, DRAM_D12, DRAM_D14, DRAM_D16	I/O	Two-way data bus with the SD-RAM
AB8	DRAM_DQM2	O	Data mask (upper byte) signal output to the SD-RAM
AB9	DRAM_D18	I/O	Two-way data bus with the SD-RAM
AB10	DRAM_SDQS3_B	O	Data strobe signal (lower byte) output to the SD-RAM
AB11	DRAM_D27	I/O	Two-way data bus with the SD-RAM
AB12	DRAM_SDBA2	O	Bank address signal output to the SD-RAM
AB13, AB14	DRAM_A8, DRAM_A1	O	Address signal output to the SD-RAM
AB15	DRAM_RAS	O	Row address signal output to the SD-RAM
AB16	DRAM_SDWE	O	Write enable signal output to the SD-RAM
AB17	DRAM_SDODT1	-	Not used
AB18	DRAM_DQM4	-	Not used
AB19 to AB23	DRAM_D38, DRAM_D41, DRAM_D42, DRAM_D52, DRAM_D60	-	Not used
AB24	GND11	-	Ground terminal
AB25	DRAM_D56	-	Not used
AC1	DRAM_D4	I/O	Two-way data bus with the SD-RAM
AC2	DRAM_VREF	I	Reference voltage input terminal
AC3	DRAM_DQM0	O	Data mask (lower byte) signal output to the SD-RAM
AC4, AC5	DRAM_D2, DRAM_D13	I/O	Two-way data bus with the SD-RAM
AC6	DRAM_DQM1	O	Data mask (upper byte) signal output to the SD-RAM
AC7 to AC9	DRAM_D15, DRAM_D22, DRAM_D28	I/O	Two-way data bus with the SD-RAM
AC10	DRAM_SDQS3	O	Data strobe signal (lower byte) output to the SD-RAM
AC11	DRAM_D31	I/O	Two-way data bus with the SD-RAM
AC12 to AC14	DRAM_A11, DRAM_A6, DRAM_A0	O	Address signal output to the SD-RAM
AC15	DRAM_SDBA0	O	Bank address signal output to the SD-RAM
AC16	DRAM_SDODT0	O	On die termination signal output to the SD-RAM
AC17	DRAM_A13	O	Address signal output to the SD-RAM
AC18, AC19	DRAM_D34, DRAM_D39	-	Not used
AC20	DRAM_DQM5	-	Not used
AC21 to AC25	DRAM_D47, DRAM_D48, DRAM_D53, DRAM_D51, DRAM_D55	-	Not used
AD1, AD2	DRAM_D5, DRAM_D0	I/O	Two-way data bus with the SD-RAM
AD3	DRAM_SDQS0_B	O	Data strobe signal (lower byte) output to the SD-RAM
AD4	GND18	-	Ground terminal
AD5	DRAM_D8	I/O	Two-way data bus with the SD-RAM
AD6	DRAM_SDQS1	O	Data strobe signal (upper byte) output to the SD-RAM

Pin No.	Pin Name	I/O	Description
AD7	GND19	-	Ground terminal
AD8	DRAM_SDQS2	O	Data strobe signal (upper byte) output to the SD-RAM
AD9	DRAM_D29	I/O	Two-way data bus with the SD-RAM
AD10	GND13	-	Ground terminal
AD11	DRAM_D30	I/O	Two-way data bus with the SD-RAM
AD12	DRAM_A12	O	Address signal output to the SD-RAM
AD13	GND14	-	Ground terminal
AD14	DRAM_SDCLK_1	O	Clock signal output to the SD-RAM
AD15	DRAM_SDCLK_0	-	Not used
AD16	GND15	-	Ground terminal
AD17	DRAM_CS1	-	Not used
AD18	DRAM_SDQS4	-	Not used
AD19	GND16	-	Ground terminal
AD20	DRAM_SDQS5	-	Not used
AD21	DRAM_D43	-	Not used
AD22	GND17	-	Ground terminal
AD23	DRAM_SDQS6	-	Not used
AD24	DRAM_DQM6	-	Not used
AD25	DRAM_D54	-	Not used
AE1	GND20	-	Ground terminal
AE2	DRAM_D1	I/O	Two-way data bus with the SD-RAM
AE3	DRAM_SDQS0	O	Data strobe signal (lower byte) output to the SD-RAM
AE4, AE5	DRAM_D7, DRAM_D9	I/O	Two-way data bus with the SD-RAM
AE6	DRAM_SDQS1_B	O	Data strobe signal (upper byte) output to the SD-RAM
AE7	DRAM_D11	I/O	Two-way data bus with the SD-RAM
AE8	DRAM_SDQS2_B	O	Data strobe signal (upper byte) output to the SD-RAM
AE9	DRAM_D24	I/O	Two-way data bus with the SD-RAM
AE10	DRAM_DQM3	O	Data mask (lower byte) signal output to the SD-RAM
AE11	DRAM_D26	I/O	Two-way data bus with the SD-RAM
AE12, AE13	DRAM_A9, DRAM_A5	O	Address signal output to the SD-RAM
AE14	DRAM_SDCLK_1_B	O	Clock signal output to the SD-RAM
AE15	DRAM_SDCLK_0_B	-	Not used
AE16	DRAM_CAS	O	Column address signal output to the SD-RAM
AE17	ZQPAD	I	External calibration resistor connection terminal
AE18	DRAM_SDQS4_B	-	Not used
AE19	DRAM_D35	-	Not used
AE20	DRAM_SDQS5_B	-	Not used
AE21, AE22	DRAM_D46, DRAM_D49	-	Not used
AE23	DRAM_SDQS6_B	-	Not used
AE24	DRAM_D50	-	Not used
AE25	GND21	-	Ground terminal

MAIN BOARD IC602 AR8035-AL1A (ETHERNET TRANSCEIVER, +1.1V/+1.8V REGULATOR)

Pin No.	Pin Name	I/O	Description
1	RSTN	I	Reset signal input from the MPU "L": reset
2	LX	O	Internal regulator output terminal (+1.1V)
3	VDD33	-	Power supply terminal (+3.3V)
4	XTALO	O	System clock output terminal (25 MHz)
5	XTALI	I	System clock input terminal (25 MHz)
6	AVDDL_6	-	Power supply terminal (+1.1V)
7	RBIAS	O	Bias current setting terminal
8	VDDH_REG	O	Internal regulator output terminal (+2.5V)
9	TRXP0	I/O	Two-way data (positive) bus with the LAN jack
10	TRXN0	I/O	Two-way data (negative) bus with the LAN jack
11	AVDDL_11	-	Power supply terminal (+1.1V)
12	TRXP1	I/O	Two-way data (positive) bus with the LAN jack
13	TRXN1	I/O	Two-way data (negative) bus with the LAN jack
14	AVDD33	-	Power supply terminal (+3.3V)
15	TRXP2	I/O	Two-way data (positive) bus with the LAN jack
16	TRXN2	I/O	Two-way data (negative) bus with the LAN jack
17	AVDDL_17	-	Power supply terminal (+1.1V)
18	TRXP3	I/O	Two-way data (positive) bus with the LAN jack
19	TRXN3	I/O	Two-way data (negative) bus with the LAN jack
20	INT	O	Interrupt signal output to the MPU
21	LED_ACT	O	LED drive signal output terminal Not used
22	LED_1000	O	LED drive signal output terminal Not used
23	CLK_25M	O	25 MHz clock signal output to the MPU
24	LED_10_100	O	LED drive signal output terminal Not used
25, 26	RXD3, RXD2	O	RGMII receive data output to the MPU
27	VDDIO_REG	O	Internal regulator output terminal (+1.8V)
28, 29	RXD1, RXD0	O	RGMII receive data output to the MPU
30	RX_DV	O	RGMII receive data valid signal output to the MPU
31	RX_CLK	O	RGMII receive clock signal output to the MPU
32	TX_EN	I	RGMII transmit data enable signal input from the MPU
33	GTX_CLK	I	RGMII transmit clock signal input from the MPU
34 to 37	TXD0 toTXD3	I	RGMII transmit data input from the MPU
38	DVDDL	-	Power supply terminal (+1.1V)
39	MDIO	I/O	Two-way management data bus with the MPU
40	MDC	I	Management data clock signal input from the MPU

FPGA DSP BOARD IC001 EP4CGX30BF14C8N (FPGA)

Pin No.	Pin Name	I/O	Description
A1	TDO	O	Data output terminal for the JTAG Not used
A2	TMS	I	Mode selection signal input terminal for the JTAG Not used
A3	TDI	I	Data input terminal for the JTAG Not used
A4	DCLK	I	Serial data transfer clock signal input from the MPU
A5	IO/DATA0	I	Configuration data input from the MPU
A6	IO/CLKUSR	I	Not used
A7	IO/PLL2_CLKOUTN	O	Not used
A8	IO/PLL2_CLKOUTP	O	Not used
A9	CLK8/DIFFCLK_5N	I	Audio data input from the audio DSP
A10	CLK9/DIFFCLK_5p	I	Not used
A11	IO/DIFFCLK_T11N/ DQ0T	O	Bit clock signal output to the audio DSP
A12	IO/DIFFCLK_T11P/ DQ0T	O	Not used
A13	IO/DIFFIO_T17N/ DQS0T/CQ0T/ DPCLK10	O	Not used
B1, B2	GND_B1, GND_B2	-	Ground terminal
B3	TCK	I	Clock signal input terminal for the JTAG Not used
B4	GND_B4	-	Ground terminal
B5	IO/ASDO	O	Not used
B6	IO/DQS1T/CQ0T#/ DPCLK13	O	Not used
B7	GND_B7	-	Ground terminal
B8	IO/DIFFCLK_T12N/ DQ0T	O	Bit clock signal output to the audio DSP
B9	GND_B9	-	Ground terminal
B10	IO/VREFB7N0	O	Audio data output to the audio DSP
B11	IO_B11	O	L/R sampling clock signal output to the audio DSP
B12	GND_B12	-	Ground terminal
B13	IO/DIFFIO_T17P/ DQ0T	O	Not used
C1	GXB_TX1n	O	Not used
C2	GXB_TX1p	O	Not used
C3	VCCIO9_C3	-	Power supply terminal (+3.3V)
C4	nCE	I	Chip enable signal input terminal Not used
C5	IO/NSCO	O	Not used
C6	IO/VREFB8N0	O	Not used
C7	VCCIO8_C7	-	Power supply terminal (+3.3V)
C8	IO/DIFFCLK_T12P/ DQ0T	O	L/R sampling clock signal output to the audio DSP
C9, C10	VCCIO7_C9, VCCIO7_C10	-	Power supply terminal (+3.3V)
C11	IO/PUP4/DQ0T	O	Not used
C12	IO/PDN4/DQ0T	O	Not used
C13	IO/DiFFIO_T1BN/ DQ0T	O	Not used
D1, D2	GND_D1, GND_D2	-	Ground terminal
D3	VCCD_PLL_D3	-	Power supply terminal (+1.2V)
D4	VCCA_D4	-	Power supply terminal (+2.5V)
D5	nCONFIG	I	Configuration signal input from the MPU
D6	GND_D6	-	Ground terminal
D7	VCC_CLKIN8A	-	Power supply terminal (+2.5V)
D8	GND_D8	-	Ground terminal
D9	VCCA_D9	-	Power supply terminal (+2.5V)
D10	IO/DIFFIO_R4n/ DEV_CLRn	I	Reset signal input from the MPU "L": reset
D11	IO/DIFFIO_R2P	O	Chip select signal output to the D/A converter
D12	IO/DIFFIO_R2N	O	Serial data output to the D/A converter
D13	IO/DIFFIO_T1BP/ DQ0T	O	Interrupt signal output to the MPU

Pin No.	Pin Name	I/O	Description
E1	GXB_RX1p	O	Not used
E2	GXB_RX1n	O	Not used
E3	GND_E3	-	Ground terminal
E4	VCCINT_E4	-	Power supply terminal (+1.2V)
E5	GND_E5	-	Ground terminal
E6	CLK10/DIFFCLK_4n/ REFCLK1n	I	Reset signal input from the MPU "L": reset
E7	CLK11/DIFFCLK_4p/ REFCLK1p	I	50 MHz clock signal input terminal
E8	VCCINT_E8	-	Power supply terminal (+1.2V)
E9	GND_E9	-	Ground terminal
E10	IO/DIFFIO_R4P/ DM0R	O	Serial data transfer clock signal output to the D/A converter
E11	VCCIO6_E11	-	Power supply terminal (+3.3V)
E12	GND_E12	-	Ground terminal
E13	IO/VREFB6N0	O	Not used
F1, F2	GND_F1, GND_F2	-	Ground terminal
F3	VCCL_GXB_F3	-	Power supply terminal (+1.2V)
F4	GND_F4	-	Ground terminal
F5	VCCINT_F5	-	Power supply terminal (+1.2V)
F6	GND_F6	-	Ground terminal
F7	VCCINT_F7	-	Power supply terminal (+1.2V)
F8	GND_F8	-	Ground terminal
F9	IO_F9	O	Chip select signal output to the audio DSP
F10	IO/DIFFIO_R5P/ DQ0R	I	Serial data input from the audio DSP
F11	IO/DIFFIO_R5N/ DQ0R	O	Serial data output to the audio DSP
F12	CLK7/DIFFCLK_3P	O	Not used
F13	CLK6/DIFFCLK_3N	O	Not used
G1	GXB_TX0n	O	Receive data (negative) output to the FPGA
G2	GXB_TX0p	O	Receive data (positive) output to the FPGA
G3	VCCH_GXB	-	Power supply terminal (+2.5V)
G4	VCCINT_G4	-	Power supply terminal (+1.2V)
G5	GND_G5	-	Ground terminal
G6	VCCINT_G6	-	Power supply terminal (+1.2V)
G7	GND_G7	-	Ground terminal
G8	VCCINT_G8	-	Power supply terminal (+1.2V)
G9	IO/DIFFIO_R6P/ DQS0R/CQ0R/ DPCLK8	O	Serial data transfer clock signal output to the audio DSP
G10	IO/DIFFIO_R6N/ DEV_OE	O	Reset signal output to the audio DSP "L": reset
G11	VCCIO6_G11	-	Power supply terminal (+3.3V)
G12	GND_G12	-	Ground terminal
G13	CLK4/DIFFCLK_2N	I	5.6448 MHz clock or 6.144 MHz clock signal input terminal
H1, H2	GND_H1, GND_H2	-	Ground terminal
H3	VCCL_GXB_H3	-	Power supply terminal (+1.2V)
H4	GND_H4	-	Ground terminal
H5	VCCINT_H5	-	Power supply terminal (+1.2V)
H6	GND_H6	-	Ground terminal
H7	VCCINT_H7	-	Power supply terminal (+1.2V)
H8	GND_H8	-	Ground terminal
H9	VCCA_H9	-	Power supply terminal (+2.5V)
H10	IO/DIQS1R/CQ0R#/ DPCLK7	O	Output enable signal output terminal for the 22.5792 MHz clock
H11	VCCIO5_H11	-	Power supply terminal (+3.3V)
H12	IO/VREFB5N0	O	Output enable signal output terminal for the 24.576 MHz clock
H13	CLK5/DIFFCLK_2P	I	Not used
J1	GXB_RX0n	I	Transmit data (negative) input from the MPU
J2	GXB_RX0P	I	Transmit data (positive) input from the MPU

Pin No.	Pin Name	I/O	Description
J3	VCCA_GXB_J3	-	Power supply terminal (+2.5V)
J4	VCCD_PLL_J4	-	Power supply terminal (+1.2V)
J5	CONF_DONE	O	CONF_DONE signal output to the MPU
J6	CLK12/DIFFCLK_7p/ REFCLK0p	I	Clock signal (positive) input from the MPU
J7	CLK13/DIFFCLK_7n/ REFCLK0n	I	Clock signal (negative) input from the MPU
J8	VCCINT_J8	-	Power supply terminal (+1.2V)
J9	GND_J9	-	Ground terminal
J10	VCCD_PLL_J10	-	Power supply terminal (+1.2V)
J11	VCCIO5_J11	-	Power supply terminal (+3.3V)
J12	GND_J12	-	Ground terminal
J13	IO/DIFFIO_R9N/ DQ0R	O	Bit clock signal output to the D/A converter
K1, K2	GND_K1, GND_K2	-	Ground terminal
K3	GND_K3	-	Ground terminal
K4	VCCA_K4	-	Power supply terminal (+2.5V)
K5	MSEL0	I	Fixed at “L”
K6	nSTATUS	O	Status signal output to the MPU
K7	VCC_CLKIN3A	-	Power supply terminal (+2.5V)
K8	IO/VREFB4N0	O	Not used
K9	IO/DQ0B	O	Not used
K10	IO/PLL3_CLKOUTP	O	Not used
K11	IO/DIFFIO_R14P/ DQ0R	O	Reset signal output to the D/A converter “L”: reset
K12	IO/DIFFIO_R14N/ DQ0R	O	Audio muting on/off control signal output terminal “L”: muting on
K13	IO/DIFFIO_R9P/ DQ0R	O	Audio data (R-ch) output to the D/A converter
L1	RREF0	I	External reference resistor connection terminal
L2	GND_L2	-	Ground terminal
L3	MSEL2	I	Fixed at “H”
L4	IO/PLL1_CLKOUTP	O	Not used
L5	IO/DQS1B/CQ0B#/ DPCLK2	O	Not used
L6	VCCIO3	-	Power supply terminal (+3.3V)
L7	IO/VREFB3N0	O	Not used
L8	VCCIO4_L8	-	Power supply terminal (+3.3V)
L9	IO/DIFFIO_B19P/ DQ0B	O	Not used
L10	VCCIO4_L10	-	Power supply terminal (+3.3V)
L11	IO/PLL3_CLKOUTN	O	Not used
L12	IO/DIFFIO_R13P/ DQ0R	O	Not used
L13	IO/DIFFIO_R13N/ DQ0R	O	Audio data (L-ch) output to the D/A converter
M1	GND_M1	-	Ground terminal
M2	VCCA_GXB_M2	-	Power supply terminal (+2.5V)
M3	NC_M3	O	Not used
M4	IO/PLL1_CLKOUTN	O	Not used
M5	GND_M5	-	Ground terminal
M6	IO/DIFFIO_B3p/ INIT_DONE	O	INIT_DONE signal output to the MPU
M7	CLK14/DIFFCLK_6P	I	Not used
M8	GND_M8	-	Ground terminal
M9	IO/DIFFIO_B19N/ DQS0B/CQ0B/ DPCLK5	O	Not used
M10	GND_M10	-	Ground terminal
M11	IO/RUP2/DQ0B	O	Not used
M12	GND_M12	-	Ground terminal
M13	IO/RDN3	O	Audio data (R-ch) output to the D/A converter

Pin No.	Pin Name	I/O	Description
N1	VCCL_GXB_N1	-	Power supply terminal (+1.2V)
N2	NC_N2	O	Not used
N3	MSEL1	I	Fixed at “L”
N4	IO/DIFFIO_B1P/ CRC_ERROR	O	Not used
N5	IO/DIFFIO_B1N/ NCEO	O	Not used
N6	IO/DIFFIO_B3N/ DQ0B	O	Not used
N7	CLK15/DIFFCLK_6N	I	Not used
N8	IO/DIFFIO_B12P/ DQ0B	O	Not used
N9	IO/DIFFIO_B12N/ DQ0B	O	Not used
N10	IO/DIFFIO_B21P/ DQ0B	O	Not used
N11	IO/DIFFIO_B21N/ DQ0B	O	Not used
N12	IO/RDN2/DM0B	O	Not used
N13	IO/RUP3	O	Audio data (L-ch) output to the D/A converter

FPGA DSP BOARD IC601 ADSP-21488KSWZ-3A1 (AUDIO DSP)

Pin No.	Pin Name	I/O	Description
1	VDD_INT	-	Power supply terminal (+1.1V) (for core)
2	CLK_CFG1	I	Core instruction rate to CLKIN (pin 12) ratio selection signal input terminal Fixed at “H” in this unit
3	BOOT_CFG0	I	Boot mode selection signal input terminal Fixed at “L” in this unit
4	VDD_EXT	-	Power supply terminal (+3.3V) (for I/O)
5	VDD_INT	-	Power supply terminal (+1.1V) (for core)
6	BOOT_CFG1	I	Boot mode selection signal input terminal Fixed at “L” in this unit
7	GND	-	Ground terminal
8, 9	NC	-	Not used
10	CLK_CFG0	I	Core instruction rate to CLKIN (pin 12) ratio selection signal input terminal Fixed at “L” in this unit
11	VDD_INT	-	Power supply terminal (+1.1V) (for core)
12	CLKIN	I	System clock input terminal (10 MHz)
13	XTAL	O	System clock output terminal (10 MHz)
14	VDD_EXT	-	Power supply terminal (+3.3V) (for I/O)
15, 16	VDD_INT	-	Power supply terminal (+1.1V) (for core)
17	nRESETOUT/ nRESETIN	I/O	Reset signal input/output terminal Not used
18	VDD_INT	-	Power supply terminal (+1.1V) (for core)
19	DPI_P01	I	Serial data input from the FPGA
20	DPI_P02	O	Serial data output to the FPGA
21	DPI_P03	I	Serial data transfer clock signal input from the FPGA
22	VDD_INT	-	Power supply terminal (+1.1V) (for core)
23	DPI_P05	I/O	Not used
24	DPI_P04	I	Chip select signal input from the FPGA
25	DPI_P06	I/O	Not used
26	VDD_EXT	-	Power supply terminal (+3.3V) (for I/O)
27, 28	DPI_P08, DPI_P07	I/O	Not used
29	VDD_INT	-	Power supply terminal (+1.1V) (for core)
30 to 36	DPI_P09 to DPI_P13, DAI_P03, DAI_P14	I/O	Not used
37 to 39	VDD_INT	-	Power supply terminal (+1.1V) (for core)
40	DAI_P13	I	Bit clock signal input from the FPGA
41	DAI_P07	I	L/R sampling clock signal input from the FPGA
42	DAI_P19	I	Audio data input from the FPGA
43, 44	DAI_P01, DAI_P02	I/O	Not used
45	VDD_INT	-	Power supply terminal (+1.1V) (for core)
46	VDD_EXT	-	Power supply terminal (+3.3V) (for I/O)
47	VDD_INT	-	Power supply terminal (+1.1V) (for core)
48	DAI_P06	I	Bit clock signal input from the FPGA
49	DAI_P05	I	L/R sampling clock signal input from the FPGA
50	DAI_P09	O	Audio data output to the FPGA
51	DAI_P10	I/O	Not used
52	VDD_INT	-	Power supply terminal (+1.1V) (for core)
53	VDD_EXT	-	Power supply terminal (+3.3V) (for I/O)
54	DAI_P20	I/O	Not used
55	VDD_INT	-	Power supply terminal (+1.1V) (for core)
56 to 63	DAI_P08, DAI_P04, DAI_P14, DAI_P18 to DAI_P15, DAI_P12	I/O	Not used
64	VDD_INT	-	Power supply terminal (+1.1V) (for core)
65	DAI_P11	I	Interrupt request signal input from the FLAG0 (pin 77)
66, 67	VDD_INT	-	Power supply terminal (+1.1V) (for core)
68	GND	-	Ground terminal
69	THD_M	O	Thermal diode cathode output terminal Not used
70	THD_P	I	Thermal diode anode input terminal Not used
71	VDD_THD	-	Power supply terminal (for thermal diode) Not used
72 to 76	VDD_INT	-	Power supply terminal (+1.1V) (for core)
77	FLAG0	O	Interrupt request signal output to the DAI_P11 (pin 65)

Pin No.	Pin Name	I/O	Description
78, 79	VDD_INT	-	Power supply terminal (+1.1V) (for core)
80 to 82	FLAG1 to FLAG3	O	Interrupt request signal output terminal Not used
83	MLBCLK	I	Media local bus clock signal input terminal Not used
84	MLBDAT	I/O	Media local bus data input/output terminal Not used
85	MLBDO	O	Media local bus data output terminal Not used
86	VDD_EXT	-	Power supply terminal (+3.3V) (for I/O)
87	MLBSIG	I/O	Media local bus signal input/output terminal Not used
88	VDD_EXT	-	Power supply terminal (+3.3V) (for I/O)
89	MVBSO	O	Media local bus signal output terminal Not used
90	!TRST!	I	Test reset signal input terminal Not used
91	!EMU!	O	Emulation status signal output terminal Not used
92	TDO	O	Test data output terminal Not used
93	VDD_EXT	-	Power supply terminal (+3.3V) (for I/O)
94	VDD_INT	-	Power supply terminal (+1.1V) (for core)
95	TDI	I	Test data input terminal Not used
96	TCK	I	Test clock signal input terminal Not used
97	VDD_INT	-	Power supply terminal (+1.1V) (for core)
98	nRESET	I	Reset signal input from the FPGA “L”: reset
99	TMS	I	Test mode selection signal input terminal Not used
100	VDD_INT	-	Power supply terminal (+1.1V) (for core)

U-COM BOARD IC7002 MB9AF156NPMC-G-JNE2 (SYSTEM CONTROLLER)

Pin No.	Pin Name	I/O	Description
1	VCC	-	Power supply terminal (+3.3V)
2 to 8	NC	-	Not used
9	AMUTE	O	Audio muting on/off control signal output terminal “L”: muting on
10	POWER_RY	O	Relay drive signal (for main power) output terminal “H”: relay on
11	AC_STOP_STBY	I	AC cut detection signal (for sub power) input terminal “H”: AC cut is detected
12	AC_STOP_MAIN	I	AC cut detection signal (for main power) input terminal “H”: AC cut is detected
13	DCDC_EN	O	DC/DC converter on/off control signal output terminal for the hard disk drive and VBUS power “H”: DC/DC converter on
14 to 24	NC	-	Not used
25	VSS	-	Ground terminal
26	VCC	-	Power supply terminal (+3.3V)
27 to 32	NC	-	Not used
33	C	-	Regulator stabilization capacitor connection terminal
34	VSS	-	Ground terminal
35	VCC	-	Power supply terminal (+3.3V)
36	NC	-	Not used
37	POWER_EN	O	Power on/off control signal output terminal for the MPU power “H”: power on
38	SYS RESET	I	System reset signal input from the MPU and reset signal generator “L”: reset For several hundreds msec. after the power supply rises, “L” is input, then it change to “H”
39	UART MPU/SYS	I	Serial data input from the MPU
40	UART SYS/MPU	O	Serial data output to the MPU
41	MPU RESET	O	Reset signal output to the MPU “L”: reset
42 to 45	NC	-	Not used
46	MD1	I	Fixed at “L”
47	MD0	I	Update signal input from the MPU “H”: update
48	XTALO	O	System clock output terminal (4 MHz)
49	XTALI	I	System clock input terminal (4 MHz)
50	VSS	-	Ground terminal
51	VCC	-	Power supply terminal (+3.3V)
52, 53	NC	-	Not used
54	SEJG+	I	Jog dial pulse (positive) input from the rotary encoder (for selector)
55	SEJG–	I	Jog dial pulse (negative) input from the rotary encoder (for selector)
56	DSEE LED	O	LED drive signal output terminal for the DSEE indicator “H”: LED on
57	NC	-	Not used
58	AD KEY1	I	BACK and ENTER key input terminal (A/D input)
59	AD KEY2	I	HOME key input terminal (A/D input)
60	AVCC	-	Power supply terminal (+3.3V)
61	AVRH	I	Reference voltage (+3.3V) input terminal
62	AVSS	-	Ground terminal
63	PWR LED GRN	O	LED drive signal output terminal for the power indicator “H”: LED on
64	NC	-	Not used
65	SIRCS IN	I	SIRCS signal input from the remote control receiver
66	PLAY KEY	I	Play key input terminal (A/D input)
67	POWER KEY	I	Power key input terminal (A/D input)
68	MODEL	I	Model setting terminal Fixed at “H” in this unit
69	DST	I	Destination setting terminal Fixed at “L”
70	NC	-	Not used
71	PROG SCK	O	Program clock signal output terminal
72	PROG SYS/JIG	O	Program data output to the MPU
73	PROG JIG/SYS	I	Program data input from the MPU
74	NC	-	Not used
75	VSS	-	Ground terminal
76	VCC	-	Power supply terminal (+3.3V)
77	NC	-	Not used
78	TCK	I	Clock signal input terminal for the JTAG
79	TMS	O	Mode selection signal output terminal for the JTAG
80	TDO	O	Data output terminal for the JTAG
81	TDI	I	Data input terminal for the JTAG
82 to 85	NC	-	Not used

Pin No.	Pin Name	I/O	Description
86, 87	TEST1, TEST2	-	Not used
88	STBY_EN	O	Power cut on/off control signal output terminal for the sub power "H": sub power cut on
89	FAN_EN	O	Fan motor on/off control signal output terminal "H": fan motor on
90	FAN_SLOW/FAST	O	Fan motor speed selection signal output terminal "L": slow, "H": fast
91	TEST3	-	Not used
92, 93	NC	-	Not used
94	E2P SCL	O	Serial data transfer clock signal output to the EEPROM
95	E2P SDA	I/O	Two-way data bus the EEPROM
96	NC	-	Not used
97	VCC	-	Power supply terminal (+3.3V)
98, 99	NC	-	Not used
100	VSS	-	Ground terminal

U-COM BOARD IC7007 µPD78F0527GB (S)-410-UET-A (REMOTE COMMANDER CODE LEARNING PROCESSOR)

Pin No.	Pin Name	I/O	Description
1	P140/PCL/INTP6	I/O	Not used
2	P120/INTP0/EXLVI	I/O	Not used
3, 4	P40, P41	I/O	Not used
5	!RESET!	I	Reset signal input from the MPU "L": reset
6	P124/XT2/EXCLKS	I/O	Not used
7	P123/XT1	I/O	Not used
8	FLMD0	I	Writing mode setting signal input from the P33/T151/INTP4 (pin 18)
9	P122/X2/EXCLK	O	System clock output terminal (5 MHz)
10	P121/X1	I	System clock input terminal (5 MHz)
11	REGC	-	External capacitor connection terminal for the internal regulator
12	VSS	-	Ground terminal
13	VDD	-	Power supply terminal (+3.3V)
14	P60/SCL0	I/O	Not used
15	P61/SDA0	I/O	Not used
16	P62/EXSCL0	I/O	Not used
17	P63	I/O	Not used
18	P33/T151/INTP4	O	Writing mode setting signal output to the FLMD0 (pin 8)
19 to 26	P77/KR7 to P70/KR0	I/O	Not used
27, 28	P32/INTP3, P31/INTP2	I/O	Not used
29	P30/INTP1	I	Interrupt signal input from the MPU
30	P17/TI50/TO50	I/O	Not used
31	P16/TOH1/INTP5	O	SIRCS signal output terminal
32	P15/TOH0	I/O	Not used
33	P14/RXD6	I	Serial data input terminal
34	P13/TXD6	O	Serial data output terminal
35	P12/SO10	I/O	Not used
36	P11/SI10/RXD0	I	Serial data input from the MPU
37	P10/!SCK10!/TXD0	O	Serial data output to the MPU
38	AVREF	I	Reference voltage (+3.3V) input terminal
39	AVSS	-	Ground terminal
40 to 46	ANI7/P27 to ANI1/P21	I/O	Not used
47	P20/ANI0	I/O	Not used
48	P130	O	Not used
49, 50	P03, P02	I/O	Not used
51	P01/TI010/TO00	O	Ground on/off control signal output terminal for learning code generator "H": ground on
52	P00/TI000	I	SIRCS (learning code) signal input terminal

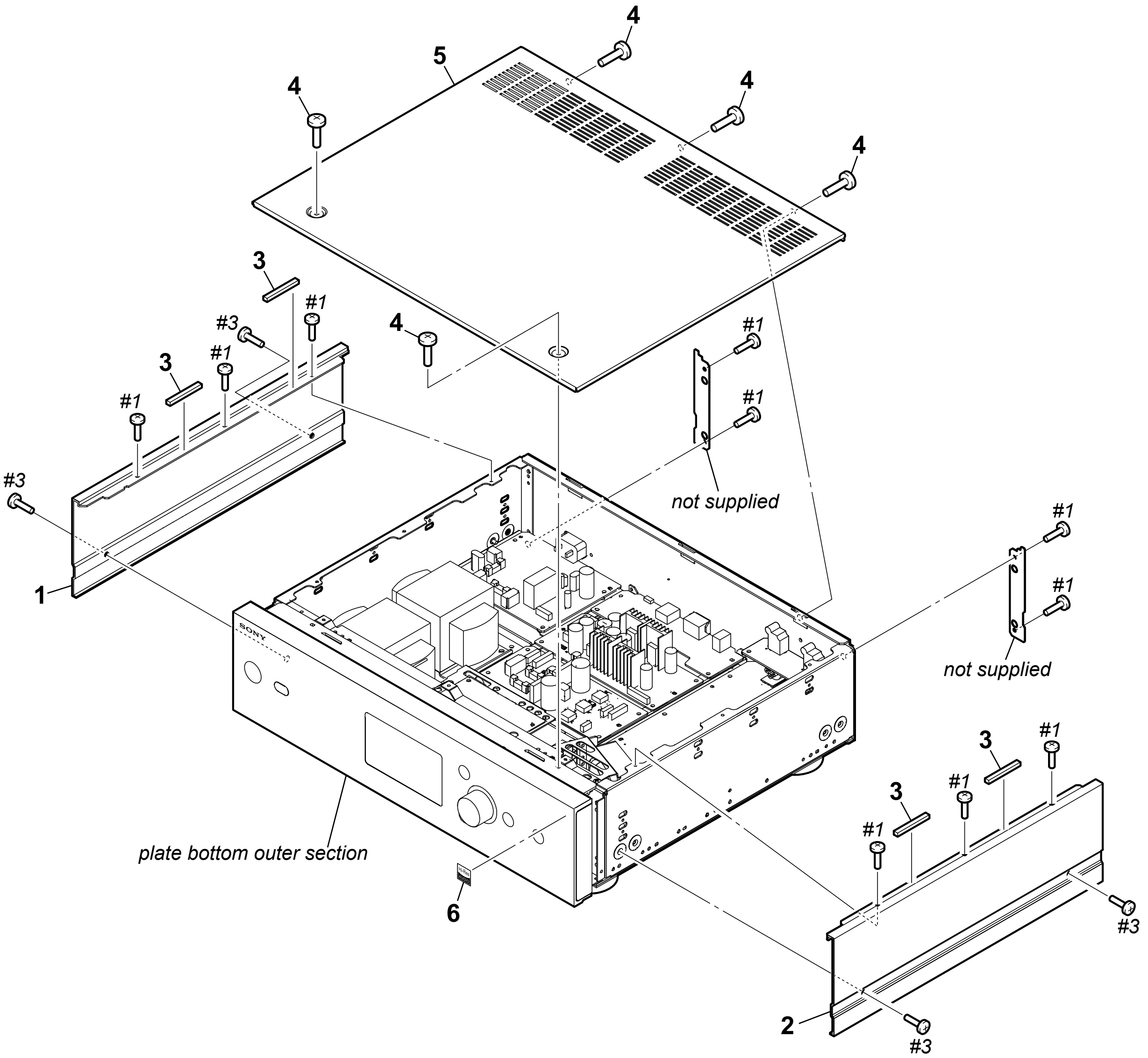
SECTION 6
EXPLODED VIEWS

Note:
- -XX and -X mean standardized parts, so they may have some difference from the original one.
- Items marked "*" are not stocked since they are seldom required for routine service. Some delay should be anticipated when ordering these items.
- The mechanical parts with no reference number in the exploded views are not supplied.
- Color Indication of Appearance Parts Example:
  KNOB, BALANCE (WHITE) . . . (RED)
  ↑ Parts Color ↑ Cabinet's Color
- Abbreviation
  CND : Canadian model

The components identified by mark ⚠ or dotted line with mark ⚠ are critical for safety.
Replace only with part number specified.

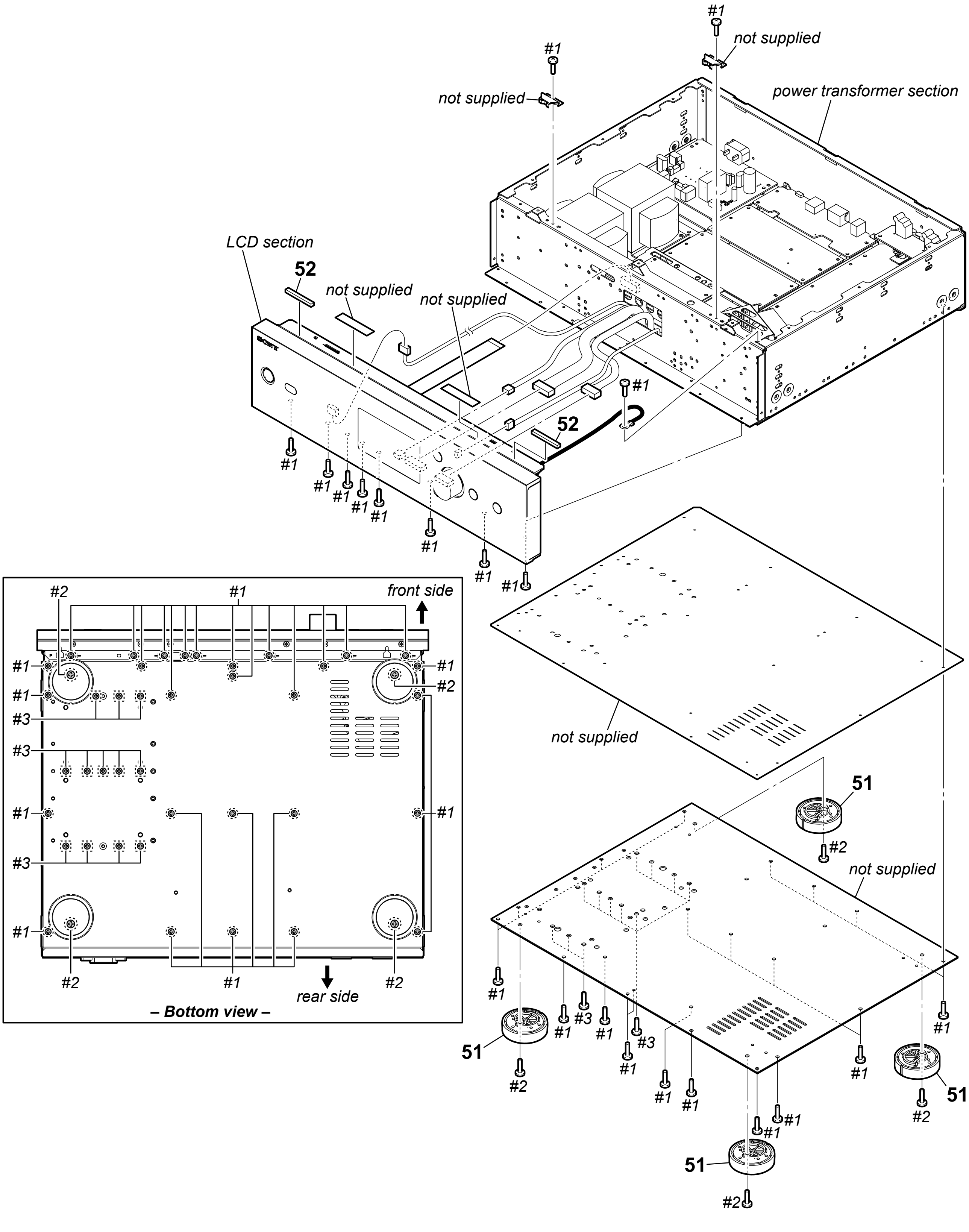
Les composants identifiés par une marque ⚠ sont critiques pour la sécurité.
Ne les remplacer que par une pièce portant le numéro spécifié.

6-1. OVERALL SECTION



Ref. No.	Part No.	Description	Remark
1	4-466-807-01	CASE L (for L-ch)	
2	4-466-808-01	CASE R (for R-ch)	
3	4-527-186-01	GASKET	
4	4-227-843-31	SCREW (TP), FLAT HEAD	
5	4-466-806-01	CASE TOP	
6	4-477-465-31	HI-RES LABEL	
#1	7-685-646-71	SCREW +BVTP 3X8 TYPE2 IT-3	
#3	7-685-881-09	SCREW +BVTT 4X8 (S)	

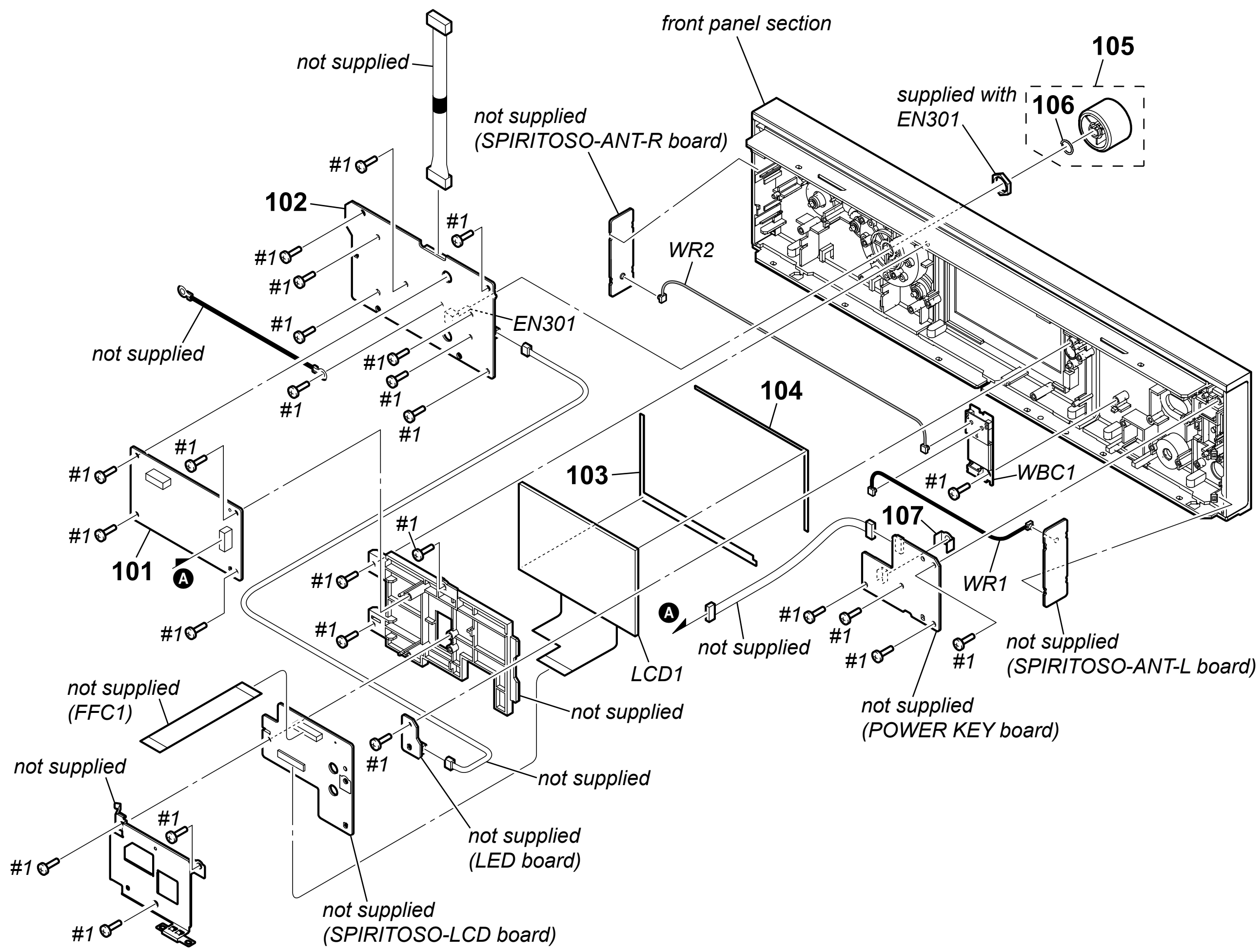
6-2. PLATE BOTTOM OUTER SECTION



Ref. No.	Part No.	Description	Remark
51	X-4955-348-1	FOOT ASSY	
52	4-527-186-01	GASKET	
#1	7-685-646-71	SCREW +BVTP 3X8 TYPE2 IT-3	
#2	7-685-886-09	SCREW +BVTT 4X20 (S)	
#3	7-685-881-09	SCREW +BVTT 4X8 (S)	

6-3. LCD SECTION

- Rear view



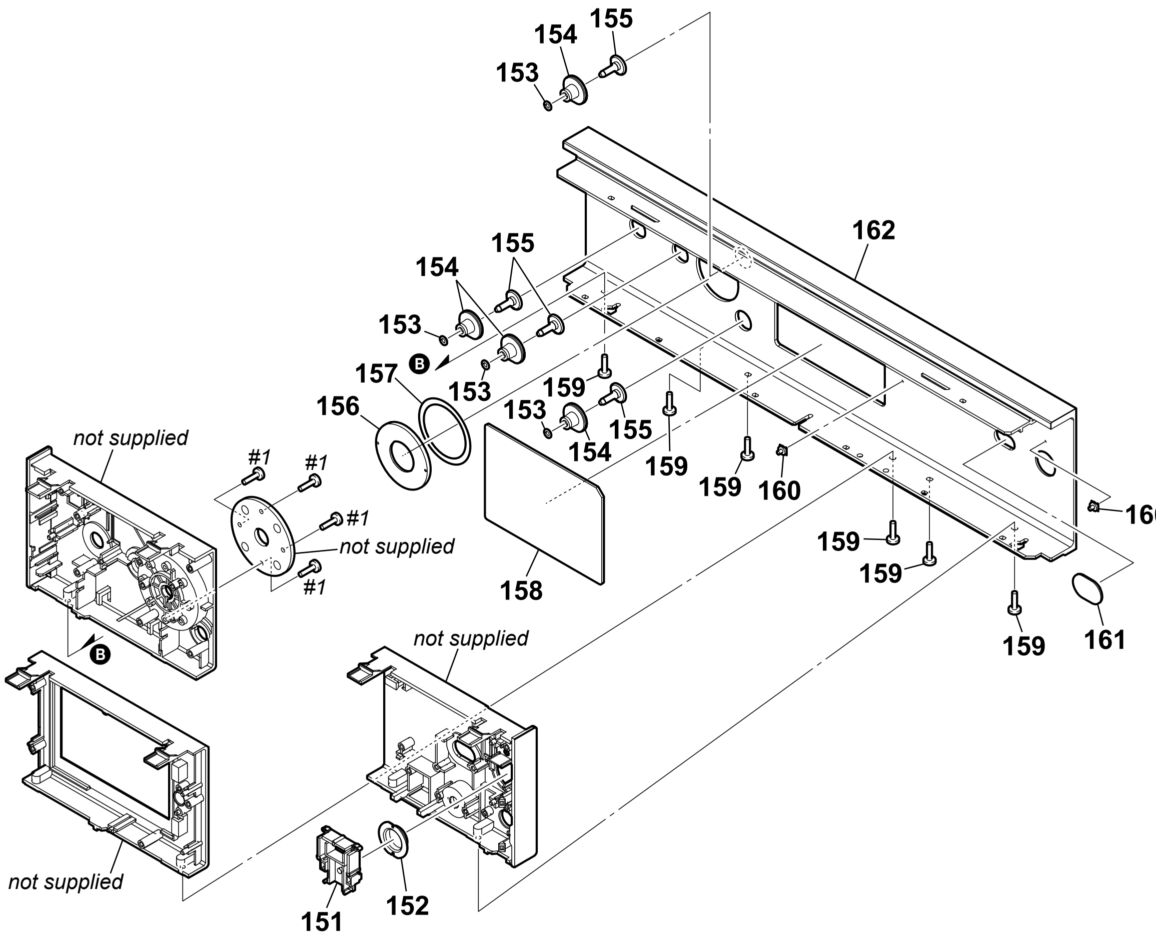
Note: When the WLAN/BT COMBO card (Ref. No. WBC1) and co-axial (U.FL connector) cables (Ref. No. WR1, WR2) are replaced, refer to "CHECKING METHOD OF NETWORK CONNECTION" on page 5.

Ref. No.	Part No.	Description	Remark
101	A-1945-111-A	U-COM BOARD, COMPLETE	
102	A-1945-123-A	KEY JOG BOARD, COMPLETE	
103	4-464-776-01	CUSHION, LCD (B)	
104	4-464-775-01	CUSHION, LCD (A)	
105	X-2587-423-2	KNOB (JOG) ASSY	
106	3-350-426-01	SPRING, RING	
107	4-530-372-01	SHEET (RM DEVICE)	
EN301	1-476-980-11	ENCODER (SELECTOR)	

Ref. No.	Part No.	Description	Remark
LCD1	1-811-873-11	PANEL, LCD	
WBC1	1-458-601-31	CARD, WLAN/BT COMBO (See Note)	
WR1	1-846-779-11	CABLE, COAXIAL (U.FL CONNECTOR) (for L-ch) (See Note)	
WR2	1-846-780-11	CABLE, COAXIAL (U.FL CONNECTOR) (for R-ch) (See Note)	
#1	7-685-646-71	SCREW +BVTP 3X8 TYPE2 IT-3	

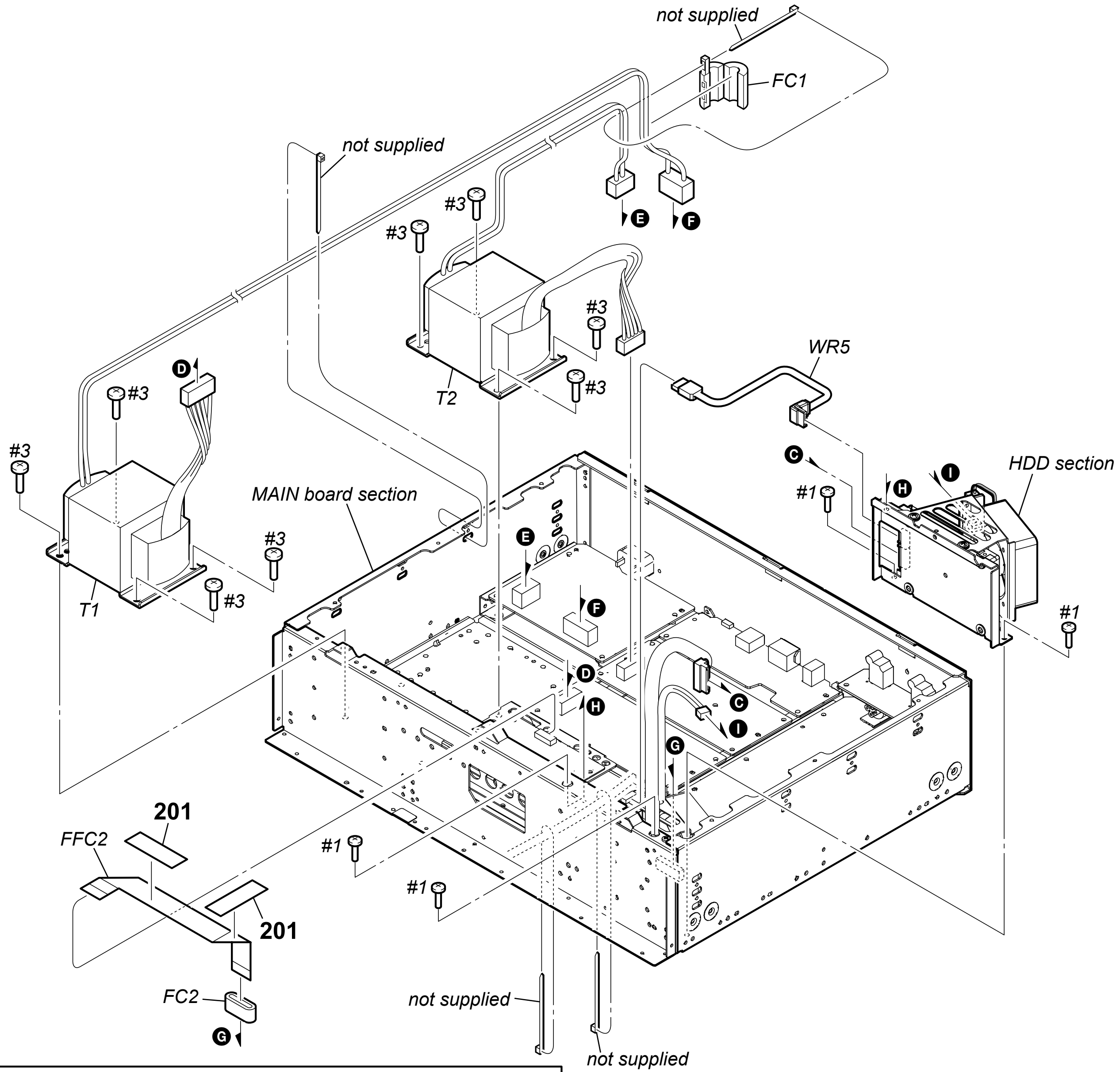
6-4. FRONT PANEL SECTION

- Rear view

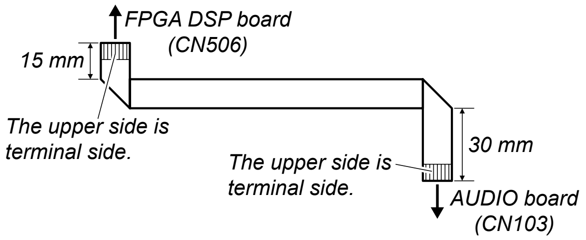


Ref. No.	Part No.	Description	Remark
151	4-465-989-01	BUTTON (POWER-BASE)	
152	4-465-988-01	BUTTON (POWER)	
153	3-325-697-21	WASHER	
154	X-2587-843-1	GUIDE (BTN) ASSY	
155	4-466-009-01	BUTTON (JOG)	
156	4-484-965-01	ESCUTCHEON_KNOB_J	
157	4-488-251-01	SHEET (ESCUTCHEON_JOG)	
158	4-466-005-01	WINDOW (LCD)	
159	3-906-887-11	SCREW (+BTP) (DIA. 3) (BZN-N)	
160	X-2587-425-1	INDICATOR (POWER) ASSY	
161	4-466-003-01	WINDOW (IR)	
162	X-2587-695-1	FRONT PANEL (PLAYER) ASSY (AEP, UK)	
162	X-2587-778-1	FRONT PANEL (PLAYER) ASSY (US, CND)	
#1	7-685-646-71	SCREW +BVTP 3X8 TYPE2 IT-3	

6-5. POWER TRANSFORMER SECTION

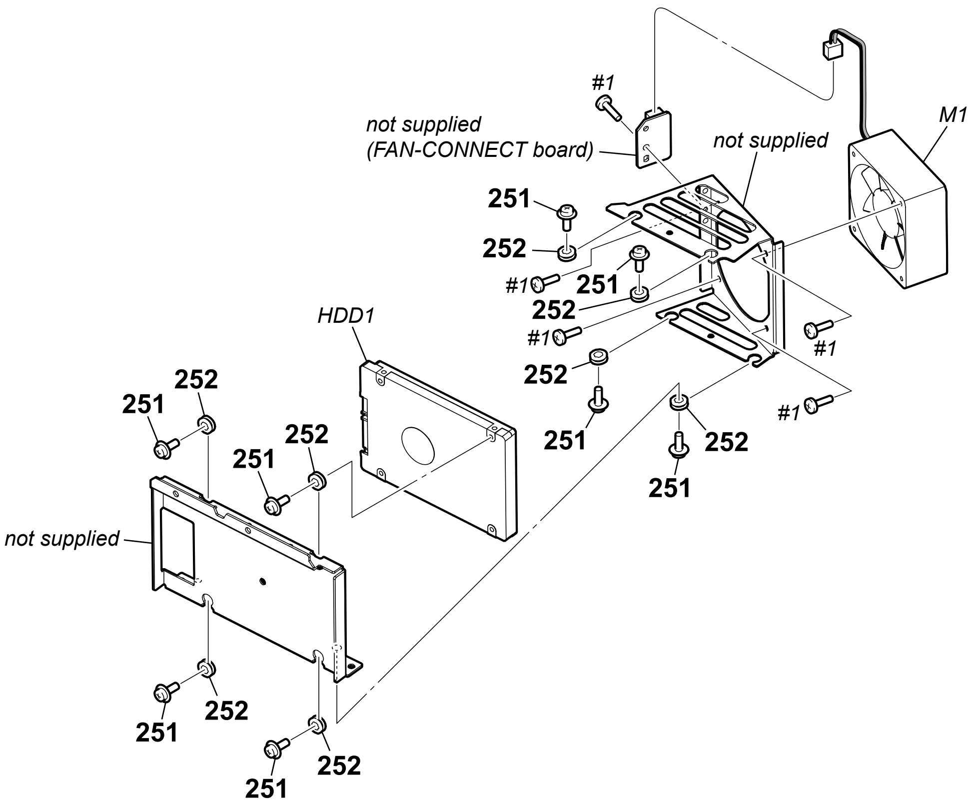


Note: Wire (flat type) (24 core) (Ref. No. FFC2) for service is all straight. When installing it, please bending it by referring to the figure below.



Ref. No.	Part No.	Description	Remark
201	4-860-518-00	PACKING, TWEETER	
FC1	1-500-386-11	FILTER, CLAMP (FERRITE CORE)	
FC2	1-400-640-11	CORE, FERRITE	
FFC2	1-828-771-51	WIRE (FLAT TYPE) (24 CORE) (See Note)	
⚠ T1	1-697-298-11	TRANSFORMER, POWER (AEP, UK)	
⚠ T1	1-697-299-11	TRANSFORMER, POWER (US, CND)	
⚠ T2	1-697-301-11	TRANSFORMER, POWER (AEP, UK)	
⚠ T2	1-697-302-11	TRANSFORMER, POWER (US, CND)	
WR5	1-969-716-11	HARNESS (S-002)	
#1	7-685-646-71	SCREW +BVTP 3X8 TYPE2 IT-3	
#3	7-685-881-09	SCREW +BVTT 4X8 (S)	

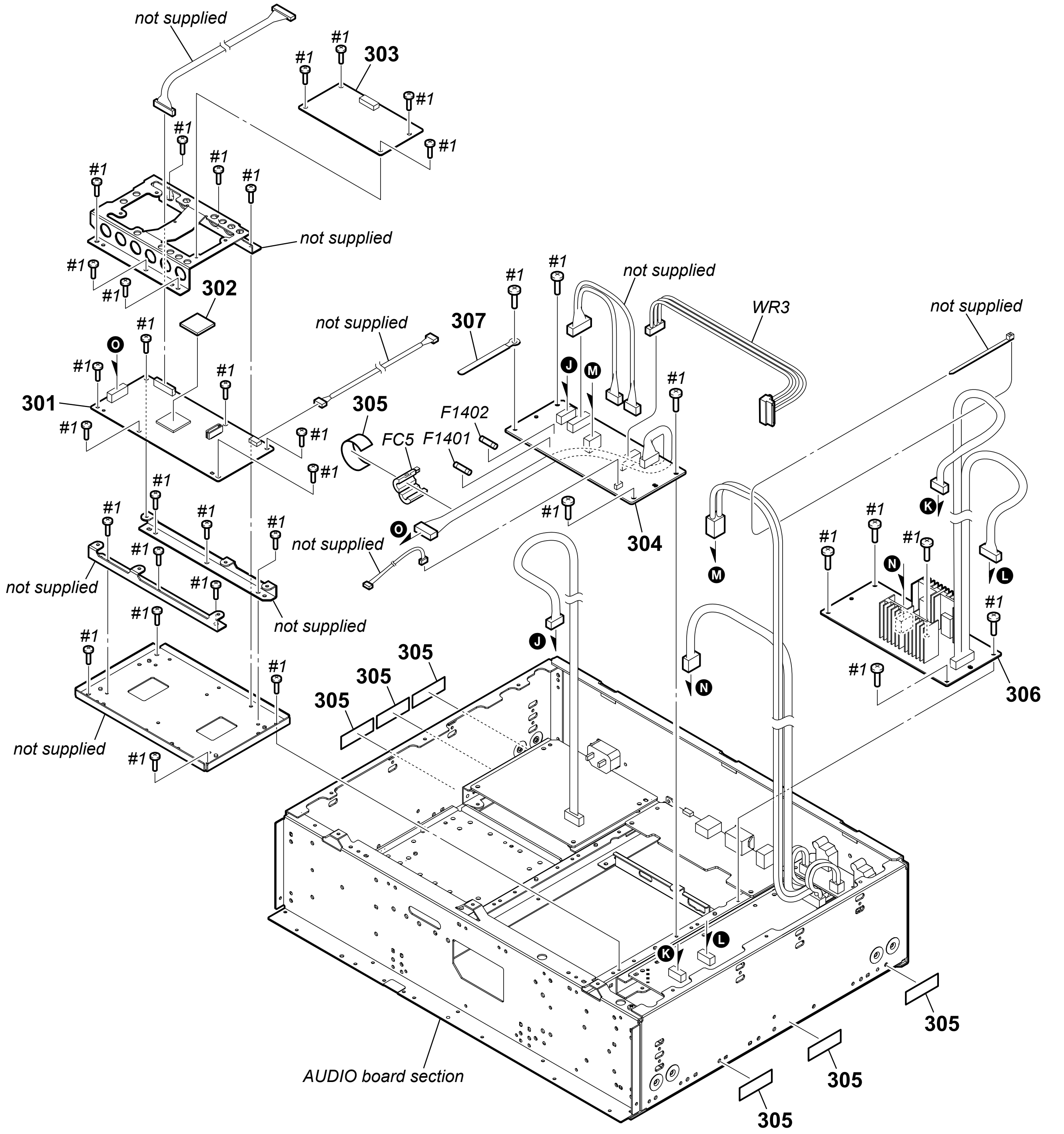
6-6. HDD SECTION



Note: When the hard disk drive (Ref. No. HDD1) is replaced, refer to "NOTE OF REPLACING THE HARD DISK DRIVE (Ref. No. HDD1)" on page 4.

Ref. No.	Part No.	Description	Remark
251	4-467-898-01	STEP SCREW	
252	4-467-899-01	DAMPER (HDD)	
⚠ HDD1	1-458-716-11	HDD/TSB-AQUARIUS (1TB) (Hard Disk Drive) (See Note)	
⚠ M1	1-787-931-11	DC FAN	
#1	7-685-646-71	SCREW +BVTP 3X8 TYPE2 IT-3	

6-7. MAIN BOARD SECTION

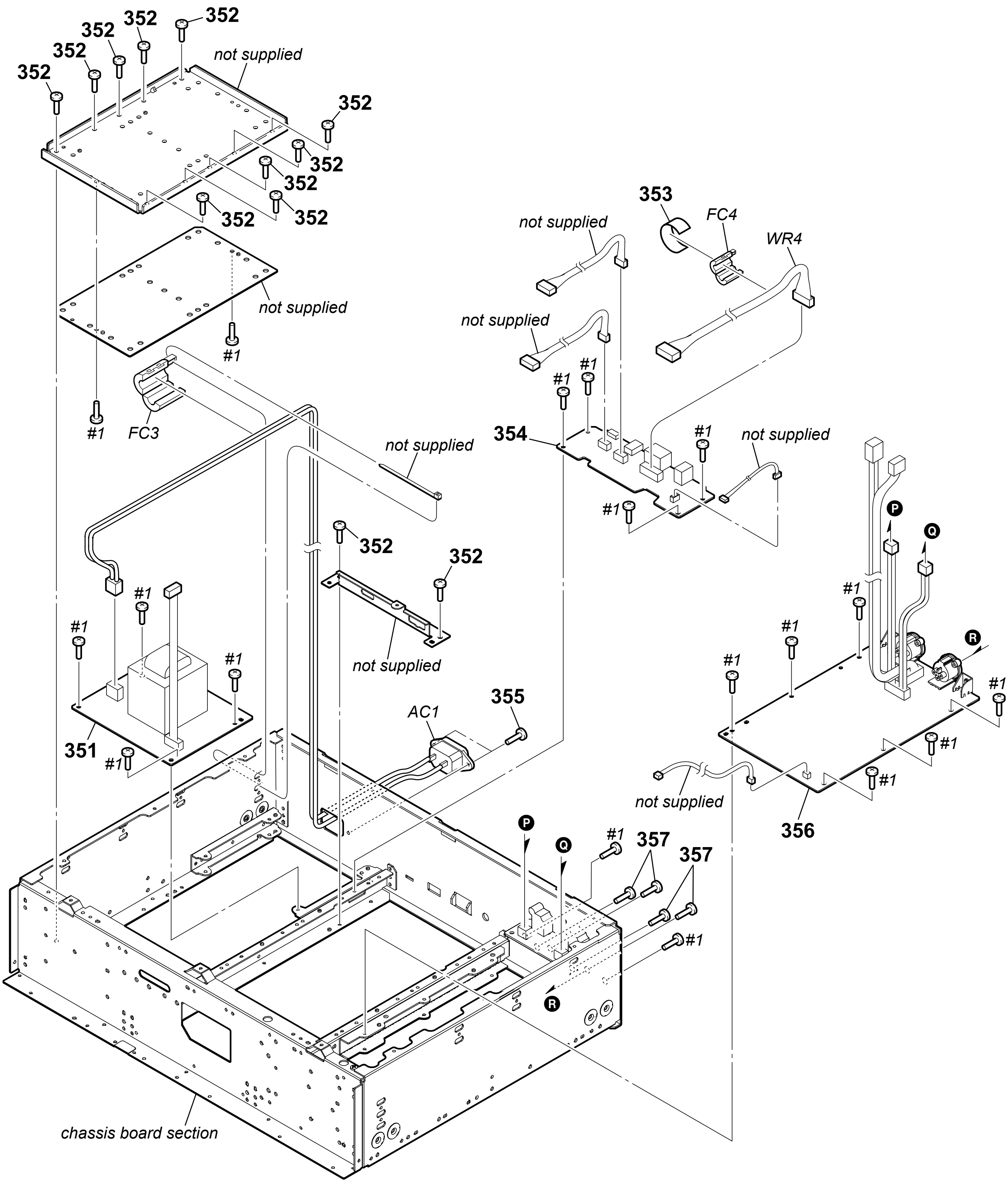


Note: When the complete MAIN board is replaced, refer to "NOTE OF REPLACING THE COMPLETE MAIN BOARD" and "CHECKING METHOD OF NETWORK CONNECTION" on page 5.

Ref. No.	Part No.	Description	Remark
301	A-1989-164-A	MAIN BOARD, COMPLETE (for SERVICE) (US, CND) (See Note)	
301	A-1989-165-A	MAIN BOARD, COMPLETE (for SERVICE) (AEP, UK) (See Note)	
302	4-478-826-01	SHEET RADIATION (20)	
303	A-1945-064-A	FPGA DSP BOARD, COMPLETE	
304	A-1945-106-A	DCDC BOARD, COMPLETE	
305	4-860-518-00	PACKING, TWEETER	

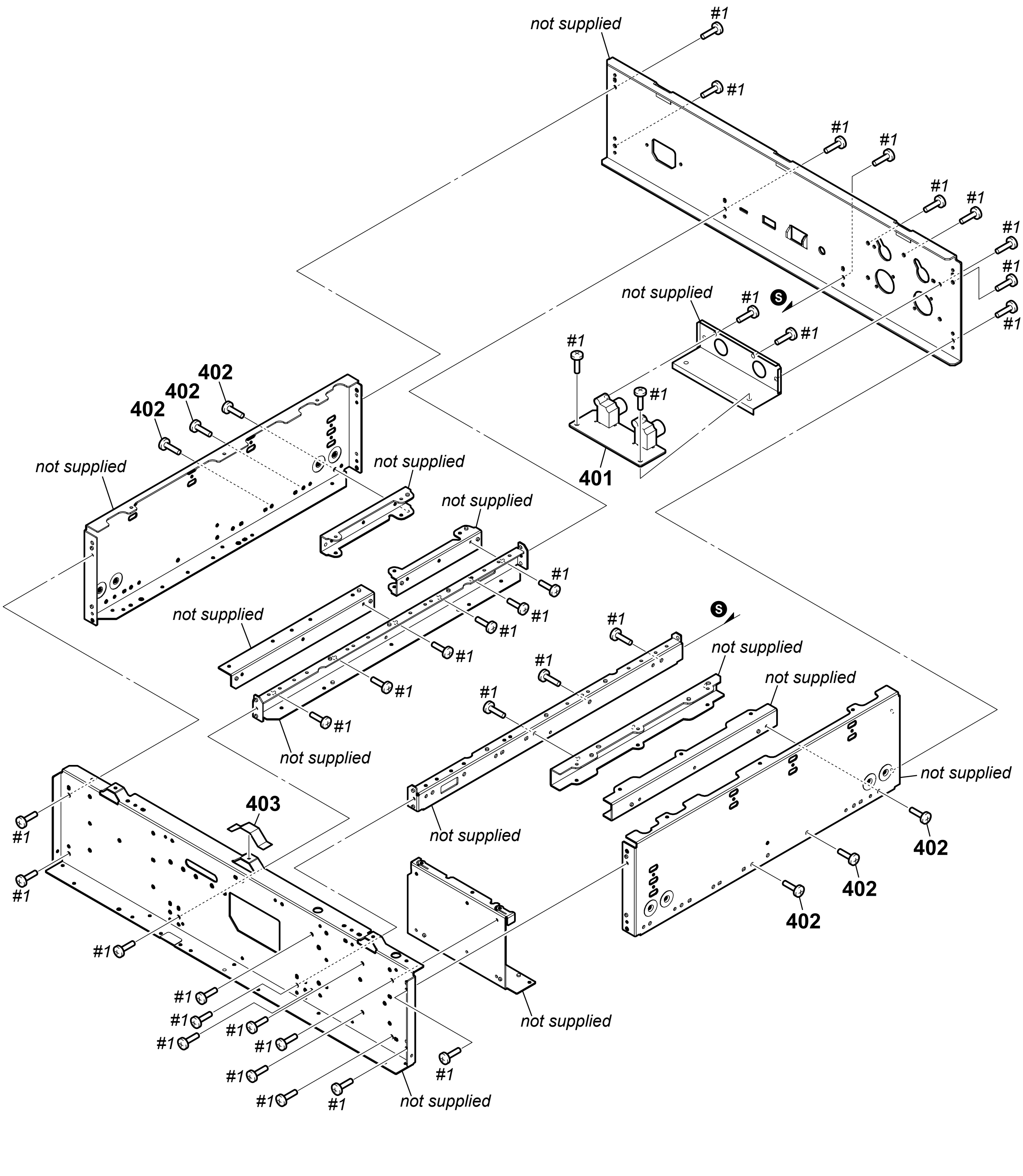
Ref. No.	Part No.	Description	Remark
306	A-1945-110-A	APOWER BOARD, COMPLETE	
* 307	3-703-150-11	CLAMP	
⚠ F1401	1-532-465-33	FUSE (T 3.15 AL/250 V)	
⚠ F1402	1-532-465-33	FUSE (T 3.15 AL/250 V)	
FC5	1-500-082-11	CLAMP, SLEEVE FERRITE	
WR3	1-969-718-11	HARNESS (S-004)	
#1	7-685-646-71	SCREW +BVTP 3X8 TYPE2 IT-3	

6-8. AUDIO BOARD SECTION



Ref. No.	Part No.	Description	Remark
351	A-1945-108-A	STANBY BOARD, COMPLETE (AEP, UK)	
351	A-1945-109-A	STANBY BOARD, COMPLETE (US, CND)	
352	3-077-331-41	+BV3 (3-CR)	
353	4-860-518-00	PACKING, TWEETER	
354	A-1945-114-A	IO BOARD, COMPLETE	
355	2-580-644-01	SCREW, +KTP2 3X8	
356	A-1945-100-A	AUDIO BOARD, COMPLETE	
357	3-087-053-01	+BVTP2.6 (3CR)	
⚠ AC1	1-821-082-41	AC INLET (2P) (AC IN)	
FC3	1-500-386-11	FILTER, CLAMP (FERRITE CORE)	
FC4	1-500-082-11	CLAMP, SLEEVE FERRITE	
WR4	1-969-720-11	HARNESS (E-002)	
#1	7-685-646-71	SCREW +BVTP 3X8 TYPE2 IT-3	

6-9. CHASSIS SECTION



Ref. No.	Part No.	Description	Remark
401	A-1945-103-A	AUUBOUT BOARD, COMPLETE	
402	3-906-887-11	SCREW (+BTP) (DIA. 3) (BZN-N)	
403	4-860-518-00	PACKING, TWEETER	
#1	7-685-646-71	SCREW +BVTP 3X8 TYPE2 IT-3	

APOWER AUDIO

SECTION 7
ELECTRICAL PARTS LIST

Note:
- Due to standardization, replacements in the parts list may be different from the parts specified in the diagrams or the components used on the set.
- -XX and -X mean standardized parts, so they may have some difference from the original one.
- Items marked "*" are not stocked since they are seldom required for routine service. Some delay should be anticipated when ordering these items.
- RESISTORS
  All resistors are in ohms.
  METAL: Metal-film resistor.
  METAL OXIDE: Metal oxide-film resistor.
  F: nonflammable
- CAPACITORS
  uF: μF
- COILS
  uH: μH
- SEMICONDUCTORS
  In each case, u: μ, for example:
  uA. . : μA. . , uPA. . , μPA. . ,
  uPB. . : μPB. . , uPC. . , μPC. . ,
  uPD. . : μPD. .
- Abbreviation
  CND : Canadian model

When indicating parts by reference number, please include the board name.

The components identified by mark ⚠ or dotted line with mark ⚠ are critical for safety.
Replace only with part number specified.

Les composants identifiés par une marque ⚠ sont critiques pour la sécurité.
Ne les remplacer que par une pièce portant le numéro spécifié.

Ref. No.	Part No.	Description				Remark
	A-1945-110-A	APOWER BOARD, COMPLETE				
		************************				
	3-905-609-13	SCREW (TRANSISTOR)				
		< CAPACITOR >				
C1501	1-114-507-11	CERAMIC CHIP	1uF	10%	50V	
C1502	1-114-507-11	CERAMIC CHIP	1uF	10%	50V	
C1503	1-112-099-11	ELECT	4.7uF	20%	50V	
C1504	1-114-507-11	CERAMIC CHIP	1uF	10%	50V	
C1505	1-114-507-11	CERAMIC CHIP	1uF	10%	50V	
C1506	1-124-703-51	ELECT	2200uF	20%	25V	
C1507	1-124-703-51	ELECT	2200uF	20%	25V	
C1509	1-112-093-11	ELECT	2200uF	20%	25V	
C1510	1-112-093-11	ELECT	2200uF	20%	25V	
C1511	1-114-507-11	CERAMIC CHIP	1uF	10%	50V	
C1512	1-114-507-11	CERAMIC CHIP	1uF	10%	50V	
C1514	1-114-507-11	CERAMIC CHIP	1uF	10%	50V	
C1515	1-114-507-11	CERAMIC CHIP	1uF	10%	50V	
C1516	1-112-093-11	ELECT	2200uF	20%	25V	
C1517	1-112-093-11	ELECT	2200uF	20%	25V	
C1519	1-112-081-11	ELECT	2200uF	20%	10V	
C1520	1-112-092-11	ELECT	1000uF	20%	25V	
		< CONNECTOR >				
CN1501	1-691-769-11	PLUG (MICRO CONNECTOR) 7P				
CN1502	1-564-320-00	PIN, CONNECTOR (3.96 mm PITCH) 2P				
		< DIODE >				
D1501	8-719-081-66	DIODE M1FS6				
D1502	8-719-081-66	DIODE M1FS6				
D1503	8-719-081-66	DIODE M1FS6				
D1504	8-719-081-66	DIODE M1FS6				
D1506	8-719-080-69	DIODE M1FS4-4063				
D1507	8-719-080-69	DIODE M1FS4-4063				
D1509	8-719-080-69	DIODE M1FS4-4063				
D1510	8-719-080-69	DIODE M1FS4-4063				
D1512	8-719-080-69	DIODE M1FS4-4063				
D1513	8-719-080-69	DIODE M1FS4-4063				
D1515	8-719-080-69	DIODE M1FS4-4063				
D1516	8-719-080-69	DIODE M1FS4-4063				
D1517	6-502-961-01	DIODE DA2J10100L				
D1518	6-502-961-01	DIODE DA2J10100L				
D1520	6-502-961-01	DIODE DA2J10100L				
D1521	6-502-961-01	DIODE DA2J10100L				
D1522	6-502-961-01	DIODE DA2J10100L				
D1523	6-502-961-01	DIODE DA2J10100L				
D1525	6-502-961-01	DIODE DA2J10100L				
D1526	6-502-961-01	DIODE DA2J10100L				
		< FUSE >				
⚠ F1501	1-523-085-11	FUSE (2.5 A/250 V)				
⚠ F1502	1-523-085-11	FUSE (2.5 A/250 V)				
⚠ F1504	1-523-085-11	FUSE (2.5 A/250 V)				
⚠ F1505	1-523-085-11	FUSE (2.5 A/250 V)				
		< IC >				
⚠ IC1501	6-712-109-01	IC KIA7815API-U/PF				
⚠ IC1502	6-712-107-01	IC KIA7915PI				
⚠ IC1504	8-759-646-52	IC KIA7805API				
⚠ IC1507	6-705-463-01	IC BA33BC0T				
		< TRANSISTOR >				
Q1501	6-553-266-01	TRANSISTOR	2PD601ART			
		< RESISTOR >				
R1505	1-250-640-11	METAL CHIP	10K	1%	1/10W	
R1506	1-216-843-11	METAL CHIP	68K	5%	1/10W	
R1507	1-250-636-11	METAL CHIP	6.8K	1%	1/10W	
**						
	A-1945-100-A	AUDIO BOARD, COMPLETE				
		*********************				
		< CAPACITOR >				
C101	1-118-386-11	CERAMIC CHIP	0.1uF	10%	16V	
C102	1-118-386-11	CERAMIC CHIP	0.1uF	10%	16V	
C103	1-118-386-11	CERAMIC CHIP	0.1uF	10%	16V	
C104	1-118-386-11	CERAMIC CHIP	0.1uF	10%	16V	
C105	1-118-386-11	CERAMIC CHIP	0.1uF	10%	16V	
C106	1-112-090-11	ELECT	100uF	20%	25V	
C107	1-118-386-11	CERAMIC CHIP	0.1uF	10%	16V	
C108	1-112-090-11	ELECT	100uF	20%	25V	
C109	1-112-090-11	ELECT	100uF	20%	25V	
C110	1-112-090-11	ELECT	100uF	20%	25V	
C111	1-116-734-11	CERAMIC CHIP	1uF	20%	16V	
C151	1-118-386-11	CERAMIC CHIP	0.1uF	10%	16V	
C152	1-124-778-00	ELECT CHIP	22uF	20%	6.3V	
C154	1-124-699-11	ELECT	220uF	20%	25V	
C155	1-116-734-11	CERAMIC CHIP	1uF	20%	16V	
C156	1-124-699-11	ELECT	220uF	20%	25V	

Ref. No.	Part No.	Description				Remark
C202	1-118-386-11	CERAMIC CHIP	0.1uF	10%	16V	
C203	1-127-956-21	FILM CHIP	0.1uF	5%	16V	
C204	1-112-090-11	ELECT	100uF	20%	25V	
C205	1-112-100-11	ELECT	10uF	20%	50V	
C206	1-127-956-21	FILM CHIP	0.1uF	5%	16V	
C207	1-112-100-11	ELECT	10uF	20%	50V	
C208	1-112-100-11	ELECT	10uF	20%	50V	
C209	1-127-956-21	FILM CHIP	0.1uF	5%	16V	
C211	1-127-956-21	FILM CHIP	0.1uF	5%	16V	
C212	1-114-907-51	FILM	0.0022uF	5%	250V	
C213	1-114-907-51	FILM	0.0022uF	5%	250V	
C214	1-127-956-21	FILM CHIP	0.1uF	5%	16V	
C215	1-127-956-21	FILM CHIP	0.1uF	5%	16V	
C216	1-130-479-91	MYLAR	0.0047uF	5%	50V	
C217	1-130-479-91	MYLAR	0.0047uF	5%	50V	
C218	1-114-901-91	FILM	680PF	5%	250V	
C219	1-114-901-91	FILM	680PF	5%	250V	
C220	1-127-956-21	FILM CHIP	0.1uF	5%	16V	
C221	1-127-956-21	FILM CHIP	0.1uF	5%	16V	
C222	1-112-857-21	ELECT	100uF	20%	100V	
C223	1-112-857-21	ELECT	100uF	20%	100V	
C226	1-100-756-91	CERAMIC CHIP	0.047uF	10%	50V	
C228	1-114-907-51	FILM	0.0022uF	5%	250V	
C229	1-114-901-91	FILM	680PF	5%	250V	
C230	1-114-901-91	FILM	680PF	5%	250V	
C231	1-127-956-21	FILM CHIP	0.1uF	5%	16V	
C232	1-128-552-11	ELECT	47uF	20%	63V	
C233	1-127-956-21	FILM CHIP	0.1uF	5%	16V	
C236	1-100-756-91	CERAMIC CHIP	0.047uF	10%	50V	
C237	1-127-961-21	FILM CHIP	220PF	5%	50V	
C238	1-127-961-21	FILM CHIP	220PF	5%	50V	
C252	1-118-386-11	CERAMIC CHIP	0.1uF	10%	16V	
C253	1-127-956-21	FILM CHIP	0.1uF	5%	16V	
C254	1-112-090-11	ELECT	100uF	20%	25V	
C255	1-112-100-11	ELECT	10uF	20%	50V	
C256	1-127-956-21	FILM CHIP	0.1uF	5%	16V	
C257	1-112-100-11	ELECT	10uF	20%	50V	
C258	1-112-100-11	ELECT	10uF	20%	50V	
C259	1-127-956-21	FILM CHIP	0.1uF	5%	16V	
C261	1-127-956-21	FILM CHIP	0.1uF	5%	16V	
C262	1-114-907-51	FILM	0.0022uF	5%	250V	
C263	1-114-907-51	FILM	0.0022uF	5%	250V	
C264	1-127-956-21	FILM CHIP	0.1uF	5%	16V	
C265	1-127-956-21	FILM CHIP	0.1uF	5%	16V	
C266	1-130-479-91	MYLAR	0.0047uF	5%	50V	
C267	1-130-479-91	MYLAR	0.0047uF	5%	50V	
C268	1-114-901-91	FILM	680PF	5%	250V	
C269	1-114-901-91	FILM	680PF	5%	250V	
C270	1-127-956-21	FILM CHIP	0.1uF	5%	16V	
C271	1-127-956-21	FILM CHIP	0.1uF	5%	16V	
C272	1-112-857-21	ELECT	100uF	20%	100V	
C273	1-112-857-21	ELECT	100uF	20%	100V	
C276	1-100-756-91	CERAMIC CHIP	0.047uF	10%	50V	
C278	1-114-907-51	FILM	0.0022uF	5%	250V	
C279	1-114-901-91	FILM	680PF	5%	250V	
C280	1-114-901-91	FILM	680PF	5%	250V	
C281	1-127-956-21	FILM CHIP	0.1uF	5%	16V	
C282	1-128-552-11	ELECT	47uF	20%	63V	
C283	1-127-956-21	FILM CHIP	0.1uF	5%	16V	
C287	1-127-961-21	FILM CHIP	220PF	5%	50V	

Ref. No.	Part No.	Description				Remark
C288	1-127-961-21	FILM CHIP	220PF	5%	50V	
		< CONNECTOR >				
CN102	1-691-766-11	PLUG (MICRO CONNECTOR) 4P				
CN103	1-815-763-32	CONNECTOR, FFC/FPC 24P				
CN151	1-691-767-11	PLUG (MICRO CONNECTOR) 5P				
CN202	1-794-097-11	CONNECTOR, ROUND TYPE (BALANCED LINE OUT L)				
CN252	1-794-097-11	CONNECTOR, ROUND TYPE (BALANCED LINE OUT R)				
* CN253	1-695-320-31	PIN, CONNECTOR (1.5 mm) (SMD) 2P				
		< DIODE >				
D193	6-502-961-01	DIODE DA2J10100L				
D194	6-502-961-01	DIODE DA2J10100L				
D201	8-719-024-71	DIODE 1SS362-TE85L				
D202	8-719-024-71	DIODE 1SS362-TE85L				
D203	8-719-024-71	DIODE 1SS362-TE85L				
D211	6-502-961-01	DIODE DA2J10100L				
D212	6-502-961-01	DIODE DA2J10100L				
D251	8-719-024-71	DIODE 1SS362-TE85L				
D252	8-719-024-71	DIODE 1SS362-TE85L				
D253	8-719-024-71	DIODE 1SS362-TE85L				
D262	6-502-961-01	DIODE DA2J10100L				
		< EARTH TERMINAL >				
* ET151	1-537-738-21	TERMINAL, EARTH				
* ET152	1-537-738-21	TERMINAL, EARTH				
		< FERRITE BEAD >				
FB101	1-469-324-21	FERRITE, EMI (SMD) (2012)				
FB102	1-469-139-21	FERRITE, EMI (SMD) (2012)				
		< IC >				
IC101	6-600-060-01	IC TC7WZ32FK				
IC102	6-712-572-01	IC 74LVC161PW-118				
IC103	6-710-376-01	IC 74LVC1G17GW-125				
IC106	6-717-694-01	IC BU33TD3WG-TR				
IC107	6-706-487-01	IC TC7SH08FU				
IC151	8-759-460-72	IC BA033FP-E2				
IC201	6-720-672-01	IC PCM1795DBR				
IC202	6-714-306-01	IC OPA2132UA/2K5G4				
IC203	8-759-447-30	IC NJM2114M-TE2				
IC204	6-714-306-01	IC OPA2132UA/2K5G4				
IC251	6-720-672-01	IC PCM1795DBR				
IC252	6-714-306-01	IC OPA2132UA/2K5G4				
IC253	8-759-447-30	IC NJM2114M-TE2				
		< TRANSISTOR >				
Q151	6-553-268-01	TRANSISTOR	PDTC114ET			
Q152	6-553-246-01	TRANSISTOR	PDTA114ET			
Q201	6-553-246-01	TRANSISTOR	PDTA114ET			
Q202	6-553-268-01	TRANSISTOR	PDTC114ET			
		< RESISTOR >				
R101	1-250-592-11	METAL CHIP	100	1%	1/10W	
R102	1-250-592-11	METAL CHIP	100	1%	1/10W	
R103	1-250-592-11	METAL CHIP	100	1%	1/10W	
R104	1-250-592-11	METAL CHIP	100	1%	1/10W	
R105	1-250-592-11	METAL CHIP	100	1%	1/10W	

AUDIO AUUBOUT DCDC

	Ref. No.	Part No.	Description			Remark	
	R106	1-250-664-11	METAL CHIP	100K	1%	1/10W	
	R107	1-250-664-11	METAL CHIP	100K	1%	1/10W	
*	R108	1-250-576-11	METAL CHIP	22	1%	1/10W	
*	R109	1-250-576-11	METAL CHIP	22	1%	1/10W	
*	R110	1-250-576-11	METAL CHIP	22	1%	1/10W	
*	R111	1-250-576-11	METAL CHIP	22	1%	1/10W	
*	R112	1-250-576-11	METAL CHIP	22	1%	1/10W	
*	R113	1-250-576-11	METAL CHIP	22	1%	1/10W	
	R114	1-250-592-11	METAL CHIP	100	1%	1/10W	
	R115	1-250-600-11	METAL CHIP	220	1%	1/10W	
	R116	1-216-864-11	SHORT CHIP	0			
	R117	1-250-664-11	METAL CHIP	100K	1%	1/10W	
	R118	1-250-664-11	METAL CHIP	100K	1%	1/10W	
	R151	1-250-640-11	METAL CHIP	10K	1%	1/10W	
	R152	1-250-640-11	METAL CHIP	10K	1%	1/10W	
	R153	1-250-640-11	METAL CHIP	10K	1%	1/10W	
	R154	1-250-640-11	METAL CHIP	10K	1%	1/10W	
	R201	1-250-592-11	METAL CHIP	100	1%	1/10W	
	R202	1-250-592-11	METAL CHIP	100	1%	1/10W	
	R203	1-250-592-11	METAL CHIP	100	1%	1/10W	
	R204	1-250-592-11	METAL CHIP	100	1%	1/10W	
⚠	R209	1-215-445-81	METAL	10K	1%	1/4W	F
	R210	1-257-426-21	CARBON	390	5%	1/2W	
	R211	1-257-426-21	CARBON	390	5%	1/2W	
	R212	1-257-371-21	CARBON	1K	5%	1/2W	
	R213	1-257-371-21	CARBON	1K	5%	1/2W	
	R214	1-257-371-21	CARBON	1K	5%	1/2W	
	R215	1-257-371-21	CARBON	1K	5%	1/2W	
	R216	1-257-371-21	CARBON	1K	5%	1/2W	
	R217	1-257-371-21	CARBON	1K	5%	1/2W	
	R218	1-257-419-21	CARBON	220	5%	1/2W	
	R219	1-257-419-21	CARBON	220	5%	1/2W	
	R220	1-257-470-21	CARBON	220K	5%	1/2W	
	R221	1-257-470-21	CARBON	220K	5%	1/2W	
	R222	1-257-395-21	CARBON	10	5%	1/2W	
	R223	1-257-395-21	CARBON	10	5%	1/2W	
	R224	1-257-371-21	CARBON	1K	5%	1/2W	
	R225	1-257-371-21	CARBON	1K	5%	1/2W	
	R226	1-257-371-21	CARBON	1K	5%	1/2W	
	R227	1-257-371-21	CARBON	1K	5%	1/2W	
	R228	1-257-371-21	CARBON	1K	5%	1/2W	
	R229	1-257-371-21	CARBON	1K	5%	1/2W	
	R230	1-257-419-21	CARBON	220	5%	1/2W	
	R231	1-257-470-21	CARBON	220K	5%	1/2W	
	R232	1-257-395-21	CARBON	10	5%	1/2W	
	R234	1-250-664-11	METAL CHIP	100K	1%	1/10W	
	R251	1-250-592-11	METAL CHIP	100	1%	1/10W	
	R252	1-250-592-11	METAL CHIP	100	1%	1/10W	
	R253	1-250-592-11	METAL CHIP	100	1%	1/10W	
	R254	1-250-592-11	METAL CHIP	100	1%	1/10W	
⚠	R259	1-215-445-81	METAL	10K	1%	1/4W	F
	R260	1-257-426-21	CARBON	390	5%	1/2W	
	R261	1-257-426-21	CARBON	390	5%	1/2W	
	R262	1-257-371-21	CARBON	1K	5%	1/2W	
	R263	1-257-371-21	CARBON	1K	5%	1/2W	
	R264	1-257-371-21	CARBON	1K	5%	1/2W	
	R265	1-257-371-21	CARBON	1K	5%	1/2W	
	R266	1-257-371-21	CARBON	1K	5%	1/2W	
	R267	1-257-371-21	CARBON	1K	5%	1/2W	
	R268	1-257-419-21	CARBON	220	5%	1/2W	

Ref. No.	Part No.	Description			Remark
R269	1-257-419-21	CARBON	220	5%	1/2W
R270	1-257-470-21	CARBON	220K	5%	1/2W
R271	1-257-470-21	CARBON	220K	5%	1/2W
R272	1-257-395-21	CARBON	10	5%	1/2W
R273	1-257-395-21	CARBON	10	5%	1/2W
R274	1-257-371-21	CARBON	1K	5%	1/2W
R275	1-257-371-21	CARBON	1K	5%	1/2W
R276	1-257-371-21	CARBON	1K	5%	1/2W
R277	1-257-371-21	CARBON	1K	5%	1/2W
R278	1-257-371-21	CARBON	1K	5%	1/2W
R279	1-257-371-21	CARBON	1K	5%	1/2W
R280	1-257-419-21	CARBON	220	5%	1/2W
R281	1-257-470-21	CARBON	220K	5%	1/2W
R282	1-257-395-21	CARBON	10	5%	1/2W
R284	1-250-664-11	METAL CHIP	100K	1%	1/10W
R291	1-257-419-21	CARBON	220	5%	1/2W
		< RELAY >			
RY201	1-755-485-11	RELAY			
RY202	1-755-485-11	RELAY			
RY252	1-755-485-11	RELAY			
		< VIBRATOR >			
X101	1-814-691-11	OSCILLATOR, CRYSTAL (22.5792 MHz)			
X102	1-814-692-11	OSCILLATOR, CRYSTAL (24.576 MHz)			

****

Ref. No.	Part No.	Description			Remark
	A-1945-103-A	AUUBOUT BOARD, COMPLETE			
		************************			
		< CAPACITOR >			
C235	1-114-896-91	FILM	270PF	5%	250V
C285	1-114-896-91	FILM	270PF	5%	250V
		< CONNECTOR >			
CN157	1-564-506-11	PLUG, CONNECTOR 3P			
CN158	1-564-507-11	PLUG, CONNECTOR 4P			
		< JACK >			
J201	1-818-094-11	JACK, PIN 1P (UNBALANCED LINE OUT L)			
J251	1-818-095-11	JACK, PIN 1P (UNBALANCED LINE OUT R)			

****

Ref. No.	Part No.	Description			Remark
	A-1945-106-A	DCDC BOARD, COMPLETE			
		*********************			
		< CAPACITOR >			
C1401	1-114-994-11	ELECT	2200uF	20%	35V
C1402	1-112-099-11	ELECT	4.7uF	20%	50V
C1405	1-114-994-11	ELECT	2200uF	20%	35V
C1406	1-116-865-11	CERAMIC CHIP	10uF	10%	25V
C1407	1-116-865-11	CERAMIC CHIP	10uF	10%	25V
C1408	1-116-865-11	CERAMIC CHIP	10uF	10%	25V
C1409	1-116-865-11	CERAMIC CHIP	10uF	10%	25V
C1410	1-118-373-11	CERAMIC CHIP	0.01uF	10%	50V
C1411	1-118-373-11	CERAMIC CHIP	0.01uF	10%	50V
C1412	1-118-347-11	CERAMIC CHIP	0.1uF	10%	25V
C1413	1-118-347-11	CERAMIC CHIP	0.1uF	10%	25V
C1414	1-118-347-11	CERAMIC CHIP	0.1uF	10%	25V
C1415	1-118-347-11	CERAMIC CHIP	0.1uF	10%	25V
C1416	1-116-734-11	CERAMIC CHIP	1uF	20%	16V

Ref. No.	Part No.	Description				Remark
C1417	1-100-158-91	CERAMIC CHIP	1000PF	5%	100V	
C1418	1-100-158-91	CERAMIC CHIP	1000PF	5%	100V	
C1419	1-118-435-11	CERAMIC CHIP	47PF	5%	50V	
C1421	1-118-418-11	CERAMIC CHIP	22uF	20%	6.3V	
C1422	1-118-418-11	CERAMIC CHIP	22uF	20%	6.3V	
C1423	1-118-418-11	CERAMIC CHIP	22uF	20%	6.3V	
C1424	1-118-418-11	CERAMIC CHIP	22uF	20%	6.3V	
C1425	1-116-865-11	CERAMIC CHIP	10uF	10%	25V	
C1426	1-116-865-11	CERAMIC CHIP	10uF	10%	25V	
C1427	1-118-373-11	CERAMIC CHIP	0.01uF	10%	50V	
C1428	1-118-347-11	CERAMIC CHIP	0.1uF	10%	25V	
C1429	1-118-347-11	CERAMIC CHIP	0.1uF	10%	25V	
C1430	1-118-347-11	CERAMIC CHIP	0.1uF	10%	25V	
C1431	1-100-158-91	CERAMIC CHIP	1000PF	5%	100V	
C1432	1-118-418-11	CERAMIC CHIP	22uF	20%	6.3V	
C1433	1-118-418-11	CERAMIC CHIP	22uF	20%	6.3V	
C1436	1-112-090-11	ELECT	100uF	20%	25V	
C1437	1-112-090-11	ELECT	100uF	20%	25V	
C1438	1-112-090-11	ELECT	100uF	20%	25V	
C1439	1-118-347-11	CERAMIC CHIP	0.1uF	10%	25V	
C1440	1-112-235-21	ELECT	330uF	20%	25V	
C1441	1-112-235-21	ELECT	330uF	20%	25V	
C1442	1-112-235-21	ELECT	330uF	20%	25V	
		< CONNECTOR >				
CN1401	1-785-102-11	PIN, CONNECTOR (3.96 mm PITCH) 4P				
* CN1402	1-564-104-00	PIN, CONNECTOR (3.96 mm PITCH) 3P				
CN1403	1-770-470-21	PIN, CONNECTOR (PC BOARD) 6P				
CN1406	1-691-765-11	PLUG (MICRO CONNECTOR) 3P				
CN1408	1-770-468-21	PIN, CONNECTOR (PC BOARD) 10P				
CN1409	1-770-469-21	PIN, CONNECTOR (PC BOARD) 2P				
		< DIODE >				
D1401	8-719-060-88	DIODE D4SBS6				
D1402	6-502-961-01	DIODE DA2J10100L				
D1403	6-502-961-01	DIODE DA2J10100L				
D1404	8-719-060-88	DIODE D4SBS6				
D1405	6-502-968-01	DIODE DZ2J062M0L				
		< FUSE HOLDER >				
⚠ FH1401	1-533-313-11	FUSE HOLDER				
⚠ FH1402	1-533-313-11	FUSE HOLDER				
⚠ FH1403	1-533-313-11	FUSE HOLDER				
⚠ FH1404	1-533-313-11	FUSE HOLDER				
		< IC >				
IC1402	6-719-190-01	IC RT7256ALZSP				
IC1403	6-719-190-01	IC RT7256ALZSP				
IC1404	6-719-190-01	IC RT7256ALZSP				
		< COIL >				
L1401	1-460-470-11	COIL, CHOKE	22uH			
L1402	1-460-470-11	COIL, CHOKE	22uH			
L1403	1-460-470-11	COIL, CHOKE	22uH			
		< TRANSISTOR >				
Q1401	6-553-266-01	TRANSISTOR	2PD601ART			
Q1402	6-553-268-01	TRANSISTOR	PDTC114ET			

Ref. No.	Part No.	Description				Remark
		< RESISTOR >				
R1401	1-218-879-11	METAL CHIP	22K	0.5%	1/10W	
R1402	1-218-863-11	METAL CHIP	4.7K	0.5%	1/10W	
R1403	1-250-640-11	METAL CHIP	10K	1%	1/10W	
R1407	1-216-845-11	METAL CHIP	100K	5%	1/10W	
R1408	1-250-592-11	METAL CHIP	100	1%	1/10W	
R1409	1-216-797-11	METAL CHIP	10	5%	1/10W	
R1410	1-216-797-11	METAL CHIP	10	5%	1/10W	
R1411	1-216-864-11	SHORT CHIP	0			
R1412	1-218-879-11	METAL CHIP	22K	0.5%	1/10W	
R1413	1-218-863-11	METAL CHIP	4.7K	0.5%	1/10W	
R1414	1-218-863-11	METAL CHIP	4.7K	0.5%	1/10W	
R1415	1-250-592-11	METAL CHIP	100	1%	1/10W	
R1416	1-216-864-11	SHORT CHIP	0			
R1417	1-218-879-11	METAL CHIP	22K	0.5%	1/10W	
R1418	1-218-863-11	METAL CHIP	4.7K	0.5%	1/10W	
R1419	1-218-863-11	METAL CHIP	4.7K	0.5%	1/10W	
R1420	1-218-831-11	METAL CHIP	220	0.5%	1/10W	
R1421	1-250-592-11	METAL CHIP	100	1%	1/10W	
R1422	1-216-797-11	METAL CHIP	10	5%	1/10W	
R1423	1-216-864-11	SHORT CHIP	0			
R1424	1-216-845-11	METAL CHIP	100K	5%	1/10W	
R1425	1-250-640-11	METAL CHIP	10K	1%	1/10W	
R1426	1-218-863-11	METAL CHIP	4.7K	0.5%	1/10W	
R1427	1-216-864-11	SHORT CHIP	0			
R1428	1-216-864-11	SHORT CHIP	0			
R1429	1-216-864-11	SHORT CHIP	0			
R1430	1-216-864-11	SHORT CHIP	0			
R1431	1-216-864-11	SHORT CHIP	0			
R1432	1-216-864-11	SHORT CHIP	0			
R1433	1-216-864-11	SHORT CHIP	0			

****

FAN-CONNECT BOARD
****

Ref. No.	Part No.	Description
		< CONNECTOR >
CN51	1-691-550-11	PIN, CONNECTOR (1.5 mm) (SMD) 3P
CN52	1-770-160-21	PIN, CONNECTOR (PC BOARD) 2P

****

A-1945-064-A FPGA DSP BOARD, COMPLETE
****

Ref. No.	Part No.	Description				Remark
		< CAPACITOR >				
C101	1-116-717-11	CERAMIC CHIP	10uF	20%	10V	
C102	1-116-717-11	CERAMIC CHIP	10uF	20%	10V	
C103	1-118-386-11	CERAMIC CHIP	0.1uF	10%	16V	
C104	1-118-386-11	CERAMIC CHIP	0.1uF	10%	16V	
C105	1-118-386-11	CERAMIC CHIP	0.1uF	10%	16V	
C106	1-118-386-11	CERAMIC CHIP	0.1uF	10%	16V	
C107	1-118-459-11	CERAMIC CHIP	0.01uF	10%	25V	
C108	1-118-459-11	CERAMIC CHIP	0.01uF	10%	25V	
C109	1-116-717-11	CERAMIC CHIP	10uF	20%	10V	
C110	1-165-908-11	CERAMIC CHIP	1uF	10%	10V	
C111	1-116-717-11	CERAMIC CHIP	10uF	20%	10V	
C112	1-116-749-11	CERAMIC CHIP	22uF	20%	6.3V	
C113	1-116-749-11	CERAMIC CHIP	22uF	20%	6.3V	
C114	1-116-749-11	CERAMIC CHIP	22uF	20%	6.3V	
C115	1-116-749-11	CERAMIC CHIP	22uF	20%	6.3V	
C116	1-118-386-11	CERAMIC CHIP	0.1uF	10%	16V	

FPGA DSP

	Ref. No.	Part No.	Description				Remark
	C117	1-118-386-11	CERAMIC CHIP	0.1uF	10%	16V	
	C118	1-118-386-11	CERAMIC CHIP	0.1uF	10%	16V	
*	C119	1-116-720-11	CERAMIC CHIP	10uF	20%	6.3V	
*	C120	1-116-720-11	CERAMIC CHIP	10uF	20%	6.3V	
	C201	1-118-386-11	CERAMIC CHIP	0.1uF	10%	16V	
	C202	1-118-386-11	CERAMIC CHIP	0.1uF	10%	16V	
	C203	1-118-386-11	CERAMIC CHIP	0.1uF	10%	16V	
	C204	1-118-386-11	CERAMIC CHIP	0.1uF	10%	16V	
	C205	1-118-386-11	CERAMIC CHIP	0.1uF	10%	16V	
	C206	1-118-386-11	CERAMIC CHIP	0.1uF	10%	16V	
	C207	1-118-386-11	CERAMIC CHIP	0.1uF	10%	16V	
	C208	1-118-386-11	CERAMIC CHIP	0.1uF	10%	16V	
	C209	1-118-386-11	CERAMIC CHIP	0.1uF	10%	16V	
	C210	1-118-386-11	CERAMIC CHIP	0.1uF	10%	16V	
	C211	1-118-386-11	CERAMIC CHIP	0.1uF	10%	16V	
	C212	1-118-386-11	CERAMIC CHIP	0.1uF	10%	16V	
	C213	1-118-386-11	CERAMIC CHIP	0.1uF	10%	16V	
	C214	1-118-459-11	CERAMIC CHIP	0.01uF	10%	25V	
	C215	1-118-459-11	CERAMIC CHIP	0.01uF	10%	25V	
	C216	1-118-459-11	CERAMIC CHIP	0.01uF	10%	25V	
	C217	1-118-459-11	CERAMIC CHIP	0.01uF	10%	25V	
	C218	1-118-459-11	CERAMIC CHIP	0.01uF	10%	25V	
	C219	1-118-459-11	CERAMIC CHIP	0.01uF	10%	25V	
	C220	1-118-459-11	CERAMIC CHIP	0.01uF	10%	25V	
	C221	1-118-459-11	CERAMIC CHIP	0.01uF	10%	25V	
	C222	1-118-459-11	CERAMIC CHIP	0.01uF	10%	25V	
	C223	1-118-459-11	CERAMIC CHIP	0.01uF	10%	25V	
	C224	1-118-459-11	CERAMIC CHIP	0.01uF	10%	25V	
	C225	1-118-459-11	CERAMIC CHIP	0.01uF	10%	25V	
	C226	1-118-459-11	CERAMIC CHIP	0.01uF	10%	25V	
	C227	1-118-459-11	CERAMIC CHIP	0.01uF	10%	25V	
	C228	1-118-459-11	CERAMIC CHIP	0.01uF	10%	25V	
	C229	1-118-459-11	CERAMIC CHIP	0.01uF	10%	25V	
	C230	1-118-459-11	CERAMIC CHIP	0.01uF	10%	25V	
	C231	1-118-459-11	CERAMIC CHIP	0.01uF	10%	25V	
	C232	1-118-459-11	CERAMIC CHIP	0.01uF	10%	25V	
	C233	1-118-459-11	CERAMIC CHIP	0.01uF	10%	25V	
	C234	1-118-459-11	CERAMIC CHIP	0.01uF	10%	25V	
	C235	1-118-459-11	CERAMIC CHIP	0.01uF	10%	25V	
	C236	1-118-459-11	CERAMIC CHIP	0.01uF	10%	25V	
	C237	1-118-459-11	CERAMIC CHIP	0.01uF	10%	25V	
	C238	1-118-459-11	CERAMIC CHIP	0.01uF	10%	25V	
	C239	1-118-459-11	CERAMIC CHIP	0.01uF	10%	25V	
	C240	1-118-459-11	CERAMIC CHIP	0.01uF	10%	25V	
	C241	1-118-459-11	CERAMIC CHIP	0.01uF	10%	25V	
	C242	1-118-459-11	CERAMIC CHIP	0.01uF	10%	25V	
	C243	1-118-459-11	CERAMIC CHIP	0.01uF	10%	25V	
	C244	1-118-459-11	CERAMIC CHIP	0.01uF	10%	25V	
	C245	1-118-386-11	CERAMIC CHIP	0.1uF	10%	16V	
	C246	1-118-386-11	CERAMIC CHIP	0.1uF	10%	16V	
	C247	1-118-386-11	CERAMIC CHIP	0.1uF	10%	16V	
	C248	1-118-386-11	CERAMIC CHIP	0.1uF	10%	16V	
	C249	1-118-386-11	CERAMIC CHIP	0.1uF	10%	16V	
	C250	1-118-386-11	CERAMIC CHIP	0.1uF	10%	16V	
	C251	1-118-386-11	CERAMIC CHIP	0.1uF	10%	16V	
	C252	1-118-386-11	CERAMIC CHIP	0.1uF	10%	16V	
	C253	1-118-386-11	CERAMIC CHIP	0.1uF	10%	16V	
	C254	1-118-386-11	CERAMIC CHIP	0.1uF	10%	16V	
	C255	1-118-386-11	CERAMIC CHIP	0.1uF	10%	16V	
	C256	1-118-386-11	CERAMIC CHIP	0.1uF	10%	16V	
	C257	1-118-386-11	CERAMIC CHIP	0.1uF	10%	16V	
	C258	1-118-386-11	CERAMIC CHIP	0.1uF	10%	16V	
	C259	1-118-386-11	CERAMIC CHIP	0.1uF	10%	16V	
	C260	1-118-386-11	CERAMIC CHIP	0.1uF	10%	16V	
	C261	1-118-386-11	CERAMIC CHIP	0.1uF	10%	16V	
	C262	1-116-151-11	CERAMIC CHIP	0.001uF	2%	50V	
	C263	1-118-386-11	CERAMIC CHIP	0.1uF	10%	16V	
	C264	1-118-459-11	CERAMIC CHIP	0.01uF	10%	25V	
	C265	1-116-151-11	CERAMIC CHIP	0.001uF	2%	50V	
	C266	1-116-151-11	CERAMIC CHIP	0.001uF	2%	50V	
	C267	1-116-151-11	CERAMIC CHIP	0.001uF	2%	50V	
	C301	1-118-459-11	CERAMIC CHIP	0.01uF	10%	25V	
	C303	1-118-459-11	CERAMIC CHIP	0.01uF	10%	25V	
	C401	1-118-386-11	CERAMIC CHIP	0.1uF	10%	16V	
	C402	1-118-386-11	CERAMIC CHIP	0.1uF	10%	16V	
*	C601	1-116-720-11	CERAMIC CHIP	10uF	20%	6.3V	
*	C602	1-116-720-11	CERAMIC CHIP	10uF	20%	6.3V	
*	C603	1-116-720-11	CERAMIC CHIP	10uF	20%	6.3V	
	C604	1-118-386-11	CERAMIC CHIP	0.1uF	10%	16V	
	C605	1-118-386-11	CERAMIC CHIP	0.1uF	10%	16V	
	C606	1-118-386-11	CERAMIC CHIP	0.1uF	10%	16V	
	C607	1-118-386-11	CERAMIC CHIP	0.1uF	10%	16V	
	C608	1-118-386-11	CERAMIC CHIP	0.1uF	10%	16V	
	C609	1-118-386-11	CERAMIC CHIP	0.1uF	10%	16V	
	C610	1-118-386-11	CERAMIC CHIP	0.1uF	10%	16V	
	C611	1-118-386-11	CERAMIC CHIP	0.1uF	10%	16V	
	C612	1-118-386-11	CERAMIC CHIP	0.1uF	10%	16V	
	C613	1-118-386-11	CERAMIC CHIP	0.1uF	10%	16V	
	C614	1-118-386-11	CERAMIC CHIP	0.1uF	10%	16V	
	C615	1-118-386-11	CERAMIC CHIP	0.1uF	10%	16V	
	C616	1-118-386-11	CERAMIC CHIP	0.1uF	10%	16V	
	C617	1-118-386-11	CERAMIC CHIP	0.1uF	10%	16V	
	C618	1-118-386-11	CERAMIC CHIP	0.1uF	10%	16V	
	C619	1-118-459-11	CERAMIC CHIP	0.01uF	10%	25V	
	C620	1-118-386-11	CERAMIC CHIP	0.1uF	10%	16V	
	C621	1-118-386-11	CERAMIC CHIP	0.1uF	10%	16V	
	C622	1-118-386-11	CERAMIC CHIP	0.1uF	10%	16V	
	C623	1-118-386-11	CERAMIC CHIP	0.1uF	10%	16V	
	C624	1-118-386-11	CERAMIC CHIP	0.1uF	10%	16V	
	C625	1-118-386-11	CERAMIC CHIP	0.1uF	10%	16V	
	C626	1-118-386-11	CERAMIC CHIP	0.1uF	10%	16V	
			< CONNECTOR >				
	CN401	1-815-136-21	CONNECTOR, BOARD TO BOARD				
	CN506	1-778-172-61	CONNECTOR, FFC/FPC (ZIF) 24P				
			< LUG TERMINAL >				
*	ET201	1-780-408-11	TERMINAL, LUG				
*	ET202	1-780-408-11	TERMINAL, LUG				
*	ET203	1-780-408-11	TERMINAL, LUG				
*	ET204	1-780-408-11	TERMINAL, LUG				
			< FERRITE BEAD >				
	FB101	1-469-094-21	FERRITE, EMI (SMD) (1608)				
	FB102	1-469-094-21	FERRITE, EMI (SMD) (1608)				
	FB103	1-469-094-21	FERRITE, EMI (SMD) (1608)				
	FB104	1-469-094-21	FERRITE, EMI (SMD) (1608)				
	FB105	1-469-094-21	FERRITE, EMI (SMD) (1608)				
	FB107	1-469-094-21	FERRITE, EMI (SMD) (1608)				
	FB108	1-469-094-21	FERRITE, EMI (SMD) (1608)				

Ref. No.	Part No.	Description			Remark
FB201	1-469-094-21	FERRITE, EMI (SMD) (1608)			
FB501	1-400-822-21	BEAD, FERRITE (1005)			
FB502	1-400-822-21	BEAD, FERRITE (1005)			
		< IC >			
IC001	(Not supplied)	IC EP4CGX30BF14C8N (See Note)			
IC101	6-718-705-01	IC TPS54295PWPR			
IC102	6-715-866-01	IC MM3464A25NRE			
* IC103	6-718-253-01	IC BD2231G-TR			
IC601	(Not supplied)	IC ADSP-21488KSWZ-3A1 (See Note)			
		< COIL >			
L101	1-457-870-11	CHOKE COIL	1.5uH		
L102	1-457-870-11	CHOKE COIL	1.5uH		
		< TRANSISTOR >			
Q301	8-729-928-90	TRANSISTOR	DTC114EE		
		< RESISTOR >			
R102	1-216-295-91	SHORT CHIP	0		
R104	1-216-295-91	SHORT CHIP	0		
R107	1-208-922-11	METAL CHIP	30K	0.5%	1/16W
R108	1-218-990-81	SHORT CHIP	0		
R110	1-208-709-11	METAL CHIP	12K	0.5%	1/16W
R111	1-208-895-81	METAL CHIP	2.2K	0.5%	1/16W
R112	1-208-709-11	METAL CHIP	12K	0.5%	1/16W
R113	1-208-898-11	METAL CHIP	3K	0.5%	1/16W
R114	1-208-918-11	METAL CHIP	20K	0.5%	1/16W
R116	1-208-905-11	METAL CHIP	5.6K	0.5%	1/16W
R117	1-216-864-11	SHORT CHIP	0		
R119	1-216-864-11	SHORT CHIP	0		
R120	1-216-864-11	SHORT CHIP	0		
R121	1-216-864-11	SHORT CHIP	0		
R122	1-218-965-11	METAL CHIP	10K	5%	1/16W
R126	1-216-864-11	SHORT CHIP	0		
R128	1-216-864-11	SHORT CHIP	0		
R129	1-216-864-11	SHORT CHIP	0		
R130	1-216-864-11	SHORT CHIP	0		
R131	1-250-495-11	METAL CHIP	1K	1%	1/16W
R132	1-250-495-11	METAL CHIP	1K	1%	1/16W
R201	1-216-864-11	SHORT CHIP	0		
R202	1-218-965-11	METAL CHIP	10K	5%	1/16W
R301	1-218-965-11	METAL CHIP	10K	5%	1/16W
R303	1-218-965-11	METAL CHIP	10K	5%	1/16W
R304	1-218-941-81	METAL CHIP	100	5%	1/16W
R305	1-218-965-11	METAL CHIP	10K	5%	1/16W
R306	1-218-965-11	METAL CHIP	10K	5%	1/16W
R308	1-218-965-11	METAL CHIP	10K	5%	1/16W
R309	1-218-965-11	METAL CHIP	10K	5%	1/16W
R310	1-218-965-11	METAL CHIP	10K	5%	1/16W
R311	1-218-953-11	METAL CHIP	1K	5%	1/16W
R312	1-218-965-11	METAL CHIP	10K	5%	1/16W
R313	1-218-953-11	METAL CHIP	1K	5%	1/16W
R315	1-218-965-11	METAL CHIP	10K	5%	1/16W
R316	1-218-965-11	METAL CHIP	10K	5%	1/16W
R325	1-218-953-11	METAL CHIP	1K	5%	1/16W
R401	1-218-941-81	METAL CHIP	100	5%	1/16W
R402	1-218-929-11	METAL CHIP	10	5%	1/16W
R403	1-218-929-11	METAL CHIP	10	5%	1/16W
R404	1-218-965-11	METAL CHIP	10K	5%	1/16W
R405	1-218-965-11	METAL CHIP	10K	5%	1/16W
R409	1-218-929-11	METAL CHIP	10	5%	1/16W
R411	1-218-929-11	METAL CHIP	10	5%	1/16W
R412	1-218-929-11	METAL CHIP	10	5%	1/16W
R413	1-218-929-11	METAL CHIP	10	5%	1/16W
R418	1-218-929-11	METAL CHIP	10	5%	1/16W
R420	1-218-929-11	METAL CHIP	10	5%	1/16W
R421	1-218-929-11	METAL CHIP	10	5%	1/16W
R422	1-218-990-81	SHORT CHIP	0		
R424	1-218-965-11	METAL CHIP	10K	5%	1/16W
R426	1-208-894-81	METAL CHIP	2K	0.5%	1/16W
R512	1-218-929-11	METAL CHIP	10	5%	1/16W
R513	1-218-929-11	METAL CHIP	10	5%	1/16W
R514	1-218-945-11	METAL CHIP	220	5%	1/16W
R515	1-218-941-81	METAL CHIP	100	5%	1/16W
R518	1-218-949-11	METAL CHIP	470	5%	1/16W
R519	1-218-929-11	METAL CHIP	10	5%	1/16W
R520	1-218-929-11	METAL CHIP	10	5%	1/16W
R521	1-218-929-11	METAL CHIP	10	5%	1/16W
R522	1-218-929-11	METAL CHIP	10	5%	1/16W
R523	1-218-929-11	METAL CHIP	10	5%	1/16W
R524	1-218-929-11	METAL CHIP	10	5%	1/16W
R526	1-218-929-11	METAL CHIP	10	5%	1/16W
R527	1-218-929-11	METAL CHIP	10	5%	1/16W
R528	1-218-929-11	METAL CHIP	10	5%	1/16W
R531	1-218-929-11	METAL CHIP	10	5%	1/16W
R532	1-218-929-11	METAL CHIP	10	5%	1/16W
R534	1-218-929-11	METAL CHIP	10	5%	1/16W
R548	1-218-941-81	METAL CHIP	100	5%	1/16W
R549	1-218-941-81	METAL CHIP	100	5%	1/16W
R550	1-218-929-11	METAL CHIP	10	5%	1/16W
R551	1-218-941-81	METAL CHIP	100	5%	1/16W
R552	1-218-929-11	METAL CHIP	10	5%	1/16W
R553	1-218-929-11	METAL CHIP	10	5%	1/16W
R554	1-218-929-11	METAL CHIP	10	5%	1/16W
R555	1-218-929-11	METAL CHIP	10	5%	1/16W
R556	1-218-929-11	METAL CHIP	10	5%	1/16W
R557	1-218-929-11	METAL CHIP	10	5%	1/16W
R558	1-218-929-11	METAL CHIP	10	5%	1/16W
R559	1-218-929-11	METAL CHIP	10	5%	1/16W
R560	1-218-929-11	METAL CHIP	10	5%	1/16W
R561	1-218-929-11	METAL CHIP	10	5%	1/16W
R562	1-218-929-11	METAL CHIP	10	5%	1/16W
R570	1-218-965-11	METAL CHIP	10K	5%	1/16W
R575	1-218-990-81	SHORT CHIP	0		
R576	1-218-957-11	METAL CHIP	2.2K	5%	1/16W
R577	1-218-957-11	METAL CHIP	2.2K	5%	1/16W
R578	1-218-957-11	METAL CHIP	2.2K	5%	1/16W
R579	1-218-957-11	METAL CHIP	2.2K	5%	1/16W
R580	1-218-965-11	METAL CHIP	10K	5%	1/16W
R581	1-218-965-11	METAL CHIP	10K	5%	1/16W
R582	1-218-965-11	METAL CHIP	10K	5%	1/16W
R601	1-216-864-11	SHORT CHIP	0		
R602	1-216-864-11	SHORT CHIP	0		
R603	1-218-935-11	METAL CHIP	33	5%	1/16W
R604	1-208-859-81	METAL CHIP	68	0.5%	1/16W
R605	1-218-935-11	METAL CHIP	33	5%	1/16W
R607	1-218-935-11	METAL CHIP	33	5%	1/16W
R608	1-218-990-81	SHORT CHIP	0		
R609	1-218-935-11	METAL CHIP	33	5%	1/16W

Note: IC001 and IC601 on the FPGA DSP board cannot exchange with single.
When these parts are damaged, exchange the complete mounted board.

FPGA DSP | **IO** | **KEY JOG**

Ref. No.	Part No.	Description			Remark
R610	1-218-990-81	SHORT CHIP	0		
R611	1-218-935-11	METAL CHIP	33	5%	1/16W
R612	1-218-990-81	SHORT CHIP	0		
R613	1-218-935-11	METAL CHIP	33	5%	1/16W
R614	1-218-990-81	SHORT CHIP	0		
R615	1-218-990-81	SHORT CHIP	0		
R617	1-218-941-81	METAL CHIP	100	5%	1/16W
R624	1-218-933-11	METAL CHIP	22	5%	1/16W
R626	1-218-933-11	METAL CHIP	22	5%	1/16W
R627	1-218-933-11	METAL CHIP	22	5%	1/16W
R628	1-218-933-11	METAL CHIP	22	5%	1/16W
R629	1-218-965-11	METAL CHIP	10K	5%	1/16W
R632	1-218-965-11	METAL CHIP	10K	5%	1/16W
R633	1-218-989-11	METAL CHIP	1M	5%	1/16W
R635	1-218-965-11	METAL CHIP	10K	5%	1/16W
R636	1-218-947-11	METAL CHIP	330	5%	1/16W
R638	1-218-965-11	METAL CHIP	10K	5%	1/16W
		< COMPOSITION CIRCUIT BLOCK >			
RB601	1-234-376-11	RES, NETWORK 2.2K (1005X4)			
		< VIBRATOR >			
X201	1-814-147-11	QUARTZ CRYSTAL OSCILLATOR (50 MHz)			
X601	1-795-244-11	VIBRATOR, CERAMIC (10 MHz)			
**					
	A-1945-114-A	IO BOARD, COMPLETE			
		******************			
		< CAPACITOR >			
C4013	1-116-151-11	CERAMIC CHIP	0.001uF	2%	50V
C4014	1-164-874-11	CERAMIC CHIP	100PF	5%	50V
C4016	1-116-151-11	CERAMIC CHIP	0.001uF	2%	50V
C4017	1-118-417-11	CERAMIC CHIP	0.1uF	10%	16V
C4018	1-128-992-21	ELECT CHIP	47uF	20%	25V
C4021	1-118-386-11	CERAMIC CHIP	0.1uF	10%	16V
		< JACK/CONNECTOR >			
CN4001	1-842-235-11	JACK, MODULAR (LAN (10/100/1000))			
CN4002	1-822-423-11	CONNECTOR, USB (A) (5V ⎓ 1A EXT)			
* CN4003	1-779-268-21	PIN, CONNECTOR (1.5 mm) (SMD) 11P			
CN4005	1-573-806-21	PIN, CONNECTOR (1.5 mm) (SMD) 6P			
CN4006	1-691-550-11	PIN, CONNECTOR (1.5 mm) (SMD) 3P			
CN4007	1-784-859-51	CONNECTOR, FFC (LIF (NON-ZIF)) 7P			
CN4008	1-573-768-21	PIN, CONNECTOR (1.5 mm) (SMD) 5P			
		< LUG TERMINAL >			
* ET4003	1-780-408-11	TERMINAL, LUG			
* ET4004	1-780-408-11	TERMINAL, LUG			
* ET4005	1-780-408-11	TERMINAL, LUG			
* ET4006	1-780-408-11	TERMINAL, LUG			
		< JACK >			
J4015	1-563-330-31	JACK (OUT IR REMOTE)			
		< TRANSISTOR >			
Q4001	8-729-026-49	TRANSISTOR	2SA1037AK-T146-R		
Q4002	6-553-268-01	TRANSISTOR	PDTC114ET		

Ref. No.	Part No.	Description			Remark
		< RESISTOR >			
R4001	1-216-295-91	SHORT CHIP	0		
R4010	1-216-864-11	SHORT CHIP	0		
R4011	1-218-990-81	SHORT CHIP	0		
R4012	1-218-990-81	SHORT CHIP	0		
R4017	1-216-864-11	SHORT CHIP	0		
R4018	1-216-864-11	SHORT CHIP	0		
R4019	1-216-825-11	METAL CHIP	2.2K	5%	1/10W
R4020	1-216-825-11	METAL CHIP	2.2K	5%	1/10W
R4021	1-216-825-11	METAL CHIP	2.2K	5%	1/10W
R4022	1-216-166-00	METAL CHIP	47	5%	1/8W
R4023	1-216-295-91	SHORT CHIP	0		
R4024	1-216-295-91	SHORT CHIP	0		
R4025	1-216-295-91	SHORT CHIP	0		
R4027	1-216-864-11	SHORT CHIP	0		
R4028	1-216-864-11	SHORT CHIP	0		
R4029	1-216-864-11	SHORT CHIP	0		
R4030	1-216-864-11	SHORT CHIP	0		
R4032	1-216-864-11	SHORT CHIP	0		
R4033	1-216-864-11	SHORT CHIP	0		
R4034	1-216-864-11	SHORT CHIP	0		
R4035	1-216-864-11	SHORT CHIP	0		
R4036	1-216-864-11	SHORT CHIP	0		
R4037	1-216-864-11	SHORT CHIP	0		
R4038	1-216-295-91	SHORT CHIP	0		
R4039	1-216-295-91	SHORT CHIP	0		
**					
	A-1945-123-A	KEY JOG BOARD, COMPLETE			
		***********************			
		< CAPACITOR >			
C301	1-118-403-11	CERAMIC CHIP	0.001uF	10%	50V
C302	1-118-403-11	CERAMIC CHIP	0.001uF	10%	50V
		< CONNECTOR >			
* CN304	1-764-007-11	PIN, CONNECTOR (SMD) 12P			
* CN305	1-580-055-21	PIN, CONNECTOR (SMD) 2P			
		< ROTARY ENCODER >			
EN301	1-476-980-11	ENCODER (SELECTOR)			
		< TRANSISTOR >			
Q301	6-553-268-01	TRANSISTOR	PDTC114ET		
		< RESISTOR >			
R307	1-216-809-11	METAL CHIP	100	5%	1/10W
R308	1-216-809-11	METAL CHIP	100	5%	1/10W
R309	1-216-833-11	METAL CHIP	10K	5%	1/10W
R310	1-216-182-00	METAL CHIP	220	5%	1/8W
		< SWITCH >			
S301	1-762-875-21	SWITCH, KEYBOARD (BACK)			
S302	1-762-875-21	SWITCH, KEYBOARD (ENTER)			
S303	1-762-875-21	SWITCH, KEYBOARD (HOME)			
S304	1-762-875-21	SWITCH, KEYBOARD (►II)			
**					

Ref. No.	Part No.	Description			Remark
		LED BOARD			
		*********			
		< CONNECTOR >			
* CN031	1-580-055-21	PIN, CONNECTOR (SMD) 2P			
		< LED >			
D031	6-502-383-11	LED CL-194S-HB8-SD-T (DSEE)			
***					
	A-1989-164-A	MAIN BOARD, COMPLETE (for SERVICE) (US, CND) (See Note)			
	A-1989-165-A	MAIN BOARD, COMPLETE (for SERVICE) (AEP, UK) (See Note)			
		********************			
		< CAPACITOR >			
C101	1-116-745-11	CERAMIC CHIP	0.22uF	10%	6.3V
C102	1-116-745-11	CERAMIC CHIP	0.22uF	10%	6.3V
C103	1-116-745-11	CERAMIC CHIP	0.22uF	10%	6.3V
C104	1-116-745-11	CERAMIC CHIP	0.22uF	10%	6.3V
C105	1-116-745-11	CERAMIC CHIP	0.22uF	10%	6.3V
C106	1-116-745-11	CERAMIC CHIP	0.22uF	10%	6.3V
C111	1-116-745-11	CERAMIC CHIP	0.22uF	10%	6.3V
C112	1-116-745-11	CERAMIC CHIP	0.22uF	10%	6.3V
C113	1-116-745-11	CERAMIC CHIP	0.22uF	10%	6.3V
C114	1-116-745-11	CERAMIC CHIP	0.22uF	10%	6.3V
C115	1-116-745-11	CERAMIC CHIP	0.22uF	10%	6.3V
C116	1-116-745-11	CERAMIC CHIP	0.22uF	10%	6.3V
C117	1-116-745-11	CERAMIC CHIP	0.22uF	10%	6.3V
C118	1-116-745-11	CERAMIC CHIP	0.22uF	10%	6.3V
C119	1-116-745-11	CERAMIC CHIP	0.22uF	10%	6.3V
C120	1-116-745-11	CERAMIC CHIP	0.22uF	10%	6.3V
C121	1-116-745-11	CERAMIC CHIP	0.22uF	10%	6.3V
C123	1-116-745-11	CERAMIC CHIP	0.22uF	10%	6.3V
C124	1-116-745-11	CERAMIC CHIP	0.22uF	10%	6.3V
C125	1-116-745-11	CERAMIC CHIP	0.22uF	10%	6.3V
C126	1-116-745-11	CERAMIC CHIP	0.22uF	10%	6.3V
C127	1-116-745-11	CERAMIC CHIP	0.22uF	10%	6.3V
C128	1-116-745-11	CERAMIC CHIP	0.22uF	10%	6.3V
C129	1-116-745-11	CERAMIC CHIP	0.22uF	10%	6.3V
C130	1-116-745-11	CERAMIC CHIP	0.22uF	10%	6.3V
C131	1-116-745-11	CERAMIC CHIP	0.22uF	10%	6.3V
C132	1-116-745-11	CERAMIC CHIP	0.22uF	10%	6.3V
C137	1-116-745-11	CERAMIC CHIP	0.22uF	10%	6.3V
C138	1-116-745-11	CERAMIC CHIP	0.22uF	10%	6.3V
C139	1-116-745-11	CERAMIC CHIP	0.22uF	10%	6.3V
C140	1-116-745-11	CERAMIC CHIP	0.22uF	10%	6.3V
C141	1-116-745-11	CERAMIC CHIP	0.22uF	10%	6.3V
C142	1-116-745-11	CERAMIC CHIP	0.22uF	10%	6.3V
C144	1-116-745-11	CERAMIC CHIP	0.22uF	10%	6.3V
C145	1-116-745-11	CERAMIC CHIP	0.22uF	10%	6.3V
C146	1-116-745-11	CERAMIC CHIP	0.22uF	10%	6.3V
C147	1-116-745-11	CERAMIC CHIP	0.22uF	10%	6.3V
C148	1-116-745-11	CERAMIC CHIP	0.22uF	10%	6.3V
C150	1-116-745-11	CERAMIC CHIP	0.22uF	10%	6.3V
C151	1-116-745-11	CERAMIC CHIP	0.22uF	10%	6.3V
C152	1-116-745-11	CERAMIC CHIP	0.22uF	10%	6.3V
C153	1-116-745-11	CERAMIC CHIP	0.22uF	10%	6.3V
C154	1-116-745-11	CERAMIC CHIP	0.22uF	10%	6.3V
C155	1-116-745-11	CERAMIC CHIP	0.22uF	10%	6.3V
C156	1-116-745-11	CERAMIC CHIP	0.22uF	10%	6.3V
C157	1-116-745-11	CERAMIC CHIP	0.22uF	10%	6.3V
C159	1-116-745-11	CERAMIC CHIP	0.22uF	10%	6.3V
C174	1-118-459-11	CERAMIC CHIP	0.01uF	10%	25V
C180	1-116-749-11	CERAMIC CHIP	22uF	20%	6.3V
C182	1-116-749-11	CERAMIC CHIP	22uF	20%	6.3V
C183	1-116-749-11	CERAMIC CHIP	22uF	20%	6.3V
C184	1-116-749-11	CERAMIC CHIP	22uF	20%	6.3V
C185	1-116-749-11	CERAMIC CHIP	22uF	20%	6.3V
C186	1-116-749-11	CERAMIC CHIP	22uF	20%	6.3V
C187	1-116-749-11	CERAMIC CHIP	22uF	20%	6.3V
C188	1-116-749-11	CERAMIC CHIP	22uF	20%	6.3V
* C189	1-116-720-11	CERAMIC CHIP	10uF	20%	6.3V
C190	1-116-749-11	CERAMIC CHIP	22uF	20%	6.3V
C192	1-116-749-11	CERAMIC CHIP	22uF	20%	6.3V
C193	1-116-749-11	CERAMIC CHIP	22uF	20%	6.3V
* C194	1-116-720-11	CERAMIC CHIP	10uF	20%	6.3V
C201	1-116-711-11	CERAMIC CHIP	22uF	20%	16V
* C202	1-116-735-11	CERAMIC CHIP	1uF	10%	16V
C203	1-116-711-11	CERAMIC CHIP	22uF	20%	16V
C204	1-118-459-11	CERAMIC CHIP	0.01uF	10%	25V
C205	1-118-386-11	CERAMIC CHIP	0.1uF	10%	16V
C206	1-118-459-11	CERAMIC CHIP	0.01uF	10%	25V
C207	1-116-711-11	CERAMIC CHIP	22uF	20%	16V
C208	1-116-711-11	CERAMIC CHIP	22uF	20%	16V
C209	1-118-386-11	CERAMIC CHIP	0.1uF	10%	16V
C210	1-118-386-11	CERAMIC CHIP	0.1uF	10%	16V
C211	1-118-459-11	CERAMIC CHIP	0.01uF	10%	25V
* C212	1-118-401-11	CERAMIC CHIP	0.0015uF	10%	50V
C216	1-116-717-11	CERAMIC CHIP	10uF	20%	10V
C217	1-116-717-11	CERAMIC CHIP	10uF	20%	10V
C219	1-116-717-11	CERAMIC CHIP	10uF	20%	10V
C220	1-116-717-11	CERAMIC CHIP	10uF	20%	10V
C221	1-116-151-11	CERAMIC CHIP	0.001uF	2%	50V
C222	1-116-151-11	CERAMIC CHIP	0.001uF	2%	50V
C223	1-116-151-11	CERAMIC CHIP	0.001uF	2%	50V
C224	1-116-151-11	CERAMIC CHIP	0.001uF	2%	50V
C229	1-116-717-11	CERAMIC CHIP	10uF	20%	10V
C230	1-116-717-11	CERAMIC CHIP	10uF	20%	10V
C231	1-116-717-11	CERAMIC CHIP	10uF	20%	10V
C233	1-165-492-21	ELECT CHIP	100uF	20%	10V
C235	1-116-151-11	CERAMIC CHIP	0.001uF	2%	50V
C236	1-116-151-11	CERAMIC CHIP	0.001uF	2%	50V
C237	1-116-711-11	CERAMIC CHIP	22uF	20%	16V
C238	1-116-711-11	CERAMIC CHIP	22uF	20%	16V
C240	1-116-745-11	CERAMIC CHIP	0.22uF	10%	6.3V
C241	1-118-386-11	CERAMIC CHIP	0.1uF	10%	16V
* C301	1-116-720-11	CERAMIC CHIP	10uF	20%	6.3V
C302	1-118-386-11	CERAMIC CHIP	0.1uF	10%	16V
C303	1-118-386-11	CERAMIC CHIP	0.1uF	10%	16V
C304	1-118-386-11	CERAMIC CHIP	0.1uF	10%	16V
C305	1-118-386-11	CERAMIC CHIP	0.1uF	10%	16V
C306	1-118-386-11	CERAMIC CHIP	0.1uF	10%	16V
C307	1-118-386-11	CERAMIC CHIP	0.1uF	10%	16V
C308	1-118-386-11	CERAMIC CHIP	0.1uF	10%	16V
C309	1-118-386-11	CERAMIC CHIP	0.1uF	10%	16V
C310	1-118-386-11	CERAMIC CHIP	0.1uF	10%	16V
C311	1-118-386-11	CERAMIC CHIP	0.1uF	10%	16V
* C312	1-116-720-11	CERAMIC CHIP	10uF	20%	6.3V
* C313	1-116-720-11	CERAMIC CHIP	10uF	20%	6.3V

Note: When the complete MAIN board is replaced, refer to "NOTE OF REPLACING THE COMPLETE MAIN BOARD" and "CHECKING METHOD OF NETWORK CONNECTION" on page 5.

MAIN

	Ref. No.	Part No.	Description			Remark
	C314	1-118-386-11	CERAMIC CHIP	0.1uF	10%	16V
	C315	1-118-386-11	CERAMIC CHIP	0.1uF	10%	16V
*	C316	1-116-720-11	CERAMIC CHIP	10uF	20%	6.3V
	C317	1-118-386-11	CERAMIC CHIP	0.1uF	10%	16V
	C318	1-118-386-11	CERAMIC CHIP	0.1uF	10%	16V
	C319	1-118-386-11	CERAMIC CHIP	0.1uF	10%	16V
	C320	1-118-386-11	CERAMIC CHIP	0.1uF	10%	16V
	C321	1-118-386-11	CERAMIC CHIP	0.1uF	10%	16V
	C322	1-118-386-11	CERAMIC CHIP	0.1uF	10%	16V
	C323	1-118-386-11	CERAMIC CHIP	0.1uF	10%	16V
	C324	1-118-386-11	CERAMIC CHIP	0.1uF	10%	16V
	C325	1-118-386-11	CERAMIC CHIP	0.1uF	10%	16V
*	C338	1-116-720-11	CERAMIC CHIP	10uF	20%	6.3V
	C340	1-118-386-11	CERAMIC CHIP	0.1uF	10%	16V
	C341	1-118-386-11	CERAMIC CHIP	0.1uF	10%	16V
*	C342	1-116-720-11	CERAMIC CHIP	10uF	20%	6.3V
	C343	1-118-386-11	CERAMIC CHIP	0.1uF	10%	16V
	C344	1-118-386-11	CERAMIC CHIP	0.1uF	10%	16V
	C345	1-118-386-11	CERAMIC CHIP	0.1uF	10%	16V
	C346	1-118-386-11	CERAMIC CHIP	0.1uF	10%	16V
	C347	1-118-386-11	CERAMIC CHIP	0.1uF	10%	16V
	C348	1-118-386-11	CERAMIC CHIP	0.1uF	10%	16V
	C349	1-118-386-11	CERAMIC CHIP	0.1uF	10%	16V
	C350	1-118-386-11	CERAMIC CHIP	0.1uF	10%	16V
	C351	1-118-386-11	CERAMIC CHIP	0.1uF	10%	16V
	C401	1-118-459-11	CERAMIC CHIP	0.01uF	10%	25V
	C402	1-118-388-11	CERAMIC CHIP	0.047uF	10%	25V
	C403	1-164-846-11	CERAMIC CHIP	6PF	0.5PF	50V
	C404	1-164-846-11	CERAMIC CHIP	6PF	0.5PF	50V
	C405	1-164-858-11	CERAMIC CHIP	22PF	5%	50V
	C406	1-164-856-81	CERAMIC CHIP	18PF	5%	50V
	C407	1-116-744-11	CERAMIC CHIP	0.22uF	10%	10V
	C408	1-116-744-11	CERAMIC CHIP	0.22uF	10%	10V
	C409	1-116-744-11	CERAMIC CHIP	0.22uF	10%	10V
	C410	1-118-386-11	CERAMIC CHIP	0.1uF	10%	16V
	C411	1-118-386-11	CERAMIC CHIP	0.1uF	10%	16V
	C412	1-118-386-11	CERAMIC CHIP	0.1uF	10%	16V
	C413	1-118-386-11	CERAMIC CHIP	0.1uF	10%	16V
	C415	1-118-386-11	CERAMIC CHIP	0.1uF	10%	16V
*	C416	1-116-720-11	CERAMIC CHIP	10uF	20%	6.3V
	C417	1-118-386-11	CERAMIC CHIP	0.1uF	10%	16V
	C418	1-118-459-11	CERAMIC CHIP	0.01uF	10%	25V
	C501	1-118-386-11	CERAMIC CHIP	0.1uF	10%	16V
	C502	1-118-386-11	CERAMIC CHIP	0.1uF	10%	16V
	C503	1-118-386-11	CERAMIC CHIP	0.1uF	10%	16V
	C504	1-118-386-11	CERAMIC CHIP	0.1uF	10%	16V
	C505	1-118-386-11	CERAMIC CHIP	0.1uF	10%	16V
	C506	1-118-386-11	CERAMIC CHIP	0.1uF	10%	16V
	C507	1-118-386-11	CERAMIC CHIP	0.1uF	10%	16V
	C508	1-118-386-11	CERAMIC CHIP	0.1uF	10%	16V
	C513	1-118-386-11	CERAMIC CHIP	0.1uF	10%	16V
	C514	1-118-386-11	CERAMIC CHIP	0.1uF	10%	16V
	C602	1-118-386-11	CERAMIC CHIP	0.1uF	10%	16V
*	C603	1-116-720-11	CERAMIC CHIP	10uF	20%	6.3V
*	C604	1-116-720-11	CERAMIC CHIP	10uF	20%	6.3V
	C605	1-164-852-11	CERAMIC CHIP	12PF	5%	50V
	C606	1-118-386-11	CERAMIC CHIP	0.1uF	10%	16V
	C608	1-118-386-11	CERAMIC CHIP	0.1uF	10%	16V
	C609	1-164-852-11	CERAMIC CHIP	12PF	5%	50V
	C610	1-118-386-11	CERAMIC CHIP	0.1uF	10%	16V
*	C611	1-116-738-11	CERAMIC CHIP	1uF	10%	6.3V
	C612	1-118-386-11	CERAMIC CHIP	0.1uF	10%	16V
	C613	1-118-386-11	CERAMIC CHIP	0.1uF	10%	16V
	C614	1-118-386-11	CERAMIC CHIP	0.1uF	10%	16V
	C615	1-118-386-11	CERAMIC CHIP	0.1uF	10%	16V
	C616	1-118-386-11	CERAMIC CHIP	0.1uF	10%	16V
	C617	1-118-386-11	CERAMIC CHIP	0.1uF	10%	16V
	C618	1-118-386-11	CERAMIC CHIP	0.1uF	10%	16V
	C619	1-116-151-11	CERAMIC CHIP	0.001uF	2%	50V
	C621	1-114-472-91	CERAMIC CHIP	0.001uF	10%	2KV
*	C622	1-116-738-11	CERAMIC CHIP	1uF	10%	6.3V
	C702	1-118-386-11	CERAMIC CHIP	0.1uF	10%	16V
	C703	1-116-078-11	CERAMIC CHIP	1uF	10%	50V
	C704	1-116-724-11	CERAMIC CHIP	4.7uF	20%	6.3V
	C705	1-118-386-11	CERAMIC CHIP	0.1uF	10%	16V
*	C706	1-116-738-11	CERAMIC CHIP	1uF	10%	6.3V
	C707	1-116-078-11	CERAMIC CHIP	1uF	10%	50V
	C708	1-116-724-11	CERAMIC CHIP	4.7uF	20%	6.3V
	C901	1-118-386-11	CERAMIC CHIP	0.1uF	10%	16V
*	C1001	1-116-720-11	CERAMIC CHIP	10uF	20%	6.3V
	C1002	1-116-744-11	CERAMIC CHIP	0.22uF	10%	10V
	C1003	1-116-744-11	CERAMIC CHIP	0.22uF	10%	10V
	C1005	1-116-744-11	CERAMIC CHIP	0.22uF	10%	10V
	C1006	1-116-724-11	CERAMIC CHIP	4.7uF	20%	6.3V
	C1007	1-116-724-11	CERAMIC CHIP	4.7uF	20%	6.3V
	C1008	1-116-744-11	CERAMIC CHIP	0.22uF	10%	10V
	C1009	1-116-744-11	CERAMIC CHIP	0.22uF	10%	10V
*	C1010	1-116-720-11	CERAMIC CHIP	10uF	20%	6.3V
*	C1011	1-116-720-11	CERAMIC CHIP	10uF	20%	6.3V
*	C1012	1-116-720-11	CERAMIC CHIP	10uF	20%	6.3V
	C1013	1-118-459-11	CERAMIC CHIP	0.01uF	10%	25V
	C1014	1-118-459-11	CERAMIC CHIP	0.01uF	10%	25V
	C1015	1-118-459-11	CERAMIC CHIP	0.01uF	10%	25V
	C1016	1-118-459-11	CERAMIC CHIP	0.01uF	10%	25V
*	C1017	1-116-720-11	CERAMIC CHIP	10uF	20%	6.3V
	C1018	1-165-492-21	ELECT CHIP	100uF	20%	10V
	C1019	1-116-151-11	CERAMIC CHIP	0.001uF	2%	50V
	C1020	1-118-386-11	CERAMIC CHIP	0.1uF	10%	16V
			< CONNECTOR >			
	CN201	1-816-296-21	PIN, CONNECTOR (PC BOARD) 9P			
	CN203	1-817-097-21	PIN, CONNECTOR (1.5 mm) (SMD) 13P			
	CN402	1-785-910-21	CONNECTOR, BOARD TO BOARD 50P			
	CN601	1-695-223-21	PIN, CONNECTOR (SMD) 10P			
	CN701	1-573-806-21	PIN, CONNECTOR (1.5 mm) (SMD) 6P			
	CN703	1-818-857-51	CONNECTOR, FFC/FPC 40P			
	CN1001	1-821-796-11	CONNECTOR, SATA SMT (7P)			
	CN1002	1-569-775-21	PIN, CONNECTOR (SMD) 5P			
	CN1003	1-580-756-21	PIN, CONNECTOR (SMD) 7P			
			< DIODE >			
	D403	6-501-199-01	DIODE NSR0320MW2T1G			
	D604	6-500-275-01	DIODE RB751S-40TE61			
	D701	6-502-454-01	DIODE RB160VA-40TR			
	D1001	8-719-049-09	DIODE 1SS367-T3SONY			
	D1002	8-719-049-09	DIODE 1SS367-T3SONY			
	D1003	8-719-049-09	DIODE 1SS367-T3SONY			
	D1004	8-719-049-09	DIODE 1SS367-T3SONY			

Ref. No.	Part No.	Description	Remark
		< LUG TERMINAL >	
* ET001	1-780-408-11	TERMINAL, LUG	
* ET002	1-780-408-11	TERMINAL, LUG	
* ET003	1-780-408-11	TERMINAL, LUG	
* ET004	1-780-408-11	TERMINAL, LUG	
* ET009	1-780-408-11	TERMINAL, LUG	
* ET010	1-780-408-11	TERMINAL, LUG	
		< FERRITE BEAD >	
FB101	1-400-462-21	FERRITE, EMI (SMD) (1005)	
FB401	1-400-462-21	FERRITE, EMI (SMD) (1005)	
FB601	1-400-462-21	FERRITE, EMI (SMD) (1005)	
FB602	1-400-462-21	FERRITE, EMI (SMD) (1005)	
FB1000	1-469-379-11	FERRITE, EMI (SMD) (2012)	
FB1001	1-469-379-11	FERRITE, EMI (SMD) (2012)	
		< FILTER >	
FL201	1-234-939-21	FILTER, EMI REMOVAL (SMD)	
FL400	1-234-939-21	FILTER, EMI REMOVAL (SMD)	
FL401	1-234-939-21	FILTER, EMI REMOVAL (SMD)	
FL501	1-234-939-21	FILTER, EMI REMOVAL (SMD)	
FL701	1-234-939-21	FILTER, EMI REMOVAL (SMD)	
		< IC >	
IC101	(Not supplied)	IC MCIMX6D5EYM10AC (See Note)	
IC201	6-717-694-01	IC BU33TD3WG-TR	
IC202	6-716-554-11	IC BD9329AEFJ-E2	
IC203	6-718-705-01	IC TPS54295PWPR	
IC204	6-715-874-01	IC MM3464A33NRE	
IC205	6-713-333-01	IC PST8429UL	
IC301	(Not supplied)	IC H5TQ1G63EFR-H9CR (See Note)	
IC302	(Not supplied)	IC H5TQ1G63EFR-H9CR (See Note)	
IC401	6-713-333-01	IC PST8429UL	
IC501	(Not supplied)	IC KLM4G1FE3B-B001T03 (See Note)	
IC602	(Not supplied)	IC AR8035-AL1A (See Note)	
IC701	6-717-669-01	IC TB62752AFUG (O, EL)	
IC1001	6-715-164-01	IC TPS2553DBVR	
		< COIL >	
L201	1-457-871-11	CHOKE COIL 2.2uH	
L202	1-457-871-11	CHOKE COIL 2.2uH	
L203	1-460-576-11	COIL, CHOKE 4.7uH	
L601	1-400-790-21	INDUCTOR 4.7uH	
L602	1-460-653-21	COIL, COMMON MODE CHOKE	
L603	1-460-653-21	COIL, COMMON MODE CHOKE	
L604	1-460-653-21	COIL, COMMON MODE CHOKE	
L605	1-460-653-21	COIL, COMMON MODE CHOKE	
L701	1-460-586-11	COIL, CHOKE 22uH	
L1001	1-460-652-21	COIL, COMMON MODE CHOKE	
L1002	1-460-652-21	COIL, COMMON MODE CHOKE	
		< TRANSISTOR >	
Q201	8-729-928-90	TRANSISTOR DTC114EE	
Q202	8-729-928-90	TRANSISTOR DTC114EE	
Q203	8-729-928-36	TRANSISTOR DTA114EE	
Q701	6-552-621-01	FET SSM3K37MFV, L3SOF	
Q702	6-552-621-01	FET SSM3K37MFV, L3SOF	
Q703	6-552-621-01	FET SSM3K37MFV, L3SOF	
Q704	6-552-029-01	FET MCH3377-S-TL-E	

Ref. No.	Part No.	Description	Remark
Q1001	6-552-621-01	FET SSM3K37MFV, L3SOF	
Q1002	6-552-029-01	FET MCH3377-S-TL-E	
		< RESISTOR/FERRITE BEAD >	
R103	1-218-990-81	SHORT CHIP 0	
R104	1-218-990-81	SHORT CHIP 0	
R105	1-218-990-81	SHORT CHIP 0	
R106	1-218-990-81	SHORT CHIP 0	
R108	1-218-990-81	SHORT CHIP 0	
R109	1-218-990-81	SHORT CHIP 0	
R116	1-218-990-81	SHORT CHIP 0	
R117	1-218-990-81	SHORT CHIP 0	
R201	1-208-695-11	METAL CHIP 3.3K 0.5% 1/16W	
R202	1-208-647-11	METAL CHIP 33 0.5% 1/16W	
R203	1-208-695-11	METAL CHIP 3.3K 0.5% 1/16W	
R204	1-218-990-81	SHORT CHIP 0	
R206	1-218-990-81	SHORT CHIP 0	
R207	1-208-905-11	METAL CHIP 5.6K 0.5% 1/16W	
R208	1-208-695-11	METAL CHIP 3.3K 0.5% 1/16W	
R209	1-250-455-11	METAL CHIP 22 1% 1/16W	
* R210	1-250-523-11	METAL CHIP 15K 1% 1/16W	
R211	1-218-990-81	SHORT CHIP 0	
R212	1-250-495-11	METAL CHIP 1K 1% 1/16W	
R213	1-250-495-11	METAL CHIP 1K 1% 1/16W	
R215	1-250-495-11	METAL CHIP 1K 1% 1/16W	
R217	1-208-695-11	METAL CHIP 3.3K 0.5% 1/16W	
R218	1-208-683-11	METAL CHIP 1K 0.5% 1/16W	
R224	1-250-519-11	METAL CHIP 10K 1% 1/16W	
R225	1-218-990-81	SHORT CHIP 0	
R242	1-216-864-11	SHORT CHIP 0	
R243	1-216-864-11	SHORT CHIP 0	
R244	1-218-990-81	SHORT CHIP 0	
R248	1-218-990-81	SHORT CHIP 0	
R253	1-218-990-81	SHORT CHIP 0	
R254	1-218-990-81	SHORT CHIP 0	
* R256	1-250-501-11	METAL CHIP 1.8K 1% 1/16W	
R257	1-218-990-81	SHORT CHIP 0	
R258	1-218-990-81	SHORT CHIP 0	
R259	1-218-990-81	SHORT CHIP 0	
R261	1-218-990-81	SHORT CHIP 0	
R301	1-250-495-11	METAL CHIP 1K 1% 1/16W	
R302	1-250-495-11	METAL CHIP 1K 1% 1/16W	
R303	1-218-965-11	METAL CHIP 10K 5% 1/16W	
R304	1-218-990-81	SHORT CHIP 0	
R305	1-218-990-81	SHORT CHIP 0	
* R306	1-250-480-11	METAL CHIP 240 1% 1/16W	
* R307	1-250-480-11	METAL CHIP 240 1% 1/16W	
R308	1-218-945-11	METAL CHIP 220 5% 1/16W	
* R309	1-250-480-11	METAL CHIP 240 1% 1/16W	
R310	1-218-945-11	METAL CHIP 220 5% 1/16W	
R404	1-220-804-11	METAL CHIP 2.2M 5% 1/16W	
R406	1-218-959-11	METAL CHIP 3.3K 5% 1/16W	
R409	1-245-604-11	METAL CHIP 10M 5% 1/16W	
R410	1-218-990-81	SHORT CHIP 0	
R411	1-218-990-81	SHORT CHIP 0	
R413	1-218-965-11	METAL CHIP 10K 5% 1/16W	
R417	1-218-965-11	METAL CHIP 10K 5% 1/16W	
R418	1-218-965-11	METAL CHIP 10K 5% 1/16W	
R422	1-218-965-11	METAL CHIP 10K 5% 1/16W	
R425	1-245-604-11	METAL CHIP 10M 5% 1/16W	

Note: IC101, IC301, IC302, IC501 and IC602 on the MAIN board cannot exchange with single. When these parts are damaged, exchange the complete mounted board.

MAIN

	Ref. No.	Part No.	Description			Remark
	R426	1-218-961-11	METAL CHIP	4.7K	5%	1/16W
	R429	1-250-471-11	METAL CHIP	100	1%	1/16W
	R430	1-250-471-11	METAL CHIP	100	1%	1/16W
	R431	1-218-990-81	SHORT CHIP	0		
	R432	1-218-990-81	SHORT CHIP	0		
	R454	1-218-990-81	SHORT CHIP	0		
	R456	1-218-990-81	SHORT CHIP	0		
	R457	1-218-990-81	SHORT CHIP	0		
	R458	1-218-990-81	SHORT CHIP	0		
	R463	1-250-471-11	METAL CHIP	100	1%	1/16W
	R466	1-218-990-81	SHORT CHIP	0		
	R468	1-218-990-81	SHORT CHIP	0		
	R469	1-218-990-81	SHORT CHIP	0		
	R470	1-250-471-11	METAL CHIP	100	1%	1/16W
	R471	1-250-471-11	METAL CHIP	100	1%	1/16W
	R472	1-250-471-11	METAL CHIP	100	1%	1/16W
	R473	1-250-471-11	METAL CHIP	100	1%	1/16W
	R501	1-218-965-11	METAL CHIP	10K	5%	1/16W
	R502	1-218-934-11	METAL CHIP	27	5%	1/16W
	R503	1-250-455-11	METAL CHIP	22	1%	1/16W
	R504	1-250-455-11	METAL CHIP	22	1%	1/16W
	R505	1-250-455-11	METAL CHIP	22	1%	1/16W
	R506	1-250-455-11	METAL CHIP	22	1%	1/16W
	R507	1-250-455-11	METAL CHIP	22	1%	1/16W
	R508	1-250-455-11	METAL CHIP	22	1%	1/16W
	R509	1-218-990-81	SHORT CHIP	0		
	R536	1-218-990-81	SHORT CHIP	0		
	R537	1-218-990-81	SHORT CHIP	0		
	R538	1-218-990-81	SHORT CHIP	0		
	R539	1-218-990-81	SHORT CHIP	0		
	R541	1-218-990-81	SHORT CHIP	0		
	R601	1-218-990-81	SHORT CHIP	0		
	R602	1-218-990-81	SHORT CHIP	0		
	R605	1-218-933-11	METAL CHIP	22	5%	1/16W
	R606	1-218-990-81	SHORT CHIP	0		
	R607	1-218-990-81	SHORT CHIP	0		
	R610	1-218-955-11	METAL CHIP	1.5K	5%	1/16W
	R611	1-218-990-81	SHORT CHIP	0		
	R612	1-218-933-11	METAL CHIP	22	5%	1/16W
	R614	1-218-990-81	SHORT CHIP	0		
	R615	1-218-990-81	SHORT CHIP	0		
	R616	1-218-990-81	SHORT CHIP	0		
	R618	1-218-965-11	METAL CHIP	10K	5%	1/16W
	R621	1-218-990-81	SHORT CHIP	0		
	R623	1-218-965-11	METAL CHIP	10K	5%	1/16W
	R624	1-218-965-11	METAL CHIP	10K	5%	1/16W
	R626	1-218-990-81	SHORT CHIP	0		
	R627	1-218-965-11	METAL CHIP	10K	5%	1/16W
*	R630	1-250-501-11	METAL CHIP	1.8K	1%	1/16W
	R632	1-250-487-11	METAL CHIP	470	1%	1/16W
	R633	1-250-471-11	METAL CHIP	100	1%	1/16W
	R637	1-218-990-81	SHORT CHIP	0		
	R638	1-218-965-11	METAL CHIP	10K	5%	1/16W
*	R639	1-250-589-11	METAL CHIP	75	1%	1/10W
*	R640	1-250-589-11	METAL CHIP	75	1%	1/10W
*	R641	1-250-589-11	METAL CHIP	75	1%	1/10W
*	R642	1-250-589-11	METAL CHIP	75	1%	1/10W
	R648	1-250-535-11	METAL CHIP	47K	1%	1/16W
	R649	1-250-535-11	METAL CHIP	47K	1%	1/16W
	R651	1-218-990-81	SHORT CHIP	0		

	Ref. No.	Part No.	Description			Remark
	R654	1-218-990-81	SHORT CHIP	0		
	R655	1-218-965-11	METAL CHIP	10K	5%	1/16W
	R657	1-218-990-81	SHORT CHIP	0		
	R701	1-218-990-81	SHORT CHIP	0		
	R702	1-218-990-81	SHORT CHIP	0		
	R705	1-218-990-81	SHORT CHIP	0		
	R726	1-250-455-11	METAL CHIP	22	1%	1/16W
*	R727	1-250-479-11	METAL CHIP	220	1%	1/16W
	R728	1-250-467-11	METAL CHIP	68	1%	1/16W
	R729	1-218-990-81	SHORT CHIP	0		
	R730	1-218-990-81	SHORT CHIP	0		
	R733	1-218-990-81	SHORT CHIP	0		
	R734	1-218-985-11	METAL CHIP	470K	5%	1/16W
	R739	1-218-990-81	SHORT CHIP	0		
	R741	1-218-990-81	SHORT CHIP	0		
	R742	1-218-990-81	SHORT CHIP	0		
	R743	1-218-990-81	SHORT CHIP	0		
	R749	1-218-990-81	SHORT CHIP	0		
	R750	1-218-990-81	SHORT CHIP	0		
	R751	1-218-990-81	SHORT CHIP	0		
	R752	1-400-462-21	FERRITE, EMI (SMD) (1005)			
	R755	1-218-990-81	SHORT CHIP	0		
	R756	1-218-990-81	SHORT CHIP	0		
	R757	1-218-990-81	SHORT CHIP	0		
	R758	1-218-990-81	SHORT CHIP	0		
	R761	1-218-965-11	METAL CHIP	10K	5%	1/16W
	R762	1-218-990-81	SHORT CHIP	0		
	R764	1-218-965-11	METAL CHIP	10K	5%	1/16W
	R801	1-250-495-11	METAL CHIP	1K	1%	1/16W
	R802	1-250-495-11	METAL CHIP	1K	1%	1/16W
	R804	1-250-495-11	METAL CHIP	1K	1%	1/16W
	R806	1-250-495-11	METAL CHIP	1K	1%	1/16W
	R807	1-250-495-11	METAL CHIP	1K	1%	1/16W
	R813	1-218-961-11	METAL CHIP	4.7K	5%	1/16W
	R814	1-218-961-11	METAL CHIP	4.7K	5%	1/16W
	R815	1-218-961-11	METAL CHIP	4.7K	5%	1/16W
	R820	1-218-961-11	METAL CHIP	4.7K	5%	1/16W
	R821	1-218-961-11	METAL CHIP	4.7K	5%	1/16W
	R822	1-218-961-11	METAL CHIP	4.7K	5%	1/16W
	R823	1-218-961-11	METAL CHIP	4.7K	5%	1/16W
	R836	1-218-961-11	METAL CHIP	4.7K	5%	1/16W
	R839	1-218-961-11	METAL CHIP	4.7K	5%	1/16W
	R845	1-218-990-81	SHORT CHIP	0		
	R846	1-218-990-81	SHORT CHIP	0		
	R853	1-218-990-81	SHORT CHIP	0		
	R854	1-218-990-81	SHORT CHIP	0		
	R901	1-218-990-81	SHORT CHIP	0		
	R1001	1-218-990-81	SHORT CHIP	0		
	R1002	1-218-965-11	METAL CHIP	10K	5%	1/16W
	R1008	1-218-990-81	SHORT CHIP	0		
	R1010	1-218-965-11	METAL CHIP	10K	5%	1/16W
	R1011	1-218-985-11	METAL CHIP	470K	5%	1/16W
	R1013	1-250-519-11	METAL CHIP	10K	1%	1/16W
	R1014	1-250-519-11	METAL CHIP	10K	1%	1/16W
	R1015	1-218-965-11	METAL CHIP	10K	5%	1/16W
*	R1017	1-250-477-11	METAL CHIP	180	1%	1/16W
	R1018	1-250-455-11	METAL CHIP	22	1%	1/16W
	R1019	1-250-455-11	METAL CHIP	22	1%	1/16W
	R1020	1-218-990-81	SHORT CHIP	0		
	R1021	1-218-990-81	SHORT CHIP	0		

	Ref. No.	Part No.	Description			Remark
	R1023	1-218-990-81	SHORT CHIP	0		
	R1024	1-218-990-81	SHORT CHIP	0		
	R1026	1-218-965-11	METAL CHIP	10K	5%	1/16W
	R1027	1-218-965-11	METAL CHIP	10K	5%	1/16W
			< COMPOSITION CIRCUIT BLOCK >			
	RB601	1-234-400-21	CONDUCTOR, NETWORK (1005X4)			
	RB602	1-234-400-21	CONDUCTOR, NETWORK (1005X4)			
	RB701	1-234-400-21	CONDUCTOR, NETWORK (1005X4)			
	RB702	1-234-400-21	CONDUCTOR, NETWORK (1005X4)			
	RB703	1-234-400-21	CONDUCTOR, NETWORK (1005X4)			
	RB704	1-234-400-21	CONDUCTOR, NETWORK (1005X4)			
	RB705	1-234-400-21	CONDUCTOR, NETWORK (1005X4)			
	RB706	1-234-400-21	CONDUCTOR, NETWORK (1005X4)			
	RB801	1-234-378-21	RES, NETWORK 10K (1005X4)			
	RB802	1-234-378-21	RES, NETWORK 10K (1005X4)			
	RB810	1-234-378-21	RES, NETWORK 10K (1005X4)			
	RB811	1-234-378-21	RES, NETWORK 10K (1005X4)			
	RB812	1-234-378-21	RES, NETWORK 10K (1005X4)			
	RB813	1-234-378-21	RES, NETWORK 10K (1005X4)			
	RB814	1-234-378-21	RES, NETWORK 10K (1005X4)			
	RB815	1-234-378-21	RES, NETWORK 10K (1005X4)			
			< TRANSFORMER >			
	T601	1-697-310-11	PULSE TRANSFORMER			
	T602	1-697-310-11	PULSE TRANSFORMER			
	T603	1-697-310-11	PULSE TRANSFORMER			
	T604	1-697-310-11	PULSE TRANSFORMER			
			< VIBRATOR >			
	X401	1-814-404-11	QUARTZ CRYSTAL UNITS (24 MHz)			
	X403	1-814-273-11	QUARTZ CRYSTAL UNIT (32.768 kHz)			
	X601	1-814-242-11	VIBRATOR, CRYSTAL (25 MHz)			

****

POWER KEY BOARD
****

	Ref. No.	Part No.	Description			Remark
			< CAPACITOR >			
	C1002	1-118-395-11	CERAMIC CHIP	0.0047uF	10%	50V
	C1003	1-118-412-11	CERAMIC CHIP	220PF	10%	50V
	C1004	1-118-388-11	CERAMIC CHIP	0.047uF	10%	25V
	C1005	1-118-388-11	CERAMIC CHIP	0.047uF	10%	25V
	C1006	1-118-388-11	CERAMIC CHIP	0.047uF	10%	25V
*	C1007	1-116-720-11	CERAMIC CHIP	10uF	20%	6.3V
			< CONNECTOR >			
*	CN1002	1-760-388-11	CONNECTOR PIN (SMD) 9P			
			< LED >			
	D1001	6-502-868-01	LED SID1050CM TP5 (LEARNING CODE TRANSMITTER)			
	D1002	6-503-970-01	LED SELU5423E-STP15 (I/⏻)			
			< IC >			
	IC1001	8-759-530-30	IC TC74VHCU04FT (EL)			
	IC1002	6-600-767-01	IC PNA4823M02S0 (R)			
			< TRANSISTOR >			
	Q1002	6-553-266-01	TRANSISTOR	2PD601ART		
			< RESISTOR >			
	R1001	1-216-837-11	METAL CHIP	22K	5%	1/10W
	R1002	1-216-829-11	METAL CHIP	4.7K	5%	1/10W
	R1004	1-216-861-11	METAL CHIP	2.2M	5%	1/10W
	R1006	1-216-833-11	METAL CHIP	10K	5%	1/10W
	R1008	1-216-841-11	METAL CHIP	47K	5%	1/10W
	R1010	1-216-821-11	METAL CHIP	1K	5%	1/10W
	R1011	1-216-817-11	METAL CHIP	470	5%	1/10W
	R1012	1-216-837-11	METAL CHIP	22K	5%	1/10W
	R1014	1-216-845-11	METAL CHIP	100K	5%	1/10W
	R1018	1-216-809-11	METAL CHIP	100	5%	1/10W
	R1019	1-216-797-11	METAL CHIP	10	5%	1/10W
	R1020	1-216-839-11	METAL CHIP	33K	5%	1/10W
			< SWITCH >			
	S1029	1-762-875-21	SWITCH, KEYBOARD (I/⏻)			

****

SPIRITOSO-ANT-L BOARD
****

The SPIRITOSO-ANT-L board is not covered in the servicing.

****

SPIRITOSO-ANT-R BOARD
****

The SPIRITOSO-ANT-R board is not covered in the servicing.

****

SPIRITOSO-LCD BOARD
****

	Ref. No.	Part No.	Description			Remark
			< CAPACITOR >			
	C001	1-118-386-11	CERAMIC CHIP	0.1uF	10%	16V
	C006	1-118-386-11	CERAMIC CHIP	0.1uF	10%	16V
			< CONNECTOR >			
	CN001	1-818-857-51	CONNECTOR, FFC/FPC 40P			
	CN002	1-843-774-81	CONNECTOR, FFC/FPC (ZIF) 40P			
			< RESISTOR >			
	R008	1-218-990-81	SHORT CHIP	0		
	R009	1-218-929-11	METAL CHIP	10	5%	1/16W
	R010	1-218-990-81	SHORT CHIP	0		
	R011	1-218-990-81	SHORT CHIP	0		
	R012	1-218-990-81	SHORT CHIP	0		
	R013	1-218-990-81	SHORT CHIP	0		
	R014	1-218-990-81	SHORT CHIP	0		
			< COMPOSITION CIRCUIT BLOCK >			
	RB007	1-234-400-21	CONDUCTOR, NETWORK (1005X4)			
	RB008	1-234-400-21	CONDUCTOR, NETWORK (1005X4)			
	RB009	1-234-400-21	CONDUCTOR, NETWORK (1005X4)			
	RB010	1-234-400-21	CONDUCTOR, NETWORK (1005X4)			
	RB011	1-234-400-21	CONDUCTOR, NETWORK (1005X4)			
	RB012	1-234-400-21	CONDUCTOR, NETWORK (1005X4)			

****

STANBY **U-COM**

Ref. No.	Part No.	Description				Remark
	A-1945-108-A	STANBY BOARD, COMPLETE (AEP, UK)				
	A-1945-109-A	STANBY BOARD, COMPLETE (US, CND)				
		************************				
		< CAPACITOR >				
C901	1-126-964-11	ELECT	10uF	20%	50V	
C903	1-126-948-11	ELECT	100uF	20%	35V	
C906	1-118-361-11	CERAMIC CHIP	0.1uF	10%	50V	
C907	1-114-329-11	CERAMIC CHIP	0.47uF	10%	50V	
⚠ C909	1-118-688-11	CERAMIC	0.0047uF	20%	300V	
		< CONNECTOR >				
⚠ * CN902	1-564-687-11	PIN, CONNECTOR (3.96 mm PITCH) 3P				
⚠ * CN903	1-565-792-11	PIN, CONNECTOR (3.96 mm PITCH) 2P				
⚠ * CN904	1-793-660-11	PIN, CONNECTOR (PC BOARD) 3P				
		< DIODE >				
D901	8-719-053-18	DIODE	1SR154-400TE-25			
D902	8-719-053-18	DIODE	1SR154-400TE-25			
D903	8-719-053-18	DIODE	1SR154-400TE-25			
D904	8-719-053-18	DIODE	1SR154-400TE-25			
D905	6-502-961-01	DIODE	DA2J10100L			
D906	6-502-961-01	DIODE	DA2J10100L			
D907	6-502-961-01	DIODE	DA2J10100L			
		< TRANSISTOR >				
Q901	6-553-266-01	TRANSISTOR	2PD601ART			
Q902	6-553-266-01	TRANSISTOR	2PD601ART			
Q903	6-553-268-01	TRANSISTOR	PDTC114ET			
Q904	6-553-266-01	TRANSISTOR	2PD601ART			
* Q905	6-551-721-01	FET	SSM6J50TU (T5RSONYF			
Q906	6-553-266-01	TRANSISTOR	2PD601ART			
		< RESISTOR >				
R901	1-218-879-11	METAL CHIP	22K	0.5%	1/10W	
R902	1-218-863-11	METAL CHIP	4.7K	0.5%	1/10W	
R903	1-218-879-11	METAL CHIP	22K	0.5%	1/10W	
R904	1-216-821-11	METAL CHIP	1K	5%	1/10W	
R905	1-250-664-11	METAL CHIP	100K	1%	1/10W	
⚠ R907	1-249-389-91	CARBON	4.7	5%	1/4W	F
R908	1-216-853-11	METAL CHIP	470K	5%	1/10W	
R909	1-216-849-11	METAL CHIP	220K	5%	1/10W	
R910	1-216-841-11	METAL CHIP	47K	5%	1/10W	
R911	1-216-841-11	METAL CHIP	47K	5%	1/10W	
R913	1-216-853-11	METAL CHIP	470K	5%	1/10W	
R914	1-216-864-11	SHORT CHIP	0			
		< RELAY >				
⚠ RY901	1-755-266-11	RELAY, AC POWER				
		< TRANSFORMER >				
⚠ T901	1-492-498-11	STANDBY TRANSFORMER (AEP, UK)				
⚠ T901	1-492-499-11	STANDBY TRANSFORMER (US, CND)				

****

Ref. No.	Part No.	Description				Remark
	A-1945-111-A	U-COM BOARD, COMPLETE				
		**********************				
		< CAPACITOR >				
C7001	1-118-418-11	CERAMIC CHIP	22uF	20%	6.3V	
C7002	1-116-734-11	CERAMIC CHIP	1uF	20%	16V	
C7003	1-118-417-11	CERAMIC CHIP	0.1uF	10%	16V	
C7004	1-118-417-11	CERAMIC CHIP	0.1uF	10%	16V	
C7005	1-116-734-11	CERAMIC CHIP	1uF	20%	16V	
C7006	1-118-418-11	CERAMIC CHIP	22uF	20%	6.3V	
* C7007	1-116-720-11	CERAMIC CHIP	10uF	20%	6.3V	
C7008	1-116-734-11	CERAMIC CHIP	1uF	20%	16V	
C7009	1-118-361-11	CERAMIC CHIP	0.1uF	10%	50V	
C7010	1-118-361-11	CERAMIC CHIP	0.1uF	10%	50V	
C7011	1-118-361-11	CERAMIC CHIP	0.1uF	10%	50V	
C7012	1-118-041-11	CERAMIC CHIP	4.7uF	10%	10V	
C7013	1-118-417-11	CERAMIC CHIP	0.1uF	10%	16V	
C7014	1-118-417-11	CERAMIC CHIP	0.1uF	10%	16V	
C7015	1-118-417-11	CERAMIC CHIP	0.1uF	10%	16V	
C7016	1-118-417-11	CERAMIC CHIP	0.1uF	10%	16V	
C7017	1-118-361-11	CERAMIC CHIP	0.1uF	10%	50V	
C7018	1-116-151-11	CERAMIC CHIP	0.001uF	2%	50V	
C7019	1-118-361-11	CERAMIC CHIP	0.1uF	10%	50V	
C7020	1-118-361-11	CERAMIC CHIP	0.1uF	10%	50V	
C7021	1-118-361-11	CERAMIC CHIP	0.1uF	10%	50V	
C7022	1-118-361-11	CERAMIC CHIP	0.1uF	10%	50V	
C7023	1-116-151-11	CERAMIC CHIP	0.001uF	2%	50V	
C7024	1-118-417-11	CERAMIC CHIP	0.1uF	10%	16V	
C7025	1-118-417-11	CERAMIC CHIP	0.1uF	10%	16V	
C7026	1-118-417-11	CERAMIC CHIP	0.1uF	10%	16V	
C7027	1-118-417-11	CERAMIC CHIP	0.1uF	10%	16V	
C7032	1-118-418-11	CERAMIC CHIP	22uF	20%	6.3V	
C7033	1-116-734-11	CERAMIC CHIP	1uF	20%	16V	
C7034	1-118-417-11	CERAMIC CHIP	0.1uF	10%	16V	
C7035	1-118-417-11	CERAMIC CHIP	0.1uF	10%	16V	
C7036	1-116-734-11	CERAMIC CHIP	1uF	20%	16V	
C7037	1-118-418-11	CERAMIC CHIP	22uF	20%	6.3V	
* C7038	1-116-720-11	CERAMIC CHIP	10uF	20%	6.3V	
C7039	1-116-734-11	CERAMIC CHIP	1uF	20%	16V	
C7040	1-118-417-11	CERAMIC CHIP	0.1uF	10%	16V	
C7041	1-118-417-11	CERAMIC CHIP	0.1uF	10%	16V	
C7042	1-116-734-11	CERAMIC CHIP	1uF	20%	16V	
C7043	1-118-417-11	CERAMIC CHIP	0.1uF	10%	16V	
C7044	1-116-151-11	CERAMIC CHIP	0.001uF	2%	50V	
C7046	1-116-151-11	CERAMIC CHIP	0.001uF	2%	50V	
C7047	1-116-151-11	CERAMIC CHIP	0.001uF	2%	50V	
C7048	1-118-361-11	CERAMIC CHIP	0.1uF	10%	50V	
C7049	1-118-361-11	CERAMIC CHIP	0.1uF	10%	50V	
C7050	1-118-391-11	CERAMIC CHIP	0.01uF	10%	50V	
C7051	1-118-391-11	CERAMIC CHIP	0.01uF	10%	50V	
		< CONNECTOR >				
CN7002	1-766-382-21	PIN, CONNECTOR (1.5 mm) (SMD) 10P				
CN7003	1-817-097-21	PIN, CONNECTOR (1.5 mm) (SMD) 13P				
CN7004	1-785-466-51	CONNECTOR, FFC/FPC 7P				
CN7005	1-785-466-51	CONNECTOR, FFC/FPC 7P				
CN7006	1-779-275-11	CONNECTOR, FFC (LIF (NON-ZIF)) 7P				
CN7007	1-691-550-11	PIN, CONNECTOR (1.5 mm) (SMD) 3P				
CN7008	1-766-376-21	PIN, CONNECTOR (1.5 mm) (SMD) 9P				
* CN7009	1-779-268-21	PIN, CONNECTOR (1.5 mm) (SMD) 11P				
* CN7013	1-695-320-21	PIN, CONNECTOR (1.5 mm) (SMD) 2P				
		< DIODE >				
D7003	6-503-704-01	DIODE	CUS521, H3SONYF			
D7004	6-503-704-01	DIODE	CUS521, H3SONYF			

Ref. No.	Part No.	Description	Remark
D7005	6-503-704-01	DIODE CUS521, H3SONYF	
D7006	6-503-704-01	DIODE CUS521, H3SONYF	
D7008	6-503-704-01	DIODE CUS521, H3SONYF	
		< FERRITE BEAD >	
FB7001	1-400-862-11	BEAD, FERRITE	
FB7002	1-400-862-11	BEAD, FERRITE	
FB7003	1-400-862-11	BEAD, FERRITE	
FB7004	1-400-862-11	BEAD, FERRITE	
		< IC >	
IC7001	6-711-653-01	IC S-1206B33-U3T1G	
IC7002	A-1945-131-A	IC MB9AF156NPMC-G-JNE2 (for SERVICE)	
IC7003	6-714-351-01	IC R1EX24016ASAS0A	
IC7004	6-719-856-01	IC BU4229F-TR	
IC7005	6-712-384-01	IC TC74VHC4052AFT (EK)	
IC7006	6-719-061-01	IC MM1839A33NRE	
IC7007	6-720-743-01	IC uPD78F0527GB (S)-410-UET-A	
		< TRANSISTOR >	
Q7001	6-553-268-01	TRANSISTOR PDTC114ET	
Q7002	6-553-267-01	TRANSISTOR 2PB709ART	
		< RESISTOR >	
R7010	1-216-295-91	SHORT CHIP 0	
R7013	1-218-965-11	METAL CHIP 10K 5% 1/16W	
R7014	1-218-965-11	METAL CHIP 10K 5% 1/16W	
R7015	1-218-941-81	METAL CHIP 100 5% 1/16W	
R7016	1-218-977-11	METAL CHIP 100K 5% 1/16W	
R7017	1-218-941-81	METAL CHIP 100 5% 1/16W	
R7018	1-218-941-81	METAL CHIP 100 5% 1/16W	
R7019	1-218-965-11	METAL CHIP 10K 5% 1/16W	
R7020	1-218-941-81	METAL CHIP 100 5% 1/16W	
R7022	1-218-957-11	METAL CHIP 2.2K 5% 1/16W	
R7023	1-218-957-11	METAL CHIP 2.2K 5% 1/16W	
R7024	1-218-941-81	METAL CHIP 100 5% 1/16W	
R7025	1-218-941-81	METAL CHIP 100 5% 1/16W	
R7026	1-218-941-81	METAL CHIP 100 5% 1/16W	
R7027	1-218-941-81	METAL CHIP 100 5% 1/16W	
R7028	1-218-941-81	METAL CHIP 100 5% 1/16W	
R7029	1-218-941-81	METAL CHIP 100 5% 1/16W	
R7030	1-218-941-81	METAL CHIP 100 5% 1/16W	
R7031	1-218-941-81	METAL CHIP 100 5% 1/16W	
R7033	1-216-809-11	METAL CHIP 100 5% 1/10W	
R7035	1-216-809-11	METAL CHIP 100 5% 1/10W	
R7036	1-216-833-11	METAL CHIP 10K 5% 1/10W	
R7037	1-216-809-11	METAL CHIP 100 5% 1/10W	
R7038	1-218-965-11	METAL CHIP 10K 5% 1/16W	
R7039	1-218-941-81	METAL CHIP 100 5% 1/16W	
R7040	1-218-957-11	METAL CHIP 2.2K 5% 1/16W	
R7041	1-218-941-81	METAL CHIP 100 5% 1/16W	
R7042	1-218-941-81	METAL CHIP 100 5% 1/16W	
R7043	1-218-941-81	METAL CHIP 100 5% 1/16W	
R7044	1-218-965-11	METAL CHIP 10K 5% 1/16W	
R7045	1-218-941-81	METAL CHIP 100 5% 1/16W	
R7049	1-218-941-81	METAL CHIP 100 5% 1/16W	
R7050	1-218-965-11	METAL CHIP 10K 5% 1/16W	
R7051	1-218-941-81	METAL CHIP 100 5% 1/16W	
R7052	1-218-941-81	METAL CHIP 100 5% 1/16W	
R7053	1-218-941-81	METAL CHIP 100 5% 1/16W	

Ref. No.	Part No.	Description	Remark
R7054	1-218-941-81	METAL CHIP 100 5% 1/16W	
R7055	1-218-941-81	METAL CHIP 100 5% 1/16W	
R7056	1-218-941-81	METAL CHIP 100 5% 1/16W	
R7058	1-218-941-81	METAL CHIP 100 5% 1/16W	
R7059	1-218-941-81	METAL CHIP 100 5% 1/16W	
R7060	1-218-941-81	METAL CHIP 100 5% 1/16W	
R7062	1-218-941-81	METAL CHIP 100 5% 1/16W	
R7063	1-218-941-81	METAL CHIP 100 5% 1/16W	
R7064	1-218-941-81	METAL CHIP 100 5% 1/16W	
R7066	1-218-990-81	SHORT CHIP 0	
R7067	1-218-941-81	METAL CHIP 100 5% 1/16W	
R7068	1-218-965-11	METAL CHIP 10K 5% 1/16W	
R7070	1-218-955-11	METAL CHIP 1.5K 5% 1/16W	
R7071	1-218-965-11	METAL CHIP 10K 5% 1/16W	
R7074	1-218-965-11	METAL CHIP 10K 5% 1/16W	
R7075	1-218-965-11	METAL CHIP 10K 5% 1/16W	
R7076	1-218-965-11	METAL CHIP 10K 5% 1/16W	
R7077	1-218-965-11	METAL CHIP 10K 5% 1/16W	
R7078	1-218-965-11	METAL CHIP 10K 5% 1/16W	
R7084	1-218-965-11	METAL CHIP 10K 5% 1/16W	
R7088	1-218-961-11	METAL CHIP 4.7K 5% 1/16W	
R7089	1-218-961-11	METAL CHIP 4.7K 5% 1/16W	
R7090	1-218-961-11	METAL CHIP 4.7K 5% 1/16W	
R7091	1-218-941-81	METAL CHIP 100 5% 1/16W	
R7092	1-216-295-91	SHORT CHIP 0	
R7093	1-218-990-81	SHORT CHIP 0	
R7095	1-218-941-81	METAL CHIP 100 5% 1/16W	
R7096	1-218-965-11	METAL CHIP 10K 5% 1/16W	
R7097	1-218-941-81	METAL CHIP 100 5% 1/16W	
R7098	1-218-941-81	METAL CHIP 100 5% 1/16W	
R7099	1-218-941-81	METAL CHIP 100 5% 1/16W	
R7101	1-218-941-81	METAL CHIP 100 5% 1/16W	
R7102	1-218-965-11	METAL CHIP 10K 5% 1/16W	
R7104	1-218-965-11	METAL CHIP 10K 5% 1/16W	
R7105	1-218-965-11	METAL CHIP 10K 5% 1/16W	
R7106	1-218-941-81	METAL CHIP 100 5% 1/16W	
R7107	1-218-941-81	METAL CHIP 100 5% 1/16W	
R7108	1-218-941-81	METAL CHIP 100 5% 1/16W	
R7109	1-218-941-81	METAL CHIP 100 5% 1/16W	
R7110	1-216-809-11	METAL CHIP 100 5% 1/10W	
R7111	1-216-821-11	METAL CHIP 1K 5% 1/10W	
R7112	1-216-825-11	METAL CHIP 2.2K 5% 1/10W	
R7113	1-216-176-11	METAL CHIP 120 5% 1/8W	
R7118	1-218-965-11	METAL CHIP 10K 5% 1/16W	
R7119	1-218-965-11	METAL CHIP 10K 5% 1/16W	
R7120	1-218-965-11	METAL CHIP 10K 5% 1/16W	
R7122	1-218-941-81	METAL CHIP 100 5% 1/16W	
R7124	1-216-833-11	METAL CHIP 10K 5% 1/10W	
R7127	1-216-825-11	METAL CHIP 2.2K 5% 1/10W	
R7128	1-216-825-11	METAL CHIP 2.2K 5% 1/10W	
R7129	1-216-809-11	METAL CHIP 100 5% 1/10W	
R7130	1-216-809-11	METAL CHIP 100 5% 1/10W	
R7131	1-216-857-11	METAL CHIP 1M 5% 1/10W	
R7132	1-216-857-11	METAL CHIP 1M 5% 1/10W	
R7133	1-218-965-11	METAL CHIP 10K 5% 1/16W	
R7134	1-218-965-11	METAL CHIP 10K 5% 1/16W	
R7135	1-218-965-11	METAL CHIP 10K 5% 1/16W	
R7136	1-216-295-91	SHORT CHIP 0	
R7137	1-216-833-11	METAL CHIP 10K 5% 1/10W	
R7138	1-216-837-11	METAL CHIP 22K 5% 1/10W	

U-COM

Ref. No.	Part No.	Description	Remark
		< COMPOSITION CIRCUIT BLOCK >	
RB7001	1-234-373-21	RES, NETWORK 220 (1005X4)	
		< VIBRATOR >	
X7001	1-781-646-21	VIBRATOR, CERAMIC (4 MHz)	
X7002	1-813-005-21	VIBRATOR, CERAMIC (5 MHz)	

****

MISCELLANEOUS
****

Ref. No.	Part No.	Description	Remark
⚠ AC1	1-821-082-41	AC INLET (2P) (AC IN)	
⚠ F1401	1-532-465-33	FUSE (T 3.15 AL/250 V)	
⚠ F1402	1-532-465-33	FUSE (T 3.15 AL/250 V)	
FC1	1-500-386-11	FILTER, CLAMP (FERRITE CORE)	
FC2	1-400-640-11	CORE, FERRITE	
FC3	1-500-386-11	FILTER, CLAMP (FERRITE CORE)	
FC4	1-500-082-11	CLAMP, SLEEVE FERRITE	
FC5	1-500-082-11	CLAMP, SLEEVE FERRITE	
FFC2	1-828-771-51	WIRE (FLAT TYPE) (24 CORE) (See Note 3)	
⚠ HDD1	1-458-716-11	HDD/TSB-AQUARIUS (1TB) (Hard Diskl Drive) (See Note 2)	
LCD1	1-811-873-11	PANEL, LCD	
⚠ M1	1-787-931-11	DC FAN	
⚠ T1	1-697-298-11	TRANSFORMER, POWER (AEP, UK)	
⚠ T1	1-697-299-11	TRANSFORMER, POWER (US, CND)	
⚠ T2	1-697-301-11	TRANSFORMER, POWER (AEP, UK)	
⚠ T2	1-697-302-11	TRANSFORMER, POWER (US, CND)	
WBC1	1-458-601-31	CARD, WLAN/BT COMBO (See Note 1)	
WR1	1-846-779-11	CABLE, COAXIAL (U.FL CONNECTOR) (for L-ch) (See Note 1)	
WR2	1-846-780-11	CABLE, COAXIAL (U.FL CONNECTOR) (for R-ch) (See Note 1)	
WR3	1-969-718-11	HARNESS (S-004)	
WR4	1-969-720-11	HARNESS (E-002)	
WR5	1-969-716-11	HARNESS (S-002)	

****

ACCESSORIES
****

Ref. No.	Part No.	Description	Remark
	1-492-558-11	REMOTE COMMANDER RM-ANU183 (S) (Remote control)	
	1-759-586-61	CONTROLLER, VIDEO (AV MOUSE) (IR blaster)	
	1-791-732-12	CORD, CONNECTION (Audio cord)	
⚠	1-837-633-12	POWER-SUPPLY CORD (AC power cord) (US, CND)	
⚠	1-839-699-12	CORD, POWER (AC power cord) (AEP, UK)	
	1-846-917-11	CABLE, PHI 3.5 mm MONAURAL 2 m (Monaural mini-plug cable)	
	1-846-946-11	CABLE, ETHERNET (LAN cable)	

Note 1: When the WLAN/BT COMBO card (Ref. No. WBC1) and coaxial (U.FL connector) cables (Ref. No. WR1, WR2) are replaced, refer to “CHECKING METHOD OF NETWORK CONNECTION” on page 5.

Note 2: When the hard disk drive (Ref. No. HDD1) is replaced, refer to “NOTE OF REPLACING THE HARD DISK DRIVE (Ref. No. HDD1)” on page 4.

Note 3: Wire (flat type) (24 core) (Ref. No. FFC2) for service is all straight.
When installing it, please bending it by referring to “6-5. POWER TRANSFORMER SECTION” on page 99.

MEMO

REVISION HISTORY

Ver.	Date	Description of Revision
1.0	2013.10	New

How to search for a contact point of signal lines or the like in DIAGRAMS SECTION
If a contact point of a BLOCK DIAGRAM, PRINTED WIRING BOARD or SCHEMATIC DIAGRAM is shown in a different page, use the PDF file search function to find one.

e.g.) If a contact point is shown as $\boxed{\text{>001Z}}$, follow the procedure below.
Procedure:
1. Press the [F] key while pressing the [Ctrl] key.
2. Input “>001Z” in the search box and press the [Enter] key.
3. The relevant part (page), where the contact point is shown, appears.

Note: If you still see the original page, press the [Enter] key again.